# Infinite dessin orders, projective resolutions and spectral determinants

Extended proofs, Schur functors and Auslander–Reiten calculations

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**Abstract**

We work throughout with infinite-dimensional locally gentle algebras and completed surface orders arising from explicit cellular clean dessins on compact Riemann surfaces. We construct their projective resolutions, Cartan determinants and dual dessins; develop determinant semi-invariants, Schur modules and cyclic-quiver character kernels; and calculate actual lattice almost-split sequences and an affine-type diamond quotient. A constrained-path Fock algebra retains the gentle relations and has a compact extension with incidence boundary map. Its degree spectral triple and an inserted arithmetic Fredholm family are explicit. Jacobi sums provide theta-dessin cellular Frobenius determinants with a positive local metric; inert primes obstruct the same real two-dimensional carrier. Recent spectral approaches are compared with their actual hypotheses. Selected power-series, module and matrix statements are verified in Lean4. A derived branch functor repairs projective determinant multiplicity; tame descent recovers a hereditary order but forces multiplicities for full-deck coefficient kernels. We also give a requested-error adaptive theta/readout construction for three fixed completed Dirichlet functions, with executed rational training and explicit sampling budgets, and derive a cutoff-uniform original-family type-I remainder envelope, retaining its unresolved exponent loss. The arithmetic positive determinant comparison remains open; no proof of GRH is claimed.

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# 1 Cellular dessins and their infinite orders

We start with the surface-order constructions of Schreiber [1, 2], retaining the original sources unchanged. The generic-resolution source [3] is background, not a substitute for exactness of a particular complex. All primary algebra examples below come from explicit clean Belyi maps. Finite-dimensional objects occur as modules or specified quotient strata of those infinite algebras, not as replacements for them. The initial arithmetic target is a primitive nonprincipal Dirichlet $L$-function. Finite-order Hecke characters, Dedekind zeta functions and general Artin representations require separate comparison theorems; Artin holomorphy is not assumed.

A clean dessin on a compact oriented surface is specified by a nonempty finite dart set $H$, a fixed-point-free involution $\alpha$, and a permutation $\sigma$ such that $\langle \alpha, \sigma \rangle$ is transitive. Contract the white vertices of degree two. The resulting cellular graph has vertices the $\sigma$-orbits, edges the $\alpha$-orbits, and faces the $\phi$-orbits, where

$$\tau = \sigma^{-1}, \qquad \phi = \alpha\sigma^{-1}, \qquad 2 - 2g = \#\operatorname{cyc}(\sigma) - |H|/2 + \#\operatorname{cyc}(\phi).$$

Write $[h] = \{h, \alpha h\}$. Our quiver has vertices these edges and arrows $a_h : [h] \to [\tau h]$. Paths are composed in travel order: $ab$ traverses $a$ first. Set

$$A = kQ/(a_h a_{\phi h} : h \in H), \quad J = A_{>0}, \quad \Lambda = \varprojlim A/J^m.$$

Each vertex has two incoming and two outgoing arrows. Each arrow has precisely one permitted successor $a_{\tau h}$ and one forbidden successor $a_{\phi h}$; these are distinct because $\alpha$ has no fixed points. The same holds for predecessors. This is the *locally gentle* condition. Every permitted cycle repeats indefinitely, so $A$ is infinite dimensional. We reserve "gentle" without qualification for the convention additionally imposing finite dimension.

**Proposition 1.1** (The actual split order)**.** *For a permitted cycle $c$ of length $\ell$, let $B_c$ be the hereditary cycle order in $M_\ell(k[[u_c]])$: diagonal and forward matrix entries are in $k[[u_c]]$, and entries crossing*

*the chosen cycle cut are in $u_c k[[u_c]]$. Then $\Lambda$ is the suborder of $B = \prod_c B_c$ whose two diagonal constant terms labelled $h, \alpha h$ agree. With $R = k[[u]]$ acting by $u \mapsto (u_c I)_c$, it is finite free of rank $\sum_c \ell(c)^2$; the conductor contains $uB$.*

*Proof.* In a cycle order a path of length less than $\ell$ is the corresponding matrix unit, multiplied by $u_c$ exactly when it crosses the cut; another full turn multiplies it by $u_c$. Thus completed paths in each permitted branch give all entries of $B_c$. An arrow belongs to only that branch, whereas $e_{[h]}$ has diagonal entries at both occurrences $h, \alpha h$. Hence the only condition on branchwise formal coefficients is the displayed equality of diagonal constants. This proves both injectivity and surjectivity of the path-normal-form identification. Multiplication respects it: switching branches across a paired edge is precisely a forbidden pair. The quotient $B/\Lambda$ is killed by $u$, so the submodule has the same generic rank as $B$. It is torsion free and finite over the DVR $R$, hence free of that rank. The path and $u$-adic topologies agree: $uA \subset J$, and if $L$ is the longest permitted cycle, any path of length at least $L+1$ is $u$ times a shorter nonempty path. Thus $J^{L+1} \subset uA$. This also justifies the completion used in the identification. □

The path-degree variable $q$ below is not $u$: one full turn has path degree $\ell(c)$ but $u$-degree one. Formal, Laurent and operator-norm completions will likewise remain distinct.

Here are five explicit global inputs. Permutations list darts starting at zero; $E = |H|/2$.

| Surface and Belyi map $\beta$ | $\sigma$ | $\alpha$ | $(g, \operatorname{rank}_R \Lambda)$ |
|---|---|---|---|
| $\mathbb{P}^1$, $4z(1-z)$ | (0)(1) | (01) | $(0, 2)$ |
| $\mathbb{P}^1$, $(z+1)^2/(4z)$ | (01) | (01) | $(0, 4)$ |
| $y^2 = x^3 - 1$, $x^3$ | (012)(345) | (03)(14)(25) | $(1, 18)$ |
| $\mathbb{P}^1$, $4z^3/(1+z^3)^2$ | (012)(345) | (03)(15)(24) | $(0, 18)$ |
| $y^2 = x^3 - x$, $x^2$ | (0123) | (02)(13) | $(1, 16)$ |

The first is the nodal interval: $A = k[x,y]/(xy)$ and $\Lambda = k[[x,y]]/(xy)$. The second has $A = k\langle x, y\rangle/(x^2, y^2)$, still infinite dimensional. The two theta dessins have the same two permitted three-cycles but different face cycles, respectively (051324) and (04)(13)(25). The elliptic square has one permitted and one forbidden four-cycle. For example $x$ has pole order two on the cubic $y^2 = x^3 - 1$; $x^3$ has ramification $(3,3)$ above zero, $(2,2,2)$ above one since $x^3 - 1 = y^2$, and (6) above infinity. The rational theta map has passport $(3,3), (2,2,2), (2,2,2)$. These ramification computations, and the analogous direct zeros/poles in the other rows, verify cellular Belyi inputs rather than merely abstract quivers.

## 2 Projective resolutions, dual dessins and determinants

Use right modules $P_i = e_i A$, $S_i = P_i/P_i J$. Internal shift $M\langle n\rangle$ places its old generator in degree $n$. Let $\mathcal{R}_n(i)$ be the *indexing words* of length $n$ from $i$ with every adjacent pair forbidden, including $e_i$ for $n = 0$. Such a word is usually zero in $A$; it still labels a summand.

**Theorem 2.1** (Exact resolutions)**.** *There is a minimal graded projective resolution, with $P_0(i) = e_i A$ and for $n \geq 1$,*

$$P_n(i) = \bigoplus_{p = a_1 \cdots a_n \in \mathcal{R}_n(i)} e_{t(a_n)} A\langle n\rangle, \qquad d_n(x)_p = a_n x \text{ in the summand } a_1 \cdots a_{n-1}.$$

*The augmentation $P_0(i) \to S_i$ is the quotient. For the completed order the identical finite-rank projective terms and formal-series maps resolve the residue simple. Ext and Tor against residue*

*simples count forbidden words with the prescribed endpoints. In the graded case Ext has matching internal shift $n$.*

*Proof.* The maps are right linear and degree zero. Adjacent differentials multiply by a forbidden pair. The augmentation kernel is the direct sum of $aA$ over initial arrows. For any arrow $a$, multiplication $e_{t(a)}A \to e_{s(a)}A$ kills exactly the permitted paths beginning with its forbidden successor $f(a)$. All other basis paths survive to distinct basis paths. Thus its kernel is $f(a)A$. For a fixed prefix, different last arrows have disjoint surviving supports; there is no cancellation between summands. This proves exactness inductively. For formal series the same assertion holds coefficient by coefficient and identifies the entire closed kernel, not just its dense polynomial subspace. Terms are idempotent projectives and all entries are arrows. Consequently applying Hom or tensor with a residue simple kills every differential, proving minimality and the Ext/Tor count. In $A$ minimality is graded; it does not require that $J$ be its ordinary Jacobson radical. □

The two arms never terminate in this closed class. Their labelled patterns follow $\phi$-cycles. An ungraded positive syzygy tail can have a shorter period by permutation of summands: the nodal interval has labelled period two but unordered tail period one; the elliptic theta has labelled period six and unordered tail period three. A simple need not be isomorphic to a positive syzygy; for the node it is $R$-torsion, whereas the positive syzygies are lattices.

Let $C_{ij}(q) = \sum_{d\geq 0} \dim_k(e_i A_d e_j) q^d$ and $E_{ij}(q) = \sum_{n\geq 0} (-1)^n \dim_k \operatorname{Ext}^n_A(S_i, S_j) q^n$. Degreewise Euler characteristic of the resolution gives $EC = I$: every coefficient is finite, because a summand in homological degree $n$ starts in internal degree $n$. Constant terms are $I$, so the inverse is two-sided. Already signed $E(q)$ must not be evaluated at $-q$ again.

**Proposition 2.2** (Dual dessin and cycle determinant)**.** *The complementary-relation algebra is another closed dessin algebra, with $\alpha^{\#} = \alpha$, $\sigma^{\#} = \phi^{-1}$, permitted successor $\phi$, and forbidden successor $\tau$. It exchanges graph vertices and faces and preserves genus. Its Cartan matrix satisfies $E(q) = C_{A^{\#}}(-q)$, with the opposite convention if required for Yoneda composition. For independent commuting arrow weights $w_a$,*

$$\det C_A(\mathbf{w}) = \frac{\prod_{f\in \mathrm{cyc}(\phi)} (1 - (-1)^{|f|} w(f))}{\prod_{c\in \mathrm{cyc}(\sigma)} (1 - w(c))}.$$

*Proof.* $\alpha(\sigma^{\#})^{-1} = \alpha\phi = \tau$, so the claimed relations and face data follow directly; the connectedness group is unchanged. To prove the determinant, put $S_{v,a} = w_a$ when $s(a) = v$, and $T_{a,v} = 1$ when $t(a) = v$, zero otherwise. On arrow states let $U_{a,b} = w_b$ for permitted adjacency and $V_{a,b} = w_b$ for forbidden adjacency. Then $TS = U + V$ and the path sum is $C = I + S(I - U)^{-1}T$. Sylvester's identity gives $\det C = \det(I + V)/\det(I - U)$. Each transition is a weighted permutation, whose cycle determinant is respectively $1 - (-1)^{\ell} w(c)$ and $1 - w(c)$. Everything holds in formal power series, hence as rational functions. This supplies a direct proof of the Bessenrodt–Holm formula in the closed dessin class [4]. □

The five univariate determinants are, in the order above,

$$\frac{1+q}{1-q}, \quad \frac{1+q}{1-q}, \quad \frac{1-q^6}{(1-q^3)^2}, \quad \frac{(1-q^2)^3}{(1-q^3)^2}, \quad 1.$$

The last cancellation does not imply finite global dimension: its forbidden resolution is infinite. These rational functions have cycle poles/zeros, not arithmetic prime Euler factors.

For clarity, a complex is a sequence with $d^2 = 0$, its homotopy category identifies maps differing by $dh + hd$, and $D(A)$ inverts maps inducing cohomology isomorphisms. A perfect complex is

represented by a bounded complex of finite projectives. Our infinite simple resolutions compute derived Hom and tensor but are generally not perfect. The two-term presentation $[P_1 \to P_0]$ is perfect even if its kernel is nonzero. Quadratic linearity is a Koszul statement; it is not by itself an equivalence between all completed derived categories. Likewise a finite-dimensional gentle graded-curve theorem [6] cannot simply be transferred to these closed infinite orders. Here the surface walks explicitly compute the resolution endpoints: $\tau$ turns round a graph vertex and $\phi$ follows a face. For longer string objects all maps can be computed from the occurrence matrices; a universal geodesic-intersection classification for this completed category remains a separate theorem.

### 2.1 Auxiliary staircase resolutions over the order

The two filtration parameters below are not dessin arrows. Let $R = k[x, y]$ for a field $k$, and let $I = (g_0, \dots, g_{m-1})$, $m \geq 2$, where $g_i = x^{a_i} y^{b_i}$,

$$a_0 > \cdots > a_{m-1} = 0, \qquad 0 = b_0 < \cdots < b_{m-1}.$$

The positive outer exponents make $R/I$ finite length. Write $R(-a, -b)$ for a free generator born in bidegree $(a, b)$.

**Proposition 2.3** (Adjacent syzygies)**.** *A minimal bigraded free resolution of $M = R/I$ is*

$$0 \longrightarrow \bigoplus_{i=0}^{m-2} R(-a_i, -b_{i+1}) \xrightarrow{d_2} \bigoplus_{i=0}^{m-1} R(-a_i, -b_i) \xrightarrow{d_1} R \longrightarrow M \longrightarrow 0,$$

*where $d_1(e_i) = g_i$ and $d_2(f_i) = -y^{b_{i+1}-b_i} e_i + x^{a_i - a_{i+1}} e_{i+1}$. Deleting row $i$ from $d_2$ gives determinant $(-1)^i g_i$. The Hilbert series numerator is $1 - \sum_i s^{a_i} t^{b_i} + \sum_{i=0}^{m-2} s^{a_i} t^{b_{i+1}}$.*

*Proof.* Each column maps to zero. In a relation $\sum h_i g_i = 0$, all terms except the last contain $x^{a_{m-2}}$. Coprimality with $y^{b_{m-1}}$ implies $h_{m-1} = x^{a_{m-2}} q$. Subtract $q d_2(f_{m-2})$ from the coefficient vector; its last component vanishes. Factor $x^{a_{m-2}}$ from the remaining generators. They form a smaller staircase with the same strict inequalities, ending in a pure $y$-power. Induction expresses the remaining relation in the preceding adjacent columns. The base of one generator has no relations, since $R$ is a domain. Hence $\ker d_1 = \operatorname{im} d_2$. If $d_2(q) = 0$, its first row gives $-y^{b_1} q_0 = 0$; successive rows then force all $q_i = 0$. This proves injectivity. Every differential entry has positive degree, so reduction modulo $(x, y)$ kills the differentials and the resolution is minimal. In the row-deleted matrix the unique nonzero permutation term uses the negative $y$ entries before row $i$ and the positive $x$ entries after it. Their exponents telescope to $b_i, a_i$, respectively, and the sign is $(-1)^i$. Finally take the degreewise Euler characteristic; the free summands have Hilbert denominator $(1 - s)(1 - t)$. □

For $I = (x^2, xy, y^2)$ the underlying sequence is also kernel checked in Lean 4 over the actual ring MvPolynomial(Fin 2, $k$). The verification proves the kernel/image equalities, the stated ideal image, surjective quotient augmentation, projective free terms, and vanishing of both induced Hom differentials to the constant-coefficient residue field. It uses regular multiplication on the free modules, distinct from the residue action on $k$. General staircases, graded minimality and the surface-order coefficient cone remain separately scoped prose arguments.

At grade $\nu \in \mathbb{N}^2$, $M_\nu$ has dimension one exactly when no generator degree is componentwise at most $\nu$. Its comparable-grade map has rank one exactly when both endpoint grades satisfy this condition. A CW realization takes one loop and attaches a disk once along it at each generator degree. The active indices form an interval: the $x$ inequality selects a final interval, the $y$ inequality an initial interval. With $c$ active disks, the cellular map $k^c \to k$ has every entry one. Thus $\dim H_1 = \mathbf{1}_{c=0}$ and

$\dim H_2 = \max(c-1, 0)$, the latter counting active adjacent syzygies. This auxiliary CW complex is not the dessin surface.

For the actual nodal order $\Lambda = k[[\xi, \eta]]/(\xi\eta)$ put $u = \xi + \eta$ and $E = \Lambda/u^2\Lambda$. The basis $(1, \xi, u, u\xi)$ of $E$ satisfies $\xi^2 = u\xi$, $u^2 = 0$, $\eta = u - \xi$. Tensoring the resolution above with $0 \to \Lambda \xrightarrow{u^2} \Lambda \to E \to 0$ gives, over $\mathcal{B} = \Lambda[x, y]$, free ranks $(1, m+1, 2m-1, m-1)$. Indeed the scalar resolution stays exact after tensoring over the field; its degree-zero module is free over $\Lambda$ on the standard monomials. Multiplication by $u^2$ is injective there and on all free terms, so the long exact homology sequence of its mapping cone gives positive-degree homology zero and degree-zero module $E \otimes_k M$. In contrast, as an $R$-module this object has ranks $4(1, m, m-1)$ in its minimal resolution. Neither finite-length $E$ nor these parameter rings replace the infinite parent dessin algebra or its lattice category.

### 2.2 A monotone quantum comparison with exact finite ranks

**Proposition 2.4** (One common channel)**.** *Let $T$ be a positive trace-preserving linear map between finite complex matrix algebras. For density matrices $\rho, \sigma$, their trace distance $d(\rho, \sigma) = \frac{1}{2}\|\rho - \sigma\|_1$ satisfies $d(T\rho, T\sigma) \le d(\rho, \sigma)$. For finitely many labelled states, keep fixed vertex birth grades $e_i$. At $(r, e)$ include vertices with $e_i \le e$ and edges with distance at most $r$, and take the flag complex. Then $T$ gives an identity-on-labels inclusion from the input to the output bifiltration, natural in $(r, e)$.*

*Proof.* Write the Hermitian difference as $A = A_+ - A_-$ with $A_\pm \ge 0$. Positivity, the trace norm triangle inequality and trace preservation give $\|TA\|_1 \le \|TA_+\|_1 + \|TA_-\|_1 = \operatorname{tr} A_+ + \operatorname{tr} A_- = \|A\|_1$. Thus every input edge is an output edge; the fixed births preserve vertices, and all cliques are preserved. Increasing either parameter preserves the simplices, and every map is the same label inclusion, so the squares commute. Complete positivity is sufficient but is not needed for this norm argument. □

For a finite inclusion $K \subset L$, work over a field and embed a basis $Z_K$ of source one-cycles into the target edge space. If $B_L$ is the matrix of target two-boundaries, the induced $H_1$ rank is

$$\rho_1(K, L) = \operatorname{rank}[B_L \; Z_K] - \operatorname{rank} B_L.$$

Its image is $(Z_K + \operatorname{im} B_L)/\operatorname{im} B_L$; source boundaries already map into target boundaries. This proves the formula, including empty complexes. Computing a flag two-skeleton suffices for $H_0, H_1$, but does not compute the full flag complex's $H_2$. Integer ceilings of scaled exact distances preserve the inclusion. For numerical distances, frozen integer labels define an exact combinatorial diagram only after their inequality is checked; this is not an interval certificate of the underlying quantum distances. None of these finite ranks identifies a completed arithmetic determinant or proves GRH.

### 2.3 A correct DG model of the infinite node

Use cohomological degrees and an uncurved differential of degree $+1$. An odd element $\varepsilon$ with $d\varepsilon = 1$ contracts the underlying $k$-complex of every unital left DG module: $h(m) = \varepsilon m$ satisfies $dh + hd = 1$ by the graded Leibniz rule. Thus this example has zero cohomology, contrary to the original ground-field-resolution claim. An isomorphism on $H^0$ alone is also insufficient for a DG quasi-isomorphism; all degrees must be checked.

**Proposition 2.5** (Nodal Koszul model)**.** *Let $k$ be a field, $R = k[\xi, \eta]$ and $f = \xi\eta$ and $K = R \oplus R\varepsilon$, with $|\varepsilon| = -1$, $\varepsilon^2 = 0$, $dR = 0$ and $d\varepsilon = f$. The augmentation $K \to A = R/(\xi\eta)$, $a + b\varepsilon \mapsto [a]$, is a DG-algebra quasi-isomorphism onto the actual infinite nodal dessin algebra. Branch rescaling $\xi \mapsto q\xi$, $\eta \mapsto q^{-1}\eta$, $\varepsilon \mapsto \varepsilon$ for $q \in k^\times$ lifts to a DG automorphism.*

*Proof.* Multiplication is $(a+b\varepsilon)(c+e\varepsilon) = ac + (ae+bc)\varepsilon$. The differential kills degree zero and sends $b\varepsilon$ to $fb$. Its square is zero. For two odd inputs the Leibniz right side is $fbc\varepsilon - bfc\varepsilon = 0$; the other homogeneous cases follow by $R$-linearity. Multiplication by $f$ is injective in the polynomial domain. Hence $H^{-1}(K) = 0$, $H^0(K) = R/(f)$, and all other groups vanish. The augmentation preserves products and kills boundaries. Rescaling fixes $f$, so its coefficientwise extension commutes with $d$ and has inverse obtained by replacing $q$ with $q^{-1}$. This is a one-relation Koszul construction [78, 79]. □

Replacing $R$ by $k[[\xi,\eta]]$ gives the completed order: setting each variable separately to zero maps onto the branch pullback, and the kernel consists precisely of mixed monomials, hence equals $(\xi\eta)$. Rescaling is generally semilinear over the coefficient ring, rather than linear over the fixed diagonal DVR $k[[u]]$, $u = \xi + \eta$. It preserves that DVR as a subring exactly when $q^2 = 1$, and fixes it pointwise exactly when $q = 1$. No Hilbert norm or arithmetic spectrum follows.

The corresponding Lean file checks the actual exterior-line algebra, differential, sign-Leibniz, homotopy and polynomial ambient-resolution identities. Full graded/category/completion objects remain separately scoped. The two theta dessins likewise require endpoint-sensitive Ext: self $\mathrm{Ext}^1$ is zero in both; for the elliptic theta self $\mathrm{Ext}^3$ has dimension two, while for the spherical theta positive even self Ext has dimension two. Outdegree alone does not count self extensions. Minimal $A_\infty$ models use additional graded data [77, §3.3]; $m_3$ is not automatically a physical curvature.

## 3 Determinant semi-invariants and Schur constructions

Assume $k$ algebraically closed of characteristic zero for invariant-ring statements. A finite-dimensional representation of the *infinite* $A$ has spaces $N_i$, arrow matrices $N(a)$ satisfying the monomial relations, and base-change group $G = \prod_i \mathrm{GL}(N_i)$. With column vectors, $N(a) \mapsto g_{t(a)} N(a) g_{s(a)}^{-1}$. Let $P_0 = \bigoplus P_i^{a_i}$, $P_1 = \bigoplus P_i^{b_i}$, and $f : P_1 \to P_0$ be a finite path-matrix map. When $\sum a_i \dim N_i = \sum b_i \dim N_i$, define

$$c_f(N) = \det D_f(N), \qquad D_f(N) = \mathrm{Hom}_A(f, N).$$

It is intrinsically a determinant-line section; chosen bases give this scalar.

**Proposition 3.1** (Weights and exact homological scope)**.** $c_f(gN) = \prod_i \det(g_i)^{b_i - a_i} c_f(N)$. *Restricted determinants of this form span all semi-invariants on a closed $G$-stable relation or nilpotence locus. Set $M = \operatorname{coker} f$. On the square locus,*

$$c_f(N) \neq 0 \quad \Longleftrightarrow \quad \mathrm{Hom}(M,N) = 0 \quad \Longleftrightarrow \quad \mathrm{Hom}(M,N) = \mathrm{Ext}^1(M,N) = 0.$$

*If $f$ is injective,* $\operatorname{coker} D_f = \mathrm{Ext}^1(M,N)$ *and higher Ext vanish. Without injectivity neither that cokernel equality nor higher vanishing follows.*

*Proof.* Naturality gives $D_f(gN) = G_1 D_f(N) G_0^{-1}$, where $G_j$ repeats the relevant $g_i$. Take determinants. The contragredient coordinate-ring action has the inverse character. Schofield–van den Bergh's spanning theorem allows oriented cycles [36, Theorem2.3]. Exactness of invariants for reductive $G$ in characteristic zero lifts a fixed-weight function from any closed stable locus to the ambient representation space; restrict the spanning determinants. Injectivity of the path map need not survive that restriction. Finally $\ker D_f = \mathrm{Hom}(M,N)$. Writing $K = \operatorname{im} f$, the exact sequence with projective $P_0$ expresses $\mathrm{Ext}^1(M,N)$ as $\mathrm{Hom}(K,N)/\operatorname{im}\mathrm{Hom}(P_0,N)$; the epimorphism $P_1 \to K$ embeds this quotient in $\operatorname{coker} D_f$. A square finite matrix with zero kernel has zero cokernel. If $f$ is injective it is the actual length-one resolution, giving the stronger assertion. □

For the node, $f = \times x$, $M = \Lambda/(x)$, $N = \Lambda/(x, y^2)$ give $\dim \operatorname{coker} D_f = 2$ but $\dim \operatorname{Ext}^1(M, N) = 1$. For the elliptic square name arrows so that $a, b : 0 \to 1$, $c, d : 1 \to 0$, permitted cycle *acbd* and forbidden cycle *adbc*. With $f = a$ and $N(a) = 1$, $N(b) = N(c) = N(d) = 0$, its Hom matrix is [1], yet the subsequent forbidden resolution gives $\operatorname{Ext}^{2,3}(M, N) = k$. Thus nonvanishing is not all-degree derived orthogonality.

**Proposition 3.2** (What completion does to invariants)**.** *A continuous $d$-dimensional representation of $\Lambda$ kills $J^d$. On the reduced nilpotent representation variety of fixed dimension vector every invariant regular function is constant. Nonzero-weight semi-invariants need not be constant.*

*Proof.* Continuity with discrete finite-dimensional target kills some $J^m$; its Loewy filtration has at most $d$ nonzero strict steps. Choose vertexwise complements to this filtration. An arrow sends a layer $p$ into layers $q \geq p+1$. Scaling layer $p$ by $z^p$ conjugates each component by $z^{q-p}$. At $z = 0$ the representation tends to the zero-arrow representation. Every invariant therefore has the same value at all points; reducedness and the Nullstellensatz give the ring assertion. In the square example the scalar $N(a)$ varies on a nilpotent locus and has weight $g_1/g_0$, so semi-invariants survive. Formal entries in a projective map are evaluated on this locus by truncating modulo $J^d$, a legitimate finite polynomial operation. □

On the node, $u = x+y$ is regular and $A$ is free of rank two over $k[u]$. The polynomial presentation $\times(u-\lambda)$ for $\lambda \neq 0$ is injective and its determinant on $N$ is $\det(u_N - \lambda I)$, the characteristic polynomial of $u_N$ up to the sign $(-1)^d$. But $u-\lambda$ is a unit in $\Lambda$, with its geometric-series inverse, so those modules do not extend continuously to the completion. An $R$-free lattice and its $u$-nilpotent quotients are different categories. On a lattice a separately specified endomorphism $F$ has a meaningful relative characteristic polynomial over $R$; the scalar base action $uI$ alone does not create Frobenius.

### 3.1 Young tableaux and a cyclic representation stratum

For a nonempty partition $\lambda \vdash m$, choose a numbering $T$ of its boxes. On $V^{\otimes m}$ let $Y_T = R_T C_T$, the unnormalized row sum followed by the signed column sum. The classical identity $Y_T^2 = h_\lambda Y_T$, where $h_\lambda$ is the hook product, makes $Y_T/h_\lambda$ an idempotent in characteristic zero. Its image is $S_\lambda(V)$, one Schur copy, rather than the whole symmetric-group isotypic component [31, 32]. Its character is $s_\lambda$, computed by semistandard tableaux (weak rows, strict columns), and

$$\dim S_\lambda(k^d) = \prod_{(i,j)\in\lambda} \frac{d+j-i}{h_{ij}}.$$

Since $f^{\otimes m}$ commutes with the place permutations, it induces $S_\lambda(f)$ and respects composition and zero maps. Applying this at every vertex and arrow therefore gives a genuine representation of the monomial dessin algebra: $S_\lambda(N(b))S_\lambda(N(a)) = S_\lambda(0) = 0$ for a forbidden pair. The empty partition does not have this property: it sends zero arrow maps to the identity of $k$.

The Littlewood–Richardson rule gives natural intertwiners

$$S_\lambda(V) \otimes S_\mu(V) \cong \bigoplus_\nu S_\nu(V)^{\oplus c^\nu_{\lambda\mu}}.$$

Here $c^\nu_{\lambda\mu}$ counts semistandard tableaux of skew shape $\nu/\lambda$ and content $\mu$ whose row-reading word is Yamanouchi. Naturality makes the same decompositions intertwine all the arrows. The tensor operation is vertexwise with synchronous arrow maps, not $\otimes_A$ and not a Hopf-algebra tensor structure. There is no automatic tensor unit preserving these zero relations. These functors are not

additive or exact; they cannot simply be applied degreewise to a projective resolution. They also need not preserve the fixed $R$-action on lattices.

In the elliptic theta set one of the two permitted three-cycle branches to zero. The other has arbitrary maps $X_1 : V_0 \to V_2$, $X_2 : V_2 \to V_1$, $X_3 : V_1 \to V_0$, with all spaces of dimension $d$. This is a closed representation stratum of our infinite dessin algebra. Put $M = X_3X_2X_1$.

**Theorem 3.3** (Cyclic invariants and a local positive kernel)**.** *The invariant ring on this stratum is $k[c_1(M), \dots, c_d(M)]$, where $c_j$ are characteristic coefficients. Its total-arrow-degree Hilbert series is $\prod_{j=1}^{d}(1 - t^{3j})^{-1}$. Common multidegree $(m, m, m)$ has basis $\chi_\lambda(M) = \operatorname{tr} S_\lambda(M)$, $\lambda \vdash m$, $\ell(\lambda) \le d$; other multidegrees have no invariants. Products obey the Littlewood–Richardson rule. Over $\mathbb{C}$, if $\|M_i\| \le R_0$ and $0 \le r$ and $rR_0^2 < 1$, the kernel*

$$K_r(M_i, M_j) = \det(I - rM_i \otimes \overline{M_j})^{-1} = \sum_\lambda r^{|\lambda|} \chi_\lambda(M_i)\overline{\chi_\lambda(M_j)}$$

*is absolutely convergent and positive semidefinite.*

*Proof.* Cycle-trace generation for quivers, including relation quotients, is the Le Bruyn–Procesi theorem [29, 30]. Every cycle here is a power of the full three-cycle, up to cyclic rotation. In characteristic zero Newton identities give the generators $c_j$. They are algebraically independent: take $X_1 = X_2 = I$ and arbitrary $X_3$. Scalar changes at the three vertices force equal arrow multidegrees. Restriction to that same section identifies degree $(m, m, m)$ with degree-$m$ symmetric polynomials in eigenvalues, whose Schur basis and product rule give the assertion. For the kernel use the ordinary Cauchy identity, first as a formal identity in eigenvalues. On Schur modules the operator norm of $S_\lambda(M)$ is at most $R_0^{|\lambda|}$, so its trace is bounded by $\dim S_\lambda(\mathbb{C}^d) R_0^{|\lambda|}$. The dimension Cauchy identity bounds the absolute series by $(1 - rR_0^2)^{-d^2}$. Each summand is a rank-one Gram kernel, proving positivity by absolute summation. □

The term for the empty partition is a constant scalar feature, not an allowed module-valued tensor unit. In dimension two,

$$(\operatorname{tr} M)^2 = \operatorname{tr} \operatorname{Sym}^2 M + \det M, \qquad \operatorname{tr} \operatorname{Sym}^2 M = \tfrac{1}{2}((\operatorname{tr} M)^2 + \operatorname{tr} M^2).$$

More generally $\det(I - zM) = \sum_j (-z)^j \operatorname{tr} \bigwedge^j M$ and its reciprocal is $\sum_m z^m \operatorname{tr} \operatorname{Sym}^m M$. These are actual determinant/Schur connections, not identifications with an arithmetic $L$-function. On continuous finite-length $\Lambda$-modules the monodromy is nilpotent and all positive-degree characters vanish. The displayed Cauchy positivity is local representation-theoretic positivity, not the global Weil form.

## 4 Lattice almost-split sequences and genuine diamond quotients

For the interval dessin let $X = (x)$, $Y = (y)$ in $\Lambda = k[[x, y]]/(xy)$, $R = k[[u]]$, $u = x + y$. A lattice is a finitely generated $\Lambda$-module free over $R$.

**Theorem 4.1** (The nodal lattice calculation)**.** *Every lattice is $\Lambda^r \oplus X^a \oplus Y^b$. The almost-split sequences in this lattice category are*

$$0 \to Y \hookrightarrow \Lambda \xrightarrow{\times x} X \to 0, \qquad 0 \to X \hookrightarrow \Lambda \xrightarrow{\times y} Y \to 0.$$

*In particular $\tau X = Y$, $\tau Y = X$. Multiplication by $u$ between two copies of either row, followed by reduction modulo $u$, gives an actual nonzero exact three-by-three diagram.*

*Proof.* Normalize $M$ *inside its total quotient module*: $\overline{M} = BM$, $B = k[[x]] \times k[[y]]$. The conductor gives $u\overline{M} \subset M \subset \overline{M}$. Both branch projections of $W = M/u\overline{M} \subset k^p \oplus k^q$ are onto. Intersect $W$ with the two factors; its remaining quotient identifies their two residual quotients. Separate bases for these pure factors and the common quotient put $W$ into pure first lines, pure second lines and diagonal pairs. Lift branch bases over the DVRs. These are respectively the summands $X, Y, \Lambda$. Their endomorphism rings are local.

Exactness of the first sequence is $\operatorname{ann} x = Y$. It does not split: a map $X \to \Lambda$ lands in $\operatorname{ann} y = X$, and its composite with $\times x$ is a nonunit in $\operatorname{End} X = k[[x]]$. Every map $\Lambda \to X$ factors through this epimorphism, $\operatorname{Hom}(Y, X) = 0$, and every nonunit endomorphism of $X$ is divisible by $x$ and factors through $X \hookrightarrow \Lambda \to X$. Classification handles all finite sums; a unit $X$ component would be a retraction. This is right almost splitness. The dual argument for nonsections out of $Y$ proves left almost splitness. Endomorphisms fixing either structure map have constant term one, hence are units, proving minimality. Swap $x, y$ for the other sequence. Finally all three terms are $R$-free. Multiplication by $u$ is injective and reduction preserves exactness, proving the diagram assertion. □

The lattice AR quiver has arrows $Y \to \Lambda \to X$ and $X \to \Lambda \to Y$; the mesh composites are zero. Its projective vertex $\Lambda$ is not the endpoint of another nonsplit AR sequence. AR meshes do not mean that all rows and columns of any drawn grid are short exact.

For a second, explicit family fix $n \geq 3$ and the rational Belyi map

$$\beta_n(z) = \frac{1}{1 - T_n(z)^2} = \frac{1}{(1 - z^2)U_{n-1}(z)^2},$$

where $T_n, U_{n-1}$ are Chebyshev polynomials. It has passport $(2n), (2^n), (2^{n-1}, 1, 1)$. Its contracted dual graph is a path, identifying its genus-zero dessin. The associated infinite algebra has vertices $1, \ldots, n$, endpoint loops $l, r$, opposite arrows $a_i : i \to i + 1$, $b_i : i + 1 \to i$, and precisely the relations $l^2, r^2, a_i b_i, b_i a_i$. Its single permitted cycle is $l a_1 \cdots a_{n-1} r b_{n-1} \cdots b_1$; the order rank is $(2n)^2$.

Retain one arrow at each edge in an orientation $\Omega$, and kill its opposite. The quotient is exactly the string algebra $H_n(\Omega)$ of Huang–Lin–Su [46]; the killed-arrow ideal contains $J^{n+2}$, so it is also an actual quotient of $\Lambda_n$. Inflation identifies its module category with the full abelian quotient-annihilator subcategory $\mathcal{C}_\Omega$ of $\Lambda_n$-modules. This is generally not extension closed in the whole order category. The imported AR property is intrinsic to $\mathcal{C}_\Omega$, not automatically an ambient lattice or finite-length AR property. The extended edition gives nine explicit string modules, all twelve maps and the four exact meshes from the supplied diamond resource, now situated in this actual dessin quotient.

The other supplied resources also have a precise home. For the interval order the subcategory annihilated by $J^2$ is $\operatorname{Mod} B_0$, $B_0 = \Lambda/J^2$. This is a domestic string quotient, not itself a gentle algebra. Inflation is full and preserves purity, and $\operatorname{Zg} B_0$ is a closed subspace of $\operatorname{Zg} \Lambda$. Ringel's infinite-word/pure-injective program and its proof by Puninski–Prest apply there [43, 44]; the isolated band $xy^{-1}$ gives Krull–Gabriel and relative Cantor–Bendixson dimension two by [45, Theorem10.19]. This is not an ambient rank calculation for all $\Lambda$-modules. The parameter in $x = J_2, y = \lambda J_2$ survives even though all cycle traces vanish; the pp condition $\exists v\ (z = xv \ \wedge \ (y - \lambda x)v = 0)$ detects it. These modules are $u$-torsion and are not nonzero order lattices. Infinite-string and locally finite AR theorems must retain their distinct finiteness hypotheses; a surviving permitted cycle has infinitely many paths at a vertex.

### 4.1 Polynomial strings, completed lattices and the Fukaya category

Crawley–Boevey's theorem [38, Theorems1.1–1.4] applies directly to the infinite polynomial dessin algebra: its finitely controlled modules decompose into strings and finite-dimensional bands; its

pointwise artinian modules additionally allow primitive injective bands. For our finite quiver, these conditions mean finitely generated and artinian respectively. The words may be infinite, but their modules have algebraic direct-sum bases. This is not a classification of every module or of arbitrary formal completions.

In fact no nonzero completed-order lattice, restricted to $A$, meets either finiteness condition. The polynomial $A$ is finite over $k[u]$: remove full turns from sufficiently long paths as in Proposition 1.1. If an $R$-free lattice were finitely generated over $A$, it would be finite torsion free, hence free, over the PID $k[u]$. Multiplication by $1-u$ would then fail to be onto, whereas it is a unit over $R = k[[u]]$. Also $M \supsetneq uM \supsetneq u^2M \supsetneq \cdots$ excludes artinianity. For a concrete indecomposable outside the ordinary string/band list take $k[[x]]$ with $y = 0$: its endomorphisms are multiplication by formal series, so its endomorphism ring is local. Its $x$ is injective and nonsurjective, while $1-x$ is invertible; these properties exclude every pure-$x$ ordinary string or band. The extended edition gives the full word argument.

The Fukaya connection requires equally precise categories. Lekili–Polishchuk explicitly allow infinite-dimensional locally gentle algebras, but their equivalence is for *perfect dg modules* of a homologically smooth algebra, with specified grading, line field and stops [66, Definition3.1.1, Lemma3.1.3, Theorem3.2.2]. Every nonempty closed dessin here has both permitted and forbidden cycles, so its polynomial algebra is neither proper nor smooth by their criterion. This does not exclude appropriate singular or completed variants; it prevents a direct use of that particular theorem. Infinite word, graded curve and geodesic are not interchangeable without a surface realization and boundary/grading data.

There is a genuine positive example inside our setting: the polynomial interval quotient $A/(y) = k[x]$ is still infinite dimensional, is smooth, and has no forbidden cycle. Its perfect category has the partially wrapped annulus model with one stop total. In degree zero $\mathrm{Perf}(k[x]) = D^b(\mathrm{mod}_{\mathrm{fg}}\, k[x])$. Inflation of this quotient is fully faithful on ordinary modules, but not on derived categories: $\mathrm{Ext}^2_A(k[x], k[x]) = k$, whereas $\mathrm{Ext}^2_{k[x]}(k[x], k[x]) = 0$. The former follows from the alternating $y, x$ resolution. Thus this quotient bridge does not identify the whole closed dessin's perfect category with the annulus category.

Finally, Leavitt path algebras *can* fail invariant basis number, but need not. The permitted-state Laurent algebras here have finite-dimensional matrix quotients, and therefore invariant basis number. The polynomial and completed dessin algebras have their vertex-simple quotients, giving the same conclusion. The complex Leavitt algebra of $G_\alpha$ below is a dense *-subalgebra of its graph C*-algebra, on that specified graph; it is not the original gentle algebra with ordinary CK1 imposed.

## 5 Leavitt relations, the constrained Fock algebra and spectrum

Work now over $\mathbb{C}$. Ordinary Leavitt and graph C* constructions [47, 48] cannot simply impose our original forbidden products: if $ab = 0$ and CK1 says $a^*a = p_{t(a)}$, then $0 = a^*ab = b$. Every arrow is a forbidden successor, hence every arrow and then every vertex is zero. CK2 is not needed. The ordinary graph-Toeplitz CK1 setting has the same collapse.

The valid permitted-arrow-state graph instead consists of the disjoint directed $\sigma$-cycles. Its Leavitt algebra is $\bigoplus_c M_{|c|}(k[z, z^{-1}])$, and its graph C* algebra is $\bigoplus_c M_{|c|}(C(\mathbb{T}))$. These forget the $\alpha$-gluing; both theta dessins give the same answer. They are not the completed original order. Nor is the ordinary Toeplitz algebra of a cycle of length $\ell > 1$ equal to $M_\ell(\mathcal{T})$: their $K_0$ ranks are $\ell$ and one respectively [50].

Let $\mathcal{H} = \ell^2(\mathcal{P})$ for all permitted finite paths, including vacua $e_v$. Define $L_a\delta_p = \delta_{ap}$ when the prefix is permitted and composable, and zero otherwise. Let $P_v$ project onto paths starting at $v$,

$Q_a = L_a L_a^*$ and $v_v$ onto the vacuum. Then

$$L_a L_{\phi(a)} = 0, \quad L_a^* L_a = P_{t(a)} - Q_{\phi(a)} = v_{t(a)} + Q_{\tau(a)}, \quad P_v - \sum_{s(a)=v} Q_a = v_v.$$

These follow on basis vectors. The different initial projection avoids CK1 collapse. Write $\mathcal{T}_A = C^*(P_v, L_a)$, a constrained-path Fock C* algebra in the monomial/subproduct setting [51, 52]. The polynomial $A$ acts faithfully: applying a finite path sum to the sum of vacua recovers its coefficients. This representation need not extend to all formal $\Lambda$ series: $\sum_{m\geq 0} c^m$ along a cycle would send its vacuum to a nonsquare-summable sum.

**Theorem 5.1** (Compact extension and incidence index)**.** *Let $\mathcal{H}_v$ be the terminal-path sector. There is an exact sequence*

$$0 \longrightarrow \bigoplus_{v\in Q_0} \mathcal{K}(\mathcal{H}_v) \longrightarrow \mathcal{T}_A \longrightarrow \bigoplus_{c\in \mathrm{cyc}(\sigma)} M_{|c|}(C(\mathbb{T})) \longrightarrow 0.$$

*With $\delta[u] = [1 - W^*W] - [1 - WW^*]$, its boundary map is $\delta(c) = -\sum_{h\in c} e_{[h]}$, minus the unsigned incidence map $B_\Gamma$ of the cellular graph. Thus, for its $E$ edges,*

| | $K_0(\mathcal{T}_A)$ | $K_1(\mathcal{T}_A)$ |
|---|---|---|
| $\Gamma$ *bipartite* | $\mathbb{Z}^{E+1}$ | $\mathbb{Z}$ |
| $\Gamma$ *nonbipartite* | $\mathbb{Z}^E \oplus \mathbb{Z}/2$ | 0. |

*Proof.* All prefixes preserve the terminal sectors. For paths $p, q$ ending at $v$, $L_p v_v L_q^* = |\delta_p\rangle\langle\delta_q|$. Therefore the vacuum ideal is exactly the displayed sum of compacts: it contains these matrix units and that sum is itself an ideal. In the quotient, $L_a$ has initial projection $Q_{\tau(a)}$ and range $Q_a$, a graph CK family for the permitted-state cycles. The $Q_a$ have infinite rank and nonzero quotient images; the length gauge action descends. Gauge-invariant uniqueness [48, Theorem2.1] identifies the quotient as claimed.

For a cycle $c$ based at arrow $a$, lift its corner unitary by $W = L_c Q_a + (1 - Q_a)$. Then $W^*W = 1$ and $1 - WW^* = Q_a - L_c Q_a L_c^*$ projects onto the first $|c|$ paths starting with $a$. Their terminal vertices count exactly its incident darts, including twice for a loop. This proves the index formula. The six-term exact sequence gives $K_1 = \ker B_\Gamma$ and $0 \to \operatorname{coker} B_\Gamma \to K_0 \to \mathbb{Z}^{\#\,\mathrm{cyc}\,\sigma} \to 0$. The free last term splits abstractly. Spanning-tree integer elimination gives Smith entries 1 repeated $V - 1$ times and one zero in the bipartite case; an odd cycle instead adds the entry two. This proves the groups. □

The nodal interval has $K_0 = \mathbb{Z}^2$, $K_1 = \mathbb{Z}$; the single-loop sphere has $\mathbb{Z} \oplus \mathbb{Z}/2$, zero. Both theta dessins have $\mathbb{Z}^4$, $\mathbb{Z}$. Coarse $K$-theory does not recover genus or the arithmetic factors.

## 5.1 A graph presentation and an actual pullback

The Fock algebra itself has a graph presentation, on a different graph from the original gentle quiver. Let $G_\alpha$ have vertices $q_h$ for $h \in H$ and sinks $b_e$ for $e \in Q_0$, and edges

$$s_h : q_h \to q_{\tau h}, \qquad t_h : q_h \to b_{[\tau h]}.$$

**Proposition 5.2.** *$C^*(G_\alpha) \cong \mathcal{T}_A$, by $q_h \mapsto Q_{a_h}$, $b_e \mapsto v_e$, $s_h \mapsto L_{a_h} Q_{a_{\tau h}}$ and $t_h \mapsto L_{a_h} v_{[\tau h]}$. The integer graph relation at $h$ is $q_h - q_{\tau h} - b_{[\tau h]}$. Eliminating these relations recovers $K_1 = \ker B_\Gamma$ and $K_0 \cong \operatorname{coker} B_\Gamma \oplus \mathbb{Z}^{\#\,\mathrm{cyc}\,\sigma}$.*

*Proof.* The two edges at $q_h$ have orthogonal initial pieces and orthogonal ranges summing to $Q_{a_h}$, by the Fock initial-projection identity. All vertex projections are nonzero. The length gauge acts with degree one on both edge families, so gauge-invariant uniqueness applies. They generate since $L_{a_h} = s_h + t_h$ and $P_e = v_e + \sum_{[h]=e} Q_{a_h}$. In the integer matrix a kernel coefficient is constant around each permitted cycle; its sink equations are exactly $B_\Gamma c = 0$. In the cokernel, eliminate all but one state per cycle; its remaining relation is the sum of its incident sink generators. The remaining cycle generators are free, proving the stated groups. □

For the nodal interval, $G_\alpha$ has two loop vertices, each with an exit to the common vacuum sink. Chirvasitu–Hajac–Tobolski's actual pullback theorem [65, Theorem3.4] gives

$$\{T \in \mathcal{T}_A : \pi(T) \in \mathbb{C}^2\} \cong C^*(E_1), \qquad \mathbb{C}^2 \hookrightarrow C(\mathbb{T})^2 \text{ by constants},$$

where $E_1$ has the same three vertices and countably many arrows from each loop vertex to the sink, with no loops. The arrow numbered $n \geq 0$ maps to loop$^n$ followed by exit. This algebra is AF, with $K_0 = \mathbb{Z}^3$, $K_1 = 0$. The extended edition verifies every hypothesis. The theorem does not turn arbitrary cycle-vertex identifications into pullbacks: its first graph is acyclic and its path functor must cover all pointed paths. Its sink-extension Theorem4.3 has further injection and path-coverage requirements.

### 5.2 A separate diagonal-fiber C* completion retaining the pairing

The formal order's residue gluing cannot be copied by taking diagonals of arbitrary full matrices: $\operatorname{diag}(E_{12}E_{21}) \neq \operatorname{diag}(E_{12})\operatorname{diag}(E_{21})$. For $\ell > 1$ there is no nonzero *-homomorphism from $M_\ell(C(\mathbb{T}))$ to a commutative C*-algebra, as restriction to its constant matrix units shows. Restrict the distinguished fiber first. Define

$$D_\ell = \{f \in C(\mathbb{T}, M_\ell(\mathbb{C})) : f(1) \text{ is diagonal}\}, \qquad \mathcal{D}_H = \Big(\prod_c D_{|c|}\Big) \underset{\mathbb{C}^H}{\times} \mathbb{C}^{Q_0},$$

where the first map evaluates diagonal entries at one and the second repeats $\lambda_e$ at the two darts paired by $\alpha$.

**Theorem 5.3** (A faithful polynomial pullback model)**.** *There is a split exact sequence*

$$0 \to \bigoplus_c C_0(\mathbb{T} \setminus \{1\}, M_{|c|}) \to \mathcal{D}_H \to \mathbb{C}^{Q_0} \to 0,$$

*so* $K_0(\mathcal{D}_H) = \mathbb{Z}^{|Q_0|}$ *and* $K_1(\mathcal{D}_H) = \mathbb{Z}^{\#\,\mathrm{cyc}\,\sigma}$*. The polynomial dessin algebra embeds faithfully and generates this C*-algebra with its adjoints. This embedding does not extend to a unital algebra map from all of* $\Lambda$*.*

*Proof.* The section consists of the constant paired diagonal projections. The kernel is the displayed vanishing fiber, a sum of matrix suspensions, giving the groups by split exactness and suspension. For an explicit embedding set $g(z) = |1 - z|^2/4$, choose one cut arrow per permitted cycle, and send $a_h$ to $g(z)z^{\epsilon_h}E_{h,\tau h}$ in its branch, where $\epsilon_h = 1$ at the cut and zero otherwise. Send $e_e$ to the constant paired diagonal projection. Every forbidden product is zero. A full cycle at $h$ maps to $w_c(z)E_{hh}$, $w_c = g^{|c|}z$. This scalar function is injective on the circle: away from its unique zero at one its argument recovers $z$. Continuous functional calculus therefore supplies every $f(z)E_{hh}$ with $f(1) = 0$. Multiplying by path images and approximating division away from one supplies every vanishing off-diagonal entry. Together with the constant paired diagonals these generate $\mathcal{D}_H$.

Path normal forms in a fixed entry are a nonzero prefactor times a polynomial in $w_c$; its infinite range proves independence. Evaluation at one first separates the zero-length coefficients. Finally for $z_0 \neq 1$, $1 - w_c(z_0)^{-1}c_h$ is a unit in the formal order, but its proposed image is singular at $z_0$. A unital map preserves units, proving nonextension. □

This is the C* algebra generated by a faithful polynomial representation and its adjoints, distinct from the completed order and from $\mathcal{T}_A$. The norm closure of the polynomial image alone need not contain those adjoints. Their $K$ groups already differ. The pairing belongs to the labelled presentation; an unmarked isomorphism class need not recover the ribbon ordering. No arithmetic descent, Frobenius action or completed-$L$ positive determinant is inferred from this additional valid gluing model.

**Proposition 5.4** (The degree spectral triple)**.** *Define $N\delta_p = |p|\delta_p$ on $\{\xi : \sum_p |p|^2|\xi_p|^2 < \infty\}$. With the polynomial creation/annihilation *-algebra this is an ungraded spectral triple, $p$-summable exactly for $p > 1$. With $N_+^{-s}$ zero on the kernel,*

$$\operatorname{Tr} e^{-tN} = E + \frac{2E}{e^t - 1}, \quad \operatorname{Tr} N_+^{-s} = 2E\zeta(s), \quad \operatorname{Tr}\big(\chi(N)N_+^{-s}\big) = 2EL(s,\chi)$$

*for $t > 0$ and $\operatorname{Re} s > 1$ in the last two formulas, where $\chi$ is an externally chosen Dirichlet character.*

*Proof.* There are $E$ vacua and $2E$ paths of every positive length: each starting arrow has a unique continuation. The real diagonal operator is self-adjoint on the stated maximal domain; finite supports are a core and its resolvent is compact. Prefixes preserve the domain and $[N, L_a] = L_a$, $[N, L_a^*] = -L_a^*$, so polynomial commutators are bounded. The trace formulas and summability reduce to the corresponding scalar series. The character multiplier is bounded; membership in $\mathcal{T}_A$ is not asserted. □

Length parity anticommutes with the creators but commutes with $N$, so it is not an even-spectral-triple grading. The ordinary odd Fredholm class of this positive $N$ is trivial: $N(1+N^2)^{-1/2} - I$ is compact. The spectrum depends only on $E$; zeros of the scalar zeta trace are not eigenvalues of $N$.

## 6 Arithmetic attachment, descent and the remaining comparison

The compact ideal gives a precise home for an arithmetic determinant, with no claim that the arithmetic is derived from the dessin. Fix one terminal sector. Its two paths of each length $n \geq 1$ give orthonormal vectors $\xi_n = (\delta_{p_{n,1}} + \delta_{p_{n,2}})/\sqrt{2}$ and an isometry $U : \ell^2(\mathbb{N}_{\geq 1}) \to \mathcal{H}_v$.

For a nonprincipal character $\chi$ put $A_n = \sum_{j \leq n} \chi(j)$, $B_0 = \sup |A_n| \leq q_\chi$, and, on $2^k \leq n < 2^{k+1}$, set

$$w_n = \sqrt{n}(k+1), \quad u_n = A_n/w_n, \quad v_n(s) = w_n(n^{-s} - (n+1)^{-s}), \quad \varphi_s(z) = \sum_n v_n(s)z_n - z_1.$$

Here $u_1 = 1$. For $\operatorname{Re} s > 0$, $u, v(s) \in \ell^2$ and the pairing is complex linear, with no conjugation of $s$. Define

$$\widetilde{K_s} = U(u \otimes \varphi_s)U^*, \qquad \det_F(I + \widetilde{K_s}) = L(s,\chi).$$

Indeed dyadic blocks give $\|u\|^2 \leq 2B_0^2$, and on a compact set $\operatorname{Re} s \geq \sigma_0 > 0$, $|s| \leq M$, the squared $v$ tail is bounded by $M^2 \sum_{k \geq J}(k+1)^2 2^{-2\sigma_0 k}$. Thus the rank-one family is trace-norm holomorphic. Its determinant is $1 + \varphi_s(u)$, equal by Abel summation to $L$. Replacing $v$ by $P_N v$ and adding $A_N(N+1)^{-s}e_1^*$ to the functional gives exactly every raw partial sum $S_N$, with compact-uniform

trace-norm convergence; $\|UXU^*\|_1 = \|X\|_1$. For primitive $\chi$, completion replaces the functional by $g(s)\sum v_n(s)z_n - z_1$, with the standard nonzero gamma factor $g$, rather than multiplying the whole old functional. The extended edition supplies the quantitative estimates.

This inserts a scalar encoding into one actual dessin compact ideal; it is not known to be $A$-linear, a derived chain map or Frobenius compatible. A common fixed positive metric cannot make this same fixed-range family symmetric for all critical heights: symmetry forces its functionals to be proportional, whereas the coefficient ratio at heights $0, t \neq 0$ is asymptotic to $(\sigma + it)n^{-it}/\sigma$, with two opposite subsequential limits. Different positive realizations remain possible.

**Proposition 6.1** (Complex dessin data do not determine arithmetic descent)**.** *The rational Belyi pairs $E_d : y^2 = x^3 - d$, $\beta_d = x^3/d$, for $d = 1, 2$, are isomorphic over $\mathbb{C}$ but have distinct Frobenius polynomials at the good prime seven: $1 - 4T + 7T^2$ and $1 - T + 7T^2$.*

*Proof.* Choose $a^3 = d$, $b^2 = d$. The projective isomorphism $(x, y) \mapsto (ax, by)$ from $E_1$ to $E_d$ preserves the Belyi map. The degree-six clean passports are the elliptic theta passport already computed. Directly counting $y$ for $x = 0, \ldots, 6$ gives respectively $(0, 1, 1, 0, 1, 0, 0)$ and $(0, 0, 0, 2, 0, 2, 2)$. Including infinity yields four and seven points, hence $a_7 = 8 - \#E_d(\mathbb{F}_7) = 4, 1$. Both discriminants $-432d^2$ are nonzero modulo seven. □

Any invariant of the complex dessin or its undecorated order/Fock algebra alone is the same for these two inputs, so cannot recover both factors without descent data. This does not obstruct a theory including a specified rational model and compatible Galois action. Elliptic $H^1$ has weight one and unshifted center $s = 1$, unlike Dirichlet center $1/2$; the example is not a counterexample to GRH.

### 6.1 A constructive arithmetic benchmark on the theta dessin

The preceding twists can be distinguished constructively once their finite-field descent data are retained. This closes the elementary local-factor computation, not the global positive-determinant problem. Let $\kappa = \mathbb{F}_q$, $\operatorname{char}\kappa \neq 2, 3$ and $d \in \kappa^*$. Put $E_d : y^2 = x^3 - d$ and $a_q(d) = q + 1 - \#E_d(\kappa)$. For $q \equiv 1 \pmod 3$, fix a nontrivial cubic character $\rho$ and the quadratic character $\eta$, both extended by zero at zero. Define

$$J = \sum_{u \in \kappa} \rho(u)\eta(1 - u), \qquad j = -\eta(-1)J, \qquad \lambda_d = \eta(d)\rho(d), \quad \pi_d = \lambda_d j.$$

**Theorem 6.2** (Uniform local factor)**.** *For every such prime power and nonzero $d$,*

$$a_q(d) = \pi_d + \overline{\pi_d}, \quad |\pi_d|^2 = q, \quad P_d(T) := 1 - a_q(d)T + qT^2 = (1 - \pi_d T)(1 - \overline{\pi_d}T).$$

*If $q \equiv 2 \pmod 3$, instead $a_q(d) = 0$. The split-case values lie in $\mathbb{Z}[\omega]$, $\omega^2 + \omega + 1 = 0$.*

*Proof.* Cube fibers have size $1 + \rho(t) + \overline{\rho(t)}$, including at zero; the first term is the literal constant one, not the zero-extended trivial character. Counting square fibers gives $a_q(d) = -\sum_x \eta(x^3 - d)$. Grouping by $t = x^3$ removes the constant term since $\sum_t \eta(t - d) = 0$. Substitution $t = du$ gives $\sum_t \rho(t)\eta(t - d) = \rho(d)\eta(-d)J$; the other term is its conjugate. This proves the signed trace formula.

For a nontrivial additive character $\psi$, write $G(\chi) = \sum_x \chi(x)\psi(x)$. Character orthogonality gives $|G(\chi)|^2 = q$ for nontrivial $\chi$: substitute $x = uy$ in its squared norm; the inner sum over $y \neq 0$ is $q - 1$ at $u = 1$ and $-1$ otherwise. Grouping $G(\rho)G(\eta)$ by $x + y$ gives $JG(\rho\eta)$, the zero-sum fiber vanishing since $\rho\eta$ is nontrivial. The three Gauss norms yield $|J|^2 = q$. Neither purity nor RH was assumed. The remaining signs and character values have modulus one. For $q \equiv 2 \pmod 3$

cubing is bijective, so the quadratic-character sum is zero. Finally all split-case summands are signs times cube roots of unity. These are the classical Gauss–Jacobi methods [68, pp.499–502], with our normalization proved explicitly. □

At $q = 7$, choose $\rho(3) = \omega$. Then $j = 3 + 2\omega$, $\lambda_2 = \omega^2$, and $\pi_2 = -1 - 3\omega$, giving the two factors already counted. At $q = 13$, $\rho(2) = \omega$ gives $J = 1 + 4\omega$ but $j = -1 - 4\omega$; this detects the essential sign hidden by the first example.

### 6.2 A genuine cellular carrier, and what still fails for the order

Assume $q \equiv 1 \pmod 3$ in this cellular construction. The elliptic theta dessin has cellular chains $C_2 = \mathbb{Q}$, $C_1 = \mathbb{Q}^3$, $C_0 = \mathbb{Q}^2$. Orient the three edges from $v_+$ to $v_-$; then $\partial_1(e_i) = v_- - v_+$ and $\partial_2 = 0$. Thus $H_1$ is the sum-zero plane, with basis $e_0 - e_1, e_1 - e_2$. Rotation $\sigma$ acts there by

$$C = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}, \qquad G = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}.$$

$G$ is the actual Euclidean edge Gram matrix in this basis. The deck involution $\alpha$ acts as $-I$. Consequently the six-element deck group $\langle \sigma, \alpha \rangle$ acts on this homology and permutes the original quiver and its entire projective resolution.

Write $\pi_d = A_d + B_d\omega$. Its multiplication action on this *actual cellular* plane is

$$M_d = A_d I + B_d C = \begin{pmatrix} A_d & -B_d \\ B_d & A_d - B_d \end{pmatrix}, \qquad \det(I - TM_d) = P_d(T), \qquad M_d^t G M_d = qG.$$

This constructs the local matrix from finite-field character sums and the dessin's deck action, rather than fitting its trace. Positivity follows from $2x^2 - 2xy + 2y^2 = (x - y)^2 + x^2 + y^2$. The normalized matrix is $G$-unitary, not asserted self-adjoint. The extended edition supplies a rational cellular chain lift, its exact determinant cancellation, and Kummer descent compatibility on homology. Its odd-over-even determinant ratio is the classical finite-field curve zeta function

$$Z(E_d/\kappa, T) = \frac{P_d(T)}{(1 - T)(1 - qT)}.$$

Identification with the geometric $\ell$-adic Frobenius polynomial uses the standard elliptic trace and determinant theorem; a canonical integral comparison or an order-derived Frobenius functor has not been constructed here.

There is also a precise obstruction to extending this same real two-dimensional carrier to $q \equiv 2 \pmod 3$. Frobenius then conjugates the cubic deck action. The relation $MC = C^2 M$ forces $M = \left(\begin{smallmatrix} a & b \\ a+b & -a \end{smallmatrix}\right)$, whose determinant is $-a^2 - ab - b^2 \le 0$, contradicting the required positive determinant $q$. This excludes that real carrier with its fixed deck action, not complex or enlarged Hilbert–Pólya constructions. The extended edition gives a compatible complex local replacement.

For the following insertion test use the characteristic-zero theta order $\Lambda_{\mathbb{Q}}$ over $R_{\mathbb{Q}} = \mathbb{Q}[[u]]$, with residue simple $S_e = \mathbb{Q}$. The limitation is concrete. The deck action has finite order and cannot alone supply the norm-$\sqrt{q}$ Jacobi coefficient. Inserting $M_d$ into $S_e \otimes \mathbb{Q}^2$ gives an actual residue-module endomorphism with determinant $P_d$, but that simple is not perfect over the order and choosing it breaks full deck symmetry. Inserting the same matrix $\Lambda_{\mathbb{Q}}$-linearly into the regular projective instead gives the $R_{\mathbb{Q}}$-determinant $P_d^{18}$, or $P_d^6$ on $e\Lambda_{\mathbb{Q}} \otimes \mathbb{Q}^2$. Taking a root is not an established determinant-line comparison; a deck-semilinear lift is a different map. These surviving

multiplicity/equivariance obligations prevent promoting the local computation to the required arithmetic order functor.

For the global comparison fix a nonzero rational $d$ and use its good reductions. The finite-field curve zeta has center $1/2$ after $T = q^{-s}$. The *global elliptic* $L$-function over $\mathbb{Q}$ has unshifted center one, whereas Dirichlet $L$-functions have center $1/2$. Local weight-one purity does not determine either global zero set. The elementary local bound proves absolute convergence of the elliptic Euler product on $\operatorname{Re} s > 3/2$; the new local identities give no determinant-preserving continuation with the desired global positive spectrum.

### 6.3 Arbitrary projective actions: an obstruction and a genuine determinant repair

The preceding tensor test has a stronger replacement. Keep the characteristic-zero completed elliptic theta order $\Lambda$ over $R = \mathbb{Q}[[u]]$, and write $K = \mathbb{Q}((u))$. All determinants in the next theorem are *ordinary underlying-$R$ determinants*; they are not reduced norms.

**Theorem 6.3** (Projective determinant multiplicity)**.** *Let $P$ be any finitely generated projective right $\Lambda$-module and $f \in \operatorname{End}_\Lambda(P)$. Then $P \simeq \bigoplus_{i=0}^{2} (e_i\Lambda)^{n_i}$. Let $A_i \in M_{n_i}(\mathbb{Q})$ be the induced maps on its three top multiplicity spaces. Then*

$$\det_R(1 - Tf)\Big|_{u=0} = \left(\prod_{i=0}^{2} \det(1 - TA_i)\right)^6.$$

*For any bounded complex of these projectives and any $\Lambda$-linear chain endomorphism, its alternating determinant ratio specializes to a sixth power in $\mathbb{Q}(T)$. Generically, the ordinary ratio is a cube in $K(T)$.*

*Proof.* The arrow ideal is the Jacobson radical: its elements admit geometric-series inverses after subtracting from one, and the quotient is $\mathbb{Q}^3$. A projective's top lifts to a map from the displayed sum; Nakayama gives surjectivity, projectivity splits it, and another application of Nakayama kills the kernel.

Put $B = \Lambda/u\Lambda$. Each $e_iB$ has path layers of dimensions $1, 2, 2, 1$: the identity, two length-one paths, two length-two paths, and one length-three path. The two full cycles based at $i$ add to $ue_i$, so become negatives; all length-four paths vanish. A positive-length entry of a projective homomorphism strictly raises path length. Consequently the associated graded map has six copies of $A = \operatorname{diag}(A_0, A_1, A_2)$, giving the formula by triangular block determinants. Base change of determinants justifies specialization. Multiplicativity along filtrations is ordinary linear algebra, consistent with [69, Tags0FJ9,0GSX]; no purity assumption enters.

Each term's determinant has constant term one in $T$, so every denominator has nonzero specialization. Taking the finite alternating product therefore preserves the sixth power, even after cancellation. Finally $\Lambda_K = M_3(K) \times M_3(K)$; a projective's branch modules are sums of three-dimensional simples. A commuting endomorphism acts on each multiplicity space, repeating every factor three times. This proves the generic assertion. □

Thus no such ordinary perfect-complex ratio can equal the constant-in-$u$ quadratic $P_d(T)$, or the full local curve zeta $P_d(T)/((1 - T)(1 - qT))$. For $P_d$, the valuation at infinity is $-2$, not a multiple of six. For the zeta ratio, $P_d(1) = q + 1 - a_q > 0$ by the local bound $|a_q| \le 2\sqrt{q} < q + 1$, so its pole at $T = 1$ is simple. This also covers square prime powers where the numerator may have a double root. It is an obstruction for the specified ordinary determinant and linear category, not all spectral or determinant theories.

There is nevertheless a *genuine integral functor* removing this multiplicity. Choose a permitted branch and its cycle cut. Its projection $\Lambda \to B_+ \subset M_3(R)$ makes the column lattice $W_+ = R^3$ a left $\Lambda$-module. Define

$$\Phi_+ : \mathrm{Perf}(\Lambda) \longrightarrow \mathrm{Perf}(R), \qquad P^\bullet \longmapsto P^\bullet \otimes_\Lambda W_+.$$

Bounded projectives compute this derived tensor functor; flatness of $W_+$ on all modules is unnecessary. Since $e_i W_+ \simeq R$, its determinant on $e_i\Lambda \otimes_{\mathbb{Q}} V$ with external action $1 \otimes M_d$ is exactly $P_d$, whereas on the regular module it is $P_d^3$. This is a defined change of determinant functor, not an arbitrary polynomial root. It does not construct the arithmetic action independently: the already computed $M_d$ is still the input.

The price of this repair is precise. Branch swap $\alpha$ exchanges $W_+$ and $W_-$; pairing them yields $P_d^2$ on that basic-projective input. The chosen branch determinant cannot intertwine $\alpha$ on all endomorphisms: the corner unit $e_i + c_i^+$ acting on $e_i\Lambda$, with $c_i^+$ a plus full cycle, acts on its determinant by $1 + u$, whereas its $\alpha$-image acts by one. Also no rank-one-per-vertex column lattice supported on one branch admits an $R$-linear lift of the rotation $\sigma$ fixing $u$. Its three nonzero arrow scalars have nonnegative $u$-valuations summing to one; an invertible cyclic lift would make all three equal. This contradiction concerns that lattice, not every determinant-level rotation comparison. Ramification $u = v^3$ *and a new lattice* with all three arrow scalars $v$ admits such a rotation, changing the stipulated base/lattice data. A descent-compatible arithmetic determinant retaining the required symmetry and a global positive realization remains open.

### 6.4 Tame descent: a surviving order and a sharper obstruction

We now test the proposed ramified repair rather than assume that its rotation descends. Extend coefficients explicitly to $k = \mathbb{Q}(\omega)$, $\omega^2 + \omega + 1 = 0$. Put $R = k[[u]]$, $S = k[[v]]$, $u = v^3$, and let $g(v) = \omega v$, fixing $k$. On the balanced plus column $W = S^3$, let $C\epsilon_i = \epsilon_{i+1}$ and

$$\Theta = Dg, \qquad D = \mathrm{diag}(1, \omega^2, \omega).$$

The plus arrows act by $vE_{i,i-1}$, and minus arrows by zero. Direct calculation gives $\Theta^3 = 1$, compatibility of $\Theta$ with this $\Lambda_R$-action, and

$$\Theta C = \omega^2 C\Theta, \qquad L = W^\Theta = R\epsilon_0 \oplus Rv\epsilon_1 \oplus Rv^2\epsilon_2.$$

Thus invariants restore precisely the original cut lattice. The extension $R \to S$ is finite *free* of rank three and faithfully flat; the group quotient is not a torsor at $v = 0$. The comparison $S \otimes_R L \to W$ has matrix $\mathrm{diag}(1, v, v^2)$, determinant $u$ and cokernel $S/(v) \oplus S/(v^2)$. Hence this semilinear group action alone is not effective descent of the column to $R$. This is the explicit local root-stack situation, consistent with the trivial-stabilizer criterion for vector-bundle descent [70, Theorem10.3]; no stack theorem is needed for the displayed calculation.

Two positive constructions survive. The determinant line $\bigwedge_S^3 W$ has trivial $D$ and $C$ actions and does descend, but the natural map

$$\det_R(L) \longrightarrow (\det_S W)^\Theta$$

has image $uR$ inside $R$, not an isomorphism. Determinant and coarse invariants cannot be interchanged while dropping this nonunit comparison. Also the *carrier* endomorphism algebra $\mathrm{End}_S(W)^\Theta$, conjugated to the invariant basis, is exactly the hereditary order with entries $R$ on and below the diagonal and $uR$ above. Its rotation descends as $\mathrm{Ad}(A)$, where

$$A = \begin{pmatrix} 0 & 0 & u \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \qquad A^3 = uI.$$

The algebra automorphism has order three although $A$ is not an integral unit. This is not $\mathrm{End}_\Lambda(W)$ or a descended arithmetic cohomology group.

The natural arithmetic attempt fails: $aI + bC$ does not commute with $\Theta$ when $b \neq 0$, as in both prime-seven Jacobi examples. The original cellular sum-zero plane is not $\Theta$-stable either. Replacing $C$ by $vC$ does give the descended operator $A$ on $L$, but

$$\det(1 - T(aI + bA)) = (1 - aT)^3 - ub^3T^3,$$

whose special fiber is $(1 - aT)^3$, not the required local quadratic. These carrier maps are not automatically $\Lambda$-linear arithmetic endomorphisms.

There is a stronger category-level conclusion.

**Theorem 6.4** (Full-deck coefficient kernels)**.** *Let $W_0$ be a finite $R$-free left $\Lambda_R$-module on which the central $u$ acts by the given scalar. Suppose it has invertible $R$-linear intertwiners lifting the order automorphisms $\sigma$ and $\alpha$. Then, for some $m \geq 0$, every $e_iW_0$ has rank $6m$. For any finite projective $P$ and $\Lambda_R$-linear $f : P \to P$, let $A_i$ be its top blocks. The genuine coefficient functor $\Phi_{W_0} = - \otimes^{\mathbf{L}}_{\Lambda} W_0$ satisfies*

$$\det_R(1 - T\Phi_{W_0}(f))\Big|_{u=0} = \left(\prod_i \det_k(1 - TA_i)\right)^{6m}.$$

*Bounded perfect ratios retain this exponent. Thus this whole class of full-deck, fixed-center finite free coefficient kernels cannot supply either the single local quadratic or the full local curve zeta.*

*Proof.* In the generic fiber intersect $W_0$ with its two central branch summands to obtain finite free integral lattices $W_\pm$; an integral direct-sum decomposition is not assumed. Rotation preserves them and branch swap exchanges them. On a branch let the common vertex rank be $r$. The three arrow matrices have cyclic product $uI_r$. Their determinant valuations are nonnegative, sum to $r$, and rotation intertwiners make them equal. Therefore $3 \mid r$. Branch swap makes the two ranks equal, so the full vertex rank is $6m$.

For the determinant formula, filter $W_0/uW_0$ by the arrow radical, whose fourth power vanishes. A positive-length entry of $f$ raises the filtration; a top block $A_i$ repeats across its $i$-th graded pieces a total of $\dim_k e_i(W_0/uW_0) = 6m$ times. Triangular determinants give the assertion. Finite alternating products preserve it. Quadratic degree and the uncancelled simple pole of the full local zeta exclude the targets as before. □

This theorem is sharp: $W_0 = \Lambda_R$ has vertex rank six. It requires neither left projectivity nor left flatness of a general kernel. It excludes this coefficient category, not arbitrary semilinear correspondences, twisted determinant lines or infinite spectral constructions. The extended edition identifies the full character-orbit repair with the hereditary normalization and explains why Schur commutator cancellation is insufficient. All determinant multiplicities here are relative to the stated $k$-valued base; restricting a rational quadratic factor to the former $\mathbb{Q}$-base squares it. No arithmetic descent or global positive determinant has been recovered by this ramification.

The recent NCG literature supplies concrete comparison targets, with strong hypotheses. Connes–Consani construct finite abelian covers of the adelic Riemann sector with prime-orbit length $\log p$ and arithmetic Frobenius monodromy [53, Theorem3.12]. The prolate/local-factor and moment constructions [54, 55] give particular finite-prime analytic models, not a uniform positive comparison for all completed $L$-functions. The zeta-spectral-triple determinant theorem [56, Theorem5.10] assumes a specified restricted Weil form, simple even bottom eigenfunction and boundary normalization. Its infinite comparison is still an additional step. The distributional real-zero theorem [57, Theorem6.1]

concerns a particular interval quadratic form, essential self-adjointness on its trigonometric core and an even simple ground state; a Fock vacuum alone verifies none of those arithmetic identifications. Deninger's periodic-packet construction and real-cohomology obstruction [60, 62] further specify, rather than remove, these issues. Full theorem-level reading coverage and exact versions are recorded alongside this draft.

A complete approach must construct arithmetic descent and prime actions on actual order/derived or analytic objects, prove determinant-line and local-factor compatibility, control regularization and conductor/gamma normalization, and establish a positive realization for that *same* function. Finite-field cohomological trace formulas and Deligne purity supply the classical model [20, 21]; alternating duality alone is not positivity or purity. In a number-field construction these are obligations, not imported conclusions.

For a sharply testable final analytic target, put $F_\chi(z) = \Lambda(1/2 + iz, \chi)$. Suppose entire functions $h_m(z)\det(zI - H_m)$, with $H_m$ finite self-adjoint and $h_m$ entire nowhere zero, converge locally uniformly on $\mathbb{C}$ to this nonzero completed function. Hurwitz on every disk disjoint from $\mathbb{R}$ then proves all its zeros real, hence the Dirichlet critical-line assertion. Establishing this convergence with independently constructed $H_m$ is exactly the missing load-bearing claim; a rank-one encoding or a reformulation does not establish it. Hecke, Dedekind and Artin extensions must state their analytic scope anew.

### 6.5 Whole components, derived endpoints and determinant relations

The full node lattice category has precisely the three indecomposables $X, \Lambda, Y$ and arrows $X \to \Lambda \to Y$, $Y \to \Lambda \to X$. Its stable category has only $X, Y$, no irreducible arrows, and suspension exchanging them; its AR triangles have zero middle term. For the polynomial node at a smooth support $x = \lambda + t$, $y = 0$, $\lambda \neq 0$, the modules $T_r = k[t]/t^r$ form an ambient homogeneous tube:

$$0 \longrightarrow T_r \xrightarrow{(\pi,t)} T_{r-1} \oplus T_{r+1} \xrightarrow{(t,-\pi)} T_r \longrightarrow 0, \quad T_0 = 0, \quad \tau T_r = T_r.$$

The shift changes the derived placement, not the quasi-length. These nonzero-central-support objects do not extend to continuous finite-length modules of the arrow-adic completion. At its singular point even the candidate pure-branch tube fails ambient almost-splitness: $T_2^y \to k$ cannot factor through $T_2^x \to k$, since every intervening map lands in the $x$-socle, killed by the last map.

The nine HLS strings above sit in an entire intrinsic quotient component $T_{p,q}$, $(p, q) \in \mathbb{Z}^2$, with arrows increasing one coordinate and $\tau T_{p,q} = T_{p-1,q-1}$ [46]. Its central object $E = M(ab)$ has infinite projective dimension: restriction to the first corner gives the dual-number simple with its infinite loop resolution, while restricted projectives are free or zero. Its module AR sequence therefore yields a distinguished derived triangle but not a derived AR triangle ending at $E$. Ringel's four-quarter wind-wheel tile [43], with infinite-word diagonals and mixed center, is a different category's quilt; it is not silently identified with that HLS component or the Belyi surface.

There is no general equation "AR mesh equals polynomial semi-invariant syzygy." The intrinsic $k(1 \to 2)$ quotient of the infinite Chebyshev algebra has AR row $0 \to S_2 \to E \to S_1 \to 0$ and a free presentation determinant $z$ on dimension $(1, 1)$, so its semi-invariant ring $k[z]$ has no generator relation. A genuine framed elliptic-square stratum has $Z = [x_1, x_2, y_1, y_2] \in \mathrm{Mat}_{2\times 4}$ with six minors and relation

$$p_{12}p_{34} - p_{13}p_{24} + p_{14}p_{23} = 0.$$

These minors have the same determinant character for the target $\mathrm{GL}_2$ with framed source; they are not all semi-invariants for the full unframed vertex group. In contrast, the nodal lattice AR row gives a genuine relative determinant identity over $R$: multiplication by $a + bx$ on the basis $(y, 1)$

has matrix $\left(\begin{smallmatrix} a & -b \\ 0 & a+ub \end{smallmatrix}\right)$ and characteristic factor $(1-Ta)(1-T(a+ub))$. The endpoints have infinite order-projective dimension but are $R$-perfect.

Finally, the three-vertex theta algebra has no unital surjection onto a six-vertex radical-square-zero circular algebra: its semisimple image has at most three primitive idempotent images. Duplicating face vertex spaces instead gives an exact faithful unfolding which is not full; it does not inherit quotient or AR-transfer claims. Zwara Hom-control means that the Hom-dimension defect is bilinear on $K_0$, not merely a covering direct-sum formula [72, 73]. The book supplies explicit fixed-dimension counterexamples, corrected circular-complex incidence resolutions, complete drawn component windows and all category hypotheses. These additions do not supply the arithmetic determinant comparison.

# 7 A genuine theta covering and its categorical limits

Use column matrices, hence covariant quiver representations (left modules); opposite categories translate the right-module convention in [74]. Let $A$ be the infinite elliptic theta dessin algebra: vertices $i \in \mathbb{Z}/3$, arrows $a_i, b_i : i \to i+1$, and all mixed length-two paths zero. For the arrow ideal $J$ and $L \geq 2$ put $B_L = A/J^L$. This is a finite string quotient of our actual infinite gentle algebra; for $L > 2$ its length-$L$ relations prevent treating it as quadratic gentle.

**Proposition 7.1** (A locally bounded integer cover)**.** *The category $\widetilde{B}_L$ with vertices $r \in \mathbb{Z}$, two arrows $a_r, b_r : r \to r+1$, and the same mixed and length-$L$ zero relations, has orbit $B_L$ under $r \mapsto r+3$. For finite-support representations, push-down $F$ is exact and*

$$\operatorname{Hom}_{B_L}(FM, FN) \simeq \bigoplus_{g \in \mathbb{Z}} \operatorname{Hom}_{\widetilde{B}_L}(M, T_g N), \qquad (T_g N)_r = N_{r-3g}.$$

*Only finitely many summands are nonzero. In general $F$ is neither full, dense, nor Hom-controlled in Zwara's sense.*

*Proof.* A nonzero path is constant-branch with length less than $L$. Thus the incoming and outgoing total Hom dimensions at a cover vertex are $2L-1$, and $\dim B_L = 3(2L-1)$. Paths and relations lift uniquely; the deck action is free and has the stated orbit. Push-down sums spaces of equal residue. A morphism decomposes uniquely into blocks with target lift index minus source index equal to $-3g$. The arrow commutation equations preserve that difference, proving the formula; finite supports bound the possible differences. Direct sums of vector spaces preserve exact sequences. The following explicit witnesses prove the three failures. □

For $L = 5$, let $P$ be the cover interval on $0, 1, 2, 3, 4$ with all its $a$-maps one and $b$-maps zero, and let $S$ be the sum of the five simples at the same vertices. They have the same $K_0$ class, but

| | $\dim \operatorname{End}_{\widetilde{B}_5}$ | $\dim \operatorname{End}_{B_5}(F(-))$ | defect |
|---|---|---|---|
| $S$ | 5 | 9 | 4 |
| $P$ | 1 | 2 | 1 |

Indeed the pushed-down simple multiplicities are $(2, 2, 1)$, whereas $\operatorname{End}_{B_5}(FP) = kI \oplus kX^3$ for its single nilpotent five-step chain $X$. The extra endomorphism $X^3$ proves non-fullness; the unequal defects at the same class exclude a bilinear $K_0$ defect. To use Zwara's finite-algebra definition literally, restrict the source to the finite convex interval $[0, 4]$, then extend by zero before push-down. This exact functor has the same endomorphism spaces and $K_0$ comparison.

**Proposition 7.2** (An indecomposable module outside this cover image)**.** *Over a characteristic-zero field, $B_3$ has a nine-dimensional indecomposable representation which is not the push-down of any finite-support $\widetilde{B}_3$ representation.*

*Proof.* Label its basis by $0, \dots, 8$ with vertex residues $(0, 1, 0, 1, 2, 1, 2, 0, 2)$, and use matrix units $E_{ij}$ acting on columns. Set

$$X = E_{10} + E_{32} + E_{43} + E_{65} + E_{76} + E_{08},$$
$$Y = E_{12} + E_{45} + E_{78}.$$

Their restrictions between adjacent vertex spaces give the $a_i, b_i$ maps. They satisfy $XY = YX = X^3 = Y^3 = 0$, hence every length-three path is zero. The endomorphism ring preserving vertex spaces is $kI \oplus \mathfrak{r}$, where

$$\mathfrak{r} = \langle E_{02} + E_{13}, E_{15}, E_{35} + E_{46}, E_{48}, E_{68} + E_{70}, E_{72} \rangle, \qquad \mathfrak{r}^3 = 0.$$

Solving the arrow commutation equations gives exactly these seven basis elements. The ring is local: an element $cI + n$ is invertible exactly when $c \neq 0$, by the finite geometric series. It has no nontrivial idempotents, so the representation is indecomposable.

A finite-support lift would define an Euler operator $D$ acting by $r$ on the summand at lift vertex $r$. Then $[D, X] = X$ and $[D, Y] = Y$. Every displayed nonzero arrow is the unique nonzero entry in its row and column in its own matrix. Its commutator entry is therefore $D_{ii} - D_{jj}$. Add the six $X$ equations and subtract the three $Y$ equations. Each diagonal term cancels, leaving $0 = 6 - 3 = 3$, a contradiction. This argument allows arbitrary $D$, not just a diagonal matrix in the displayed basis. □

*Corollary* 7.3 (Higher-genus and derived non-density)*.* For the clean dessin with rotations $\sigma = (0 \cdots m-1)(m \cdots 2m-1)$ and $\alpha = \prod_{i=0}^{m-1}(i\, i+m)$, odd $m \geq 3$, and genus $(m-1)/2$, the analogous $A_m/J^3$ has a $3m$-dimensional representation outside its integer-cover image in characteristic zero. Its bounded-derived push-down is not dense either.

*Proof.* Use basis $v_0, \dots, v_{3m-1}$, assign vertices $i, i+1, i$ to $v_{3i}, v_{3i+1}, v_{3i+2}$ modulo $m$, and set

$$Xv_{3i} = v_{3i+1}, \quad Xv_{3i+2} = v_{3i+3}, \quad Yv_{3i+2} = v_{3i+1},$$

with indices modulo $3m$ and all other values zero. Again mixed products and cubes vanish. An Euler operator would yield $2m - m = m = 0$ by the same telescoping calculation, impossible in characteristic zero. If a bounded finite-support complex pushed to this module in degree zero, exactness would make its zeroth cohomology a module lift, a contradiction. □

The source's derived result is a *precovering* in the stated bounded categories [74, Theorem 4.28]; density remains an additional condition. Neither this result nor the GLS equivariant classification [75] identifies our completed order with a GLS algebra. Graded surface/Fukaya models [76] likewise require their own dg category, stops and grading. These categorical tools do not construct the independent positive arithmetic determinant.

# 8 Verification and research frontier

The two nodal/invariant Lean files construct the node as an actual subring of $k[[X]] \times k[[X]]$ with equal constants, prove its annihilators, regular common parameter, augmentation, alternating

projective resolution and a bijective pair of power-series coordinates. A second file proves CK1 collapse under the stated permutation/coverage hypotheses, determinant base change, actual $2 \times 2$ symmetric-square functoriality and relation preservation, a Schur character identity and the fixed-parameter lattice failure. These thirty printed theorem types have only standard Lean foundations. They do not formalize general dessins, abstract Ext/AR/Schur–Weyl, C* exactness, Fredholm determinants, arithmetic descent or GRH. Earlier formal layers remain preserved with their literal scopes.

Exact-rational diagnostics independently check the displayed dessins, path identities, Schur and Young examples and the nine-object grid. These finite checks supplement the proofs, rather than prove infinite spectral or arithmetic assertions. The next arithmetic target is an exact descent/Frobenius-compatible construction on the labelled order/Fock data, with a determinant-preserving positive comparison. The local Schur kernel and the simple degree spectrum alone do not supply it. All source clones and earlier research are retained; the earlier general focused PDFs are archived at the preceding commit.

Ten further declarations in `DessinPullbackComponents.lean` check actual matrix-unit identities, nonmultiplicativity of raw diagonal extraction, impossibility of a unital two-state matrix residue into a nonzero commutative ring, multiplicativity on diagonal fibers, and genuine equal-image pullback subring/projection components. The analytic suite then contained187printed declarations at unchanged pins. The circle-function C* model, functional calculus, generation, graph uniqueness and K-theory remain prose/external-theorem arguments, not hidden assumptions of these component checks.

Sixteen additional declarations in `ThetaJacobi.lean` check genuine finite-field MulChar Jacobi conjugation, norm and rescaling with explicit nontriviality hypotheses; cubic–quadratic noncancellation; the cyclotomic multiplication determinant and positive Gram identity; and actual theta cellular boundary, homology-inclusion, restriction and rational chain-map matrices. The full analytic suite now contains203declarations. The point-count proof, deck/Kummer comparison, cohomological trace interpretation and global arithmetic comparison are not thereby fully formalized.

The final three `ThetaJacobi.lean` declarations verify the actual conjugation matrix identities and the real two-dimensional inert-prime determinant obstruction. They do not assume any unknown zero location or exclude complex/enlarged spectral realizations.

`ThetaProjectiveMultiplicity.lean` adds five declarations. Arbitrary upper-block matrices with six identical diagonal blocks have the sixth-power determinant and characteristic-factor formulas. A literal power-series constant-coefficient map specializes matrix determinants, for each scalar parameter; the coefficientwise polynomial/order interface remains prose. A nonzero quadratic cannot be a rational-function sixth power, proved by polynomial degrees. Three genuine power series with product $X$ cannot have the cyclically equivalent residue-vanishing conditions forced by invertible rank-one arrow intertwiners. The derivation of the path blocks from the full theta order, projective decomposition, derived branch functor and deck intertwiners is not fully formalized. The complete analytic suite now has208printed declarations; no hidden arithmetic positivity assumption occurs.

`ThetaTameDescent.lean` adds twelve printed component declarations (and one auxiliary polynomial lemma). They use genuine formal-power-series rescaling and its coefficient operator, proving the coefficient action, invariant-coefficient vanishing and the tame operator's cube. Actual three-state matrices satisfy the clock/rotation commutator, comparison determinant and closed-fiber equations, weighted-cycle cube and full characteristic polynomial, noncommutation of a nonconstant rotation polynomial, trivial whole-column determinant actions, and the same-operator comparison for $vC$. The classification of all fixed series by residues, full order action and hereditary invariant algebra, stack interpretation, valuation/rank theorem, general coefficient-functor filtration and GRH are not

thereby fully formalized. No unknown positivity premise is assumed.

# 9 Why an absolute coprime pair-square estimate cannot close the short rows

Keep the original sextic family

$$M_u(D) = D^{-1/2} \sum_n \mu(n)\nu(n)W(Nn/D)\chi_n(u), \qquad \chi_n(u) = (u/n)_6,$$

with squarefree primary column ideals outside the fixed excluded set, their full annular coefficients (fix $W$ supported in $[1,2]$), and all nonzero Eisenstein element rows $Nu \le H$. The cubic symbol is $\chi_n^2$, not $\chi_n$. The following classical frame inequality [80] gives a new obstruction to one proposed estimate in this arithmetic program.

**Proposition 9.1** (Coprime pair-square obstruction)**.** *Let $A$ be an $R$-by-$N$ complex matrix with $|A_{ui}| = 1$, and put $K_{ij} = \sum_u A_{ui}\overline{A_{uj}}$. Then*

$$\sum_{i \ne j} |K_{ij}|^2 \ge NR(N-R).$$

*For the original character kernels, let $B_1(H,D)$ be the sum of $|K_H(n,m)|^2$ over all distinct coprime squarefree column ideals in $[D,2D]$. At $H = D^{3/4}$,*

$$B_1(H,D) \gg \frac{D^{11/4}}{(\log D)^2}.$$

*In particular the proposed sufficient input $B_1(H,D) \ll H^2 D^{\varepsilon}$ fails for each fixed $\varepsilon < 5/4$. This does not refute the signed Möbius mean square.*

*Proof.* Set $F_{uv} = \sum_i A_{ui}\overline{A_{vi}}$. Reordering four finite sums proves $\sum_{i,j} |K_{ij}|^2 = \sum_{u,v} |F_{uv}|^2$. The row diagonal contributes $RN^2$; the column diagonal contributes $NR^2$. Removing the latter and retaining the former gives the inequality.

Restrict the column pairs in the nonnegative sum $B_1$ to distinct prime ideals $p, q$ of norm in $[D,2D]$, outside the same fixed set. If $H < D$, no such prime divides a nonzero row element: divisibility would imply $Nu \ge Np$. Thus all original zero masks are exactly one on this ball, $|\chi_p(u)| = 1$, and the residual primitive finite character conductor is $pq$, of norm between $D^2$ and $4D^2$. All these pairs have common divisor1. The lattice count gives $R = (2\pi/\sqrt{3})H + O(\sqrt{H}+1)$. The classical unconditional fixed-field prime-ideal theorem gives $N \sim D/\log D$; this is an external analytic input, not a GRH hypothesis. At the stated endpoint $R = o(N)$, and the bound is therefore $(1-o(1))N^2R \gg D^{11/4}/(\log D)^2$. □

To see the exact failed step, write $c_n = \mu(n)\nu(n)W(Nn/D)$ and $C_D = \sum_n |c_n|^2 = O(D)$. Cauchy over coprime pairs bounds the absolute signed contribution by $D^{-1}C_D B_1^{1/2}$. The hoped-for pair-square upper bound would have closed this particular route, but is false. A large absolute pair-square mass is not a lower bound on the quadratic form with the fixed $c_n$. Nor is the prime part a positive summand of the full Möbius energy: prime–composite cross terms can cancel. The original low-gcd estimate and the positive arithmetic determinant bridge remain open.

## 10 A signed Möbius decomposition and its termwise obstruction

Keep the original squarefree good-ideal family, its fixed $\nu$, its full $W(Nn/D)$ and every zero-extended sextic symbol. In ideal Dirichlet convolution, write $\zeta_0(n) = 1$ for every good ideal and let $\delta$ be the convolution unit. For $1 \le U < D$ put $a_U = \mu \mathbf{1}_{Nn \le U}$ and $b_U = \mu - a_U$. The classical Vaughan algebra gives

$$\mu = 2a_U - a_U * a_U * \zeta_0 + b_U * b_U * \zeta_0. \tag{1}$$

Indeed $\mu * \zeta_0 = \delta$, and expansion of the last product proves the identity without an analytic assumption. This is a classical decomposition, not a new cancellation theorem; compare [87, Lemma4.1].

On squarefree $n$ all factors in $n = dc = abc$ are pairwise coprime. Define the actual, nonarbitrary coefficient functions

$$\beta_U(d) = \sum_{\substack{ab=d \\ Na, Nb \le U}} \mu(a)\mu(b), \qquad g_U(s) = \sum_{\substack{b \mid s \\ Nb > U}} \mu(b).$$

For squarefree $d$, $\beta_U(d)$ is $\mu(d)$ times the number of the displayed ordered factorizations. For $Ns > U$, $g_U(s) = -\sum_{b \mid s,\, Nb \le U} \mu(b)$. The two regrouped pieces are

$$I_u = D^{-1/2} \sum_{\substack{d,c \text{ squarefree} \\ (d,c)=1}} \beta_U(d)\nu(dc)\chi_d(u)\chi_c(u)W(NdNc/D),$$

$$II_u = D^{-1/2} \sum_{\substack{a,s \text{ squarefree} \\ Na, Ns > U,\, (a,s)=1}} \mu(a)g_U(s)\nu(as)\chi_a(u)\chi_s(u)W(NaNs/D).$$

Then $M_u = -I_u + II_u$. All ideals here remain outside the same fixed excluded set. The short $2a_U$ term vanishes because $U < D$ and $W$ is supported in $[1, 2]$. No phase or coprimality mask was removed.

**Proposition 10.1** (An all-cutoff separate-piece bound is false)**.** *Take the allowed principal $\nu$, $U = 1$, and a fixed nonnegative smooth $W$ supported in $[1, 2]$, bounded below by a positive constant on a closed subinterval of $(1, 2)$ of positive length. For each of the two pieces,*

$$\sum_{0 < Nu \le H} |I_u|^2 \gg_W \frac{D}{\log^2 D}, \qquad \sum_{0 < Nu \le H} |II_u|^2 \gg_W \frac{D}{\log^2 D} \quad (H \ge 1).$$

*Consequently a proposed bound $HD^{\varepsilon}$ for each piece at $H = D^{3/4}$ fails for $\varepsilon < 1/4$. In the exact original notation $H(HD)^{\varepsilon}$, it fails for $\varepsilon < 1/7$. This refutes an all-cutoff termwise strengthening, not the original signed mean square or a specified moving-cutoff argument.*

*Proof.* At $U = 1$, $a_U = \delta$, so $a_U * a_U * \zeta_0 = \zeta_0$ and $b_U * b_U * \zeta_0 = \mu + \zeta_0 - 2\delta$. On the annulus the two coefficients are 1 and $1 + \mu(n)$, respectively. On squarefree ideals the latter is zero or two. At the actual unit row $u = 1$, every original character equals one. A fixed-field prime-ideal theorem gives $\gg D/\log D$ good prime columns in the subinterval, so $I_1 \gg_W \sqrt{D}/\log D$. Fix a good prime ideal $q_0$. Products $q_0 p$ with varying distinct good primes in the rescaled subinterval have $\mu(q_0 p) = 1$. Their nonnegative contributions give $II_1 \gg_W \sqrt{D}/\log D$. Keeping that one row in each nonnegative energy proves the result. The prime-ideal theorem is a classical unconditional external input, with the field and $q_0$ fixed. □

The factorization does not shorten the primitive row conductor: for squarefree coprime $a, s$ it is $as$, not $a$ or $s$ separately. For two resulting columns $n, n'$ with common factor $h$, the oscillatory conductor is $nn'/h^2$ and a zero-preserving period is $nn'/h$. The signed low-gcd estimate and the independent positive arithmetic determinant comparison therefore remain open.

## 11 A full-mask Hecke expansion for the type-I piece

We now continue the signed decomposition through its actual squarefree inner sum. Fix $K = \mathbb{Q}(\sqrt{-3})$, the finite-order Hecke character $\nu$, and the same finite excluded prime set $S$, containing primes above 6 and the conductor of $\nu$. For $u \in \mathcal{O}_K \setminus \{0\}$, let $\psi_u$ be the primitive Hecke character induced by $\nu(n)(u/n)_6$ away from $S \cup \operatorname{supp}(u)$. It is defined before, and independently of, the type-I divisor $d$. The abelian Kummer extension $K(u^{1/6})/K$ constructs the sextic factor; multiplying by $\nu$ gives the stated character. This uses classical abelian reciprocity, not general Artin holomorphy. Write $\psi_{u,2}$ for the primitive character inducing its square and $\mathfrak{f}_u, \mathfrak{f}_{u,2}$ for the two primitive conductors. For a squarefree good ideal $d$ put

$$T = T(u,d) = S \cup \operatorname{supp}(u) \cup \operatorname{supp}(d), \qquad R_T = \prod_{p \in T} Np, \quad Q_u = |D_K| N\mathfrak{f}_u.$$

All primes in $T$ are deleted even if absent from either primitive conductor.

**Theorem 11.1** (Restoration and a uniform scalar contour bound)**.** *For $\Re s > 1$ the original type-I inner Dirichlet series is*

$$F_{u,d}(s) = \sum_{\substack{c \text{ squarefree} \\ (c,T)=1}} \frac{\psi_u(c)}{(Nc)^s} = \frac{L(s,\psi_u)}{L(2s,\psi_{u,2})}\, C_T(s), \tag{2}$$

$$C_T(s) = \frac{\displaystyle\prod_{p \in T,\ p \nmid \mathfrak{f}_u} (1 - \psi_u(p)(Np)^{-s})}{\displaystyle\prod_{p \in T,\ p \nmid \mathfrak{f}_{u,2}} (1 - \psi_{u,2}(p)(Np)^{-2s})}. \tag{3}$$

*It continues meromorphically to $\Re s > 1/2$, with only a possible simple pole at $s = 1$ when $\psi_u = 1$. In that case its residue is*

$$\rho_T = \frac{\operatorname{res}_{s=1} \zeta_K(s)}{\zeta_K(2)} \prod_{p \in T} (1 + (Np)^{-1})^{-1}.$$

*Fix $0 < \delta < 1/2$, $\varepsilon > 0$ and $W \in C_c^\infty((0,\infty))$ supported in $[1,2]$. Set*

$$\mathcal{M}_4(W) = \sum_{j=0}^{4} \sup_x |(x\partial_x)^j W(x)|.$$

*Uniformly for $Y \geq 1$, $u$ and $d$,*

$$\sum_{\substack{c \text{ squarefree} \\ (c,T)=1}} \psi_u(c) W(Nc/Y) = \mathbf{1}_{\psi_u = 1} \rho_T Y \widehat{W}(1) + O_{\nu,S,\delta,\varepsilon}\Big(\mathcal{M}_4(W) Y^{1/2+\delta} Q_u^{1/4-\delta/2+\varepsilon} R_T^{\varepsilon}\Big). \tag{4}$$

*Here $\widehat{W}(s) = \int_0^\infty W(x) x^{s-1}\, dx$. The continuation and functional equation of degree-one Hecke functions are classical external inputs; the contour argument is given below.*

At a good prime with $v_p(u) = 3$, $\psi_u$ is ramified while its square is unramified; the second product in (3) must restore a factor invisible to squaring the zero-extended primitive value. At $v_p(u) = 6$

both characters can forget the prime, but the original row still vanishes on columns divisible by it. In particular, for principal $\nu$, $u=\pi^3$ and $u=\pi^6$ at $N\pi=7$, the deleted local factors are

| | primitive ratio at $s=1$ | restoration | full masked factor |
|---|---|---|---|
| $v_p(u)=3$ | 48/49 | 49/48 | 1 |
| $v_p(u)=6$ | 8/7 | 7/8 | 1 |

These are exact local calculations, not numerical evidence about zeros.

For $1\le U$, $U^2\le D$, insert $Y=D/Nd$ into (4) and retain the actual $\beta_U(d)$ and outer $\nu(d)\chi_d(u)$. Classical tame Kummer conductors give $Q_u\ll_\nu Nu$. Thus the normalized $I_u$ equals its displayed principal-residue sum plus an error $E_u$ with

$$|E_u|\ll_{\nu,S,\delta,\varepsilon}\mathcal{M}_4(W)D^{\delta}U^{1-2\delta+\varepsilon}(Nu)^{1/4-\delta/2+\varepsilon}. \tag{5}$$

The resulting elementary energy envelope is only

$$\sum_{0<Nu\le H}|E_u|^2\ll\mathcal{M}_4(W)^2D^{2\delta}U^{2-4\delta+\varepsilon}H^{3/2-\delta+\varepsilon}. \tag{6}$$

It does not prove the short-row target of order $H$ up to small losses. This is an insufficient upper bound, not a lower bound on the actual error or a counterexample to coefficient-sensitive averaging. The original signed mean square and positive determinant bridge stay open.

## 12 Actual renormalization retains the effective action

A compression score does not specify an RG transformation. Following the real-space formulation of Wilson and Kogut [81], we specify the eliminated variables, effective action, normalization and scale map. Our exact finite decimation is distinct from an approximate tensor-network truncation such as Levin–Nave TRG [82].

**Definition 12.1** (Finite classical RG step)**.** Let the fine configuration be $(s,t)\in S\times E$, with finite nonempty sets $S,E$, and dimensionless log Boltzmann weight $V(s,t)=-\beta H(s,t)$. Define

$$(\mathcal{R}V)(s)=\log\sum_{t\in E}e^{V(s,t)},\qquad Z(V)=\sum_{s,t}e^{V(s,t)}.$$

The full action includes its additive constant. A truncation or a learned approximation is a further operation with its own error, not this identity.

**Proposition 12.2** (Partition, observables and composition)**.** *One has $Z(\mathcal{R}V)=Z(V)$, and the normalized expectation of every observable depending only on $s$ agrees in the two descriptions. Eliminating $E_1$ and then $E_2$ equals eliminating $E_1\times E_2$ at once, with the same tensor labels and inverse temperature.*

*Proof.* Exponentiating the definition gives $e^{\mathcal{R}V(s)}=\sum_t e^{V(s,t)}$. Sum over $s$, optionally multiplying by an observable $O(s)$, and divide both sides by the same strictly positive $Z$. Reordering finite sums proves the composition statement. No projection onto pair interactions is used. □

For an even homogeneous periodic Ising chain, write the transfer matrix as $T_{ab}=\exp(C+Kab+h(a+b)/2)$, $a,b\in\{-1,1\}$. Its partition function is $\operatorname{tr}T^N$. Summing alternate spins replaces $T$ by $T^2$ and $N$ by $N/2$; the physical spacing doubles. Writing $P=T^2$, its positive entries determine

$$K'=\tfrac14\log\frac{P_{++}P_{--}}{P_{+-}^2},\quad h'=\tfrac12\log\frac{P_{++}}{P_{--}},\quad C'=\tfrac14\log(P_{++}P_{--}P_{+-}^2).$$

These three parameters reproduce every entry of $P$, so no normalization is lost. At $h = 0$, $\tanh K' = \tanh^2 K$ and the eigenvalue-ratio correlation length of the homogeneous extension in blocked units halves. Its product with the spacing stays fixed. The fixed points in $t = \tanh K$ are $t = 0, 1$; the latter is zero temperature, not a positive-temperature transition. Odd dessin examples below use the full-action definition, not this even-chain shortcut.

### 12.1 Genuine dessin incidence models

For odd $m$, the clean dessin $y^2 = x^m - 1$, $\beta = x^m$ has two black vertices, $m$ white vertices and one face; its genus is $(m-1)/2$. Its incidence graph is $K_{2,m}$ with $2m$ clean edges. Its infinite locally gentle order quiver instead has $m$ vertices and completed DVR-rank $2m^2$. We place physical spins $B_+, B_-, s_0, \ldots, s_{m-1}$ on the incidence vertices; this is a specified finite spin model, not a faithful representation of that infinite algebra or order. The abstract incidence graph alone does not recover the ribbon embedding or genus.

With $\eta_j = (-1)^j$ and fixed nonuniform fields $h_j$, choose

$$V = J_+ B_+ \sum_j s_j + J_- B_- \sum_j \eta_j s_j + b B_+ - \tfrac{3}{5} b B_- + \sum_j h_j s_j.$$

Summing the two black spins gives the *entire* effective action

$$V'(s) = \sum_j h_j s_j + \log\left[2\cosh\left(b + J_+ \sum_j s_j\right)\right] + \log\left[2\cosh\left(-\tfrac{3}{5} b + J_- \sum_j \eta_j s_j\right)\right].$$

Every one of the $2m$ physical edges is present, although its couplings belong to a tied three-parameter family. The Walsh expansion $V'(s) = \sum_{A \subseteq \{0,\ldots,m-1\}} c_A \prod_{j \in A} s_j$, with $c_A = 2^{-m} \sum_s V'(s) \prod_{j \in A} s_j$, is unique by Boolean orthogonality. It retains the constant $c_\varnothing$ and genuinely generated higher-body interactions. Our genus-two and genus-four examples use $m = 5, 9$ and continue by exact nested marginalization. Dropping all terms of degree greater than two is displayed as a measured approximation.

### 12.2 Quantum thermal effective Hamiltonians

For a finite Hermitian Hamiltonian $H$ on nonzero finite-dimensional $\mathcal{H}_S \otimes \mathcal{H}_E$ and $\beta > 0$, define

$$W' = \operatorname{Tr}_E e^{-\beta H}, \qquad H_{\text{eff}}(\beta) = -\beta^{-1} \log W'.$$

The matrix $W'$ is strictly positive: for $v \neq 0$, sum the strictly positive quadratic forms of $v \otimes e_j$ over an orthonormal basis of $\mathcal{H}_E$. Its Hermitian spectral logarithm is therefore defined. The identity $e^{-\beta H_{\text{eff}}} = W'$ proves partition and retained *equal-time* observable preservation by the partial-trace property. Repeated traces compose. The full Pauli expansion, including its identity coefficient, is retained. It is generally nonlocal and depends on $\beta$; this statement does not preserve all fine real-time dynamics or supply a local zero-temperature/thermodynamic RG fixed point.

`FiniteRG.lean` checks nine actual finite scalar marginal, exponential-action, partition, observable, composition, error-sandwich and transfer-trace statements. The quantum spectral-log and complete thermodynamic/field-theory interfaces are not formalized by these lemmas. These physics constructions give no arithmetic determinant comparison or new GRH conclusion.

## 13 Linear relations and functorial branch filtrations

The original linear-relations chapter mixes graphs, partial relations and stable kernels. We keep these different objects separate. For a field $k$, a linear relation is a subspace $C \subseteq V \oplus V$; its inverse swaps coordinates and $DC = \{(x,z) : \exists y, (x,y) \in C, (y,z) \in D\}$. The full graph $\Gamma_T = \{(x, Tx) : x \in V\}$ always has dimension $\dim V$, including pairs with zero output. In particular $\Gamma_T^n(0) = 0$, whereas $(\Gamma_T^{-1})^n(0) = \ker T^n$. Thus the source's $\kappa(\Gamma_T) = \bigcup_n \Gamma_T^n(0)$ is zero, not the ordinary stable kernel.

For odd $m \geq 3$, use the actual clean dessin of $y^2 = x^m - 1$, $\beta = x^m$, on the compact smooth curve, in characteristic zero. Its genus is $(m-1)/2$; its infinite locally gentle algebra $A_m$ has vertices $i \in \mathbb{Z}/m$, arrows $a_i, b_i : i \to i-1$, and mixed consecutive branch products zero. Below paths act in their direction of travel on left representations over any coefficient field $k$. Let $\mathcal{E}_m$ be the full category of finite-dimensional representations with every $b_i = 0$ and $a_i$ invertible for $i \neq 0$.

**Theorem 13.1** (A concrete relation category on an infinite dessin algebra)**.** *The category $\mathcal{E}_m$ is equivalent to finite-dimensional pairs $(V,T)$ with intertwining maps. A normal form $N_m(T)$ puts $V$ at each vertex, $a_0 = T$, and all other $a_i = 1$. For $n = \dim V$,*

$$V = K(T) \oplus I(T), \qquad K(T) = \ker T^n, \quad I(T) = \operatorname{im} T^n.$$

*These summands and their kernel/image filtrations are functorial. The first restriction is nilpotent and the second invertible. The central full-turn parameter $u$ acts as $T$. A finite $N_m(T)$ extends continuously to the $u$-adically completed order precisely when $T$ is nilpotent. Such modules are torsion, not nonzero order lattices.*

*Proof.* Transport all vertex spaces to vertex zero along the invertible arrows; the remaining map is the full-turn monodromy $T$. Arrow-compatible maps become exactly $hT = T'h$. Finite kernel/image chains stabilize by step $n$. If $v = T^n w$ and $T^n v = 0$, then $w \in \ker T^{2n} = \ker T^n$, hence $v = 0$; rank-nullity proves the direct sum. Stability gives the two restrictions, and $hT^r = T'^r h$ proves functoriality. Continuity on a finite-dimensional discrete space means some $u^r$ kills it. Conversely nilpotence makes every formal series act by a finite sum. This proves the completion assertion. □

Over an algebraically closed $k$, the indecomposable branch pairs are $J_r(\lambda)$. The value $\lambda = 1$ is permitted for a polynomial band; it is not a forbidden deformation parameter. The nilpotent blocks are strings, but are not all rigid: $N_m(J_2(0))$ is a nonsplit self-extension of $N_m(J_1(0))$. The Hom dimension between two such Jordan pairs is $\min(r,s)$ for equal eigenvalues and zero otherwise. These assertions concern this specified full subcategory, not all infinite modules.

If $L$ transports once along the $a$-arrows on $\bigoplus_i V$, then

$$\det(1 - zL) = \det(1 - z^m T). \tag{7}$$

Eliminating the identity arrow blocks leaves the final monodromy block; the eliminated identity blocks have determinant one. For $J_r(\lambda)$ the answer is $(1 - \lambda z^m)^r$. Nilpotent strings therefore contribute one, and arbitrary bands have no forced unit-circle zero location. This determinant is an actual cyclic-quiver invariant; it is not the missing arithmetic positive determinant comparison.

## 14 A kernel-checked extension over the infinite dessin algebra

The preceding branch example can be realized on every element of the infinite algebra, rather than verified only on sampled arrow matrices. Use the complex genus-two clean dessin $y^2 = x^5 - 1$,

$\beta = x^5$, and any coefficient field $k$. Put $H \subseteq M_5(k[t])$ equal to the matrices whose reduction modulo $t$ is upper triangular. Let

$$B = H \times_{k^5} H,$$

where each map to $k^5$ takes the constant diagonal. Equivalently, $B$ is generated inside $M_5(k[t])^2$ by

$$e_i = (E_{ii}, E_{ii}), \quad a_i = (w_i E_{i-1,i}, 0), \quad b_i = (0, w_i E_{i-1,i}), \qquad w_0 = t, \quad w_i = 1 \ (i \neq 0).$$

Indices are modulo five. Matrix multiplication acts on left column modules. The path-normal-form proof below identifies $B$ with $A_5$; this is a polynomial algebra, not an arrow-truncated finite quotient.

**Theorem 14.1** (An ambient module extension)**.** *Let $D = k[\epsilon]/(\epsilon^2)$, $M_1 = k^5$ and $M_2 = D^5$. For $(C(t), D'(t)) \in B$, act on $M_1$ by $C(0)$ and on $M_2$ by $C(\epsilon)$. These are unital $B$-modules, and*

$$0 \longrightarrow M_1 \xrightarrow{i} M_2 \xrightarrow{p} M_1 \longrightarrow 0, \qquad i(x) = \epsilon x, \quad p(x + \epsilon y) = x,$$

*is exact and nonsplit. Moreover $B$ is infinite dimensional over $k$. The assertions hold in every characteristic, including two.*

*Proof.* Evaluation of polynomials gives both actions. Multiplying $C(\epsilon) = C_0 + \epsilon C_1$ proves $B$-linearity of $i, p$; their kernel and image follow coordinatewise. If $s$ were a $B$-linear section, the constant coefficient of $(se_0)_0$ would be one. But $a_0$ kills $M_1$, while $(a_0 se_0)_4 = \epsilon(se_0)_0$ has nonzero $\epsilon$ coefficient, a contradiction. Finally the permitted turn $c = a_1a_2a_3a_4a_0$ has first component $tE_{00}$. The unital algebra map $k[t] \to B$, $f \mapsto f(c)$, is injective: its $(0,0)$ first-component entry is $f(t)$. Thus $B$ cannot be finite dimensional. □

In the preceding notation these modules are $N_5(J_1(0))$ and $N_5(J_2(0))$. They are finite torsion modules over an infinite algebra; they are not nonzero lattices over its completed surface order. The nonsplit sequence defines a nonzero Yoneda extension class in the ambient module category. No assertion that it is almost split in that entire category is needed or inferred. In particular it does not supply an arithmetic spectral determinant or a zero-location theorem.

**Theorem 14.2** (Unbounded normal-image faithfulness)**.** *In $M_5(k[t])^2$, index the stationary images by $i$ and the positive images by $(b, i, j, q)$, where $b \in \{a, b\}$, $0 \leq i, j < 5$, $q \geq 0$ and $i < j$ or $q > 0$. The positive image is $t^q E_{ij}$ in factor $b$, with zero in the other factor. Its path length is $5q + j - i > 0$. All these images, including the shared stationary images, are linearly independent over every field. Every finite normal expansion has uniquely recoverable coefficients, without a bound on degree or path length.*

*Proof.* Read a stationary coefficient from the first factor's constant diagonal; read a positive coefficient from the indicated factor, entry and degree. The index restriction ensures that no positive image contributes to a constant diagonal. These linear selectors are biorthogonal to the images, so applying each selector to a finite expansion recovers its coefficient. □

This theorem is now kernel checked for actual polynomial matrix pairs. The membership and spanning steps are supplied by the next theorem. Comparison with a separately constructed path quotient remains a distinct algebraic formalization task.

**Theorem 14.3** (Normal coordinates in the generated algebra)**.** *Let $A \subseteq M_5(k[t])^2$ be the algebra generated by the displayed vertices and arrows, and let $\mathcal{K}$ be the normal index set of Theorem 14.2. Every normal image belongs to $A$. The actual $A$-valued finite expansion*

$$E : k^{(\mathcal{K})} \longrightarrow A, \qquad c \longmapsto \sum_{x \in \mathcal{K}} c_x v_x$$

*is a linear isomorphism, whose inverse is* $(E^{-1}a)_x = \lambda_x(a)$*. Thus the normal images form a basis of the infinite generated algebra, not merely an independent family in its ambient matrix space.*

*Proof.* Write $P_b(i,j)$ for the shortest positive branch walk from $j$ to $i$, using a full turn when $i = j$. Its image is $t^d E_{ij}$ in branch $b$, where $d = 0$ if $i < j$ and $d = 1$ otherwise. The actual arrow product $P_b(i,i)^n P_b(i,j)$ has image $t^{n+d}E_{ij}$ for every $n \geq 0$. Admissibility gives $q \geq d$, so $n = q - d$ proves membership.

The normal span contains the generators and $1 = \sum e_i$ and is multiplicatively closed: stationary factors select endpoints; positive factors either multiply to zero or to $t^{q+r}E_{il}$ in the same branch. When $q + r = 0$, the two strict upper inequalities imply $i < l$, so this output slot is still permitted. Adjoin induction gives spanning of $A$. The previous coefficient selectors prove uniqueness and the inverse formula. □

All membership, spanning and finite-expansion assertions in this theorem are now Lean checked over arbitrary fields. The formal linear equivalence is not asserted to be an algebra map from an independently defined path quotient, nor a statement about completed orders, derived categories or GRH.

## 15 Learned fields, width limits and quantum geometry

Learned functions enter this program as candidate generators, physical surrogates and control laws. A network may suggest an arithmetic metric, operator or gluing datum; a successful fit does not establish its positivity, analytic domain or identification with a completed $L$-function. Those are independent mathematical obligations. This section gives explicit interfaces and executed examples rather than treating "quantum", "tropical" or "infinite width" as a substitute for that comparison.

**Proposition 15.1** (A specified quantum-feature width limit)**.** *Let* $x \in \mathbb{R}^d$*,* $w \sim N(0, I_d)$ *and* $b \sim \mathrm{Unif}[0, 2\pi]$ *be independent, and set* $\phi_{w,b}(x) = \sqrt{2}\langle Z\rangle_{R_Y(w\cdot x+b)|0\rangle} = \sqrt{2}\cos(w \cdot x + b)$*. For independent features,*

$$K_m(x,y) = m^{-1}\sum_{r=1}^{m} \phi_r(x)\phi_r(y) \longrightarrow K(x,y) = e^{-\|x-y\|^2/2}$$

*almost surely on every specified finite set of pairs. Every* $K_m$ *is a positive semidefinite Gram kernel. For a readout* $f_a(x) = m^{-1/2}\sum_r a_r\phi_r(x)$ *with only a trained, its tangent kernel is exactly* $K_m$*. These statements do not assert the same limit for a jointly trained finite quantum transformer.*

*Proof.* Uniform-phase integration gives $\mathbb{E}_b[2\cos(u + b)\cos(v + b)] = \cos(u - v)$. The Gaussian characteristic function then gives $K$. Products are bounded by two, so the strong law applies to each pair and hence any finite union. For any finite coefficient vector $c$, the Gram quadratic form is $m^{-1}\sum_r |\sum_i c_i\phi_r(x_i)|^2 \geq 0$. Differentiating with respect to $a_r$ gives the final assertion. □

The frozen-kernel squared-loss flow has residual $r(t) = \exp(-K_m t)r(0)$ when the loss normalization is absorbed into time. Null directions do not decay. A complex feature space instead uses the Hermitian kernel $\mathbb{E}[\phi(x)\overline{\phi(y)}]$. A finite coefficient field has neither this Gaussian law nor an ordered positive Hilbert metric; changing algebraic coefficients is a separate operation.

**Proposition 15.2** (Tropical limit and a cancellation boundary)**.** *For real* $s_1, \ldots, s_n$ *and* $\tau > 0$*,*

$$0 \leq \tau \log \sum_j e^{s_j/\tau} - \max_j s_j \leq \tau \log n.$$

*An attention maximum therefore has a controlled log-sum-exp approximation. It does not by itself define a tropicalization of a variety or an HMS functor. In a valued coefficient field, valuation of a sum can increase by cancellation.*

*Proof.* After subtracting $M = \max s_j$, the exponential sum lies between one and $n$. For the second statement, both $1+T$ and $-1+T$ have valuation zero in $\mathbb{Q}((T))$, whereas their sum $2T$ has valuation one. In characteristic two that sum is zero. Thus replacing every addition by its tropical minimum without tracking leading coefficients loses information. □

Kontsevich–Soibelman constructions require valued analytic and integral-affine gluing data; appropriate gentle/Fukaya comparisons require specified categories, gradings, stops and smoothness hypotheses [92, 91, 93]. Learned coefficients can propose such data, but the compatibility equations and arithmetic comparison still need proofs.

**Executed physical examples.** On the actual clean genus-two dessin, put seven quantum spins on its $K_{2,5}$ incidence graph and compute a normalized Gibbs state. Exact white-spin partial trace gives a two-spin quantum RG marginal. Twelve RTX4090 runs train four-qubit-feature diffusion and flow-matching models for its smoothed $Z$-measurement law: two objectives, entangled/product circuit controls, and three seeds. They have 5,810 nominal parameters and 2,000 steps each. A separate four-qubit variational purification flow learns the full known density interpolation, with 48 controls. Its minimum fidelity on a 33-point grid is 0.999903; this is not a uniform theorem or universal noise inverse. The matched controls show no consistent entanglement advantage.

Actual cuStateVec/cuTensorNet execution also covers 7-, 11- and 23-qubit finite circuits on genus-two, genus-four and genus-ten dessin incidence graphs. The 23-qubit case has 8,388,608 amplitudes and 447 gates. These larger circuits are untrained; the small cases have independent NumPy comparisons, while the largest has normalization and inverse-circuit checks. The exact environment and all failed/preliminary records are preserved.

**Geometric and hardware interfaces.** For the contracted $m$-edge dessin, the graph fundamental group is $F_{m-1}$, and its universal graph cover is the $m$-regular tree. The theta case $m = 3$ has a three-regular cover, distinct from the four-regular Cayley graph of $F_2$. Neither is the coefficient field $\mathbb{F}_2$ or the universal cover of the filled compact surface. Explicit path modules over the infinite genus-two algebra give Hom, Ext and Tor computations; Frobenius pants tensors give exact TQFT gluing, with a heralded physical implementation. CHSH gives a finite game benchmark, distinct from the unrestricted correlation separation in MIP*=RE [99]. Three executable tropical hardware references and a source-pinned chip-design survey complement the circuits. They are arithmetic/control designs, not synthesized chips, validated CFT/string dualities or a proof of GRH.

## 16 Complementary physical datasets and checked dynamics

The learned program now has two additional, separately constructed corpora: 16,384 effective-material response records and22,951 molecular conformer/force records in their default configurations. Time samples are not counted as independent systems. Source-linked and dynamical configurations are additional. The active HolonForge/Aster and Mora-3 repositories were read only; no data, vocabulary, checkpoint, source enrollment or training process was changed.

The material corpus covers SSH, Kitaev, QWZ, Rice–Mele, Harper and actual clean-dessin classical RG models. The molecular corpus includes designed amides and peptides, with explicit-H MMFF94 calculations, geometric persistence, and method-labelled conformations. Forty-eight

existing Aster MP20 structures, 32 existing Mora PubChem identities and eight public PDB structures supply separately labelled anchors. Their new computed responses are not new experimental observations, DFT labels, binding affinities or CISS measurements.

All conformer, gauge, intervention and task descendants share a partition. Specified physical/chemical similarity edges are closed transitively before component-level stratification. Large components are kept intact even when 80/10/10 proportions or rare-class coverage become impossible. This gives a finite, threshold-relative separation statement; it cannot guarantee every kind of statistical independence or prevent overfitting. Source anchors are train only. Molecular anchors are also checked against every default heldout using the same chemical and applicable geometric thresholds; conflicting parent groups are excluded. Global exposure to the models' previous corpora is a separate evaluation gate.

For a number-conserving one-particle Hamiltonian $H$, write $P_{ij} = \langle c_j^\dagger c_i \rangle$ and $U(t) = e^{-itH}$. The data use $P(t) = U(t)P(0)U(t)^*$, occupations $n_i = P_{ii}$ and outgoing currents $J_{ij} = -2\,\mathrm{Im}(H_{ij}P_{ji})$. Then $\dot{n}_i = -\sum_j J_{ij}$, directly from $i\dot{P} = [H, P]$. Paired BdG evolution includes its independently computed condensate source; it is not falsely treated as electron-number-conserving normal transport. An exact interacting Hubbard dimer supplements Gaussian exchange correlations. Molecular nuclear dynamics uses MMFF94, with a separately labelled, uncalibrated effective $\pi$-orbital model for charge modulation.

Parquet configurations retain condition, target and evaluation roles, source versions/hashes, units, failures and audit evidence. The local dataset explorer shows actual structures, currents, time evolution, graph wavelets and checked circuit compilations. These computations support candidate discovery and model validation; the independent arithmetic determinant/zero-location bridge remains open. No new whole-chapter Lean formalization is asserted.

# S1 Expanded dessin, resolution and derived calculations

## S1.1 The cycle order in coordinates

Number a permitted cycle in travel order $0, \dots, \ell - 1$. Map an arrow $i \to i+1$ to $E_{i,i+1}$ for $i < \ell - 1$, and the final arrow to $u_c E_{\ell-1,0}$. A path from $i$ to $j$ then gives $u_c^r E_{ij}$, where $r$ counts cut crossings. These are linearly independent path normal forms, and every positive-length permitted path has exactly one such expression on exactly one branch. The full-cycle sum in a branch is $u_c I$. At a paired edge its two diagonal occurrences come from a single original idempotent; no positive-length coefficients are identified. This proves the order description including self-pairings within one cycle. Although the suborder is generically a product of full matrix algebras, it is not their unglued product at the special fiber.

For the interval this becomes the literal fiber product

$$\Lambda = \{(f, g) \in k[[X]]^2 : f(0) = g(0)\}, \qquad x = (X, 0),\ y = (0, X),\ u = (X, X).$$

Every element has a unique expression $a(u) + b(u)x$: take $a = g$ and $b = (f - g)/X$. Thus it is $R$-free on $1, x$ with $x^2 = ux$ and $y = u - x$. The annihilator of $x$ consists of pairs $(0, g)$ with $g(0) = 0$, exactly $y\Lambda$; the other branch is symmetric. Multiplication by $u$ is injective, coefficientwise. These are actual formal power-series statements, not conclusions from finite truncations.

## S1.2 The two-arm complex and its calculations

For an arrow $a$ write $f(a)$ for its unique forbidden successor. The summand in $P_n(i)$ indexed by the initial arrow $a$ has endpoint $t(f^{n-1}(a))$. Its map multiplies by that last arrow. Thus the

resolution can be computed by iterating a finite permutation, while the projectives themselves remain infinite-dimensional. The Hom complex against a module $N$ is the direct sum of the endpoint spaces with differentials given by the same arrow matrices. Tensor against a left module instead uses the same path entries acting on the left. No rank or exactness claim for an arbitrary $N$ is inferred by counting endpoints alone.

For the nodal residue field the maps are particularly explicit:

$$\cdots \xrightarrow{\operatorname{diag}(x,y)} \Lambda^2 \xrightarrow{\operatorname{diag}(y,x)} \Lambda^2 \xrightarrow{(x \quad y)} \Lambda \longrightarrow k \longrightarrow 0.$$

If $xz_1 + yz_2 = 0$, its first branch says the first branch of $z_1$ is zero, and its second branch says the second branch of $z_2$ is zero. Hence $z_1 = yw_1$, $z_2 = xw_2$. The same annihilator calculation proves the alternating kernels in all degrees. Projectivity is genuine: the terms are regular modules or their finite direct sums. It is not a finite free resolution, despite the period of its positive tail.

For the elliptic theta the quiver vertices are $v_0 = \{0, 3\}$, $v_1 = \{1, 4\}$, $v_2 = \{2, 5\}$. The arrows $a_0, a_3$ go $v_0 \to v_2$, $a_2, a_5$ go $v_2 \to v_1$, and $a_1, a_4$ go $v_1 \to v_0$. The two forbidden arms from $v_0$ are

$$a_0, a_5, a_1, a_3, a_2, a_4, \ldots, \qquad a_3, a_2, a_4, a_0, a_5, a_1, \ldots.$$

Their top pairs are $(v_2, v_2), (v_1, v_1), (v_0, v_0)$, repeated. The three steps exchange the arms. Their distinct top pairs exclude shorter unordered positive periods. The spherical theta instead has forbidden two-cycles and successive top pairs $(v_2, v_1), (v_0, v_0)$, giving period two. Thus even identical hereditary overorders need not give identical resolutions.

Quadratic Koszul duality here has a directly checkable local meaning: relations complement the two possible successors. The dual dessin replaces the graph rotation by $\phi^{-1}$. In the Ext algebra against $A/J$, concatenating the forbidden-word basis gives the product when the join is forbidden; a permitted join gives zero. One can see this by lifting a length-$r$ basis class to the resolution: discard the prescribed initial $r$ letters and identify the remaining summand with its suffix resolution, taking zero on other prefixes. These prefix-discard comparison maps commute with the suffix differentials, with the standard shift signs when viewed as shifted complexes. The concatenated indexing word exists exactly when its join is forbidden; evaluation of the two basis classes on that word gives one. If the join is permitted there is no such indexing word and the product is zero. Composition therefore concatenates indexing words, with the chosen right-module/Yoneda opposite convention. This is a graded algebra statement; it neither makes the unbounded resolution perfect nor constructs an equivalence of all formal-completion derived categories.

For an arbitrary bounded projective complex $P^\bullet$, its determinant line over a commutative coefficient ring is $\bigotimes_n (\det P^n)^{(-1)^n}$, with determinant duals for negative exponents. A chain automorphism acts on this line; a quasi-isomorphism gives the corresponding determinant-functor isomorphism. For a general endomorphism $F$, the expression for $1 - zF$ uses the locus where it is invertible and gives a rational determinant-line action/trivialization, not an everywhere invertible line map at singular parameters. Additivity is compatible with exact triangles under the usual determinant-functor conventions [17, 18]. In our order context an $R$-linear determinant uses the finite free underlying $R$-modules; it must specify a separate endomorphism. Determinants of a differential presentation, alternating determinants of a chain action, regularized analytic determinants and the Cartan path matrix are four distinct constructions. Their equality requires a proof, not their shared name.

The geometric model literature [6, 8, 9] is useful for identifying which crossings and gradings would have to be represented by a functor. To apply it here one must specify the category of polynomial strings, completed lattices or finite-length quotients; construct its surface/curve objects; and prove that degree-$r$ intersections compute the Hom group into shift $r$, with local systems and

boundary exceptions included. We have supplied explicit walks and maps in the examples, not claimed this general completed-order theorem. On a finite word, a map can already be computed rigorously without such a theorem: assign a coefficient matrix between equal-vertex occurrences and solve $F_{t(a)}N(a) = N'(a)F_{s(a)}$ for every arrow. The diamond calculations below implement exactly this system.

# S2 Semi-invariants, tableaux and explicit intertwiners

## S2.1 Derived information retained by a presentation

Let $K = \operatorname{im} f$. The injection $\operatorname{Hom}(K, N) \to \operatorname{Hom}(P_1, N)$ descends to the Ext injection used in Theorem 3.1; its kernel modulo $\operatorname{im}\operatorname{Hom}(P_0, N)$ is zero because the map $P_1 \to K$ is onto. The missing part of the cokernel detects maps on $P_1$ not descending through $K$. This explains exactly why an injective resolution map is needed for its full Euler interpretation.

In the node example the resolution of $M = \Lambda/(x)$ has maps $x, y, x, y, \ldots$. On $N = k[y]/(y^2)$ the Hom cochain maps are $0, y, 0, y, \ldots$. Thus $\operatorname{Hom}(M, N) = N$ and $\operatorname{Ext}^1(M, N) = \ker y$ have dimensions two and one; the cokernel of the first zero map is two-dimensional. In the elliptic-square example the first four Hom maps are $1, 0, 0, 0$, periodically. Hence degrees zero and one vanish, degrees two and three are $k$, and degree four vanishes. This also makes clear why a nonzero scalar semi-invariant does not give a derived orthogonality theorem.

With coordinate action $(gF)(N) = F(g^{-1}N)$ the Schofield weight is $a - b$, the negative of the point-evaluation weight. For a single arrow it is the same elementary distinction as the action on $\operatorname{Hom}(V, W)$ versus on its coordinate functions $V \otimes W^*$. The Cauchy decomposition for coordinate functions is

$$\operatorname{Sym}^m(V \otimes W^*) \cong \bigoplus_{\lambda \vdash m} S_\lambda(V) \otimes S_\lambda(W)^*.$$

A relation quotient can kill these components; its ambient multiplicities do not descend unchanged. Acyclic-quiver semi-invariant/LR results such as [35] retain their acyclicity assumptions. The explicit two-arrow slice below is acyclic, whereas the whole theta quiver is not.

## S2.2 One Young operator and one LR decomposition in coordinates

For $\lambda = (2, 1)$ number the first row $1, 2$ and second row $3$. Then

$$Y = (I + P_{12})(I - P_{13}), \quad Y^2 = 3Y, \quad \dim \operatorname{im}(Y \mid (k^2)^{\otimes 3}) = 2.$$

The hook product is three; its Specht dimension is $3!/3 = 2$, also two here for a different reason. The normalized projector extracts one $S_{(2,1)}$ copy. The trace $\operatorname{tr}((Y/3)M^{\otimes 3})$ is therefore the Schur character, not twice that character. This fixes the common row/column normalization ambiguity.

Write $t = \operatorname{tr} M$, $\Delta = \det M$ for a two-dimensional endomorphism. Its complete characters obey $h_0 = 1$, $h_1 = t$, $h_m = t h_{m-1} - \Delta h_{m-2}$, and

$$\chi_{(a,b)}(M) = \Delta^b h_{a-b}(M), \quad a \geq b \geq 0.$$

In particular $h_2 = t^2 - \Delta$, $h_3 = t^3 - 2t\Delta$, and $\chi_{(2,1)} = t\Delta$. Thus $t h_2 = h_3 + \chi_{(2,1)}$ is the LR identity $V \otimes \operatorname{Sym}^2 V \cong \operatorname{Sym}^3 V \oplus S_{(2,1)}V$, with $S_{(2,1)}V \cong \det V \otimes V$.

An explicit natural splitting avoids relying just on characters. In characteristic zero multiplication $m : V \otimes \operatorname{Sym}^2 V \to \operatorname{Sym}^3 V$ has natural section

$$s(v_1 v_2 v_3) = \tfrac{1}{3} \sum_{i=1}^{3} v_i \otimes \prod_{j \neq i} v_j.$$

The map $j : \bigwedge^2 V \otimes V \to V \otimes \operatorname{Sym}^2 V$ sends $(v \wedge w) \otimes z$ to $v \otimes wz - w \otimes vz$. In dimension two it is injective: for a basis $e_1, e_2$ its images at $z = e_1, e_2$ have distinct independent coordinates. They lie in $\ker m$, whose dimension is $6 - 4 = 2$. Hence $(s, j)$ is the required isomorphism. Tensorial definitions make both maps commute with every linear endomorphism, including singular arrow maps. This is a genuine LR intertwiner on our dessin representation stratum.

For a concrete theta representation take

$$X_1 = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad X_2 = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}, \quad X_3 = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}.$$

Then $M = X_3X_2X_1 = \left(\begin{smallmatrix} 2 & 2 \\ 2 & 3 \end{smallmatrix}\right)$, $t = 5$, $\Delta = 2$, $h_2 = 23$, $h_3 = 105$ and $\chi_{(2,1)} = 10$. The identity is $115 = 105 + 10$. Since the other permitted branch is zero, every forbidden product is zero; this is an actual representation of the infinite elliptic dessin algebra.

In the invertible-arrow open stratum one can gauge $X_1, X_2$ to the identity, leaving conjugation of $M$. The invariant-ring proof in the core works also at singular arrows by cycle-trace generation, not by assuming that this gauge choice extends there. The degree-one monodromy coefficients have total arrow degree three; this is why the Hilbert-series denominator uses $3j$, not $j$.

For the Cauchy kernel choose the Schur subspace inside the tensor power with its inherited Hilbert norm. $M^{\otimes m}$ preserves it and has norm at most $\|M\|^m$. This proves the trace bound for nonnormal matrices as well as diagonal ones. Formal Cauchy identities extend by polynomial continuity in matrix entries; the stated absolute bound then justifies analytic summation. Gram positivity does not assert entrywise positivity or positivity of an unrelated arithmetic form. For three exact test matrices $M$, $\left(\begin{smallmatrix} 1 & 1 \\ 0 & 2 \end{smallmatrix}\right)$, $\left(\begin{smallmatrix} 0 & 1 \\ -1 & 1 \end{smallmatrix}\right)$, the choice $r = 1/100$ satisfies $rR_0^2 \le 21/100 < 1$ using Frobenius norms, and all seven principal minors of the rational kernel matrix are nonnegative.

### S2.3 Two precise functorial limitations

At any actual vertex simple $S$ of $A$, the split exact sequence $0 \to S \to S \oplus S \to S \to 0$ uses first inclusion and second projection. Termwise $\operatorname{Sym}^2$ has dimensions $1, 3, 1$ at that vertex; its middle kernel contains the mixed monomial as well as the first square, but its first image contains only the first square. Middle homology is $S$. Thus even split exactness is not preserved. Genuine derived Schur constructions instead use mixed tensor degrees, signed total differentials and suitable flat/projective replacements [32, 33]; their membership in a particular order category must still be checked.

The $R$-lattice $M = Re_x \oplus Re_y$ with $x = \operatorname{diag}(u, 0)$, $y = \operatorname{diag}(0, u)$ has $x + y = uI$. On $\operatorname{Sym}^2_R M$ the images are $\operatorname{diag}(u^2, 0, 0)$ and $\operatorname{diag}(0, 0, u^2)$. Their sum kills the mixed vector, hence equals neither $uI$ nor $u^2I$. This is a specific failure to retain the original central action, not a failure of the polynomial monomial representation functor. Even a common ramified base substitution does not repair this mixed vector.

Semi-invariants retain useful moduli on nilpotent modules. In the theta quiver keep only $a_0, a_3 : v_0 \to v_2$, put their scalar values $(x, y) = (1, \lambda)$ and all other arrows zero. The $v_1$ simple splits off. Every length-two path acts by zero, so these are continuous completed modules. The two scalars have the same weight $g_{v_2}/g_{v_0}$, and $y/x$ is a rational invariant on $x \neq 0$, giving the projective pencil $[x : y]$. It is not a regular invariant on the whole affine locus, whose orbit closures meet zero.

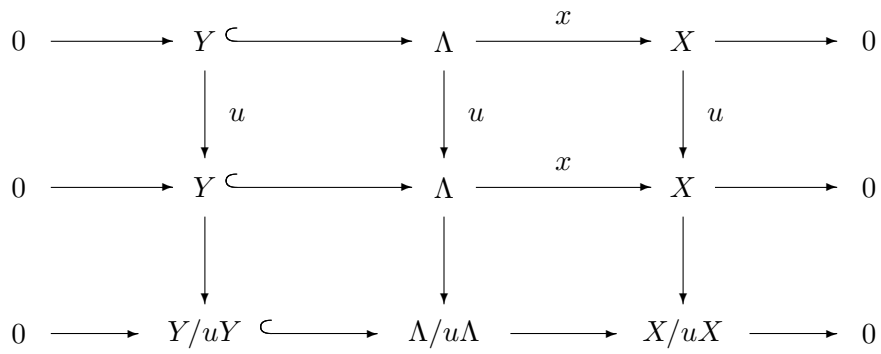
For the interval quotient take $x = J_2$, $y = \lambda J_2$. With its top and socle specified, the separated Kronecker coordinates are again $(1, \lambda)$; the relation to the same-weight ratio is exact. The pp formula in the core is stronger than merely noticing a ratio: on $M_\lambda$ its solution set is $\operatorname{im} x$, whereas on $M_\mu$, $\mu \neq \lambda$, the equation $(\mu - \lambda)xv = 0$ forces $z = xv = 0$. Different parameters are therefore distinguishable by actual positive-primitive conditions even though every cycle trace is zero.

# S3 Full lattice and diamond diagrams

## S3.1 The nodal exact diagram

The normalization proof of Theorem 4.1 uses $BM$ inside the generic module. Replacing it without justification by $B \otimes_\Lambda M$ could introduce torsion and would not prove the same statement. Once the subdirect residue subspace is decomposed, the branch basis changes lift over complete DVRs; taking inverse images modulo the conductor gives the direct decomposition, not just a decomposition of residues.

Here is the promised diagram, with the terminal short exact column for every position:




Each vertical column is completed by zero above and below. In the bottom row identify $Y/uY = k$ via generator $y$, $X/uX = k$ via $x$, and $\Lambda/u\Lambda = k[\epsilon]/(\epsilon^2)$ with $x = \epsilon$, $y = -\epsilon$. Its first map is $1 \mapsto -\epsilon$, its second the augmentation. It is nonsplit as a dual-number sequence. This explicit order reduction, not an unrelated finite algebra, explains its appearance.

For a further Ext calculation, resolve $X \cong \Lambda/(y)$ with successive maps $y, x, \ldots$. On $Y$ the first is multiplication by $y$ and the second zero, so $\mathrm{Ext}^1(X, Y) = Y/yY = k$; on $X$ the first is zero and the second injective, so $\mathrm{Ext}^1(X, X) = 0$. Interchanging branches gives the other two cases. The two nonsplit lattice rows realize the nonzero cross classes. Their Hom spaces are finite over $R$, generally infinite over $k$.

## S3.2 Chebyshev geometry and the exact quotient

The identity $1 - T_n(z)^2 = (1 - z^2)U_{n-1}(z)^2$ follows from $z = \cos\theta$ and the sine/cosine multiple-angle formulas, hence as a polynomial identity. $T_n' = nU_{n-1}$. Therefore $\beta_n$ has its only zero at infinity, of order $2n$; its $n$ points over one are double since $\beta_n - 1 = T_n^2/(1 - T_n^2)$; and its poles are simple at $\pm 1$ and double at the $n - 1$ interior roots of $U_{n-1}$. The derivative has no additional critical values. The total ramification is $(2n - 1) + n + (n - 1) = 4n - 2$, as required for a degree-$2n$ rational Belyi map. The dual graph has $n + 1$ vertices of degrees $1, 1, 2, \ldots, 2$, and $n$ edges; it is a connected path. Its planar ribbon embedding is unique, so this identifies the particular nested-loop dessin, not only its passport.

In the quotient retaining orientation $\Omega$, any surviving path traverses each chain edge at most once and either endpoint loop at most once. Its length is at most $n + 1$; thus the killed-arrow ideal contains $J^{n+2}$. The quotient of $A_n$ and of its completion is the same $H_n(\Omega)$. For $n = 3$ with forward

chain arrows, its path counts in lengths zero through four are $3, 4, 3, 2, 1$, total thirteen. These counts describe a quotient representation category of the infinite order, not a change of primary object.

Inflation is full because the two algebras have exactly the same actions on these annihilated objects. It preserves internal exact sequences. It does not preserve arbitrary ambient extension closure: the two adjacent simples are annihilated by the killed reverse arrow, but the two-dimensional extension having only that arrow nonzero is not. Likewise endpoint-local freeness over $k[l]/l^2$ or $k[r]/r^2$ in [46] is not $R$-freeness. To transfer an almost-split property to a larger category one must prove the requisite approximation/factorization statements; locally finite quiver criteria [40, 41, 42] do not apply just because each vertex has two arrows. The permitted cycles violate interval/Hom finiteness.

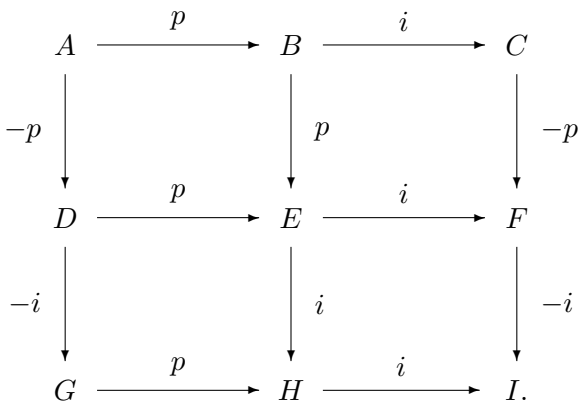
### S3.3 Nine actual strings and all maps

Take $n = 3$ with $1 \xrightarrow{a} 2 \xrightarrow{b} 3$, loops $l, r$ and $l^2 = r^2 = 0$ in the quotient. Here $b$ denotes the retained forward $a_2$, not a reverse arrow of the original doubled chain. The following words are in travel order. A superscript $-$ means a formal inverse in a word, not an inverse algebra element.

| Object | word | start | dimension vector |
|---|---|---|---|
| $A$ | $l^-abrb^-a^-l^-$ | 1 | $(4, 2, 2)$ |
| $B$ | $l^-ab$ | 1 | $(2, 1, 1)$ |
| $C$ | $l^-abr^-$ | 1 | $(2, 1, 2)$ |
| $D$ | $abrb^-a^-l^-$ | 1 | $(3, 2, 2)$ |
| $E$ | $ab$ | 1 | $(1, 1, 1)$ |
| $F$ | $abr^-$ | 1 | $(1, 1, 2)$ |
| $G$ | $r^-b^-a^-labrb^-a^-l^-$ | 3 | $(4, 3, 4)$ |
| $H$ | $r^-b^-a^-lab$ | 3 | $(2, 2, 3)$ |
| $I$ | $r^-b^-a^-labr^-$ | 3 | $(2, 2, 4)$ |

A basis vector is assigned to each visited position $0, \ldots, \ell(w)$ at its vertex. A direct letter sends its preceding vector to its following vector; an inverse letter sends the following vector to the preceding vector. Sum these elementary maps for each arrow and put all unmentioned entries zero. This defines every matrix, including repeated loops, and verifies the relations. Set the killed reverse arrows to zero to obtain actual $\Lambda_3$-modules.

Their signed grid is



Horizontal projections keep the initial word positions and kill the remaining positions. Horizontal inclusions append the final $r^-$, keeping old positions. Upper vertical projections delete the initial $l^-$: old position $j \geq 1$ maps to position $j-1$, and old position zero maps to zero. Lower vertical inclusions prepend $r^-b^-a^-l$: position $j$ maps to $j+4$. Multiply these maps by the indicated signs. For every arrow these formulas commute with the string action, including the two positions adjacent to a cut or join; those are exactly where the loop inverse and chosen orientation matter.

Every small square has two length-two composites whose sum is zero. The exact sequences are

$$0 \to A \to B \oplus D \to E \to 0, \qquad 0 \to B \to C \oplus E \to F \to 0,$$
$$0 \to D \to E \oplus G \to H \to 0, \qquad 0 \to E \to F \oplus H \to I \to 0.$$

For example the maps out of $A$ retain its initial positions $0,1,2,3$ in $B$ and delete its first position in $D$, so their joint kernel is zero. The maps to $E$ are onto, and the dimension vectors give $\dim A + \dim E = \dim B + \dim D$ vertexwise. The zero composite plus this equality proves exactness. The other three use precisely the same keep/delete or append/prepend rules; their vertexwise dimension equalities and jointly injective/surjective maps prove exactness. The shipped exact checker constructs all matrices and verifies these assertions independently over $\mathbb{Q}$.

The stronger almost-split property is imported by identifying $E = w = ab$ as the minimal string of type $(2,2)$ and these exact hook/cohook words with [46, Proposition3.12], using that paper's Propositions2.6/2.9 and Lemma3.10. Opposite-module conventions are reconciled by reversing travel/composition conventions; the closed doubled quiver is isomorphic to its opposite by exchanging the two arrows at each chain edge. Thus intrinsically $\tau E = A$, $\tau F = B$, $\tau H = D$, $\tau I = E$. The rows and columns of this grid are *not* generally short exact, unlike the nodal diagram. Two small cautions in that source are not needed here: a tube-rank label in Example3.9(3) is inconsistent with its listed mouth, and a general homogeneous-tube statement must not be read as a converse classifying every such object as a band. Neither is used to import these four meshes.

### S3.4 The pure-injective quotient and what its dimension means

For $B_0 = \Lambda/J^2 = k\langle x,y\rangle/(x,y)^2$, words alternate direct and inverse letters because every direct/direct and inverse/inverse adjacent pair is forbidden. Up to rotation and inverse its only primitive band is $xy^{-1}$; the bridge quiver has two oriented band vertices with no connecting nontrivial bridge. The longest path has length zero. The domestic string results [44, 45] therefore give relative dimension $0+2=2$, not an automatic dimension three for every one-domestic algebra. Finite-dimensional points have relative CB rank zero; the appropriate Prüfer/adic and one-sided periodic string points have rank one; the unique generic point has rank two. These statements concern the quotient category with its inherited pp structure.

Indeed a pp formula is a finite system of linear equations with existentially quantified unknowns. On a module annihilated by $J^2$ every coefficient reduces to $B_0$, so purity and pure-injectivity are preserved in both directions under inflation. The condition $MJ^2 = 0$ is a definable closed condition, giving the closed Ziegler subspace. Its relative ranks need not be ranks in the entire $\Lambda$ spectrum. This subcategory is not extension closed: $\Lambda/(x^3, y)$ gives an extension of objects killed by $J^2$ that is not itself killed by $J^2$. Since $u^2$ annihilates every object of the subcategory, it contains no nonzero $R$-free lattice. Ringel's representation-theoretic node construction is also not a Morita equivalence identifying the geometric nodal order with the Kronecker algebra. Infinite-dimensional polynomial string classifications [38] retain finitely-controlled or pointwise-artinian assumptions; an arbitrary complete lattice does not automatically satisfy them as an algebraic $A$-module.

# S4 The operator extension in detail

## S4.1 A corrected cycle lift and the integer boundary map

In the permitted-state graph each length-$\ell$ cycle has a full matrix-corner Laurent/circle coordinate. Its chosen $K_1$ generator is the full cycle at one state plus the complementary identity. In Fock space $L_c^* L_c = Q_a + v_{s(a)}$. Thus the tempting lift $L_c + (1 - Q_a)$ has overlapping initial pieces and is not an isometry. The corrected lift in Theorem 5.1 first restricts to $Q_a$:

$$W = L_c Q_a + (1 - Q_a), \quad W^* W = I, \quad I - W W^* = Q_a - L_c Q_a L_c^*.$$

A path starting with $a$ and length greater than $\ell$ has uniquely the full prefix $c$ followed by another path starting with $a$. The missing range consists exactly of lengths $1, \ldots, \ell$. Its terminal multiset is the original graph's incident-edge multiset at that cycle vertex. These finite defect vectors all lie in terminal compact ideals; their classes give the claimed unsigned incidence entries. The sign convention affects only the overall sign of the boundary map.

Here is the Smith calculation without spectral assumptions. Choose a spanning tree of the connected underlying graph and one root. By changing signs on one bipartition of the tree, its $E \times V$ incidence submatrix becomes the ordinary oriented tree incidence, with $V - 1$ unit pivots. After eliminating these pivots, a remaining column represents the common root coordinate, and each nontree edge evaluates on it as zero when its fundamental cycle is even, or $\pm 2$ when odd. A loop is the latter case. If all are even the graph is bipartite and this column is zero; otherwise its entries have gcd two, giving the final pivot two. Remaining rows are free. Consequently the kernel/cokernel, then the six-term sequence, give the $K$ groups stated in the core. No theorem reconstructing a cyclic embedding from those groups is asserted.

The compact ideal is block diagonal, not $\mathcal{K}(\mathcal{H})$: prefixes never move terminal vertices, so cross-terminal rank-one maps do not belong to the displayed ideal. This is why the arithmetic insertion must choose one terminal sector. A graph/Cuntz boundary quotient kills the vacua and hence this ideal, so that insertion need not survive passage to the quotient.

## S4.2 Spectral domain and arithmetic trace

A vector belongs to the adjoint domain of the diagonal $N$ precisely when its coefficientwise product by the real sequence $|p|$ is square summable; this proves self-adjointness on the given domain. The truncations by length prove finite supports are a core. Resolvent coefficients $(|p| - i)^{-1}$ tend to zero and each length space is finite dimensional, giving compactness. Prefix and deletion shift the length by at most one, so preserve the domain and prove the commutator formulas there. Leibniz then handles the dense polynomial *-algebra.

Every terminal sector has one vacuum and two paths at each length $n \geq 1$, even when some graph endpoints coincide: distinct formal paths remain distinct basis vectors. Thus restriction to a sector still has a simple vacuum, but it need not be a faithful representation of the whole concrete algebra. The trace of $e^{-tN^2}$ is $E + 2E \sum_{n \geq 1} e^{-tn^2}$, distinct from the semigroup trace of $N$ in the core. The trace identity with $\chi(N)$ is only on $\operatorname{Re} s > 1$, where the product is trace class. Analytic continuation of the scalar $L$-function does not make that same operator product trace class on $\operatorname{Re} s > 0$.

## S5 Abel determinant estimates and the arithmetic program

For completeness the inserted family has a full elementary convergence proof. Since complete nonprincipal character periods sum to zero, $|A_n| \le B_0 \le q_\chi$. On $2^k \le n < 2^{k+1}$,

$$\sum |u_n|^2 \le \frac{B_0^2}{(k+1)^2}, \qquad \sum |v_n(s)|^2 \le M^2 (k+1)^2 r^k, \quad r = 2^{-2\sigma_0} < 1.$$

The second follows from $n^{-s} - (n+1)^{-s} = s \int_n^{n+1} x^{-s-1}\, dx$. Define

$$Q_J(r) = r^J \left( \frac{(J+1)^2}{1-r} + \frac{2(J+1)r}{(1-r)^2} + \frac{r(1+r)}{(1-r)^3} \right).$$

Expanding $(J+j+1)^2$ in the three geometric moment sums proves $Q_J = \sum_{k \ge J} (k+1)^2 r^k \to 0$. Locally uniform $\ell^2$ convergence of the holomorphic finite-coordinate vectors proves Banach holomorphy. The linear functional $z \mapsto \sum v_n z_n$ has norm $\|v\|$; this is the complex-linear coefficient pairing, not a conjugate-linear dependence on $s$.

Write $\mathcal{I}(v)(z) = \sum_n v_n z_n$ for the coefficient functional, distinct from the arrow ideal $J$. For $N_J = 2^J - 1$, $J \ge 1$, let $K_N = u \otimes (\mathcal{I}(P_N v) - e_1^*)$. Then

$$\|K - K_{N_J}\|_1 \le \sqrt{2} B_0 M \sqrt{Q_J(r)}.$$

Finite summation by parts gives

$$T_N = \det_F(I + K_N) = \sum_{n \le N} A_n (n^{-s} - (n+1)^{-s}) = S_N - A_N (N+1)^{-s}.$$

Adding the last endpoint times $e_1^*$ to the functional recovers $S_N$ exactly because $u_1 = 1$. Its trace-norm error is at most

$$\sqrt{2} B_0 (M \sqrt{Q_J(r)} + B_0 (N_J + 1)^{-\sigma_0}).$$

For arbitrary cutoffs use the last whole dyadic block for an enclosing tail. Galerkin compressions also converge: $\|u - P_{N_J} u\| \le B_0 / \sqrt{J}$. With $C_K = 1 + M\sqrt{(1+r)/(1-r)^3}$,

$$\|K - P_{N_J} K P_{N_J}\|_1 \le \frac{B_0 C_K}{\sqrt{J}} + \sqrt{2} B_0 M \sqrt{Q_J(r)}.$$

The raw endpoint adds the same endpoint bound as before. Isometric insertion by $U$ preserves nonzero singular values, every norm estimate, and the Fredholm determinant. Everything is in one compact ideal of the actual infinite dessin Fock algebra.

For primitive parity $a$ use $g(s) = (q_\chi/\pi)^{(s+a)/2} \Gamma((s+a)/2)$, holomorphic nonzero on $\operatorname{Re} s > 0$. The completed functional is $g(s)\mathcal{I}(v) - e_1^*$, and the completed raw functional is $g(s)(\mathcal{I}(P_N v) + A_N (N+1)^{-s} e_1^*) - e_1^*$. Their determinants are exactly $gL$ and $gS_N$. Scaling the entire old perturbation would instead give $1 + g(L-1)$, the wrong normalization. Trace-class background is standard [26], while these special rank-one identities follow directly from $\det_F(I + u \otimes \varphi) = 1 + \varphi(u)$.

The metric obstruction is likewise elementary. With inner product conjugate-linear in its first variable, symmetry of $T = u \otimes \varphi$ implies

$$\varphi(z) \langle u, u \rangle = \overline{\varphi(u)} \langle u, z \rangle.$$

A nonzero $\varphi$ cannot have $\varphi(u) = 0$; thus all symmetric maps with that range have proportional functionals. For $n \geq 2$ the coefficient ratios at $\sigma + it$ and $\sigma$ satisfy

$$\frac{v_n(\sigma + it)}{v_n(\sigma)} = \frac{\sigma + it}{\sigma}\, n^{-it}(1 + O_{\sigma,t}(1/n)).$$

Integers nearest $\exp(2\pi k/|t|)$ and $\exp((2\pi k + \pi)/|t|)$ give opposite nonzero subsequential limits. Hence no common positive metric symmetrizes this family for all critical heights. Restricting any proposed metric on the inserted family to $U\ell^2$ gives the same contradiction. An independently constructed positive comparison is still a possible different route.

### S5.1 Precisely how classical and recent sources enter

Over a finite field the Grothendieck trace-formula architecture writes a zeta function as alternating determinants on finite-dimensional cohomology; Deligne's purity supplies the weights of eigenvalues [22, 20, 21]. If a weight-$w$ Frobenius is $q^{w/2}$ times a positive-metric unitary, each eigenvalue has modulus $q^{w/2}$. The norm conclusion is immediate; constructing that form or proving purity is the deep step. Symplectic/alternating duality $F^t J F = q^w J$ alone only pairs reciprocal eigenvalues and does not give their individual moduli. An order resolution or Schur-positive character expansion alone cannot replace this step in number-field cohomology.

The recent primary comparisons used here are specific:

- [53, Theorem3.12, pp.15–17] gives finite abelian covers with local orbit length $\log p$ and genuine arithmetic monodromy. It suggests what a descent-equipped order functor should match; it does not identify our $u$ or a face cycle with a prime without such a functor.
- [54, Propositions4.1–4.3, Theorem4.6] builds finite-prime local models with archimedean Fourier factors. Its Hilbertian isomorphisms are not an asserted uniformly controlled common isometry for our arithmetic approximants. [55, Proposition3.1] gives determinacy of a particular local moment problem; positive finite Hankel matrices are not the missing global Weil inequality.
- [56, Theorem3.6] gives discreteness/lower boundedness for the stated restricted Weil form. Theorem5.10, pp.23–24, additionally assumes a finite truncation, $\lambda > 1$, a simple even bottom eigenfunction $\xi$ and $\delta_N \xi = 1$. Its quotient metric is positive and its regularized determinant is $-i\lambda^{-iz}\widehat{\xi}(z)$. Passing to the infinite arithmetic comparison, including the ground-state and convergence conditions in its §8, is not automatic.
- [57, Theorem6.1, p.13] assumes a specified real distribution on an interval, the corresponding trigonometric-core form, essential self-adjointness, and an isolated simple lowest eigenvalue with reflection-even eigenfunction. Its Fourier transform then has only real zeros. Our degree operator's vacuum and length parity are not those data.
- [60, Theorem6.1] gives prime-indexed periodic packets in its dynamical construction, not one proved orbit per prime with the full desired cohomological trace formula. [62] isolates an obstruction to a particular real-coefficient cohomology with full functorial Artin multiplicities; it is not a theorem excluding every Hilbert–Pólya operator. The current survey [58] distinguishes remaining steps. The recent Picard-monoid construction [59] is not already a polarized abelian variety with a number-field purity theorem. [63] uses RH as an input in the cited spectral identification, so cannot be used circularly as its proof.

These are targeted theorem/proof readings, not claims that every page of every cited work has been audited. The preserved review records specify actual page coverage and hashes, including inaccessible original PDFs where the modern primary proof was used instead.

### S5.2 A concrete sequence of remaining constructions

First choose a number-field model of a Belyi pair, its finite covers and conductor/ramification data. Construct actions or exact correspondences on the labelled order/module or analytic categories that recover the local prime Frobenius polynomials. The elliptic twists show why omitting this datum loses information. Test at more than one arithmetic model of the same complex dessin, not just its topological passport.

Second specify a finite projective or otherwise determinant-admissible complex and its *independent* arithmetic endomorphism. Prove exact-sequence/determinant-line compatibility. Use semi-invariants to encode presentation degeneracies, and Schur/exterior characters to calculate local factors, retaining every central-action and nilpotence distinction above. General derived Schur replacements require their own construction; termwise Schur on the resolution fails the exhibited split sequence test.

Third prove the same local factors, gamma terms and conductor give a globally convergent or correctly regularized determinant. Trace-norm convergence must be established for the chosen analytic perturbations; strong convergence is insufficient in general. The inserted Abel family supplies this topology for an elementary scalar encoding, but has neither the independent arithmetic spectrum nor the positive metric needed for zero location.

Finally produce an independently proved positive form or self-adjoint determinant-preserving realization. One acceptance criterion is the Hurwitz target in the core; another is equality with the precise arithmetic quadratic form in the recent spectral theorems, with all their domain, ground-state, normalization and limit hypotheses verified. The local Cauchy kernel, $K$-group computation and graph spectral dimension alone meet none of these final comparison hypotheses. No remaining condition is silently asserted true here. Dirichlet, Hecke, Dedekind and Artin scopes must remain individually recorded.

## S6 Graph pullbacks and completions with specified fibers

### S6.1 The complete Fock graph presentation

For graph C* conventions the edge operator has initial projection at its range vertex and final projection under its source vertex. Put

$$S_h = L_{a_h} Q_{a_{\tau h}}, \qquad T_h = L_{a_h} v_{[\tau h]}.$$

The two right-hand projections are orthogonal subprojections of $L_{a_h}^* L_{a_h} = Q_{a_{\tau h}} + v_{[\tau h]}$. Thus $S_h^* S_h = Q_{a_{\tau h}}$, $T_h^* T_h = v_{[\tau h]}$, and $S_h^* T_h = 0$. For distinct $h$ the prefix ranges are orthogonal, so every other cross-edge CK1 product is zero. Their final projections sum to $Q_{a_h}$. The vacua have no outgoing edges. The resulting graph has $|H| + |Q_0|$ vertices and $2|H|$ edges, with cycles only among its state vertices and exits to the paired sinks. Gauge covariance and nonzero vertices prove faithfulness as well as surjectivity. This explains precisely how a graph C* algebra is obtained without imposing the incompatible original-quiver CK1 relations.

The graph integer map has one column for each regular vertex $h$ and no sink columns:

$$d : \mathbb{Z}^H \longrightarrow \mathbb{Z}^H \oplus \mathbb{Z}^{Q_0}, \qquad d(e_h) = e_h - e_{\tau h} - b_{[\tau h]}.$$

The state part of $d(x) = 0$ says $x_h = x_{\tau^{-1} h}$; the sink part then says the constant cycle coefficients are in $\ker B_\Gamma$. For the cokernel choose a base state on each cycle and successively use $q_h = q_{\tau h} + b_{[\tau h]}$ to eliminate all other states. Closing the cycle leaves $\sum_{h \in c} b_{[h]} = 0$, while its base state remains free. Hence

$$\operatorname{coker} d \cong \mathbb{Z}^{\#\,\mathrm{cyc}\,\sigma} \oplus \operatorname{coker} B_\Gamma.$$

This gives an independent integral check of the previous compact-extension computation. The full labelled graph, diagonal projections and edge families carry more information than those coarse groups.

### S6.2 A genuine application of the supplied pullback theorem

For the interval dessin let $E_2 = G_\alpha$ have loop vertices $q_x, q_y$, common sink $b$, loops $s_x, s_y$ and exits $t_x, t_y$. Remove $b$ to obtain $F_2$, two disjoint loop graphs. Its quotient is exactly $C(\mathbb{T})^2$. Let $E_1$ have the same three vertices and edges $e_{x,n} : q_x \to b$, $e_{y,n} : q_y \to b$, $n \geq 0$, and no other edges. Remove $b$ to obtain the edgeless $F_1$, whose algebra is $\mathbb{C}^2$.

The inclusions $F_i \subset E_i$ are admissible in the precise sense of [65, Definition3.1]: the removed sink set is hereditary/saturated; all edges ending at retained vertices are retained; no vertex emits infinitely many edges into the removed set and a positive finite number outside it. For $E_1$ the latter outside count is zero, which is allowed. Define a path functor identity on objects by

$$e_{x,n} \longmapsto s_x^n t_x, \qquad e_{y,n} \longmapsto s_y^n t_y.$$

Images of distinct positive edges are prefix incomparable: at the first different length one has an exit and the other a loop. Source vertices and identities cause no exception. The image consists of all positive pointed paths in $E_2$, whose last edge is not a short loop; vertices are included separately. The first graph is acyclic and the removed sink has no short loop in the second. All hypotheses of Theorem3.4 are now verified, giving

$$C^*(E_1) \cong \mathcal{T}_A \underset{C(\mathbb{T})^2}{\times} \mathbb{C}^2.$$

The quotient of $C^*(E_1)$ by its sink ideal is $\mathbb{C}^2$, and its sink ideal is compact operators on the paths ending at $b$. The graph is acyclic, hence its algebra is AF. There are no regular vertices in $E_1$: the two sources are infinite emitters and $b$ is a sink. The graph $K$ matrix has no columns and three rows, yielding $K_0 = \mathbb{Z}^3$ and $K_1 = 0$.

This is a positive instance of the new source inside our infinite dessin framework. It is a constants-symbol subalgebra of the Fock model, not the entire Fock algebra or the formal order. For a permitted cycle of length greater than one, its full closed path is itself pointed in this paper's convention, so an object-preserving acyclic path functor cannot cover all such paths. That simple obstruction prevents copying this example to every cycle by analogy. The sink-extension Theorem4.3 also requires a specified original pullback, injective gluing maps, annihilation of glued vertex projections and coverage of all paths ending at the gluing sinks. A two-to-one identification of sink occurrences within one branch does not satisfy these assumptions without another construction.

### S6.3 Diagonal residue obstruction and the corrected pullback

Suppose one tries to glue full matrix-circle blocks by their diagonal values. In $M_2$, $E_{12}E_{21} = E_{11}$, although both factors have zero diagonal. Thus the proposed boundary map is not a homomorphism. For $\ell > 1$, more generally any *-homomorphism from the full matrix-circle algebra to a commutative C*-algebra has equal images of $E_{ii} = E_{ij}E_{ji}$ and $E_{jj} = E_{ji}E_{ij}$. These images are both orthogonal and equal, hence zero; summing the diagonal units kills the image of one and then the whole homomorphism. This works for nonunital *-maps as well: a nonzero image of one would give a nonzero unital corner map. The failure concerns full matrix-circle blocks, not the already-restricted fiber algebra $D_\ell$.

Evaluation $D_\ell \to \mathbb{C}^\ell$ is a genuine surjective *-homomorphism since at one both factors are diagonal. Its kernel is $C_0(\mathbb{T} \setminus \{1\}, M_\ell)$. Pulling it back along the repeated-coordinate embedding $\Delta_\alpha : \mathbb{C}^{Q_0} \to \mathbb{C}^H$ gives exactly the fiber conditions of $\mathcal{D}_H$. The paired constant diagonal matrices give a unital *-section. Hence this is a split C*-extension, not a claim deduced from a merely commutative diagram of nonmultiplicative maps. Since $\mathbb{T} \setminus \{1\} \cong \mathbb{R}$, each kernel block is a matrix suspension of $\mathbb{C}$, with $K_0 = 0$, $K_1 = \mathbb{Z}$. Split exactness now proves Theorem 5.3.

The pairing is retained in this specified presentation, not recovered from the unmarked isomorphism class. In particular arbitrary constant permutation matrices can relabel a branch's diagonal positions, so the spherical and elliptic theta examples have isomorphic unmarked $\mathcal{D}_H$ despite different ribbon orderings. Keep the labelled arrows, cycle cuts and distinguished fiber as additional data when using it as a dessin model. Also the central branch unitary $zI_\ell$ represents $\ell$ times the primitive corner generator in $K_1$, not a new independent generator per matrix entry.

## S6.4 Generation, faithfulness and the exact failure of extension

Let $\rho$ denote the polynomial representation in Theorem 5.3. Its vertex images are orthogonal projections summing to one. Every arrow image lies under the correct source and range vertex projections. A forbidden product vanishes because the intermediate state indices are $\tau h$ and $\alpha\tau h$, which are distinct. Thus $\rho$ is an actual algebra map, not a fitted characteristic polynomial.

For a full turn starting at $h$, $\rho(c_h) = w_c E_{hh}$ with $w_c(z) = g(z)^{|c|}z$. It is a normal element of $\mathcal{D}_H$. The function $w_c$ vanishes exactly at one; away from that point $w_c/|w_c| = z$. Therefore it is injective on all of $\mathbb{T}$, and a homeomorphism to its image. Nonunital continuous functional calculus at $\rho(c_h)$, using functions zero at zero, gives every $fE_{hh}$ with $f \in C(\mathbb{T})$, $f(1) = 0$. The zero condition makes the result vanish in all other matrix entries and branches.

To obtain an off-diagonal entry choose the unique shortest permitted path from state $h$ to $j$ within its cycle, of length $r > 0$ and cut-crossing count $\kappa \in \{0, 1\}$. Its image is $g^r z^\kappa E_{hj}$. Given $f(1) = 0$, multiply it on the left by the already available diagonal element

$$\frac{f(z)z^{-\kappa}}{g(z)^r + \varepsilon} E_{hh}, \qquad \varepsilon > 0.$$

The product is $fg^r/(g^r + \varepsilon)E_{hj}$ and converges uniformly to $fE_{hj}$. Indeed near one $|f|$ is small, and on the complement of that neighborhood $g^r$ has a positive minimum. These two bounds prove uniform convergence without assuming division by a function vanishing at the fiber. Thus the generated algebra contains the whole kernel ideal. Its quotient contains the entire paired diagonal algebra, so it is $\mathcal{D}_H$.

For injectivity let a finite linear combination of path images vanish. At one, all positive paths vanish, forcing the coefficients of the original vertex idempotents to vanish. Fix a state-matrix entry in a branch. Its remaining paths have lengths $r + m\ell$ and images

$$g^r z^\kappa \sum_m a_m (g^\ell z)^m E_{hj},$$

with a possible initial zero coefficient if $r = 0$. On $\mathbb{T} \setminus \{1\}$ the prefactor is nonzero, and $w_c$ has infinite range, so the polynomial has infinitely many distinct zeros. All coefficients are zero. Repeating by entries proves faithfulness.

No boundedness improvement repairs full formal extension. Pick any $z_0 \neq 1$, let $\lambda = w_c(z_0)^{-1}$, and consider $1 - \lambda c_h$. Its inverse $\sum_{m \geq 0} \lambda^m c_h^m$ exists in the arrow-adic order. The matrix image at $z_0$ has a zero eigenvalue on state $h$, so cannot be invertible. A unital algebra homomorphism would preserve the inverse identity, an immediate contradiction. This proves nonextension without any

continuity assumption. It does not rely on the false claim that a norm greater than one alone forces a geometric-series denominator to be singular. Scaling $g$ to have norm at most one still does not allow arbitrary formal series.

The two valid operator completions are structurally different. In the bipartite case $K_0(\mathcal{T}_A) = \mathbb{Z}^{E+1}$ differs from $K_0(\mathcal{D}_H) = \mathbb{Z}^E$; in the nonbipartite case the former has the extra $\mathbb{Z}/2$ torsion. Neither is automatically the norm completion of the arrow-adic topology. No invariant constructed solely from these complex objects distinguishes the rational twists already exhibited; arithmetic descent remains a separate input.

### S6.5 Invariant basis number and the Leavitt comparison

If a unital $k$-algebra $R_1$ has a unital representation on a nonzero finite-dimensional vector space $V$, an isomorphism $R_1^m \cong R_1^n$ induces an isomorphism $V^m \cong V^n$ by tensoring. Comparing dimensions gives $m = n$. This proves invariant basis number without requiring the representation to be faithful. Every original dessin algebra and completed order has a vertex-simple representation, and each permitted-state Laurent algebra has evaluation onto a nonzero matrix block. The Leavitt algebra of $G_\alpha$ has the same finite-dimensional representation after killing the vacuum sinks and evaluating one boundary cycle. Thus all of these examples have invariant basis number. Leavitt's free-rank phenomena are important elsewhere but cannot be assumed in this class simply from the name LPA.

For the specified finite graph $G_\alpha$, the usual complex Leavitt path algebra embeds in $C^*(G_\alpha)$ with norm-dense image, using its canonical generators and involution [67, Theorem7.3]. Those generators are the split $S_h, T_h$, not the unsplit original gentle arrows equipped with ordinary CK1. This is the exact algebraic/analytic twin relationship applicable here.

## S7 String finiteness and the smooth quotient surface

For a finite quiver, finitely controlled implies finitely generated: choose one finitely generated submodule containing each vertex space and sum the finitely many choices. Conversely the module itself works. Stabilization at each of finitely many vertices likewise gives ordinary artinianity. Crawley–Boevey's polynomial theorem therefore applies directly to our finitely generated and artinian polynomial modules, with its primitive injective bands in the latter case. It does not assert a classification of every possible algebraic module.

The formal branch $M = k[[x]]$, $y = 0$, gives a precise counterexample to that stronger assertion. An $A$-endomorphism $F$ preserves each $x^n M$. For the polynomial truncation $f_n$ of $f$, one has $F(f) - f_n F(1) \in x^n M$. Separatedness and all $n$ imply $F(f) = fF(1)$. Hence $\mathrm{End}_A(M) = k[[x]]$ is local, so $M$ is indecomposable. If an ordinary word module has $y = 0$, its word cannot contain $y$ or $y^{-1}$, since such a letter would create a nonzero basis map. A reduced word using only $x^{\pm 1}$ cannot change direction. Its nonperiodic choices are finite strings, the free ray $k[x]$, and the opposite locally nilpotent ray. The first and last fail injectivity of $x$, and the free ray fails invertibility of $1 - x$. A two-sided pure-$x$ word is periodic, and in every associated band $x$ acts invertibly. On $M$, $x$ is injective but not onto. None of these possibilities equals $M$. This includes arbitrary nonzero indecomposable Laurent coefficient modules, not just finite bands.

For a general nonzero completed lattice, finite generation over $A$ would imply finite generation over $k[u]$. Every nonzero polynomial in $u$ is a power of $u$ times a unit of $k[[u]]$, so it acts injectively. The lattice would be a nonzero finite free $k[u]$-module. Multiplication by $1 - u$ on such a free module has a nonzero cokernel, but the same map is invertible over $k[[u]]$. This proves the contradiction in the core. The strictly descending $u$-adic chain separately excludes artinianity. These are actual failed hypotheses, not merely unverified applicability.

Lekili–Polishchuk's precise convention takes $D(A)$ to mean perfect dg modules. Their locally gentle definition allows infinite dimension, and their smoothness criterion says no forbidden cycle; properness says no permitted cycle. Both fail for every nonempty closed dessin here. Their Theorem3.2.2 therefore does not apply to the full polynomial dessin algebra, even though it applies to suitable infinite-dimensional quotients. The OPS finite-dimensional geometric-intersection theorem has its own grading, local-system, boundary and exceptional coincident-curve conditions. An unrestricted statement that every gentle/string derived object is an ungraded geodesic crossing is not justified by either source.

For the interval quotient $B = A/(y) = k[x]$, global dimension one follows from its PID structure: every submodule of a finite free module is free, giving finite projective resolutions of all finitely generated modules. Thus $\mathrm{Perf}(B) = D^b(\mathrm{mod}_{\mathrm{fg}}\, B)$ in grading zero. Its permitted loop gives one unmarked boundary component and its length-one forbidden thread $x$ gives one stop on the other boundary, the annulus of the smooth partially wrapped model. This construction uses the infinite polynomial quotient, not the completed $k[[x]]$ category.

Finally the quotient does not give a derived fully faithful embedding by ordinary inflation. Resolve $B$ over $A$ by the alternating maps $y, x, y, x, \ldots$. Hom into $B$ gives $0, x, 0, x, \ldots$, so $\mathrm{Ext}^{2r}_A(B, B) = B/xB = k$ for every $r \geq 1$, whereas all positive self-Ext over $B$ vanish. Consequently its inflation is not derived fully faithful, and the perfect $B$-module $B$ becomes an infinite-projective-dimension $A$-module. A surface functor transferring these categories must account for that difference explicitly.

# S8 Arithmetic character sums and the cellular determinant

## S8.1 The uniform calculation and its exceptional cases

In the finite-field calculation write $k = \mathbb{F}_q$; the final order-insertion paragraphs specify a separate characteristic-zero coefficient field. The curve is nonsingular because its discriminant is $-432d^2 \neq 0$, and has one point at infinity. Thus counting $1 + \eta(x^3 - d)$ in each affine fiber proves the trace formula in Theorem 6.2 directly. The multiplicative group of $k$ is cyclic. When $3 \mid q - 1$, a nonzero cube has three roots and a noncube none, detected by $1 + \rho(t) + \bar{\rho}(t)$. At zero this expression is one, using the *constant* one. This convention differs from mathlib's zero-extended trivial multiplicative character, whose value there is zero. The proof and formal components never replace the constant by that character.

The substitution is valid including $u = 0, 1$ because the characters are zero extended:

$$\sum_t \rho(t)\eta(t - d) = \sum_u \rho(du)\eta(du - d) = \rho(d)\eta(-d) \sum_u \rho(u)\eta(1 - u).$$

Multiplying by the initial minus sign gives exactly $\pi_d + \bar{\pi}_d$. Changing $\rho$ to $\rho^{-1}$ conjugates both $j$ and $\lambda_d$, so does not change the unordered roots or the integer trace. Characteristics two and three and $d = 0$ are excluded throughout. For a residue characteristic $p \equiv 2 \pmod 3$, odd extension degrees have trace zero by cube bijectivity, whereas even degrees lie in the split case; no claim of zero trace for all extensions of that characteristic is made.

For completeness, choose a nonzero $\mathbb{F}_p$-linear functional on $k$ and compose it with $t \mapsto e^{2\pi i t/p}$ to obtain $\psi$. For a nontrivial multiplicative character $\chi$,

$$\begin{aligned} |G(\chi)|^2 &= \sum_{u \neq 0} \chi(u) \sum_{y \neq 0} \psi((u - 1)y) \\ &= (q - 1) - \sum_{u \neq 0,1} \chi(u) = q. \end{aligned}$$

Both orthogonality identities follow by translating a finite character sum by an element on which its character is not one. The $z = 0$ terms in $G(\chi)G(\theta)$ sum to $\theta(-1)\sum_x(\chi\theta)(x)$, zero if the product is nontrivial. For $z \neq 0$ set $x = zu$, $y = z(1-u)$ to obtain $J(\chi,\theta)G(\chi\theta)$. Coprime orders three and two ensure our product character is nontrivial. Taking norms proves $|J|^2 = q$, rather than presuming a desired positivity. This is the classical calculation reviewed directly in [68]; no claim of novelty is made.

At seven the five nonzero summands are $-\omega^2, -\omega, \omega, -\omega^2, 1$, so $J = 3 + 2\omega$. Multiplication by $\lambda_2 = \omega^2$ gives $-1 - 3\omega$. Their matrices are

$$\begin{pmatrix} 3 & -2 \\ 2 & 1 \end{pmatrix}, \qquad \begin{pmatrix} -1 & 3 \\ -3 & 2 \end{pmatrix}.$$

Both have determinant seven and the same positive Gram form; their traces are four and one. The arithmetic distinction between the rational models now comes from explicit finite-field characters, while their complex dessin is unchanged. Exact prime-field diagnostics test every nonzero twist for the listed nineteen primes; the theorem itself is uniform in prime powers and rests on the proof, not that finite list.

## S8.2 Deck descent and the actual cellular basis

The CM cellular construction in this and the next subsection assumes $q \equiv 1 \pmod 3$. The cover $\beta = x^3/d$ has deck transformations $(x,y) \mapsto (\zeta_3 x, y)$ and $(x,y) \mapsto (x,-y)$, generating $C_3 \times C_2 = C_6$. Label their actions on darts by $\sigma = (012)(345)$ and $\alpha = (03)(14)(25)$; they commute. Their product is $(0\,4\,2\,3\,1\,5)$. They preserve the edge pairing and both successor permutations. Consequently they define algebra automorphisms, preserve $J$ and its completion and fix the common $u$. The projective-resolution maps are compatible since $\theta_g(a_h x) = a_{gh}\theta_g(x)$. Twisting module actions gives an exact functor and the corresponding relabelled resolution, without an assumption of arithmetic Frobenius.

Over an algebraic closure choose $a^3 = d$, $b^2 = d$ and $\varphi(x,y) = (ax, by)$ from $E_1$ to $E_d$. For the $q$-power Frobenius morphisms,

$$\varphi^{-1} \circ F_d \circ \varphi = h_d \circ F_1, \qquad h_d(x,y) = (a^{q-1}x, b^{q-1}y).$$

When $q \equiv 1 \pmod 6$, the first coordinate is the cubic residue $d^{(q-1)/3}$ and the second $\eta(d)$. Choose the embedding of cube roots defining $\rho$ consistently. On the deck eigenspace labelled by $dx/y$, the *deck* character is their ratio, $\rho(d)/\eta(d) = \lambda_d$. This labels the twist, not a nonzero Frobenius action on differentials in characteristic $p$: that pullback is zero. An inverse convention for arithmetic Galois Frobenius changes the action convention; our displayed formula uses the geometric curve morphism explicitly. The character-sum polynomial fixes its trace without an unstated convention.

Orient the three geometric edges as in the core. Each occurs twice with opposite orientation in the single face, so $\partial_2 = 0$; $\partial_1$ has columns $(-1,1)^t$. The homology basis $b_0 = e_0 - e_1$, $b_1 = e_1 - e_2$ has Gram matrix $G$ inherited from the orthonormal cellular edges. Rotation sends $b_0 \mapsto b_1$, $b_1 \mapsto -b_0 - b_1$, exactly $C$. The involution swaps vertices and reverses each edge, hence acts as $-I$ on this plane. Thus multiplication by $A + B\omega$ is literally $AI + BC$, not merely an abstract matrix with the correct trace. Expanding its determinant gives the norm $A^2 - AB + B^2$, and multiplication of this norm proves $M^t G M = qG$. The displayed positive quadratic decomposition proves the metric independently of unknown zeros.

### S8.3 An explicit chain lift and determinant cancellation

Let $P$ be the three-cycle permutation on $C_1$. Write $\pi_d = A + B\omega$ and $\gamma = A + B$. Define

$$F_1 = AI + BP, \qquad F_0 = \frac{1}{2}\begin{pmatrix} 1+\gamma & 1-\gamma \\ 1-\gamma & 1+\gamma \end{pmatrix}, \qquad F_2 = q.$$

These are rational cellular maps. The sum-zero plane of $F_1$ has restriction $M_d$, while the edge-sum line has eigenvalue $\gamma$. On $C_0$ the vertex-sum line has eigenvalue one and the boundary line eigenvalue $\gamma$. It follows directly that $F_0\partial_1 = \partial_1 F_1$; the other chain condition is zero. The induced maps on $H_0, H_1, H_2$ have characteristic factors $1 - T$, $P_d(T)$, $1 - qT$. In particular

$$\det(I - TF_0) = (1 - T)(1 - \gamma T), \qquad \det(I - TF_1) = (1 - \gamma T)P_d(T).$$

The contractible boundary factor cancels, giving

$$\frac{\det(I - TF_1)}{\det(I - TF_0)\det(I - TF_2)} = \frac{P_d(T)}{(1 - T)(1 - qT)}.$$

This equality holds as a formal rational identity and by cancellation at apparent intermediate singularities. The alternating determinant with exponent $(-1)^i$ is its inverse; signs are not changed silently. The standard elliptic trace and determinant theorem identifies the right side with the finite-field curve zeta function [22, 20]. The elementary proof above supplies its actual trace and norm here; it does not independently reprove the general cohomological trace formalism.

This lift also has the required descent character on $H_1$. Write the base Jacobi value as $j = A_1 + B_1\omega$. Then $h_d$ acts there by $\eta(d)C^r$ when $\rho(d) = \omega^r$, so twisting $A_1 I + B_1 C$ multiplies its eigenvalue $j$ by $\lambda_d$, giving $M_d = A_d I + B_d C$. Reduction modulo $C^2 + C + I = 0$ gives the coefficients of $\pi_d$. If the corresponding full three-edge polynomials differ by $c(I + P + P^2)$, the coefficients are $c = 0, \eta(d)B_1, \eta(d)A_1$ for $r = 0, 1, 2$ respectively; they have the same homology action and differ only on the contractible boundary pair. An explicit homotopy sends $v_- - v_+$ to $c(e_0 + e_1 + e_2)$ and the vertex sum to zero. Its $\partial H + H\partial$ is the edge difference and the boundary scalar difference $3c$. Thus this normalization is compatible up to rational chain homotopy. It is not asserted to arise from a continuous Frobenius map of the complex Riemann surface, or to give a canonical integral Tate-lattice comparison.

### S8.4 A decisive restriction at inert primes

For $q \equiv 2 \pmod 3$, Frobenius sends $\zeta_3$ to $\zeta_3^{-1}$. Thus its cohomological relation with the cubic deck action requires $MC = C^2 M$ (the pullback convention yields the equivalent reversed relation). On a real two-dimensional carrier with our fixed $C$, comparing entries forces

$$M = \begin{pmatrix} a & b \\ a+b & -a \end{pmatrix}, \qquad \det M = -a^2 - ab - b^2 = -\tfrac{1}{2}(a^2 + b^2 + (a+b)^2) \le 0.$$

The genuine elliptic Frobenius determinant is $q > 0$, so such a real carrier is impossible. This is an elementary instance of the real-coefficient cohomology issue discussed by Deninger [62]. It does not rule out a complex realization, a larger real space, a different coefficient theory or an infinite-dimensional height operator. In particular it is not a refutation of an unrestricted Hilbert–Pólya approach.

There is a local complex replacement, with its limitations explicit. The conjugation matrix

$$K = \begin{pmatrix} 1 & -1 \\ 0 & -1 \end{pmatrix} \quad \text{satisfies} \quad K^2 = I,\ KC = C^2K,\ K^tGK = G.$$

Set $F = i\sqrt{q}\,K$ on the complexification. Then $\operatorname{tr} F = 0$, $\det F = q$, $F^2 = -qI$ and $F^*GF = qG$, giving the correct trace-zero local factor. This is a two-dimensional complex, hence four-dimensional real, local model. This is a one-$q$ model; independently choosing $i\sqrt{q}K$ for each extension field does not give Frobenius-power compatibility. Such a family would need to start from one $F_p$ and prove the comparison for its powers. The inserted $i\sqrt{q}$ is not a canonical rational or integral Tate comparison or a construction of a global positive determinant. The fixed real two-dimensional model has genuinely failed; its surviving split-prime construction and this complex alternative must not be conflated.

### S8.5 The remaining order and global spectral obligations

The deck automorphism has finite order, so over characteristic zero its eigenvalues are roots of unity. Its permutation action alone cannot supply a weight-one norm $\sqrt{q} > 1$. The independently computed Jacobi coefficient is essential. The above construction closes the specified local cellular benchmark, not a reconstruction of arithmetic from bare complex dessin data.

For these order-insertion tests take $\Lambda = \Lambda_{\mathbb{Q}}$ over $R = \mathbb{Q}[[u]]$, with rational residue simples and $V_d = \mathbb{Q}^2$. This is distinct from reducing the matrix in characteristic $p$, where its determinant $p$ becomes zero. There are two concrete failures of a naive order lift. First, a residue simple $S_e$ is annihilated by all arrows; $S_e \otimes V_d$ with independent endomorphism $I \otimes M_d$ has the desired underlying finite-$\mathbb{Q}$ determinant. But its projective dimension is infinite by the nonterminating forbidden resolution, and $\sigma$ permutes all three vertex labels, so choosing $S_e$ does not preserve full deck symmetry. This is not a perfect-order determinant realization.

Second, $\Lambda \otimes V_d$ is free of rank two over $\Lambda$, but the underlying $R$-rank of $\Lambda$ is eighteen. In an $R$-basis, the $\Lambda$-*linear* $I \otimes M_d$ is eighteen copies of $M_d$, hence its determinant is $P_d^{18}$. On $e\Lambda$ the rank is six: each of its two hereditary branches contributes a row of length three, with finite conductor gluing changing no rank. This gives $P_d^6$. Restoring all three residue vertices similarly gives $P_d^3$ for this untwisted tensor action. A genuine deck-semilinear map permutes factors as well and need not have these powers; it requires its own comparison. No justified determinant-line root or multiplicity-removing functor is supplied here.

Finally fix a nonzero rational $d$ when forming its global Euler product from good reductions. Local purity cannot be multiplied into a global zero-location proof. Since each reciprocal root has modulus $\sqrt{p}$, the product of good elliptic factors is absolutely convergent on $\operatorname{Re} s > 3/2$. The local zeros at $\operatorname{Re} s = 1/2$ are outside that initial product argument. The finite-field curve zeta's center is $1/2$; the global elliptic $L$-function's unshifted center is one, or $1/2$ only after a half-unit shift. Neither is an identification with the primitive Dirichlet function targeted earlier. Ramified factors, archimedean normalization, adelic descent and the global determinant-preserving positive spectral limit remain separate obligations.

## S9 The theta order interface beyond tensor insertion

We prove the multiplicity theorem and construct its actual functorial escape, retaining the completed infinite order as the primary algebra. The finite algebra used below is its specified special fiber, not a replacement primary example.

## S9.1 Normal forms and the special fiber

Index the three quiver vertices by $i \in \mathbb{Z}/3$. Let $a_i^{\pm} : i \to i-1$ be the two permitted branches. Only paths of constant sign survive. Write $p_{i,r}^{\pm}$ for their length-$r$ paths and $c_i^{\pm} = p_{i,3}^{\pm}$. The split order model gives

$$ue_i = c_i^+ + c_i^-, \qquad p_{i,r+3}^{\pm} = up_{i,r}^{\pm} \quad (r \geq 1).$$

The second identity follows because multiplying by the opposite full cycle switches branch and vanishes. An $R$-basis of $e_i\Lambda$ is

$$e_i,\ a_i^+, a_i^-,\ p_{i,2}^+, p_{i,2}^-,\ c_i^+.$$

These span by the two identities and completion. They are independent in the two hereditary branch orders: off-diagonal entries distinguish the four shorter paths; the remaining combination $xe_i + yc_i^+$ has paired diagonal entries $(x + uy, x)$, forcing $x = y = 0$. This is an actual power-series basis, not just a generic rank calculation.

In $B = \Lambda/u\Lambda$, the same eighteen basis vectors remain. A product of nonempty paths is zero if branches switch or the total length exceeds three; the length-three minus path is the negative of the plus path. A length-four path is $u$ times a length-one path, hence zero. The radical $\mathfrak{r}$ is the positive-length ideal, with $\mathfrak{r}^4 = 0$ and $B/\mathfrak{r} = \mathbb{Q}^3$. Each $e_iB$ has radical-layer dimensions $(1, 2, 2, 1)$ and has two composition factors at each vertex, so the Cartan matrix is

$$C_B = \begin{pmatrix} 2 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}.$$

The Cartan matrix alone does not establish the endomorphism formula; the path filtration supplies that step.

## S9.2 All projectives and all linear endomorphisms

Let $J$ be the completed arrow ideal. Every $1 - x$, $x \in J$, has its convergent formal inverse; the same applies to $1 - yx$ for arbitrary $y \in \Lambda$. Thus $J$ is contained in the Jacobson radical, and $\Lambda/J = \mathbb{Q}^3$ implies equality. If $P$ is finitely generated projective, choose a basis of each multiplicity space in $P/PJ$ and lift its vectors into $Pe_i$. This gives

$$Q = \bigoplus_i (e_i\Lambda)^{n_i} \longrightarrow P$$

inducing an isomorphism on tops. Nakayama makes this surjective. Its split kernel $N$ is finitely generated and satisfies $N/NJ = 0$, hence vanishes by Nakayama. This proves the decomposition for every such projective without an unproved freeness assumption over $\Lambda$.

An entry $e_i\Lambda \to e_j\Lambda$ is left multiplication by an element of $e_j\Lambda e_i$. Its degree-zero part is possible only when $i = j$ and is a scalar multiple of $e_i$. For $f$ on $P$, write its top blocks $A_i$ and set $A = \operatorname{diag}_i A_i$. The remainder has positive path length. On $P/uP$ it strictly raises the path filtration. At each of the two length-one and two length-two slots, and the length-zero and length-three slots, the degree-preserving map is exactly $A$. In an increasing-degree basis the matrix is lower block triangular. For the formal upper-block convention order the six slots instead as

$$3+,\ 2+,\ 2-,\ 1+,\ 1-,\ 0.$$

There are six diagonal copies of $A$, allowing arbitrary positive-length off-diagonal entries. Hence

$$\det_{\mathbb{Q}}(1 - T\overline{f}) = \det_{\mathbb{Q}}(1 - TA)^6.$$

Specializing the finite $R$-matrix determinant coefficientwise gives Theorem 6.3. No simultaneous diagonalizability, tensor form of $f$, arithmetic purity or deck equivariance is required.

For a bounded complex $P^\bullet$ with a linear chain map $f^\bullet$, fix either parity convention and write

$$D_R(P^\bullet, f^\bullet; T) = \prod_j \det_R(1 - Tf^j)^{(-1)^j}.$$

The inverse convention gives the reciprocal and the same obstruction. Each factor has constant coefficient one, so its reduction is nonzero. Specialization of this displayed rational expression is defined and equals

$$\left[\prod_{j,i} \det_{\mathbb{Q}}(1 - TA_i^j)^{(-1)^j}\right]^6.$$

Cancellations cannot change the divisibility of divisor multiplicities. In particular adding contractible projective pairs cannot repair it. Here a perfect endomorphism means a bounded-projective chain representative; no statement about an arbitrary infinite resolution or nonperfect lattice is inferred.

Generically put $N = \sum_i n_i$. As a right module over $M_3(K) \times M_3(K)$, $P_K$ is the sum of $N$ copies of each branch's three-dimensional row simple. Any commuting map has matrices $F_+, F_- \in M_N(K)$ on their multiplicity spaces, so

$$\det_K(1 - Tf_K) = \big(\det(1 - TF_+)\det(1 - TF_-)\big)^3.$$

For example, left multiplication by $c_i^+$ on $e_i\Lambda$ has ordinary characteristic determinant $(1 - uT)^3$, so a sixth power is forced only at the special fiber. This also shows why generic reduced determinants and ordinary determinants must not be conflated. We do not need an integral choice of either generic matrix to derive the cube obstruction.

For the local numerator, suppose a sixth power $A(T)^6/B(T)^6$ equals a nonzero quadratic $P(T)$ with $B \neq 0$. Polynomial degrees would give $6 \deg A = 2 + 6 \deg B$, impossible. This includes repeated roots. For the complete curve zeta, $q > 1$ and $|a_q| \leq 2\sqrt{q}$ imply $P(1) > 0$ and $P(1/q) = P(1)/q > 0$. Its two denominator roots are distinct uncancelled simple poles, again impossible for a sixth power. These arguments do not assume global RH or Artin holomorphy.

## S9.3 An actual integral determinant functor of multiplicity one

Project to one hereditary cycle order $B_+ \subset M_3(R)$. Its standard column $W_+ = R^3$ is a left $\Lambda$-module with $e_iW_+ = R\epsilon_i$. More explicitly $a_i^+$ acts by $u^{w_i}E_{i,i-1}$, with $(w_0, w_1, w_2) = (1, 0, 0)$ for one cut, while every $a_i^-$ acts by zero. These matrices respect travel-order multiplication as written: $E_{ij}E_{jk} = E_{ik}$. Each plus full cycle is $uE_{ii}$, so the common central $u$ acts by $uI$. Nonempty paths tend to zero $u$-adically, giving the action of the *full completed order*, not just its polynomial subalgebra.

This $(\Lambda, R)$-bimodule defines $- \otimes^{\mathbf{L}}_{\Lambda} W_+$. On a bounded complex of finite projectives it is computed termwise, and its terms are finite free over $R$. An acyclic bounded projective complex is split contractible, and tensor preserves that contraction. Thus it descends to a genuine triangulated functor $\Phi_+ : \mathrm{Perf}(\Lambda) \to \mathrm{Perf}(R)$. No assertion of exactness on all un-derived modules or flatness of $W_+$ is needed. In fact $W_+$ has infinite projective dimension as a left $\Lambda$-module: $\Lambda$ is noetherian, being finite over the central noetherian ring $R$, and $W_+$ is finitely generated. A finite projective resolution could thus be chosen with finite terms. Every left projective term has $R$-rank divisible by six by the same top argument, whereas $W_+$ has rank three, contradicting the alternating rank identity. This does not prevent tensoring bounded right-projective complexes.

The isomorphism $e_i\Lambda \otimes_\Lambda W_+ \to e_iW_+$ sends $e_i\lambda \otimes w \mapsto e_i\lambda w$. Its inverse is $e_iw \mapsto e_i \otimes e_iw$, proving rank one directly. Consequently $e_i\Lambda \otimes_{\mathbb{Q}} V$, with action $1 \otimes M$, maps to $R \otimes_{\mathbb{Q}} V$ with action $M$. Its characteristic determinant is $\det(1 - TM)$ *once*. On the regular input, the three vertex lines survive and the factor appears three times. A full symmetric vertex orbit therefore is not silently replaced by a chosen vertex.

Different cycle cuts give different integral column lattices. Their generic modules are isomorphic, and branch endomorphism characteristic determinants agree by generic conjugation; equality over $K$ implies equality over $R$. Nevertheless an equivariant lattice is stronger data. Branch selection also changes the functor: for the unit $f = e_i + c_i^+$ on $e_i\Lambda$, the plus determinant is $1 + u$ and its characteristic factor is $1 - (1 + u)T$. Under $\alpha$, $c_i^+$ becomes $c_i^-$, whose plus action is zero; the determinant becomes one and the characteristic factor $1 - T$. Since these scalar automorphisms differ, no natural isomorphism of determinant lines can make this chosen branch functor $\alpha$-invariant for all endomorphisms. Pairing both branch functors makes this branch-swap data available but doubles the local factor on the basic-projective tensor input.

### S9.4 The exact rotation obstruction and its allowed escape

Consider any left $\Lambda$-lattice supported on the plus branch, free of rank one at every $e_i$, and with central $u$ acting as multiplication by $u$. Write its plus arrow scalars as $a_0, a_1, a_2 \in R$. Their product is $u$, so they are nonzero and their nonnegative valuations sum to one. A strict $R$-linear lift of $\sigma$ that intertwines the order rotation permutes the three vertex lines by units. Its arrow-intertwining identities make the three arrow valuations equal. Thus $3m = 1$ for an integer $m \geq 0$, impossible. The argument already excludes an invertible intertwiner; imposing that its cube be one cannot help.

There is an elementary coefficient proof of the same obstruction. Intertwining by units makes vanishing of the three arrow constant coefficients equivalent. Their product has constant coefficient zero, so all three vanish. Factoring $u$ from two of them would make $a_0a_1a_2$ divisible by $u^2$, contradicting that it equals $u$. This is the literal power-series component checked in Lean.

After the explicit ramified change $R' = \mathbb{Q}[[v]]$, $u = v^3$, *choose a new column lattice* with all three plus arrow maps $vE_{i,i-1}$. Cyclic permutation of the basis is an order-three intertwiner fixing $v$. The full-cycle relation is now $v^3 = u$ as required. Merely base-changing the old cut lattice gives arrow valuations $(3, 0, 0)$ and does not produce that linearization. This construction changes the base and lattice and does not establish an unramified descent comparison. It also still needs both branches to address $\alpha$.

The rotation obstruction is about actual central-fixed rank-one branch lattices, not every possible determinant-level natural transformation. The branch characteristic determinants may be rotation-invariant through their generic reduced interpretation even without a lattice lift. Neither this theorem nor the ordinary-determinant obstruction excludes semilinear correspondences, other determinant categories or infinite analytic models.

The next arithmetic task is therefore narrower and more concrete: choose a branch functor, a ramified lattice or a different equivariant replacement only after proving the required descent comparison, and construct its prime action from arithmetic data rather than feed in the already known $M_d$. Bad primes, integral/Tate comparison, local-to-global conductor and gamma factors, and a positive spectral limit for the same completed function remain unresolved.

## S10 Explicit tame descent and the full-deck coefficient test

The preceding ramified column is a precisely specified object of the original infinite theta-order framework. We compute its descent instead of treating a semilinear group action as automatically

effective integral descent. All constructions below are over the new *mathematical* coefficient field $k = \mathbb{Q}(\omega)$; they do not replace the original rational model or its source history.

## S10.1 The action on genuine formal series

Write $R = k[[u]]$, $S = k[[v]]$ with $u = v^3$. Every $f(v) \in S$ has a unique expansion $f_0(u) + vf_1(u) + v^2 f_2(u)$. Hence $S$ is finite free of rank three over $R$. Let $g(f)(v) = f(\omega v)$. On $W = S^3$ with basis $\epsilon_0, \epsilon_1, \epsilon_2$, put

$$\Theta(f_i\epsilon_i) = \omega^{-i} f_i(\omega v)\epsilon_i, \qquad C\epsilon_i = \epsilon_{i+1} \quad (i \bmod 3).$$

The plus arrow $a_i^+$ acts by $vE_{i,i-1}$, the minus arrow by zero. The cyclic product is $uE_{ii}$ and arbitrarily long permitted paths tend to zero $v$-adically, so this is a module over the full completed order. For $i = 0$, the coefficient in the target under $\Theta a_i^+$ is $\omega v$ and in the source under $a_i^+\Theta$ it is $v\omega^{-2}$; these agree. The same calculation for $i = 1, 2$ proves compatibility with every arrow and therefore the completed algebra. Also $\Theta^3 = 1$.

For all coefficients, not just a truncated diagnostic,

$$[v^n](\Theta w)_i = \omega^{n-i}[v^n]w_i.$$

Thus invariance is equivalent to support in $n \equiv i \pmod 3$, and

$$L = W^\Theta = \bigoplus_{i=0}^{2} Rb_i, \qquad b_i = v^i\epsilon_i.$$

The arrows on this basis have scalar valuations $(1, 0, 0)$, since $a_0^+ b_2 = ub_0$, $a_1^+ b_0 = b_1$ and $a_2^+ b_1 = b_2$. Tame invariants return the old cut column rather than a new strict unramified rotation carrier.

The extension $R \subset S$ is faithfully flat. Its group quotient has a fixed point at $v = 0$, however, and is not a $\mu_3$-torsor there: $S \otimes_R S$ is not the product of the three group graphs at that point. Semilinear $\mu_3$ data are therefore not by themselves the full effective fpqc descent data for a vector bundle on $\operatorname{Spec} R$. The adjunction comparison is explicitly

$$S \otimes_R L \longrightarrow W, \qquad \operatorname{diag}(1, v, v^2),$$

with determinant $v^3 = u$, cokernel $S/(v) \oplus S/(v^2)$ and closed-fiber rank one. This proves its failure to be an isomorphism directly. The quotient stack $[\operatorname{Spec} S/\mu_3]$ retains the stabilizer at zero; its coarse space is $\operatorname{Spec} R$. This local root construction is described in [71]. Alper's vector-bundle criterion [70, Theorem10.3, pp.2386–2387] distinguishes precisely this nontrivial closed-fiber action from effective coarse descent. We use these sources for interpretation; all displayed module comparisons have already been calculated explicitly.

## S10.2 A projective symmetry survives on the endomorphism order

Directly on the three basis vectors,

$$\Theta C = \omega^{-1} C\Theta = \omega^2 C\Theta.$$

Thus the rotation does not preserve $L$. The corrected $vC$ does preserve it, since the factor $\omega$ from $g(v)$ cancels the commutator. In the invariant basis it has matrix $A$ from the core, with $A^3 = uI$ and $\det A = u$. This is generically invertible and gives an order-three projective rotation after passing to the fraction field; it is not an invertible integral lattice intertwiner.

There is nevertheless a true algebra-level escape. Let $F = (f_{ij})$ be an arbitrary $S$-linear endomorphism of the *carrier* $W$, not assumed $\Lambda$-linear. Its tame conjugate has entries $\omega^{j-i} g(f_{ij})$. Invariance therefore means

$$f_{ij} \in v^{[i-j]_3} R,$$

where $[r]_3 \in \{0, 1, 2\}$ is the least residue. After replacing $\epsilon_i$ by $b_i = v^i \epsilon_i$, the entry becomes

$$v^{[i-j]_3+j-i} R = \begin{cases} R & i \geq j, \\ uR & i < j. \end{cases}$$

Consequently $\mathrm{End}_S(W)^\Theta$ is exactly the hereditary three-cycle order $B_+$ in the previously fixed cut convention. Multiplication is preserved by conjugation in the generic matrix algebra; no entrywise projection is being misused as a ring map.

The scalar commutator cancels under conjugation. Hence $\mathrm{Ad}(C)$ commutes with tame conjugation, descends to $B_+$, and is represented there by $\mathrm{Ad}(A)$ because $v$ is central. Since $A^3 = uI$, this algebra automorphism has order three and preserves the hereditary order despite $A^{-1}$ not lying in its unit group. This explains the surviving rotation of the branch order; it does not construct a rotated integral column or a Frobenius endomorphism. The other branch is identical and the paired-diagonal residue gluing still defines the original $\Lambda_R$ inside $B_+ \times B_-$. Its label automorphisms remain the already constructed dessin symmetries.

### S10.3 Determinants, Schur weights and the boundary correction

On $\det_S W = \bigwedge_S^3 W$, both $\det D = 1$ and $\det C = 1$. Thus this line has the scalar action $g$ and genuinely descends to the trivial $R$-line with trivial rotation. This is a real positive determinant-level result. On the other hand,

$$b_0 \wedge b_1 \wedge b_2 = v^3(\epsilon_0 \wedge \epsilon_1 \wedge \epsilon_2),$$

so the natural comparison

$$\det_R(W^\Theta) \longrightarrow (\det_S W)^\Theta$$

is multiplication by $u$, with cokernel $R/(u)$. Both lines are abstractly free of rank one, but their given comparison is not an isomorphism. It records the divisor of the ramified lattice change and cannot be discarded when claiming a determinant-preserving integral comparison. For the *same* operator $vC$, the scalar on both determinant lines is $u$; naturality is respected. Comparing $\det C = 1$ with $\det A = u$ without the extra factor $v$ would instead compare different operators.

For an induced nonzero homogeneous tensor/Schur carrier of degree $d$, the same functorial calculation gives commutator $\omega^{-d}$. The two induced invertible operators commute exactly when $3 \mid d$. This is a group-action statement, not sufficient for coarse vector-bundle descent: $\mathrm{Sym}^3(S^3)$ has closed-fiber inertia multiplicities $(4, 3, 3)$ for the three characters, so its inertia is still nontrivial even though the commutator vanishes. A mixed tensor $W^{\otimes a} \otimes (W^*)^{\otimes b}$ has net exponent $a - b$; $\mathrm{End}_S(W)$ is the concrete net-zero escape computed above.

These carrier functors are not automatically functors on unital $\Lambda$-modules. Applying global $\bigwedge^3$ to the rank-one vertex idempotents kills all three, although their sum is the identity before applying this nonadditive functor. Vertexwise $\bigwedge^3$ of the rank-one column is zero. Vertexwise $\mathrm{Sym}^d$ for $d > 0$ does retain the monomial zero relations but changes the full-cycle scalar from $u$ to $u^d$. It leaves the fixed central-action problem unless that change of base is included. None of the Schur cancellations supplies the missing two-dimensional arithmetic carrier.

### S10.4 Why the local Jacobi plane does not descend by this repair

On constant vectors $\Theta$ acts by $D$. It sends the sum-zero cycle $\epsilon_0 - \epsilon_1$ to $\epsilon_0 - \omega^2\epsilon_1$, whose coordinate sum is nonzero. It also does not preserve the edge-sum line. Thus this order column is not already a tame descent of the earlier cellular chain complex or its $H_1$.

For the split local coefficient $\pi = a + b\omega$, the natural rotation polynomial on the balanced carrier is $F = aI + bC$. Since $g$ fixes $a, b \in k$, its commutator is

$$\Theta F - F\Theta = b(\omega^2 - 1)C\Theta.$$

It is nonzero when $b \neq 0$. In particular both prime-seven twists $(a, b) = (3, 2), (-1, -3)$ fail this test. Prime powers with $b = 0$ are not excluded by this particular calculation; that exceptional commutation does not prove a global comparison.

Replacing $C$ by $vC$ gives an invariant carrier map, whose matrix on $L$ is $aI + bA$. Direct determinant expansion gives

$$\det(1 - T(aI + bA)) = (1 - aT)^3 - ub^3T^3.$$

At $u = 0$ this is $(1 - aT)^3$, not $P_d(T)$, which has degree two. The weighted replacement therefore does not recover the established local factor. Neither $C$ nor $vC$ is generally $\Lambda$-linear: they intertwine its rotation. Their sums with the identity have not thereby become endomorphisms in the ordinary order-module category. They are legitimate carrier tests, with exactly the failures just exhibited.

### S10.5 The full orbit is the normalization, and arbitrary kernels still fail

Twist the tame action by the three characters: $\Theta_r = \omega^r\Theta$, $r \in \mathbb{Z}/3$. Its fixed lattice is

$$L_r = \bigoplus_i Rv^{[i-r]_3}\epsilon_i,$$

with the cut at arrow $r$. Rotation gives an actual $R$-linear isomorphism $C : L_r \to L_{r+1}$, taking each displayed basis vector to the corresponding next one. Thus the direct sum of all three cuts has a strict rotation, at the cost of vertex rank three. Before invariants the orbit is $S$-rank nine and underlying $R$-rank twenty-seven; its invariant lattice is $R$-rank nine. The original orbit $S$-lattice still has nontrivial inertia at zero and need not be the base change of these invariants.

There is an explicit integral identification of the invariant orbit. In the standard hereditary order $B_+$, the $j$-th column has entries $u^{t_{ij}}R$ with $t_{ij} = 1$ for $i < j$ and zero otherwise. In its natural column basis the arrow coefficient has exponent

$$\delta_{i0} + t_{i-1,j} - t_{ij} = \delta_{ij}.$$

Its cut is therefore exactly $j$, identifying $B_+e_j$ with $L_j$. Consequently $\bigoplus_r L_r \simeq B_+$ as left $\Lambda_R$-modules. Adding the exchanged minus-branch orbit gives the hereditary normalization $\widetilde{\Lambda} = B_+ \times B_-$, with full deck action and vertex rank six. On a basic projective with external arithmetic matrix this yields $P_d^6$.

In fact this normalization replacement restores the *ordinary determinant for every linear endomorphism*. For a finite projective $P$, the inclusion $P \to P \otimes_\Lambda \widetilde{\Lambda}$ becomes an isomorphism over $K = k((u))$, since $\widetilde{\Lambda}_K = \Lambda_K$. Its extended endomorphism and the original $f$ therefore have the same generic characteristic determinant. Both polynomials have coefficients in $R$, which injects into $K$, so they agree already over $R$. The bounded-complex ratios also agree. This symmetry repair returns the previously refuted ordinary determinant, not merely its external tensor exponent.

For completeness we now prove Theorem 6.4 for arbitrary finite free kernels, without assuming they split integrally into branches or are projective over the order. Let

$$p_\pm = u^{-1} \sum_i c_i^\pm \in \Lambda_K.$$

These are central generic branch idempotents. Put $W_\pm = W_0 \cap p_\pm(W_0 \otimes_R K)$. They are finite free $R$-lattices spanning those generic summands, stable under $\Lambda_R$. Rotation fixes them, branch swap exchanges them. Their vertex ranks are constant on each branch: the arrow maps become invertible over $K$ because a full cycle acts by $u$. Write these ranks $r_+, r_-$. No equality $W_0 = W_+ \oplus W_-$ over $R$ has been assumed.

On the plus branch choose integral vertex bases. The three square arrow matrices $X_0, X_1, X_2$ have product $uI_{r_+}$ around every cycle. Thus

$$\sum_i \mathrm{val}_u(\det X_i) = r_+.$$

The rotation lift is an invertible matrix between each pair of vertex lattices, hence its determinant is an $R$-unit. Its intertwining identities make all three arrow determinant valuations equal. Therefore $r_+ = 3m_+$; similarly $r_- = 3m_-$. The branch-swap lift gives $r_+ = r_-$, so generic rank additivity gives $\mathrm{rank}_R\, e_i W_0 = 6m$. Only invertible intertwiners were used; imposing the full deck group relations is stronger than necessary.

For the general determinant statement reduce modulo $u$ and filter each vertex by $e_i J^a(W_0/uW_0)$. Since $J^4 \subset u\Lambda$, this is a finite filtration. In $\Phi_{W_0}(P) = \bigoplus_i (e_i W_0)^{n_i}$, positive-length entries of $f$ raise the filtration, while the degree-zero matrix $A_i$ repeats on its $i$-th graded pieces. Their total dimension is $6m$. Multiplying the graded characteristic determinants proves the asserted power for *every* $f$. Flatness of the kernel as a left $\Lambda$-module was not used; bounded right projectives compute the derived functor and give finite free $R$-outputs. Denominator constant terms one permit specialization of every bounded alternating ratio. The regular left kernel $\Lambda_R$ attains $m = 1$, making the rank bound sharp.

The remaining arithmetic problem therefore requires a different hypothesis: a genuinely twisted determinant target, a specified non-linear/semilinear correspondence with its own comparison, or an infinite analytic realization. One cannot continue searching inside this closed finite-free, fixed-center, full-deck kernel class for a missing small estimate. The local purity and descent computations supply no global positive spectral limit. Moreover $k/\mathbb{Q}$ has degree two: ordinary restriction of scalars sends a rational local factor $P_d$ to $P_d^2$. A relative $k$-determinant must not silently be identified with the former rational determinant. Finite-field curve zeta, global elliptic $L$-functions and Dirichlet GRH keep their distinct scopes and centers.

## S11 Literal formal and computational coverage

`DessinNodeOrder.lean` defines the fiber-product subring of two genuine power-series rings. Its twenty printed declarations establish diagonal injectivity, nonzero branches, branch zero products, factorizations and annihilator identities, regular $u$, augmentation surjectivity and exactness, bijective power-series coordinates, square-zero alternating maps, three kernel-equals-range statements and projectivity of the actual regular/product terms. The coordinate statement is a bijection; the file does not yet package it as an $R$-linear equivalence or declare the general order freeness theorem.

`DessinInvariantComponents.lean` has ten printed declarations: three CK1-collapse components, determinant change of bases with an actual inverse, symmetric-square zero, identity,

composition and relation preservation, the degree-two Schur character identity, and the nodal central-parameter obstruction. Its symmetric-square matrix is

$$\mathrm{Sym}^2\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a^2 & ab & b^2 \\ 2ac & ad+bc & 2bd \\ c^2 & cd & d^2 \end{pmatrix}.$$

This is the ordinary monomial basis convention; composition is proved universally over any commutative ring. The full project suite prints types and foundational dependencies; no theorem about general LR decompositions, abstract derived equivalences, C* $K$-theory or GRH is inferred from these component proofs.

The exact Python checks use fixed rational matrices and exhaustive finite sets, with no random seeds. They construct the displayed permutations and paths, evaluate rational Cartan identities, check formal Euler products through a stated degree, verify unrestricted-output Fock basis identities on bounded input sets, compute incidence Smith data, construct all nine string representations and signed maps, and test Young/LR/Cauchy examples and the seven-element-field counts. Those diagnostics are reproducible finite corroboration. The universal infinite claims rest on the proofs or explicitly cited theorems, and the arithmetic comparison remains open.

The ten additional `DessinPullbackComponents.lean` statements use literal two-state matrix units and a subring of a product of arbitrary rings, with equal images under two ring homomorphisms. The section hypothesis of its projection-surjectivity theorem is explicit. It does not construct a topological C*-pullback or assert that a commutative diagram is automatically a pullback. The no-residue theorem is universal algebra, with a nontrivial commutative target; it is not an assumed arithmetic positivity premise.

`ThetaJacobi.lean` adds sixteen declarations, with actual finite fields and MulChar rather than a hypothetical arithmetic kernel. Its norm theorem invokes mathlib's proved Jacobi inverse-product identity and derives characteristic inequality from finiteness; all three character nontrivialities are explicit. Other declarations verify the rescaling substitution, CM multiplication/polynomial/Gram positivity components, exact prime-seven matrix polynomials, and actual 2-by-3 boundary/3-by-2 cellular inclusion, homology restriction, three-edge determinant and rational vertex chain-map equations. It does not formalize elliptic point counting or an order-to-Frobenius functor.

The final three `ThetaJacobi.lean` declarations verify the actual conjugation matrix identities and the real two-dimensional inert-prime determinant obstruction. They do not assume any unknown zero location or exclude complex/enlarged spectral realizations.

`ThetaProjectiveMultiplicity.lean` adds five declarations. Arbitrary upper-block matrices with six identical diagonal blocks have the sixth-power determinant and characteristic-factor formulas. A literal power-series constant-coefficient map specializes matrix determinants, for each scalar parameter; the coefficientwise polynomial/order interface remains prose. A nonzero quadratic cannot be a rational-function sixth power, proved by polynomial degrees. Three genuine power series with product $X$ cannot have the cyclically equivalent residue-vanishing conditions forced by invertible rank-one arrow intertwiners. The derivation of the path blocks from the full theta order, projective decomposition, derived branch functor and deck intertwiners is not fully formalized. The complete analytic suite now has208printed declarations; no hidden arithmetic positivity assumption occurs.

`ThetaTameDescent.lean` adds twelve printed component declarations (and one auxiliary polynomial lemma). They use genuine formal-power-series rescaling and its coefficient operator, proving the coefficient action, invariant-coefficient vanishing and the tame operator's cube. Actual three-state matrices satisfy the clock/rotation commutator, comparison determinant and closed-fiber equations,

weighted-cycle cube and full characteristic polynomial, noncommutation of a nonconstant rotation polynomial, trivial whole-column determinant actions, and the same-operator comparison for $vC$. The classification of all fixed series by residues, full order action and hereditary invariant algebra, stack interpretation, valuation/rank theorem, general coefficient-functor filtration and GRH are not thereby fully formalized. No unknown positivity premise is assumed.

## S12 Explicit covering calculations and derived transport

Here are the linear details behind Proposition 7.2. Write $x_i = F_{ii}$ for diagonal entries of an endomorphism $F$ preserving the three vertex spaces. Solving $FX = XF$, $FY = YF$ leaves a common diagonal $c$ and six free entries. More explicitly its only off-diagonal entries are

$$F_{02} = F_{13} = t_0, \quad F_{15} = t_1, \quad F_{35} = F_{46} = t_2, \quad F_{48} = t_3, \quad F_{68} = F_{70} = t_4, \quad F_{72} = t_5.$$

All others vanish. This can be checked column by column using the displayed partial permutation matrices $X, Y$: their nonzero columns specify all constraints, and the zero columns force the remaining entries to zero. If $r_0, \dots, r_5$ are the six radical generators in the displayed order, the only nonzero products are

$$r_0 r_2 = r_1, \qquad r_2 r_4 = r_3, \qquad r_4 r_0 = r_5.$$

Every product of three radical generators vanishes. Thus the listed space is a local algebra, not merely a seven-dimensional vector-space count.

For clarity, the Euler equations used in the obstruction are

$$\begin{aligned} &d_1 - d_0 = 1, && d_3 - d_2 = 1, && d_4 - d_3 = 1,\\ &d_6 - d_5 = 1, && d_7 - d_6 = 1, && d_0 - d_8 = 1,\\ &d_1 - d_2 = 1, && d_4 - d_5 = 1, && d_7 - d_8 = 1, \end{aligned}$$

where $d_i = D_{ii}$. Subtract the last row from the first two rows after summing: the left side is zero and the right side is three. This is an identity in every characteristic-zero coefficient field. In characteristic three this particular obstruction disappears; no conclusion there is asserted. The closed walk

$$a_0 b_0^{-1} a_0\, a_1 b_1^{-1} a_1\, a_2 b_2^{-1} a_2$$

explains the computation: six forward steps and three backward steps have net lift displacement three. Arbitrary changes of basis cannot remove the Euler contradiction. This is an algebraic lift obstruction, not an assertion about the topological fundamental group of the compact torus.

The Hom formula also has a constructive perfect-complex version. Each cover representable is finite support and pushes to its corresponding vertex projective. For a bounded complex $P^\bullet$ of finite sums of such representables and a bounded finite-support complex $N^\bullet$, decompose the entire Hom complex by deck displacement. The differential preserves that displacement, and only finitely many displacements occur across the finitely many degrees. Taking cohomology gives

$$\operatorname{Hom}_{D(B_L)}(FP^\bullet, FN^\bullet[q]) \simeq \bigoplus_g \operatorname{Hom}_{D(\widetilde{B}_L)}(P^\bullet, T_g N^\bullet[q]).$$

Bounded projective complexes compute these derived Hom spaces, which justifies the equality rather than assuming every module perfect. This elementary proof does not assert an equivalence of entire bounded derived categories or cover objects of infinite projective dimension. The broader precovering statement has its exact hypotheses in [74, Theorem 4.28].

Finally the branch involution $\sigma(a_i) = b_i$, $\sigma(b_i) = a_i$ preserves $B_L$. In characteristic different from two one can study $B_L \# k[\mathbb{Z}/2]$ through equivariant representations. For the nine-dimensional module above, $\operatorname{rank} X = 6$ and $\operatorname{rank} Y = 3$, so $M \not\cong \sigma M$. The sum $M \oplus \sigma M$ has canonical swap equivariance. This is an explicit orbit construction; it neither chooses a nonexistent fixed-object linearization nor identifies this skew group algebra with the particular GLS algebras in [75].

`CoveringBand.lean` verifies the actual matrices' mixed products, cubic nilpotence and absence of an arbitrary Euler matrix. The general covering, module endomorphism classification, derived functors and the surface/Fukaya equivalence are not fully formalized by these three matrix statements. Exact rational code additionally checks thirty-six Hom pairs, the full radical multiplication and the two Hom-control defects.

## S13 Weighted profiles, exact arithmetic examples and the surviving task

The finite matrix proof also retains an arbitrary row profile. Suppose $|A_{ui}|^2 = w_u$ for every column, set $S = \sum_u w_u$ and $Q = \sum_u w_u^2$. The row diagonals are $Nw_u$ and column diagonals are $S$, so the same identity proves

$$\sum_{i \neq j} |K_{ij}|^2 \geq N^2 Q - NS^2.$$

This is a finite statement. Applying it to an infinite Schwartz tail would require its convergence and every newly encountered prime collision; these are not discarded. If a weighted pair-square attempt includes the annular $W$, one can instead choose a closed subannulus on which a nonzero $|W|$ is bounded below. Its prime count still has order $D/\log D$. The fixed $\nu$ has unit modulus outside its conductor. These facts do not replace the original coefficients in a signed estimate.

Here are exact sextic computations with all nonzero elements of the Eisenstein norm ball, and both prime ideals above each split rational prime in the column annulus. These kernel diagnostics use $\nu = 1$ and exclude only primes above2,3; they do not compute a signed Möbius energy. Values are integer norms in $\mathbb{Z}[\omega]$.

| $D$ | $H$ | $R$ | $N$ | $\sum_{i \neq j} \lvert K_{ij} \rvert^2$ | $NR(N-R)$ |
|---|---|---|---|---|---|
| 128 | 4 | 18 | 22 | 6840 | 1584 |
| 1024 | 9 | 36 | 138 | 651672 | 506736 |
| 4096 | 81 | 300 | 456 | 60080328 | 21340800 |
| 256 | 64 | 240 | 40 | 363672 | −1920000 |

The final row has $H = D^{3/4}$ but $N < R$; its negative lower bound is correctly vacuous. The asymptotic theorem follows from the proof, not an extrapolation of these examples. For $p \equiv 1 \pmod 3$ and $w^2 + w + 1 = 0$ in $\mathbf{F}_p$, evaluate $a + b\omega$ as $a + bw$ and take its $(p-1)/6$ power, lifting the six roots via $\zeta_6 = 1 + \omega$. Every zero is tested before a unit value is assigned. For instance $1 + 3\omega$ has norm7 and vanishes at $\omega = 2$ modulo7, so the unimodularity hypothesis cannot be used beyond its stated domain.

`ShortRowPairBarrier.lean` verifies the finite frame identity, row-diagonal bound, weighted inequality and unimodular specialization, with actual complex sums and explicit hypotheses. It does not formalize the ideal-character comparison or prime-ideal theorem. The deterministic integer checker also verifies weighted profiles and256 scalar pair sums independently of its matrix computation. No numerical model or learned coefficient is used.

The next signed route must retain $\mu, \nu, W$ before Cauchy. An exact starting identity in ideal Dirichlet convolution is

$$\mu = 2a - a * a * \mathbf{1} + b * b * \mathbf{1}, \qquad a = \mu\, \mathbf{1}_{Nn \le U}, \quad b = \mu - a.$$

Indeed $\mu * \mathbf{1} = \delta$ implies $(\mu - a) * (\mu - a) * \mathbf{1} = \mu - 2a + a * a * \mathbf{1}$. On squarefree $n$, each divisor factorization retains pairwise coprimality; the original row zeros still multiply along it. Estimating these type-I/type-II kernels with their true conductors is a new open arithmetic task. The identity alone supplies no cancellation or GRH proof.

# S14 Exact factorization, retained zeros and the failed Cauchy route

## S14.1 Proof of the regrouping with original coefficients

The monoid consists of nonzero integral ideals prime to the fixed excluded set $S$. Every divisor of a good ideal is good, so the identity $\mu * \zeta_0 = \delta$ holds on this same monoid. Expanding convolution in (1) gives finite divisor sums. Multiply the identity by $\mu(n)^2 \nu(n) \chi_n(u) W(Nn/D)$ and sum. The extra squarefree indicator does not change $\mu(n)$, since $\mu(n)^3 = \mu(n)$. On its support $n = abc$ forces each factor to be squarefree and any two to be coprime. Sextic multiplicativity, including zeros, gives $\chi_n(u) = \chi_a(u)\chi_b(u)\chi_c(u)$; a collision with any factor makes both sides zero. The fixed character also satisfies $\nu(n) = \nu(a)\nu(b)\nu(c)$ on these good ideals.

In the small-factor piece group $d = ab$. The remaining variable $c$ is squarefree and coprime to $d$; $Nd \le U^2$ is necessary, but not sufficient for a nonzero $\beta_U(d)$: its specified truncated factorizations must still exist. For each such factorization on squarefree $d$, $\mu(a)\mu(b) = \mu(d)$, proving the coefficient formula. In the large-factor piece group $s = bc$. The condition $Nb > U$ implies $Ns > U$, and summing that divisor gives exactly $g_U(s)$. Since $Ns > U \ge 1$, $s$ is not the unit ideal; its complete Möbius divisor sum vanishes, proving the alternative formula for $g_U$. This regrouping never replaces the two-variable weight by $W(Ns/D)$, never deletes $(a, s) = 1$, and never replaces the resulting coefficients by a favorable arbitrary family.

One can instead keep the unrestricted convolution identity on nonsquarefree columns, but then both individual pieces and their local conductor description change. For $U = 1$ and a good prime $p$, the column $p^2$ contributes one to each of $a_U * a_U * \zeta_0$ and $b_U * b_U * \zeta_0$, even though $\mu(p^2) = 0$; these cancel only after the pieces are recombined. Its squared sextic symbol has a cubic local component. Thus dropping total squarefreeness is not a harmless piecewise simplification, although the full unrestricted algebraic identity still holds.

## S14.2 The two conductor scales are unchanged

For a squarefree $n = as$ with $(a, s) = 1$, every prime of $n$ has one nontrivial sextic local component in the row variable. Hence its primitive finite conductor is the ideal $as$; $NaNs \asymp D$ on the column support. On multiplying a column by the conjugate of another, the local exponent is zero at common primes, 1 at a prime of $n/h$, and $-1$ at a prime of $n'/h$. The common prime still imposes its original zero on rows. Writing $n = hr$, $n' = ht$ therefore gives exactly

$$\chi_n(u)\overline{\chi_{n'}(u)} = \mathbf{1}_{(u,h)=1}\chi_r(u)\overline{\chi_t(u)}, \qquad q = rt, \quad \mathfrak{l} = hrt.$$

Here $q$ is the primitive oscillatory conductor and $\mathfrak{l}$ a period of the full zero extension. These are ideals, not necessarily squarefree rational norms. In particular, conjugate primes over one split rational prime are distinct coprime ideal factors. Full defining moduli of $\nu$ and primary conventions remain fixed in $S$; this row-conductor calculation does not assert a new uniform reciprocity theorem in the column variable. In the coprime bulk $Nq \asymp D^2$ remains large.

### S14.3 Why a first generic Cauchy step does not supply the saving

On one type-II dyad let $Na \asymp A$, $Ns \asymp B$, $AB \asymp D$. Keep the full profile and coprimality restrictions in

$$X_a(u) = \nu(a)\chi_a(u) \sum_{\substack{s \text{ squarefree} \\ Ns \asymp B,\,(a,s)=1}} g_U(s)\nu(s)\chi_s(u)W(NaNs/D).$$

The exact elementary Cauchy inequality is

$$\sum_{0<Nu\le H} \left| D^{-1/2} \sum_{Na\asymp A} \mu(a)X_a(u) \right|^2 \le \frac{\sum_{Na\asymp A}|\mu(a)|^2}{D} \sum_{Na\asymp A} \sum_{0<Nu\le H} |X_a(u)|^2.$$

The $a$-mask has been kept inside $X_a$. Even if one granted an optimistic uniform slice bound $(H+B)\sum_{Ns\asymp B}|g_U(s)|^2 D^\varepsilon$, the ordinary divisor estimate $|g_U(s)| \le \tau(s)$ and the fixed-field ideal count would only yield the envelope

$$A(H+B)D^\varepsilon \asymp (AH+D)D^\varepsilon.$$

That hypothetical slice estimate is not established here. This is an insufficiency of the generic upper-bound calculation, not a lower bound on the signed energy. At balanced $A, B \asymp D^{1/2}$ and $H = D^{3/4}$, the $AH$ term loses $D^{1/2}$ relative to the target; the $D$ term also exceeds $H$. A sharper use of the actual $g_U$, cross-factor constraints or covariance between the original pieces is required. This calculation does not refute all bilinear, moving-cutoff or dispersion methods.

The fixed-$U = 1$ lower bounds in Proposition 10.1 are different: they are genuine energy lower bounds for the separate pieces. If $W \ge w_0 > 0$ on $[c_1, c_2] \subset (1, 2)$, count primes in $[c_1 D, c_2 D]$ for $I_1$ and in $[c_1 D/Nq_0, c_2 D/Nq_0]$ for $II_1$. Removing $S$ and $q_0$ removes finitely many primes. The prime-ideal theorem gives positive constants times $D/\log D$ in both cases; the latter contribution has coefficient two. Because every coefficient of $II_1$ is nonnegative, keeping these products is valid. This is unlike isolating the prime part of the full signed Möbius energy, where uncontrolled cross terms remain. For the exact requested bound $H(HD)^\varepsilon$ at $H = D^{3/4}$, the competing power is $D^{3/4+7\varepsilon/4}$, giving the threshold $\varepsilon < 1/7$. Replacing $(HD)^\varepsilon$ by $D^\varepsilon$ reparametrizes the loss and gives the equivalent $1/4$ formulation. The positivity argument is specific to $U = 1$. For a concrete control, the primary element $7 - 21\omega$ has norm $637 = 7 \cdot 7 \cdot 13$ and three distinct ideal prime factors. At $U = 7$ its type-I coefficient is $-1$, its type-II coefficient is $-2$, and its Möbius value is $-1$. Thus the separate pieces need not have nonnegative coefficients at another cutoff; no moving-cutoff obstruction follows from the preceding positivity proof.

### S14.4 Exact finite checks and formal boundaries

The deterministic checker represents every good ideal of norm at most 1024 once by its primary generator, including inert prime ideals and squarefree products of three distinct prime ideals. It checks 1535 coefficient/regrouping cases at five cutoffs, retains 19680 encountered original zeros in actual sextic sums, and verifies 900 pair-conductor exponent ledgers. It evaluates principal $\nu$ and the fixed Kummer character $\nu(n) = \chi_n(2)$ on the same good ideals. The finite diagnostic weight is the rational profile $((Nn - D)(2D - Nn))^2/D^4$ on $[D, 2D]$, zero elsewhere; it is not claimed to be the smooth analytic weight of the theorem. Every identity is algebraic for arbitrary finite weights; these samples do not prove seminorm or scale-supremum estimates.

For example, at $D = 128$, $H = 16$, principal $\nu$ and $U = 1$, the unit-row normalized sums have the following exact common scaling:

$$\begin{array}{c|r} D^{9/2}M_1 & -122721831 \\ D^{9/2}I_1 & 314726661 \\ D^{9/2}II_1 & 192004830 \end{array}$$

Their exact difference is $II_1 - I_1 = M_1$. The lower-bound theorem uses the prime-ideal proof, not extrapolation from these small numbers. An independent no-production-import oracle also checks convolution, squarefree regrouping and weighted sextic sums.

`MobiusSplit.lean` proves eight actual statements: the inverse-ring identity, natural-number arithmetic truncation/tail, the Möbius Dirichlet identity, the unit cutoff, its exact tail convolution and coefficient, and nonnegativity. The natural-number Dirichlet ring is an actual arithmetic object; it is not the ideal convolution ring. Ideal arithmetic, sextic/Hecke character construction, analytic prime counting and the unproved signed mean square remain prose or named classical inputs. No new zero-free region, full GRH proof or positive determinant bridge has been obtained.

# S15 Primitive conductors, deleted factors and the scalar contour

## S15.1 Constructing the column character without losing its zeros

Let $\alpha^6 = u$. Since $K$ contains $\mu_6$, the extension $K(\alpha)/K$ is abelian, with character $g \mapsto g(\alpha)/\alpha$. At $p \nmid 6u$, arithmetic Frobenius gives the residue $u^{(Np-1)/6}$, exactly the source's sextic symbol. Abelian reciprocity makes it a finite-order Hecke character; multiply by $\nu$ and take its primitive inducing character. The classical reciprocity input is [89, Theorem 0.8 and Chapter I,§1]. The finite-order archimedean character is trivial, since $\mathbb{C}^\times$ is connected.

At $p \nmid 6$, write $u = \pi^e v$ with $v$ a local unit. A sixth root of $v$ lies in an unramified extension: its residue root is separable and lifts by Hensel's lemma. Inertia acts on a sixth root of $\pi$ through $\mu_6$, so on $\alpha$ the inertia character has order $6/\gcd(6, e)$. It is trivial exactly when $6 \mid e$; its square is trivial exactly when $3 \mid e$. A nontrivial tame one-dimensional character has conductor exponent one. Since $\nu$ is unramified outside $S$, it cannot cancel this inertia there. At primes above 6, the groups $K_p^\times/(K_p^\times)^6$ are finite; their finitely many characters have a uniform maximum conductor. Consequently

$$\mathfrak{f}_u \mid \mathfrak{c}_\nu \prod_{p \mid u,\ p \nmid 6} p, \qquad Q_u \ll_\nu Nu.$$

The fixed ideal $\mathfrak{c}_\nu$ includes the wild and fixed $\nu$ factors. No assertion identifies $\mathfrak{f}_u$ with the larger full zero modulus.

For $p \notin T$ put $z = \psi_u(p)(Np)^{-s}$. Its squarefree factor $1 + z = (1 - z^2)/(1 - z)$ proves the ratio using the *same* deleted set in numerator and denominator. Restoring the primitive Euler factors yields (3). At a prime where the primitive square becomes unramified its value cannot be obtained by squaring the sentinel zero $\psi_u(p) = 0$. For principal $\nu$, $u = \pi^3$ has a quadratic Kummer character whose primitive square is globally principal. Its local value in that square is one, not zero. For $u = \pi^6$ both primitive characters are principal, although the full family still deletes $p$. These two cases prove the local table directly.

## S15.2 Why the half-plane above one half is available

For a character $\eta$ write $L^T(s, \eta)$ for its Euler product with all primes of $T$ removed. For $\sigma > 1/2$ the product $1/L^T(2s, \psi_{u,2})$ converges absolutely and is bounded by $\zeta_K(2\sigma)$, uniformly in the character,

$T$ and $t = \Im s$. Every deleted finite factor is nonzero there. Therefore

$$F_{u,d}(s) = L(s,\psi_u) \prod_{p\in T, p\nmid \mathfrak{f}_u} (1-\psi_u(p)(Np)^{-s})/L^T(2s,\psi_{u,2})$$

has just the stated possible principal pole. This uses absolute convergence at $2\sigma > 1$, not a conjectural zero-free assertion on the critical line. For any fixed $\sigma > 0$ and $\eta > 0$,

$$\prod_{p\in T}(1+(Np)^{-\sigma}) \ll_{\sigma,\eta} R_T^{\eta}.$$

Indeed $1 + q^{-\sigma} \le q^{\eta}$ for all sufficiently large $q$; the remaining finitely many prime ideals give a uniform fixed constant. This controls every moving deleted prime; the denominator has already been bounded by the full convergent zeta product.

We use the classical completed Hecke functional equation in the form

$$\Lambda(s,\psi_u) = Q_u^{s/2}(2\pi)^{-s}\Gamma(s)L(s,\psi_u), \qquad \Lambda(s,\psi_u) = w_u\Lambda(1-s,\overline{\psi_u}), \quad |w_u| = 1.$$

This is a named external theorem, not a newly formalized analytic result; compare [88, Theorem 4.3, Corollary 4.4]. The cited notes sketch some of Tate's calculations; their incomplete displayed epsilon formulas are not used as independently checked constants here. At $\Re s = 1+\eta$, absolute convergence bounds $L$ by $\zeta_K(1+\eta)$. The functional equation and Stirling's formula give on $\Re s = -\eta$ the bound $Q_u^{1/2+\eta}(1+|t|)^{1+2\eta}$ up to a fixed constant. Phragmén–Lindelöf on this strip therefore gives, for fixed $0 < \sigma < 1$,

$$|L(\sigma+it,\psi_u)| \ll_{\sigma,\eta} Q_u^{(1-\sigma)/2+\eta}(1+|t|)^{1-\sigma+2\eta}.$$

For the principal character apply the strip argument to $(s-1)L(s,1)/(s+1)$, then divide back on the fixed line $\sigma < 1$. The constants need not be uniform as $\sigma \to 1$. Polynomial vertical strip growth and finite order, as supplied by classical Hecke theory, justify this application. No Lindelöf or subconvexity bound is inserted.

### S15.3 Mellin inversion, residues and profile uniformity

Initially, absolute convergence and compact smooth support give

$$S_{u,d}(Y) = \frac{1}{2\pi i}\int_{(1+\eta)} \widehat{W}(s) Y^s F_{u,d}(s)\,ds.$$

Four integrations by parts using $x\partial_x$ give $|\widehat{W}(\sigma+it)| \ll_{\sigma} \mathcal{M}_4(W)(1+|t|)^{-4}$. Choose $0 < \eta < \min(\varepsilon/8, 1/8)$, decreasing it further when distributing the small losses, and shift to $\sigma = 1/2+\delta$. The convergent reciprocal Euler product, the finite factor bound and convexity above control the complete integrand. Their polynomial $t$ growth is integrable against this Mellin decay and makes the horizontal edges tend to zero. The only crossed pole is $s = 1$ when $\psi_u = 1$; evaluating its finite factors gives exactly $\rho_T$. Taking absolute values on the new vertical line proves (4), after renaming the small losses. This also proves uniformity over any family of profiles with bounded $\mathcal{M}_4$; it does not interchange a profile supremum with an arithmetic mean square.

In the original $I_u$, the inner profile is exactly $W(Nc/Y)$ with $Y = D/Nd$; no $d$-dependent derivative enlargement occurs. On the support of $\beta_U$, $Nd \le U^2 \le D$, hence $Y \ge 1$. Also $R_T \le C_S Nu\,Nd$. The principal term is explicitly

$$P_u = \mathbf{1}_{\psi_u=1}\sqrt{D}\,\widehat{W}(1) \sum_{\substack{d \text{ squarefree}\\(d,S)=1}} \frac{\beta_U(d)\nu(d)\chi_d(u)}{Nd}\rho_{T(u,d)}.$$

This sum retains all original phases and zeros; no positivity of it is assumed. For the error use the actual truncated factorizations, not a free coefficient bound:

$$\sum_{d \text{ squarefree}} \frac{|\beta_U(d)|}{(Nd)^{1/2+\delta-\eta}} \le \left( \sum_{Na \le U} (Na)^{-1/2-\delta+\eta} \right)^2 \ll_{\delta,\eta} U^{1-2\delta+2\eta}.$$

The fixed-field ideal count is $O_K(X)$; dyadic summation proves the last inequality for sufficiently small $\eta < \delta + 1/2$. Inserting this and the conductor/radical bounds proves (5). The elementary lattice count $\#\{u : 0 < Nu \le H\} = O_K(H)$ then gives (6). Rows are not divided by units.

At $H = D^{3/4}$ and $U = D^a$ with $0 \le a \le 1/2$, the power of the resulting error envelope *relative to* $H$ is

$$\frac{3}{8} + \frac{5}{4}\delta + (2 - 4\delta)a$$

up to the arbitrarily small losses. It remains positive for every $0 < \delta < 1/2$. Thus this scalar convexity argument alone supplies no short-row saving. This is a limitation of the estimate just derived, not an obstruction to averaging the numerators over Kummer rows, retaining signed coefficients, or exploiting covariance with type II.

The seminorm uniformity is not a proof of the original scale supremum. For each independently chosen scale $D'$ the displayed outer argument requires $U^2 \le D'$. If a fixed cutoff is retained while $D'$ shrinks, then eventually even $D' \le U$: the short $2a_U$ term in the Vaughan identity can meet the support and must be restored. Neither that term nor the covariance with type II may be omitted in a supremum or reassembly.

### S15.4 Verification boundaries

`HeckeTypeI.lean` checks seven actual complex local/finite-product statements, including independent primitive coefficient arrays and their full deleted-factor restoration. It does not identify an ideal Hecke series with an abstract finite product or formalize a Mellin contour. The exact checker includes split and inert prime ideals, actual sextic phases, conductor drops, sixth-power rows and original mask zeros. The analytic theorem above uses classical abelian reciprocity, Hecke continuation/functional equation and standard complex-analytic tools as external inputs. Full ideal-character/formal analytic objects, the original signed low-gcd mean square and every GRH completion obligation remain open.

## S16 Controlled errors and a trained full-action RG approximation

**Proposition S16.1** (Classical action-error propagation)**.** *If $|\widehat{V}(s,t) - V(s,t)| \le \varepsilon$ at every finite configuration, then $|\mathcal{R}\widehat{V}(s) - \mathcal{R}V(s)| \le \varepsilon$. The partition ratio lies in $[e^{-\varepsilon}, e^{\varepsilon}]$. Every further* exact *marginalization preserves the same action bound.*

*Proof.* The pointwise inequalities $e^{-\varepsilon}e^{V} \le e^{\widehat{V}} \le e^{\varepsilon}e^{V}$ survive finite summation. Taking logarithms proves the first assertion; summing also over the retained variables proves the second. Iterate the same argument. For normalized measures the likelihood ratio lies between $e^{-2\varepsilon}$ and $e^{2\varepsilon}$, so a bounded observable error is at most $\|O\|_\infty \min(2, e^{2\varepsilon} - 1)$. Per-step learned errors would add; a single initial error does not grow under subsequent exact elimination. This scalar proof is not an assertion that the matrix exponential is operator-monotone. □

### S16.1 Executed finite calculations

The implementation checks full enumeration against the closed black-spin formula, Walsh reconstruction, raw partition equality, retained-spin magnetizations and two-stage versus direct elimination. At the quantum $\Gamma = 0$, $\beta = 1$ limit, every Pauli coefficient agrees with the classical effective action, with the required minus sign between energy and logweight. At nonzero transverse field the model is

$$H = -J_+ Z_+ \sum_j Z_j - J_- Z_- \sum_j \eta_j Z_j - bZ_+ + \tfrac{3}{5} bZ_- - \sum_j h_j Z_j - \Gamma \sum_i X_i.$$

At general $\beta$, the classical zero-transverse-field limit uses $\beta V$, including the fixed white fields, rather than scaling only the three varied controls. For $m = 5$, its full dimension is 128 and the first retained dimension is 32. Every one of the 1024 five-spin Pauli coefficients is computed; later retained dimensions are 8,4,2. Spectral-log/exponential and full Pauli reconstruction residuals, exact-trace composition and retained-observable comparisons are recorded separately. Floating-point residuals are numerical checks of the proved finite identities, not interval certificates.

The local laboratory exposes every scale, coupling/field controls for the classical model, selected generated interaction supports on actual dessin radial lifts, the even-chain $K, h, C$ trajectory and physical correlation length, and three fixed quantum matrix calculations. Live quantum parameter sliders are not offered for uncomputed Hamiltonians. Pair truncation carries the identity term and reports both partition and probability/density errors. A late-scale floating-point loss of the small eigenvalue is avoided by transporting the exact log-eigenratio coordinate $\log r \mapsto 2 \log r$; this is explicitly the transfer RG identity, not a newly diagonalized value.

### S16.2 A quantum-feature model trained on actual RG

The learner targets all 32 unnormalized logweights of the $7 \to 5$ spin $m = 5$ decimation. Its three inputs are $(J_+, J_-, b)$, with ranges $[0.15, 1.1]^2 \times [-0.6, 0.6]$. Fixed fields are $h_j = 0.045 \cos(1.7j) + 0.02j/5$. Seed 20261103 fixes 160 input samples, a 120/40 train/test split, and 40 OOD samples with $J_\pm \in [1.15, 1.5]$, $b \in [-0.8, 0.8]$. The three feature qubits encode the couplings by $R_y(1.5J_+) \otimes R_y(1.5J_-) \otimes R_y(2b)|000\rangle$. Two local $R_y, R_z$ layers, with a CZ triangle after each, contain 12 nominal optimizer angles. The final three $R_z$ phases commute with the measured $Z$ products and are structurally inactive for this loss; they still change the displayed state phases. We count them as nominal parameters, not additional effective expressive directions. Seven nonempty Pauli-$Z$ products feed an affine $8 \times 32$ readout. There are 268 parameters. Feature registers are not the seven physical spin sites or five quiver vertices.

The training objective is full-action MSE, including the constant; KL alone would ignore that constant. A training-only ridge fit initializes the head, then 120 Adam updates optimize both circuit and head. Actual PennyLane analytic expectation calls implement the feature circuit. Parameter-shift derivatives of each feature are combined with the classical head gradient; full-loss finite differences agree to $6.46 \times 10^{-11}$. All data, parameters, intermediate checkpoints and complete loss curves are preserved.

The held-out full-action MSE is approximately 0.00985, mean $D_{\mathrm{KL}}(p_{\mathrm{exact}} \| p_{\mathrm{learned}}) = 0.00300$, mean TV 0.0290, and mean absolute log-partition error 0.0610. A classical quadratic-feature baseline with the same 256 readout parameters obtains 0.00613,0.00240,0.0233 and 0.0138, respectively. OOD action MSE increases to 0.267 for the quantum hybrid, versus 0.0555 for the baseline. The quantum-feature constant MAE is 0.0444 and higher-body coefficient RMSE is 0.00900, versus 0.0124 and 0.0130 for the classical baseline; different metrics favor different aspects. The aggregate results show no quantum advantage. They expose genuine full-action approximation errors rather than substituting state-compression success for RG.

Each saved example computes its maximum error over all 32 coarse states; the proposition then applies to its subsequent exact marginalizations. These per-example finite bounds are not uniform over the input parameter box. The synchronized architecture inspector shows literal gate outputs, Pauli features, head, logweights, generated interactions and partition value, linked to the exact genus-two dessin/order context. No full order representation, continuum universality, material-specific accuracy or arithmetic zero-location bridge is inferred.

### S16.3 Quantum TokenGT and the geometric error-correction boundary

A further executed node/edge-token hybrid uses the same exact RG teacher, with frozen three-qubit features, trained Q/K/V and a full-action head. Softmax, componentwise max-plus and hybrid variants are compared with a matched classical feature model; the latter has lower held-out action error. An explicit intervention-code/change variational MI objective learns concept directions; a 61-candidate latent forest and a held-out analogy transfer are scored on exact physical observables. Rips $H_1$ is actually reduced over $\mathbf{F}_2$, and a continuous Gaussian GFlowNet trains a trajectory-balance objective with specified root/Lebesgue measures. Empirical residuals do not prove exact terminal sampling, semantic reasoning or categorical equivalence.

The physical bit-flip repetition channel is separately executed and has $p_L = 3p^2 - 2p^3$ under ideal recovery. A genuine cellular CSS calculation on the genus-two one-face dessin gives $[[10, 4, 1]]$: vertex-incidence rank 6, zero face boundary, X-distance 1 and Z-distance 4. Thus this cellulation provides no useful protection merely from its genus. These finite layers, full formulas, proofs, parameters and scope are developed in the research book and interactive laboratories; the physical models are not faithful representations of the infinite completed order and supply no GRH bridge.

### S16.4 An actual finite gentle cut for the categorical interface

On the genuine $m = 5$ dessin quiver write $a_i, b_i : i \to i - 1$ with indices modulo 5; cross-branch consecutive pairs are forbidden. Remove both arrows with source 0. The surviving algebra $B$ has vertices 0 through 4, arrows $a_i, b_i$ for $i = 1, \dots, 4$, and relations $a_i b_{i-1} = b_i a_{i-1} = 0$ for $i = 2, \dots, 4$. It is an explicit finite gentle quotient, not the completed order itself.

Every surviving positive path stays on one branch and has length at most 4. For length $\ell = 1, \dots, 4$ there are $2(5 - \ell)$ paths, while length 0 contributes five idempotents. Therefore $\dim_k B = 5 + 2(4 + 3 + 2 + 1) = 25$. The forbidden-chain resolution already proved for monomial gentle algebras has two alternating chains of length $n$ from vertex $i$ to $i - n$ when $1 \le n \le i$, and none for $n > i$. Applying Hom to simples kills its arrow differentials. Hence $\mathrm{pd}_B S_i = i$ and $\mathrm{Ext}^4_B(S_4, S_0) \cong k^2$. The split finite-dimensional simple-module criterion gives $\mathrm{gldim}\, B = 4$.

Every full branch cycle crosses a removed arrow, so the common completed order parameter $u$ maps to zero. Thus $B$ is a finite $u$-torsion quotient, not a DVR-free surface order. The difference is visible in derived Hom: $S_0$ is projective over $B$, whereas in the original completed order the two forbidden arms of length 5 return to vertex 0 and give $\mathrm{Ext}^5_\Lambda(S_0, S_0) \cong k^2$. The exact restriction along $\Lambda \twoheadrightarrow B$ is consequently not derived fully faithful. This supplies a concrete cut on which to investigate graded gentle/Fukaya models, while proving why its equivalence cannot simply be transported to the original completed order. Boundary/stops/line-field data and an actual module-valued neural decoder would still have to be supplied. Exact path and endpoint enumeration accompanies the prose proof; this cut/category comparison is not claimed wholly Lean formalized.

# S17 Relation calculations, splitting functors and determinant limits

## S17.1 Correct graphs and stable spaces

In the original basis let $fe_1 = e_2$, $fe_2 = e_3$, $fe_3 = 0$ and $ge_1 = 0$, $ge_2 = e_1$, $ge_3 = e_2$. Then

$$\Gamma_f = \langle (e_1, e_2), (e_2, e_3), (e_3, 0) \rangle,$$
$$\Gamma_{f^2} = \langle (e_1, e_3), (e_2, 0), (e_3, 0) \rangle,$$
$$\Gamma_{gf^2} = \langle (e_1, e_2), (e_2, 0), (e_3, 0) \rangle.$$

Each graph has dimension three. Omitting zero-output generators changes the relation into a partial graph and changes its powers and domain. The following are subspaces of $V$, not lists of pairs in $V \oplus V$:

| $T$ | $\bigcup_r \ker T^r$ | $\bigcap_r \operatorname{im} T^r$ |
|---|---|---|
| $f, g$ | $V$ | $0$ |
| $gf = \operatorname{diag}(1,1,0)$ | $ke_3$ | $ke_1 \oplus ke_2$ |
| $fg = \operatorname{diag}(0,1,1)$ | $ke_1$ | $ke_2 \oplus ke_3$ |

For all four full graphs $\bigcup_r \Gamma_T^r(0) = 0$. This corrects both the source's graph convention and its interchanged $gf, fg$ kernel entries. For an arbitrary multivalued relation there need not even be a linear map on its domain; its regular-part theorem requires the appropriate relation subquotients and finiteness conditions. We do not replace those hypotheses by the graph calculation. For comparison, in [38, Section 4, Lemmas 4.2,4.5–4.6] write $C' = \bigcup_n C^n(0)$ and let $C''$ consist of vectors with a compatible infinite backward chain. In finite dimension $C'' = \bigcap_n C^n V$; this equality is not automatic for arbitrary infinite-dimensional spaces. The regular quotient is $C^\sharp / C^\flat$, where

$$C^\sharp = C'' \cap (C^{-1})'', \qquad C^\flat = (C'' \cap (C^{-1})') + (C' \cap (C^{-1})'').$$

The automorphism statement uses this quotient for an arbitrary relation. The splitting statement additionally requires its finite-dimensionality, rather than the source's unqualified $\iota/\kappa$ expression.

## S17.2 Proof of the dessin-category equivalence

Write $a_i : V_i \to V_{i-1}$ for $i = 1, \dots, m-1$ and $p_i = a_1 \cdots a_i : V_i \to V_0$, with $p_0 = 1$; the rightmost map acts first. Every $p_i$ is an isomorphism. Conjugating the arrow maps by these identifications makes $a_i = 1$ for $i \neq 0$, and makes $T = p_{m-1} a_0$ at the remaining arrow. All $b$-relations vanish. An intertwiner is transported to maps $h_i$ with $h_i = h_{i-1}$ for $i \neq 0$ and $h_{m-1} T = T' h_0$. Hence every $h_i = h$ and $hT = T'h$. This proves full faithfulness and essential surjectivity, with inverse given by $(V, T) \mapsto N_m(T)$. It is also an exact equivalence: kernels and cokernels of intertwiners are the usual invariant kernels/cokernels; the identity arrow maps remain identities. Finite dimension is a condition on modules, while $A_m$ itself is infinite dimensional through its arbitrarily long permitted branch paths.

The sum of the full permitted turns on both branches is central in this closed dessin algebra. On the chosen modules the $b$-turns vanish and each $a$-turn equals $T$, so $u$ acts as $T$ at every vertex. If $T^n = 0$, the action factors through $A_m/(u^n)$ and extends to the completion. If $T$ has a nonzero invertible Fitting summand, no power of $u$ kills that summand, precluding continuity. For $T = J_r(\lambda)$ with $\lambda \neq 0$, the obstruction is stronger: $u - \lambda$ is a unit of $k[[u]]$ but its $r$th power annihilates the proposed module, impossible for a nonzero unital module. This rules out even an unqualified algebraic extension of that band, not just a chosen topology.

## S17.3 Functoriality and the actual filtration

Kernel dimensions are nondecreasing and at most $n$. Once two successive kernels agree, applying $T^{-1}$ shows that all later kernels agree. Thus they stabilize by $n$; rank-nullity gives image stabilization as well. On $K(T)$, $T^n = 0$. On $I(T)$, $T(I(T)) = I(T)$; finite dimension makes this a bijection. If $hT = T'h$, induction gives $hT^r = T'^r h$, hence

$$h(\ker T^r) \subseteq \ker T'^r, \qquad h(\operatorname{im} T^r) \subseteq \operatorname{im} T'^r.$$

This proves functoriality even when $\dim V \neq \dim V'$: choose a common stabilization exponent for the two finite objects. The decomposition projection onto $K(T)$ along $I(T)$ is canonical and commutes with all intertwiners; the existence of a Jordan basis is not needed. Finite dimension is essential. On $V = k[t]$, multiplication by $t$ has stable kernel zero and stable image $\bigcap_r t^r k[t] = 0$, although $V \neq 0$. The same construction $N_m(V, t)$ is an actual infinite-dimensional module of $A_m$. Thus the finite Fitting split cannot be extended to every module of our infinite algebra merely by repeating its notation.

Let $k_r = \dim \ker T^r$, with $k_0 = 0$, and form $G_r = \ker T^r / \ker T^{r-1}$. For a nilpotent Jordan block of size $s$, $k_r = \min(r, s)$, so $\dim G_r$ counts the blocks of size at least $r$, and $2k_r - k_{r-1} - k_{r+1}$ counts blocks of size exactly $r$. For $J_3(0) \oplus J_1(0) \oplus J_2(1)$ the stable dimensions are 4 and 2; the kernel dimensions begin $0, 2, 3, 4, 4$. The $G_r$ are functorial subquotients. Their dimensions do not specify a natural choice of basis or split each filtration step.

**Proposition S17.1** (What the source's splitting-functor assumptions imply)**.** *Let $\mathcal{A}, \mathcal{B}_i$ be additive categories with finite biproducts. Suppose additive functors $S_i : \mathcal{B}_i \to \mathcal{A}$ and $F_i : \mathcal{A} \to \mathcal{B}_i$ satisfy $F_i S_i \cong 1$, $F_i S_j \cong 0$ for $i \neq j$; only finitely many $F_i M$ are nonzero for each $M$; the $F_i$ jointly reflect isomorphisms; and maps $\gamma_{i,M} : S_i F_i M \to M$ have invertible $F_i$-images. Then $M \cong \bigoplus_i S_i F_i M$. Every nonzero indecomposable is $S_i B$ for a unique index and an indecomposable $B$ up to isomorphism.*

*Proof.* Sum the finitely many nonzero $\gamma_{i,M}$ to a map $\gamma_M$. After applying $F_j$, all off-diagonal summands vanish, leaving the assumed isomorphism $F_j \gamma_{j,M}$, or $0 \to 0$. Joint reflection therefore makes $\gamma_M$ invertible. For an indecomposable $M$, exactly one of the nonzero summands can occur. Conversely, if $S_i B = X \oplus Y$ and $B$ is indecomposable, applying $F_i$ makes one summand vanish, say $F_i X = 0$; every other $F_j X$ also vanishes. Joint reflection applied to $0 \to X$ gives $X = 0$ up to isomorphism. Finally apply $F_i$ to an isomorphism $S_i B \cong S_j B'$ to recover both index and object. □

The direction of $\gamma$ matters: it is into $M$, contrary to one source diagram. Local finiteness is objectwise in $i$, not a finiteness assertion over all objects of a category. Abelian categories require kernels, cokernels and the usual exactness axioms; addition is on Hom groups, not on an alleged group of quiver vertices. The abstract assumptions above do not assert that the family $F_i$ is faithful on arbitrary morphisms. Our Fitting example is concrete: $\mathcal{B}_0$ consists of nilpotent pairs, $\mathcal{B}_\times$ of invertible pairs, $S$ are inclusions and $F_0, F_\times$ the two stable summands.

## S17.4 Strings, bands, Hom and one non-rigid string

In this branch category, Jordan normal form classifies indecomposables over an algebraically closed field. Over a general field replace Jordan eigenvalues by powers of monic irreducible polynomials. A Jordan block $J_r(0)$ is one directed string with $mr$ basis visits and $mr - 1$ nonzero successor arrows; $J_r(\lambda)$ for $\lambda \neq 0$ is a permitted cycle band. Formal inverse letters are reverse traversals, not inverse elements of $A_m$. Strings and their reverses describe the same string module under the usual isomorphism convention. Arbitrary infinite modules are not all classified by this finite calculation; the actual finiteness hypotheses in [38] remain essential.

Identify a Jordan pair with $k[t]/(t-\lambda)^r$. An intertwiner from size $r$ to size $s$ is determined by the image of 1, annihilated by $(t-\lambda)^r$. For equal eigenvalues this subspace has dimension $\min(r,s)$. For different eigenvalues, Bezout's identity makes $t-\lambda$ invertible on the target, so only zero remains. Full faithfulness above transfers this Hom computation to $A_m$. The sequence $0 \to J_1(0) \to J_2(0) \to J_1(0) \to 0$ is not split: with $J_2(0)e_1 = e_2$, a lift of the quotient generator has nonzero first coordinate and is not killed by $J_2(0)$. Its image under $N_m$ is a nonsplit ambient-$A_m$ self-extension. Thus even the string $N_m(J_1(0))$ is not rigid. No assertion that this exact full subcategory is derived fully faithful is needed for the nonsplitting argument.

For this particular closed $A_m$, every oriented or inverse letter has exactly two admissible successors: continue on its branch in the same orientation, or switch branch and reverse orientation. Cancellation and forbidden pairs, including inverse forbidden pairs, remove the other choices. A fixed ordering of these choices identifies each first-letter cylinder with $\{0,1\}^{\mathbb{N}}$. The whole infinite-word space is a finite disjoint union of Cantor spaces, hence itself Cantor. There are no terminating words inside that space to remove. Arbitrary gentle quivers can have isolated or eventually forced paths, so this conclusion is not universal. Algebraic covers retain the lifted zero-relations and deck equivariance data; arbitrary periodic cover representations are not automatically bands or a unitary time evolution.

### S17.5 A determinant comparison with its exact limits

On $\bigoplus_i V_i$, $L^m = \operatorname{diag}(T,\dots,T)$. For a direct determinant proof of (7), successively eliminate the blocks 1 at vertices $1,\dots,m-1$ in $1-zL$. Block row additions preserve determinant and leave $1-z^mT$ at vertex zero. This works over $k[z]$ without inverting $z$ or $T$; the zero-size case gives one on both sides. The Fitting splitting separates a nilpotent factor 1 and the determinant on the invertible summand. For instance $m=5$, $T=J_2(2)$ gives $(1-2z^5)^2$, whose complex roots have modulus $2^{-1/5}$, despite being a genuine genus-two dessin module.

Under conjugation of $T$ this is an invariant polynomial. With independent source/target bases a presentation determinant is instead a semi-invariant with its determinant-character weight; those actions must not be conflated. The polynomial equality does not construct a positive Hermitian metric, an arithmetic Euler factor, a regularized determinant or a zero-preserving map to completed $L$-functions. Functorial kernel filtrations similarly do not construct an equivalence of derived/Fukaya categories. A fixed graded surface category and a determinant-compatible arithmetic realization remain separate obligations.

The exact rational checker evaluates actual genus-two/four quivers, Hom equations, full-turn matrices, both filtration directions and the nonsplit extension. `lean-analytic/LinearRelations.lean` proves full graph identities, semiconjugacy and filtration preservation, actual finite-space Fitting splitting/stabilization and unit-power annihilation. It does not package the entire dessin category, Jordan classification, general splitting-functor proposition or determinant proof in Lean.

## S18 Normal forms and the full module action

### S18.1 Why the polynomial matrix algebra is the path algebra

With our convention a path from $j$ to $i$ acts as $E_{ij}$; the rightmost arrow acts first. Every nonzero positive path uses only the $a$-branch or only the $b$-branch, because mixed consecutive arrows are the defining quadratic zero relations. Along either branch the path of length $\ell$ starting at $j$ is unique. Write

$$\ell = j - i + 5q, \qquad 0 \le i, j < 5.$$

Its image is $t^q E_{ij}$ in the chosen factor and zero in the other. For $i < j$, $q \geq 0$; for $i > j$, $q \geq 1$; for $i = j$ and positive length, $q \geq 1$. The stationary path is the shared $(E_{ii}, E_{ii})$.

These images are linearly independent: off-diagonal matrix units, positive diagonal monomials, branch factors and degrees are independent; the shared constant diagonal units are independent of all positive paths. They span $H \times_{k^5} H$, since upper entries permit all polynomials, lower entries are divisible by $t$, and only the constant diagonals of the two factors must agree. Products of paths agree with matrix products. Thus the arrow assignment induces a bijective algebra map from $kQ/(a_i b_{i+1}, b_i a_{i+1})$ to $B$. This argument includes arbitrarily long paths; it imposes no nilpotence relation on a full turn.

The central parameter $u$, the sum of all full turns on both branches, has image $(tI_5, tI_5)$. Completing in this parameter is a separate operation. The polynomial algebra and its completion are not identified by the normal-form calculation. The finite modules below are annihilated by $u^2$, so their continuous completed actions require only a finite Taylor polynomial.

## S18.2 Coefficient selectors without a truncation

Here is an explicit proof of Theorem 14.2 and the coordinate convention used by the implementation. Let

$$\mathcal{S} = \{(i, j, q) : 0 \leq i, j < 5,\ q \in \mathbb{N},\ i < j \text{ or } q > 0\}, \qquad \mathcal{K} = \{0, \ldots, 4\} \sqcup (\{a, b\} \times \mathcal{S}).$$

For a stationary index set $v_i = (E_{ii}, E_{ii})$; for a positive index set $v_{a,i,j,q} = (t^q E_{ij}, 0)$ and $v_{b,i,j,q} = (0, t^q E_{ij})$. These are elements of the ambient polynomial matrix-pair space, with no degree cutoff. For $M = (M_a, M_b)$ define

$$\lambda_i(M) = [t^0](M_a)_{ii}, \qquad \lambda_{b,i,j,q}(M) = [t^q](M_b)_{ij}.$$

Here the subscript $b$ in the second formula denotes either branch. Every $\lambda_x$ is $k$-linear, since extraction of a fixed matrix entry and polynomial coefficient is linear.

We verify $\lambda_x(v_y) = \delta_{xy}$, including the cases at degree zero. For two stationary indices this is the ordinary diagonal matrix-unit identity. A stationary selector vanishes on a positive image in the other factor. In its own factor it could see a constant diagonal only if $i = j$ and $q = 0$, which the definition of $\mathcal{S}$ excludes. Conversely, a positive selector applied to a stationary image could be nonzero only at precisely the same forbidden slot. For two positive indices, different branches have disjoint supports. In one branch the matrix entry forces both endpoints to agree and the monomial coefficient forces the degrees to agree. These conditions give equality of the full indices. The witness of the condition defining $\mathcal{S}$ carries no extra algebraic data. This proves biorthogonality in every characteristic, using no division.

For any finitely supported $c : \mathcal{K} \to k$, linearity now gives

$$\lambda_x \left( \sum_{y \in \mathcal{K}} c_y v_y \right) = c_x.$$

If the sum vanishes, every $c_x$ vanishes; if two sums agree, their coefficients agree. This proves linear independence and injectivity of normal expansion on the actual $k$-module of finitely supported functions. It does not replace finite support by a convergent or formal infinite sum.

The length formula also contains useful information beyond independence. If $i < j$ then $5q + j > i$ even at $q = 0$; otherwise $q > 0$ and $i \leq 4$ give the same strict inequality. Hence $\ell = 5q + j - i$ is a positive integer, and $\ell + i = 5q + j$. For fixed starting vertex $j$ and length $\ell$, two such equations

imply $i - i' = 5(q - q')$. Since $i, i' \in \{0, \dots, 4\}$, this forces $i = i'$ and then $q = q'$. In Lean the subtraction is natural subtraction; positivity is proved before using the rearranged equation.

For example the $a$-path of length 12 from vertex 4 ends at vertex 2 and has image $(t^2 E_{24}, 0)$. The length-4 $a$-path from 3 to 4 has image $(tE_{43}, 0)$. Their product, with the rightmost path first, is $(t^3 E_{23}, 0)$, of length 16. Replacing the second path by its $b$-branch version makes the product zero. Degree-zero upper entries such as $(E_{24}, 0)$ are permitted, but $(E_{22}, 0)$ is not a stationary image: the stationary element is the shared pair $(E_{22}, E_{22})$. This last distinction prevents a spurious extra idempotent in the proposed basis.

The selector theorem alone gives the faithfulness part of the intended quotient comparison. The next construction proves membership and spanning inside the generated algebra without assuming that comparison. A separately defined path algebra, mixed-zero quotient and its universal algebra map still require formal packaging. Completion, module-category and almost-split universal properties require their own subsequent constructions. None of these finite-support identities supplies an arithmetic self-adjoint determinant or controls zeta zeros.

### S18.3 Constructing every normal image from actual arrows

We now prove Theorem 14.3. This closes the membership and spanning gap left by ambient coefficient recovery. Let $p(j) = j - 1$ modulo five and, inside the actual generated algebra, define

$$W_b(0, j) = e_j, \qquad W_b(n + 1, j) = W_b(n, p(j))\, b_j.$$

In this formula $b_j$ denotes the arrow in the chosen branch; it may be an $a$-arrow. The rightmost arrow acts first. This recursive definition is an element of $A$ for every $n$, because it uses only its generators and multiplication. Set

$$d_{ij} = \begin{cases} 0 & i < j, \\ 1 & i \geq j, \end{cases} \quad L_{ij} = j - i + 5d_{ij}, \quad P_b(i, j) = W_b(L_{ij}, j), \quad C_b(i) = P_b(i, i).$$

The integer $L_{ij}$ lies between one and five. In particular $P_b(i, i)$ is a positive full turn, whereas $W_b(0, i)$ is the shared stationary idempotent. These two elements are not interchangeable.

Directly following the arrows, the path $P_b(i, j)$ crosses the weighted arrow $b_0$ exactly $d_{ij}$ times. Its image is therefore $t^{d_{ij}} E_{ij}$ in factor $b$, with zero in the other factor. The Lean proof checks all five-by-five endpoint pairs for each of the two branches; this is a finite verification of a finite combinatorial case split, not a bound on permitted path lengths. For all $n \in \mathbb{N}$, induction gives

$$\mathrm{view}\big(C_b(i)^n P_b(i, j)\big) = t^{n + d_{ij}} E_{ij} \quad \text{in factor } b.$$

For $n = 0$ the power is the unit of the whole two-factor algebra, so multiplication by $P_b(i, j)$ still selects the correct branch. The induction step uses $C_b(i)$ with image $tE_{ii}$ and $E_{ii} E_{ij} = E_{ij}$. No assertion that a branch-only degree-zero diagonal belongs to the algebra is needed. Since a permitted slot satisfies $q \geq d_{ij}$, the choice $n = q - d_{ij}$ constructs its preimage in $A$. This proves membership at every polynomial degree.

For clarity, if $i = 2$, $j = 4$, $q = 2$, then $P_a(2, 4) = a_3 a_4$ and $C_a(2)^2 P_a(2, 4)$ realizes $(t^2 E_{24}, 0)$ by a length-12 arrow word. For $i = 4$, $j = 0$, $q = 1$, we have $P_a(4, 0) = a_0$ and the exponent $q - d_{40}$ is zero. The corresponding exponent must never be replaced by a negative power or by an inverse arrow. The construction works over $\mathbb{F}_2$ as well as characteristic zero.

## S18.4 Spanning, multiplication and the inverse coordinate map

Let $S$ be the $k$-linear span of all normal images in the ambient space. Membership just proved gives $S \subseteq A$. We prove the reverse inclusion by showing that $S$ is closed under the operations used to generate $A$. The product rules are

$$e_i e_j = \delta_{ij} e_i, \qquad e_r v_{b,i,j,q} = \delta_{ri} v_{b,i,j,q}, \qquad v_{b,i,j,q} e_r = \delta_{jr} v_{b,i,j,q},$$

and for two positive images

$$v_{b,i,j,q} v_{c,r,l,s} = \begin{cases} v_{b,i,l,q+s} & b = c \text{ and } j = r, \\ 0 & \text{otherwise.} \end{cases}$$

The displayed nonzero output is always an admissible positive image. If $q + s > 0$ this follows from its degree. If $q + s = 0$, nonnegativity gives $q = s = 0$, so the input conditions imply $i < j = r < l$. Thus $i < l$ and the output is admissible even at degree zero. This explains why closure cannot be justified merely by ignoring the lower-triangular constraints.

By bilinearity the product of two finite linear combinations lies in $S$. Every vertex is one of its normal images, and each arrow is a permitted image with degree zero except the wrap arrow, which has degree one. The sum of the shared vertices is the unit. Scalar multiples of this unit belong to $S$. Induction on the algebra generated by the arrows and vertices therefore gives $A \subseteq S$. In Lean this is actual `Algebra.adjoin_induction`, not a predicate that assumes the desired normal form for every element. Consequently $A = S$ as submodules of the polynomial matrix-pair space.

Now lift each $v_x$ to an element $\widetilde{v}_x \in A$ using its proved membership. Their images under the injective inclusion are the already independent ambient images, so the lifts are independent. The equality $A = S$ gives surjectivity of finite expansion; independence gives injectivity. This constructs the linear equivalence $E$ on the genuine generated algebra. For any $a \in A$, apply the ambient coefficient recovery identity to $E^{-1}a$ to obtain

$$(E^{-1}a)_x = \lambda_x(\mathrm{view}(a)).$$

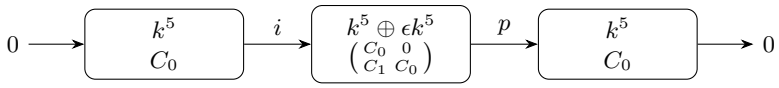
The formula holds for every element of $A$, not only on a sampled list of paths or generators. Its finite support follows from the equivalence with finitely supported functions. No convergence of an infinite expansion is asserted. Although the product table has been used to prove span closure, we have not yet supplied the formal universal map from a separately constructed bound-quiver path quotient. The distinction between a linear coordinate equivalence and that algebraic quotient interface is retained.

## S18.5 Exactness for arbitrary algebra elements

In coordinates $M_2 = k^5 \oplus \epsilon k^5$ let $C(t) = \sum_{r \geq 0} C_r t^r$ be the first component of an arbitrary $B$-element. Its action is

$$(x, y) \longmapsto (C_0 x, C_1 x + C_0 y), \qquad \rho_2(C) = \begin{pmatrix} C_0 & 0 \\ C_1 & C_0 \end{pmatrix}.$$

Evaluation is multiplicative: the constant coefficient of a product is $C_0 C_0'$ and the linear coefficient is $C_0 C_1' + C_1 C_0'$. Consequently the formula is a representation of the entire algebra. The second branch acts by zero but has not been removed from the algebra. Both maps in the sequence commute with this general block action, not merely with a finite test list of generators.




Every vector in $\ker p$ is $(0, y) = i(y)$, $i$ is injective, and $p$ is surjective by $x = p(x, 0)$. The section obstruction uses only $a_0$: it is zero on $M_1$, while on $M_2$ it sends the coordinate-zero constant coefficient to the coordinate-four infinitesimal part. A section commuting with this arrow would force $1 = 0$. No algebraic closure, eigenvalue classification, division by two or positivity premise occurs.

### S18.6 Formal and computational coverage

The file `lean-analytic/BranchExtension.lean` defines $B$ as the actual generated subalgebra of polynomial matrix pairs, proves its infinite dimensionality through injective polynomial evaluation, and constructs $i, p$ as $B$-linear maps. The original 21 printed theorem types and axiom reports cover the generators, infinite embedding, actions, exactness and absence of a module section. Eight additional reports in `NormalPath` cover slot extensionality, positive length, the length equation, recovery from start and length, biorthogonality, coefficient recovery, linear independence and injective normal expansion. Ten further printed reports now establish actual generated-algebra membership, equality with the normal span, lifted independence, bijective finite expansion and its explicit inverse coefficients. The normal-form isomorphism with the separately constructed path quotient above is proved in prose; it is not packaged as a Lean categorical equivalence. Nor is a Yoneda-Ext object or the universal factorization property of an almost-split sequence formalized by these maps alone.

The exact checker `examples/relations/check_extension.py` independently uses unbounded normal paths and polynomial matrix pairs. It checks 405 distinct paths through length 40, 7,225 products, 30 distinct loop powers and 80 whole-polynomial module actions. The full generator system for a hypothetical section has coefficient rank 49 and augmented rank 50 over $\mathbb{Q}, \mathbb{F}_2, \mathbb{F}_3, \mathbb{F}_5$. These finite diagnostics complement the symbolic proof; their finite cutoffs do not establish infinite dimensionality by themselves.

The separate exact checker `examples/relations/check_normal_paths.py` checks 475 normal images at degrees up to 1,000,003 (length 5,000,019), all 225,625 selector pairs, 1,030 direct arrow steps, 15,625 composable-or-zero product comparisons and 200 finite expansions over $\mathbb{Q}, \mathbb{F}_2, \mathbb{F}_3, \mathbb{F}_5$. Invalid constant slots are rejected. These are finite diagnostics of the code and conventions; the unbounded assertion is established by the Lean theorem, not by the size of the tests.

A further deterministic checker, `examples/relations/check_normal_algebra.py`, constructs all 50 shortest arrow words and 375 actual preimages, including degree 1,000,003, using sparse binary powers of the full turns. It checks 5,625 product round trips and 200 exact expansions over the four stated fields. Three ambient matrices with forbidden constant shapes fail reconstruction, as required. These diagnostics test the implementation independently of finite degree bounds; the general membership/span/equivalence result is the Lean theorem, not an extrapolation from these samples.

## S19 Physical generative models and their verification

### S19.1 From a dessin to an actual quantum RG target

Take $y^2 = x^5 - 1$, $\beta = x^5$. Its clean incidence graph has two black vertices and five white vertices, hence seven physical spin sites and ten incidence edges. These differ from the five quiver vertices

and the infinite-dimensional polynomial algebra. The physical Hilbert space below is finite; no faithful action of that whole algebra on it is assumed.

Let $X_q, Z_q$ be Pauli matrices on physical site $q$, with the black sites numbered zero and one and the white sites $2 + j$. Our executed Hamiltonian is

$$\begin{aligned} H = & -.50 \sum_{j=0}^{4} Z_0 Z_{2+j} - .35 \sum_{j=0}^{4} (-1)^j Z_1 Z_{2+j} \\ & - .55 \sum_{q=0}^{6} X_q - .12 Z_0 + .072 Z_1 - \sum_{j=0}^{4} h_j Z_{2+j}. \end{aligned}$$

Here $h_j = .045 \cos(1.7j) + .02j/5$. All coefficients are dimensionless and inverse temperature is one. Diagonalizing the actual $128 \times 128$ Hermitian matrix gives $\rho = e^{-H}/\operatorname{tr}(e^{-H})$. We retain the two black spins:

$$\rho_R = \operatorname{Tr}_{\text{white}} \rho, \qquad (\rho_R)_{ab} = \sum_{j=0}^{31} \rho_{aj,bj}.$$

This is an actual coarse quantum state, with trace one and positive spectrum. An independent Pauli–Kronecker NumPy reconstruction agrees with the saved matrix to $2.04 \cdot 10^{-10}$. Its minimum eigenvalue is approximately 0.0806414, and its off-diagonal Frobenius norm is 0.300551. There is therefore quantum coherence that a probability histogram alone cannot describe. No chemical or experimental material dataset is involved: this is an exactly specified finite quantum-matter benchmark.

Let $p_s = (\rho_R)_{ss}$ for $s \in \{\pm 1\}^2$. The hybrid samplers learn

$$p_\sigma(x) = \sum_s p_s \mathcal{N}(x; s, \sigma^2 I_2), \qquad \sigma = .2.$$

This is a continuous dequantization of the measured spin law. A sign measurement of these samples differs from $p$ by at most $2\Phi(-1/\sigma)$ in total variation, by a coordinatewise sign-flip coupling and the union bound. This is below $6 \cdot 10^{-7}$ here. It is small compared with finite-sampling error, but it is not zero. The samplers do not recover phases or the entire $\rho_R$. For example, $\rho_R$ and $(Z \otimes I)\rho_R(Z \otimes I)$ have the same diagonal but different density matrices.

## S19.2 Diffusion and exact versus approximate reverse controls

For $T = 64$ use the cosine cumulative noise schedule, with endpoint offset .008, clipped step variances between .0001 and .999. Write $\alpha_k = 1 - \beta_k$ and $\bar{\alpha}_k = \prod_{j \le k} \alpha_j$. The forward noising law is

$$x_k = \sqrt{\bar{\alpha}_k}\, x_0 + \sqrt{1 - \bar{\alpha}_k}\, \epsilon, \qquad \epsilon \sim N(0, I_2).$$

A network $\epsilon_\theta(x_k, k/T)$ minimizes the ordinary mean squared noise-prediction error. It is a classical stochastic sampler containing quantum-circuit features, not an implemented coherent quantum channel. The executed reverse update uses

$$\mu_\theta = \frac{x_k - \beta_k \epsilon_\theta / \sqrt{1 - \bar{\alpha}_k}}{\sqrt{\alpha_k}}, \qquad \widetilde{\beta}_k = \beta_k \frac{1 - \bar{\alpha}_{k-1}}{1 - \bar{\alpha}_k}, \qquad x_{k-1} = \mu_\theta + \sqrt{\widetilde{\beta}_k}\, z.$$

The last step is deterministic. Even an exact conditional noise predictor does not make this Gaussian reverse kernel the exact reverse mixture at finite $T$: posterior uncertainty of the mixture component

remains. The oracle-noise control therefore retains a reverse-kernel approximation, finite schedule and Monte Carlo error, not just a numerical integration error.

There is also an exact, known-teacher reverse-mixture control. Conditional on center $s$, put $v_k = \bar{\alpha}_k\sigma^2 + 1 - \bar{\alpha}_k$ and $m_{k,s} = \sqrt{\bar{\alpha}_k}s$. At a supplied $x_k$, draw the center with posterior weights proportional to $p_s \exp(-\|x_k - m_{k,s}\|^2/(2v_k))$. Then draw

$$x_{k-1} \mid x_k, s \sim N\left(m_{k-1,s} + \frac{\sqrt{\alpha_k}v_{k-1}}{v_k}(x_k - \sqrt{\alpha_k}m_{k-1,s}), \frac{v_{k-1}\beta_k}{v_k}I_2\right).$$

These formulas follow by conditioning a jointly Gaussian pair with covariance $\sqrt{\alpha_k}v_{k-1}I_2$. Starting from the exact forward terminal mixture and composing these reverse conditionals recovers $p_\sigma$ exactly in distribution, up to numerical and sampling error. It uses the full teacher, so it is a diagnostic rather than a learned solution.

### S19.3 Flow matching with a measurable target velocity

Independently sample $z \sim N(0, I_2)$ and $x_1 \sim p_\sigma$, set $x_t = (1-t)z + tx_1$, and train $v_\theta(x_t, t)$ toward $x_1 - z$. This is the conditional flow-matching regression objective. Conditional on center $s$, its time-$t$ mean and variance are

$$ts, \qquad v_t = (1-t)^2 + t^2\sigma^2.$$

Let $\pi_s(x,t)$ be the corresponding mixture posterior. The exact conditional velocity is

$$v_*(x,t) = \sum_s \pi_s(x,t)\left[s + \frac{t\sigma^2 - (1-t)}{v_t}(x - ts)\right].$$

Indeed $\mathrm{Cov}(x_1 - z, x_t \mid s) = (t\sigma^2 - (1-t))I_2$; Gaussian conditioning gives the displayed formula. The positive variance $v_t$ includes both endpoints. The code verifies this formula and the diffusion-score formula by independent finite differences of a separately written mixture log density. Learned sampling uses 128 Heun steps. Oracle-velocity sampling isolates finite integration and sampling error for this known smooth density; it is not an upstream QFM reproduction [94].

### S19.4 Architecture, matched controls and actual GPU results

The neural input is $[x_0, x_1, t, \sin 2\pi t, \cos 2\pi t, \sin 4\pi t, \cos 4\pi t]$. An affine encoder maps its seven channels to four angles through $\pi\tanh$. A two-layer four-qubit circuit uses angle-reuploading $R_Y, R_Z$ gates and, for the entangled architecture, a CNOT ring. Fifteen commuting $Z$-string expectations are concatenated with the seven input channels. A $22 \to 64 \to 64 \to 2$ SiLU head predicts noise or velocity. The product control uses the same nominal parameters and removes the entanglers. Final diagonal phase rotations are invisible to the all-$Z$ readout; equality of nominal parameter counts does not establish equality of effective Fisher capacity.

Every run has 5,810 nominal parameters, batch size 512 and 2,000 Adam steps. There are 16,384 training draws and 4,096 disjoint held-out draws, with seeds 20261104, 20261105 and 20261106. The four architectures/objectives were rerun at every seed. The following figures are mean and sample standard deviation across those three runs, not confidence intervals or asymptotic theorems.

| Objective | circuit | sign-law TV | sliced $W_1$ |
|---|---|---|---|
| Diffusion | product | $.0380 \pm .0113$ | $.0768 \pm .0306$ |
| Diffusion | entangled | $.0451 \pm .0133$ | $.0939 \pm .0183$ |
| Flow | product | $.0217 \pm .0079$ | $.0460 \pm .0077$ |
| Flow | entangled | $.0293 \pm .0130$ | $.0522 \pm .0159$ |

The second metric uses sixteen fixed projection directions and an independent teacher draw. The experiment demonstrates learning but no consistent quantum or entanglement advantage; the product control has lower mean error on both reported metrics. No physical quantum processor, shot noise, industrial chemical prediction, OOD material transfer or complexity speedup was tested. All checkpoints, seeds, histories, runtimes and teacher controls are retained.

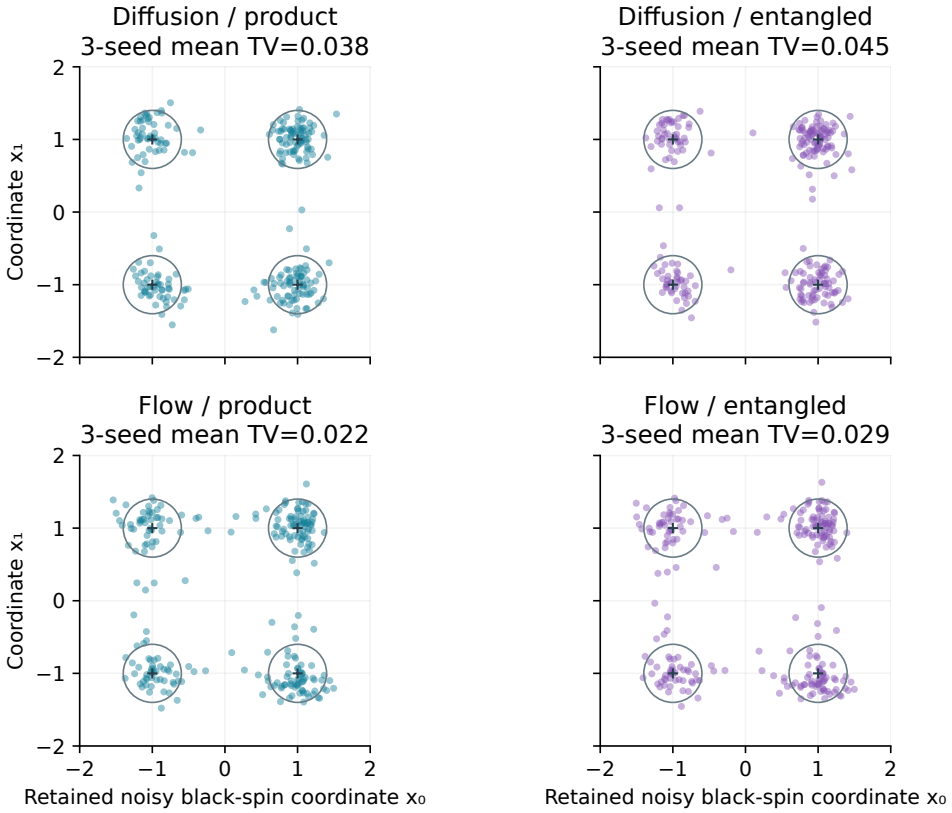



Figure 1: Actual final samples from seed20261104, with exact mixture centers and two-standard-deviation circles. Titles report mean sign-law TV over three seeds. The product control is competitive; no entanglement advantage is inferred.

### S19.5 A coherent learned purification flow

We separately implement quantum evolution, without identifying it with the classical samplers. Exact depolarizing diffusion is

$$D_t(\rho_R) = e^{-t}\rho_R + (1 - e^{-t})I_4/4.$$

It preserves trace and positivity and satisfies $D_s D_t = D_{s+t}$ by direct substitution. Its linear inverse on all matrices is not positive: $D_t^{-1}(P) = e^t P - (e^t - 1)I_4/4$ has a negative eigenvalue on the orthogonal complement of any rank-one projection $P$ when $t > 0$. Thus a learned inverse cannot be a universal deterministic physical channel.

For a constructive state-preparation benchmark, take the family $R_t = (1 - t)I_4/4 + t\rho_R$, $0 \le t \le 1$, and its canonical purification $|\Psi_t\rangle = \mathrm{vec}\sqrt{R_t}$ in $\mathbb{C}^4 \otimes \mathbb{C}^4$. It is normalized because $\|\sqrt{R_t}\|_F^2 = \mathrm{tr}\, R_t = 1$. The initial purification is two Bell pairs. We train a six-block four-qubit

circuit: each block conjugates local $R_Y$ rotations by a fixed CNOT network and its inverse; angles are $t\theta + t^2\omega$. All blocks are identity at the control time zero. The 48 learned controls are trained against full known purification states, using the fidelity loss. This ideal supervision is stronger than access to measured samples and is accounted for explicitly.

An RTX4090 run of 2,000 steps, batch 64, gives minimum fidelity 0.999903 on a 33-point grid and maximum reduced-density Frobenius error 0.009642. The reduced state is obtained by an actual ancilla partial trace and passes trace/positivity checks. These are grid diagnostics, not a uniform or out-of-family guarantee. The construction is a learned coherent control path for known targets, not a universally learned inverse to depolarization. Its full 76-gate replay and intermediate complex states are exported to the same numerical architecture inspector as the other models.

### S19.6 Width, fields, RG and arithmetic obligations

Proposition 15.1 fixes which width is being increased: the number of independent one-qubit features, with only their readout trained. At initialization let $a_r \sim N(0,1)$ be independent standard centered readout coefficients, independent of the frozen features. Conditional finite-dimensional outputs are then exactly centered Gaussian with covariance $K_m$. The almost-sure kernel convergence implies convergence of these finite-dimensional characteristic functions to the Gaussian law with covariance $K$. This proves the specified finite-dimensional GP limit. It does not prove tightness in a function-space norm, parameter-feature learning, a many-qubit limit or a continuum QFT. For frozen $K_m$, squared-loss gradient flow gives $\dot{r} = -K_m r$, whose eigencomponents decay at their kernel eigenvalues. A zero eigenvalue leaves a nonzero residual unchanged.

The executable width experiment fits the existing exact 32-configuration *classical* black-spin-decimated action on genus two, including its constant. This is a second true RG teacher, distinct from the quantum white-spin partial trace above. With 96 parameter triples for training, 32 for testing, eight feature seeds and ridge $10^{-5}$, widths 16 and 4,096 give mean relative kernel errors 0.2990 and 0.01726 and mean full-action test MSE 0.02005 and 0.000743. The limiting RBF readout is also evaluated. Finite numerical convergence supports the implementation; the proposition supplies the limit proof. Quantum NTK literature requires its own parameterization, ensemble and lazy/large-width hypotheses [91, 90].

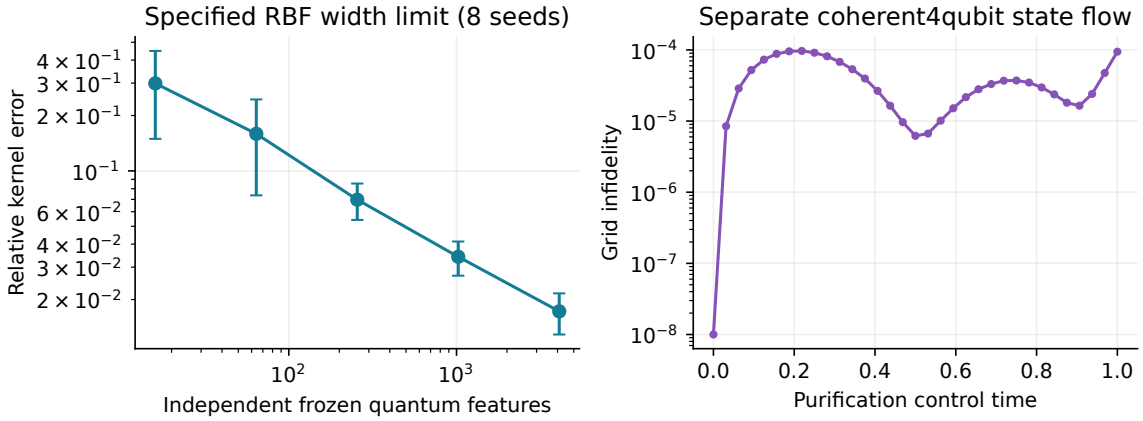



Figure 2: Left: eight-seed frozen-feature kernel errors, with sample standard deviations. Right: the separate learned coherent purification flow, evaluated on33time points. These are distinct width and quantum-control experiments.

For complex embeddings the Hermitian Gram form, realification $A + iB \mapsto \left(\begin{smallmatrix} A & -B \\ B & A \end{smallmatrix}\right)$, and conjugation must be respected. An arbitrary real network on twice as many channels need not be complex linear. In characteristic two, $1+1 = 0$; an embedding into real vectors cannot be a unital field homomorphism preserving that addition. Finite-field path algebras and cellular/CSS computations are still meaningful, but Gaussian integration, ordered Gibbs weights and real backpropagation do not transfer by renaming the field. An explicit discrete learner over $\mathbb{F}_p$, $p = 2, 3, 5$, uses features $1, x, y, xy$ and coefficients searched in the finite field to learn $f(x, y) = xy$. All $p^2$ inputs are verified; an affine-only learner has nonzero minimum error. The trained map compiles to the permutation unitary $U_f|x, y, z\rangle = |x, y, z + f(x, y) \bmod p\rangle$ on a complex Hilbert space of dimension $p^3$. Retaining $x, y$ makes its inverse subtraction explicit. This is reversible encoding of a field-valued polynomial learner, not a field embedding into complex amplitudes or backpropagation in $\mathbb{F}_p$. Its training is exact finite coefficient search. A tropical semiring is not a field; coefficient cancellation, valuation signs and the underlying valued algebra must be specified.

For GRH, each architecture offers an experimentally testable role: quantum features can parameterize candidate metrics or probe observables; graph tokens can encode actual local incidence, masks and monodromy; tropical attention can search extremal valuations or active constraints; flow/diffusion can propose continuous operator/control parameters; RG can organize a genuine hierarchy of coarse actions. None of these roles proves that its Gram matrix is the arithmetic positive metric. One must still construct the arithmetic object with correct conductor, gamma factors, zero divisor and domain, prove compact-uniform control and zero location, and keep Dirichlet, Hecke, Dedekind and Artin hypotheses separate. A learned candidate is accepted only after its exact defining equations and analytic estimates are supplied.

# S20 Covers, paths, pants and quantum games

## S20.1 Which free group and which universal cover?

Write $F_2 = \langle x, y\rangle$, $\mathbb{F}_2$ and $\widehat{F}_2$ for the free group, coefficient field and profinite completion, respectively. For a theta graph with vertices $A, B$ and edges $a, b, c$, an explicit cover has vertices $(g, A), (g, B)$, $g \in F_2$, and edges

$$a : (g, A) - (g, B), \qquad b : (g, A) - (gx, B), \qquad c : (g, A) - (gy, B).$$

Left multiplication is the deck action. Contracting all $a$-edges produces the usual four-regular Cayley tree of $F_2$; before contraction the graph is three-regular. Equivalently, vertices are reduced nonbacktracking edge paths from $A$, with an immediately reversed edge cancelled. This gives a tree and unique path lifting in every length. Its radius-$r$ patch has $1 + 3\sum_{k=0}^{r-1} 2^k$ vertices, hence 766 at radius eight. With a planar ribbon structure the abstract graph is a spine of a topological pair of pants. The actual $m = 3$ dessin instead has the ribbon structure of a once-punctured torus. Forgetting the cyclic orders identifies the abstract graphs, not these thickenings. Graph distance is not automatically hyperbolic geodesic length.

For the contracted clean $m$-edge dessin there are two vertices and $m$ parallel edges, so $\pi_1$ is $F_{m-1}$ and the analogous universal tree is $m$-regular. The new examples include $m = 5, 9$, of genus two and four. Reinserting clean white bivalent vertices subdivides its edges. Attaching the face to recover the compact surface introduces a relation; the closed genus-two surface has its surface group and a disk universal cover, not one of these graph trees. Separately, $\mathbb{P}^1 \setminus \{0, 1, \infty\}$ has fundamental group $F_2$. The unbranched cover induced by a clean dessin removes all preimages of those three values. For our degree-$2m$ map there are $m + 3$ punctures; its Euler characteristic is $-2m$, not that

of the dessin's one-skeleton. Thus the occurrences of $F_2$ in monodromy, graph covers and arithmetic profinite theory must not be silently identified.

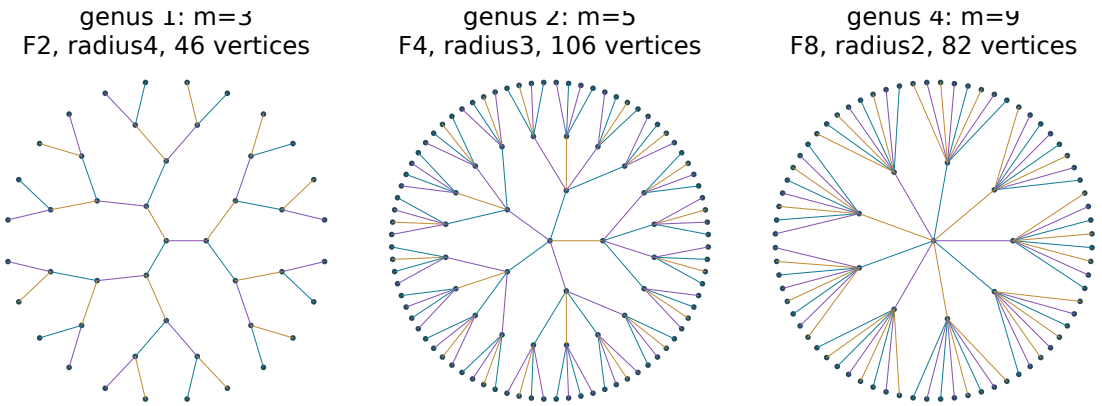



Figure 3: Finite windows of universal covers of contracted actual clean dessins. These are schematic graph trees, not uniformizing embeddings of the compact surfaces. Clean white bivalent vertices are suppressed.

For finite-supported representations of a suitably graded bound quiver, pushdown maps split into deck-homogeneous blocks. The arrow equations preserve the degree, and monomial zero relations are homogeneous. The resulting Hom direct sum controls all deck pieces, not just degree one. For an explicit graph-only diagnostic two lifted edge intervals have upstairs endomorphism dimension two, but downstairs endomorphism dimension four: the deck pieces have dimensions $2+1+1$. This graph-quiver calculation is *not* substituted for a dessin-algebra module example. The actual infinite polynomial module examples below supply that part. A Hom-controlled functor needs its specified exceptional/source/target subcategory and control statement; exactness, a deck-sum formula or equal dimensions alone is not an equivalence or an AR universality theorem. A derived comparison must control all $\mathrm{Hom}(X, Y[n])$ and composition, together with an actual derived/enhanced functor.

## S20.2 Actual infinite-algebra path modules and unbounded resolutions

To fix conventions, let $A_{\mathrm{col}}$ be the earlier rightmost-arrow-first polynomial matrix algebra acting on left columns. Here we use travel-order path multiplication and write $A_{\mathrm{trav}} = A_{\mathrm{col}}^{\mathrm{op}}$ for the prose-identified bound-quiver algebra. Right $A_{\mathrm{trav}}$ representations therefore encode the earlier covariant left-column representations. There are vertices $i \in \mathbb{Z}/5$ and arrows $a_i, b_i : i \to i-1$, with

$$a_i b_{i-1} = b_i a_{i-1} = 0.$$

Both pure branch paths survive at every length. No arrow-truncated finite algebra is being used. We work with ordinary polynomial modules, not with all completed lattices or all infinite strings.

Put $P_i = e_i A_{\mathrm{trav}}$ and let $S_i$ be its residue simple. The augmentation kernel is $a_i A \oplus b_i A$. Multiplication by an arrow kills exactly the opposite-branch positive paths at its target, while each surviving path has a distinct normal image. Its kernel is therefore the right ideal generated by its unique forbidden successor, without cancellation. Iteration yields a projective resolution with

$$P_0 = P_i, \qquad P_n = P_{i-n} \oplus P_{i-n} \quad (n \geq 1).$$

The two arms are indexed by alternating-branch relation words. The *differential entry is the last arrow*, not the product of the zero relation word. This kernel description proves exactness in every

degree. Applying Hom to a simple kills every arrow entry, hence

$$\dim \mathrm{Hom}(S_i, S_j) = \mathbf{1}_{i=j}, \qquad \dim \mathrm{Ext}^n(S_i, S_j) = 2\mathbf{1}_{j=i-n} \quad (n \geq 1).$$

Tensoring with the left residue simple gives the same Tor dimensions. A graded version needs its internal degree shifts. In particular the residue simples have infinite projective dimension in this ambient algebra.

Let $B$ be the one-arrow string supported at vertices zero and four, with $a_0 = 1$ and every other arrow zero. Its presentation is

$$P_4 \oplus P_3 \xrightarrow{(b_0,\ a_0 a_4)} P_0 \longrightarrow B \longrightarrow 0.$$

The two images have disjoint normal-path support and span the kernel, because $B$ retains exactly $e_0, a_0$. The next entries are $a_4$ on the first arm and $b_3$ on the second; continue by their unique forbidden successors. This gives an unbounded exact resolution by the same normal-path argument. Full arrow-commuting equations give

$$\dim \mathrm{Hom}(B, B) = 1, \quad \dim \mathrm{Hom}(S_0, B) = 0, \quad \dim \mathrm{Hom}(B, S_0) = 1, \quad \dim \mathrm{Hom}(S_4, B) = 1.$$

In the Hom complex for $\mathrm{Ext}^1(B, B)$ both adjacent differentials vanish, so $\mathrm{Ext}^1(B, B) \simeq \mathbb{Q}$. A concrete nonsplit self-extension has $\mathbb{Q}^2$ at zero and four, $a_0 = I_2$, $b_0 = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$, all other arrows zero. The first-coordinate submodule and quotient are $B$. A section at vertex zero would lift the quotient basis to a vector with second coordinate one; $b_0$ sends it to a nonzero first-coordinate vector, contradicting commutation with the zero $b_0$ action on $B$. Thus $B$ is not rigid in this actual ambient category. No almost-split universality is inferred from this nonsplitting.

For finite-dimensional $N$ let $DN = \mathrm{Hom}_{\mathbb{Q}}(N, \mathbb{Q})$. On each finite idempotent projective the evaluation pairing identifies $P \otimes_A DN$ with $D\,\mathrm{Hom}_A(P, N)$. Dualization is exact on these finite-dimensional terms, so the resulting complexes give $\mathrm{Tor}_n^A(M, DN) \simeq D\,\mathrm{Ext}_A^n(M, N)$ for the stated finitely generated resolutions. In particular $\mathrm{Tor}_1^A(B, DB) \simeq \mathbb{Q}$. The exact program checks six source/target pairs through degree twelve, all five simple-target formulas and 250 normal-kernel samples. The unbounded resolution assertions rest on the normal-form proof, not on the finite cutoff. These new homological proofs are prose/exact calculations; full corresponding resolution objects are not yet Lean packaged.

A geometric path may label a string after specifying the ribbon structure, relations and category. Turning it into a geodesic needs a metric and appropriate boundary/stop data. Intersections compute morphisms only under the hypotheses of an actual graded/Fukaya comparison. The homologically smooth graded-gentle perfect-category theorems do not identify our entire ordinary or completed module category with the Fukaya category by inspection [93]. In particular the residue resolutions warn against silently imposing smoothness or finite projective dimension.

### S20.3 Pants tensors, physical dilation and sewing boundaries

For $C = \mathbb{Q}[\epsilon]/\epsilon^2$, take $\varepsilon(1) = 0$, $\varepsilon(\epsilon) = 1$. In the ordered bases $(1, \epsilon)$ and $(1 \otimes 1, 1 \otimes \epsilon, \epsilon \otimes 1, \epsilon \otimes \epsilon)$, multiplication and comultiplication are

$$\mu = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \end{pmatrix}, \qquad \Delta = \begin{pmatrix} 0 & 0 \\ 1 & 0 \\ 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

The pairing is $\left(\begin{smallmatrix}0&1\\1&0\end{smallmatrix}\right)$, with determinant $-1$. Associativity follows from $\epsilon^2 = 0$; coassociativity sends one to the three placements of a single one among two $\epsilon$'s and sends $\epsilon$ to $\epsilon^{\otimes 3}$. On the four tensor basis vectors, all three maps in

$$\Delta\mu = (\mu \otimes I)(I \otimes \Delta) = (I \otimes \mu)(\Delta \otimes I)$$

give, respectively, $1 \otimes \epsilon + \epsilon \otimes 1$, $\epsilon \otimes \epsilon$, $\epsilon \otimes \epsilon$, and zero. This proves the Frobenius identity. The unit and counit identities follow directly on the same basis. These are exact oriented 2D TQFT gluing data [95], not a presumed positive Hilbert structure.

The handle map $H = \mu\Delta$ sends $1 \mapsto 2\epsilon$ and $\epsilon \mapsto 0$. Closed genus-$g$ contractions $\varepsilon H^g \eta$ are $0, 2, 0, 0, \ldots$. The genus-two value really vanishes in this example. In characteristic two the torus value also vanishes, illustrating a meaningful coefficient change rather than an embedding-space relabelling. After extending to $\mathbb{C}$ with the Euclidean metric, $\Delta^*\Delta = \mathrm{diag}(2, 1)$ and $\Delta \neq \mu^*$; $\mu : \mathbb{C}^4 \to \mathbb{C}^2$ is not an isometry. The exact code constructs $K = \mu/\sqrt{2}$ and a $4 \times 4$ failure block $L$ with $K^*K + L^*L = I_4$. Stacking them gives an isometry into $\mathbb{C}^2 \oplus \mathbb{C}^4$, extendable to a unitary with ancillas. Its success probabilities on the four basis states are $1/2, 1/2, 1/2, 0$. Postselection implements $K$ with state-dependent success; normalizing the selected outcome is not the original linear map $\mu$.

For a closed genus-$g \geq 2$ surface a pants decomposition has $2g - 2$ pieces and $3g - 3$ sewn curves. Conformal sewing adds moduli-dependent propagators and analytic consistency [96]; string amplitudes need worldsheet, ghost and integration data beyond this finite tensor. A pants network or a hyperbolic graph is likewise not an AdS/CFT duality or a Ryu–Takayanagi derivation [97]. Such interpretations remain specified research targets until their extra structure is constructed. Arithmetic/anabelian reconstruction additionally requires arithmetic profinite fundamental-group data and its Galois structure under the proper field/curve hypotheses. Neither a discrete $F_2$ graph nor a coefficient $\mathbb{F}_2$ module supplies that arithmetic extension [98].

### S20.4 Quantum games and the Connes-embedding distinction

For uniform input bits $x, y$, CHSH accepts outputs $u, v$ when $u \oplus v = xy$. With $\pm 1$ observables its success probability is $1/2 + \langle T \rangle/8$, where

$$T = A_0 \otimes (B_0 + B_1) + A_1 \otimes (B_0 - B_1).$$

For a deterministic classical assignment one bracket vanishes and the other is $\pm 2$, giving success at most $3/4$; all sixteen assignments are checked. On $|\Phi^+\rangle$, take $A_0 = Z, A_1 = X$, $B_0 = (Z + X)/\sqrt{2}, B_1 = (Z - X)/\sqrt{2}$. Then $T = \sqrt{2}(Z \otimes Z + X \otimes X)$ gives success $(2 + \sqrt{2})/4$. For arbitrary self-adjoint involutions, $T^2 = 4I - [A_0, A_1] \otimes [B_0, B_1]$ and each commutator has norm at most two. Hence $\|T\| \leq 2\sqrt{2}$, proving optimality. The exact matrix checker verifies the entire spectrum $\{-2\sqrt{2}, 0, 0, 2\sqrt{2}\}$, not just a numerical sample.

MIP*=RE, in its pinned v4, gives an external unrestricted correlation separation and a negative answer to Connes embedding [99]. We do not rerun its external formalization or claim a new proof of it. CHSH has its optimum on two qubits and the same commuting optimum by the preceding bound; it does not exhibit the general separation. A learned game strategy is useful for finite witness search and checked bounds, but unrestricted computability and dimension quantifiers cannot be settled by a finite game simulation.

### S20.5 Tropical accelerator reference designs and reproducibility

The chip-design campaign pins twenty public repositories, twenty-four focused document ranges and thirteen primary papers. It covers AlphaChip/Circuit Training, independent macro-placement

benchmarks, OpenROAD-family flows, quantum layout and low-level control. The source ledger records what was actually read, not a full audit or reproduction of every model. An exact public identity for the phrase "bare-metal Qiskit" was not established; no repository was invented.

Three original executable references implement a masked signed max-plus tile, an ordered mixed MAC/max-plus scheduler and a fail-closed synthetic IQ steering controller. Wide addition precedes winner selection and final saturation; earliest-index tie handling is explicit. Empty masks remain invalid, not an extra candidate of value zero. The controller requires a current calibration epoch, enough samples and a valid margin/deadline before allowlisting a steering action. Checks include all 65,536 signed eight-bit additions, 262,144 masked two-lane cases, reordered completions and twelve negative controls. These are digital reference models, not synthesized RTL, routed chips, measured RF feedback, power/area estimates or universal hardware speedups. A tropical context block can avoid multiplications inside that primitive while its projections, scores and other layers still require their own arithmetic.

The Linux snapshot preserves the Mac project, source clones, histories, checkpoints and environment archives. Linux GPU execution uses the existing Torch CUDA environment and a separately approved isolated cuQuantum prefix; macOS binaries are not treated as a Linux installation. The conda lock, setup instructions, gate/head replay records and all failures are retained. The on-page book contains one canonical laboratory for this chapter and links all five new checkpoint architectures into the existing tensor inspector. New physics, width and homological results have their explicit prose/numerical/external status; selected earlier Lean components do not constitute a full formalization of this extension or a proof of GRH.

# S21 Construction, curation and interpretation

## S21.1 A record is an evidence-bearing physical task

Each default Parquet row has a persistent record identity, parent/similarity group, partition, evidence class, lossless scientific payload and typed numeric columns. Forward inputs have the prefix `condition_`; supervision uses `target_`; `evaluation_` fields are excluded from inputs. The full payload, source hash, record identifier and audit text are not automatically fed into a model. Otherwise an apparent condition/target mask could still leak the target through metadata or a duplicated representation. An inverse task needs an explicit desired-condition contract and role change.

The origin fields distinguish a new simulation from a reused planned-source identity or geometry with new labels. The source-anchor configuration preserves existing train assignments. New public PDB coordinates retain their own source identity rather than being called members of PDBbind or newly acquired experimental quantum measurements. All anchors are train only: they do not establish blind performance on a model that may have encountered the parent.

The actual native interfaces are tested without changing either model. Aster accepts condition-only Hamiltonian records through its existing tokenizer. Changing a hidden target leaves those input tokens unchanged. Mora-3's scalar graphs preserve computed/predicted evidence. Its existing scalar/array supervision interfaces reject those targets under their observed-only policy. That rejection is recorded, not bypassed by changing the evidence label. Source construction is therefore distinct from the owning model's future admission, scaling, objective and evaluation decisions.

## S21.2 Effective material families and gap guards

The two-band QWZ model uses

$$H(k_x, k_y) = \sin k_x\, \sigma_x + \sin k_y\, \sigma_y + (m + \cos k_x + \cos k_y)\sigma_z.$$

Its continuum band gap is $2\min\{|m+2|, |m|, |m-2|\}$. To see this, put $u = \cos k_x$, $v = \cos k_y$. The squared positive energy is $2 + m^2 + 2m(u+v) + 2uv$, a bilinear function on $[-1,1]^2$; its minimum occurs at a corner. Thus an odd mesh that misses $\pi$ cannot certify a gap near $m = 0$ or2. The corrected release uses this analytic guard, even32/64 meshes, random gauge checks and an unavailable label near closure. The occupied-band integer uses the oriented link-product convention of Fukui–Hatsugai–Suzuki [100]; the sign convention is retained in the row.

For the Rice–Mele pump, take $t_1 = 1 + \delta\cos\theta$, $t_2 = 1 - \delta\cos\theta$ and $\Delta = r\sin\theta$. With positive $t_1, t_2$, minimization over $k$ first gives $(t_1 - t_2)^2 + \Delta^2$ at $k = \pi$; minimization over $\theta$ gives the exact gap $2\min(2\delta, r)$. Harper's direct gap has a conservative mesh/Lipschitz guard, explicitly distinguished from interval certification. The displayed bulk slice has $k_y = 0$; an open strip is evaluated separately at $k_y = .37$. A single strip is not a proof of its entire edge dispersion.

The SSH winding, Kitaev sign invariant and classical dessin RG retain their own hypotheses. Surface genus is not an electronic phase index. In the Kitaev row, a physical onsite defect changes particle and hole diagonals with opposite signs. Its gauge change has the form $\mathrm{diag}(G, \bar{G})$. These preserve the recorded particle–hole symmetry; arbitrary independent diagonal perturbations would not. A quarantined draft revealed this error before release. No failed draft replaces a verified artifact.

## S21.3 Molecular geometry and forces

The molecular set contains explicit chemical graphs, atom order, hydrogens, stereochemistry, method-labelled optimized references and bounded coordinate perturbations. MMFF94 supplies empirical molecular-mechanics energies, analytic forces and partial charges [101]. It supplies neither ab-initio electron density nor experimental affinity, ADMET, redox or biological efficacy. An optimization convergence code is not a theorem of a unique global minimum. Absolute MMFF energies are not cross-molecule atomization energies.

The checker reconstructs the stored bond graph in its *actual atom order*. A canonical SMILES string determines chemical identity but need not give the coordinate-array order. Geometry-assigned stereochemistry is checked against that identity; mirror controls fail. Energies and forces are recalculated on every accepted record. Central energy differences check force signs/units, and proper rotations/translations check the expected scalar/vector behavior. Missing or unsupported labels are not imputed.

For the heavy-atom point cloud, the complete finite Vietoris–Rips2-skeleton gives $H_0, H_1$ over $\mathbb{F}_2$. All edges and triangles, including ties, are ordered explicitly. Independent boundary-rank calculations verify selected Betti numbers against the barcodes. These are geometric distance-filtration features; a molecular loop or persistence bar is not an electronic topological phase. MMFF and the idealized $\pi$-orbital Hamiltonian retain separate labels.

## S21.4 What the split certificate proves

Let $V$ be the finite parent set and connect two parents whenever a declared similarity test passes. Add edges for all shared source/variant lineages. Assign a partition to each connected component and inherit it for every row. If an accepted similarity edge joined two partitions, its endpoints would

lie in one component with two assignments, a contradiction. This proves the exact finite no-crossing property for the declared relation.

For materials, normalized parameter and relative finite-spectrum thresholds are family/size-specific. SSH hopping swaps, Kitaev chemical-potential sign and the classical RG global-sign symmetry are folded conservatively. For molecules, stereo-agnostic constitution, nonempty Murcko scaffold and radius-two Morgan Tanimoto threshold .65 form chemical components. A second all-conformer pass adds edges for equal heavy-element multisets with sorted pair-distance RMS at most .10Å. This descriptor is rotation/permutation invariant but is not a complete congruence invariant or a certified RMSD optimizer; conservative false-positive grouping is possible.

Whole-component allocation balances row counts and declared family/topology or molecular-size/persistence strata. It never breaks a component to manufacture an80/10/10 result. The large peptide component makes the molecular training partition much larger than80%; rare/held-out peptide coverage can consequently be absent. Exact distributions and cross-partition nearest witnesses accompany the release. Protein TM-score, homology separation and global overlap with the models' complete historical corpora are not certified by small-molecule fingerprints. Threshold separation cannot itself guarantee freedom from all leakage or overfitting.

### S21.5 Motion, charge modulation and exchange

For normal free fermions, $P(t) = UP(0)U^*$ preserves Hermiticity, trace and the spectrum: $P^* = P$ follows by adjoints, trace follows by cyclicity, and $P(t) = UP(0)U^{-1}$ is a similarity. Thus $0 \le P \le I$ and projector occupancy remain valid. Computing the diagonal of $-i[H, P]$ gives

$$\dot{n}_i = 2\,\mathrm{Im}\sum_j H_{ij}P_{ji}, \qquad J_{ij} = -2\,\mathrm{Im}(H_{ij}P_{ji}) = -J_{ji}.$$

The currents therefore obey site continuity and conserve total particle number. Their arrows represent probability/particle flux, not measured trajectories of individual electrons. Charge modulation is $-[n_i(t) - n_i(0)]$ in the declared electron-charge convention.

For a paired model, put $\Psi = (c, c^\dagger)$ and $R = \langle \Psi\Psi^\dagger \rangle$. From an initial Slater projector use $R_0 = \mathrm{diag}(I - P_0, P_0^{\mathsf{T}})$. Evolve with the full BdG Hamiltonian and recover $P = I - R_{11}$, $F = R_{12}$. The independently computed pairing source is

$$S_i = -2\,\mathrm{Im}(\Delta F^*)_{ii}, \qquad \dot{n}_i = -\sum_j J_{ij} + S_i.$$

The commutator derivative and this separate source formula agree in the checks; the source is not defined as the residual of the desired equation. For a Gaussian state, Wick's rule gives, for $i \neq j$, $\mathrm{Cov}(n_i, n_j) = |F_{ij}|^2 - |P_{ij}|^2$; on the diagonal it is $n_i(1 - n_i)$. The negative term is an exchange hole, not a computed Coulomb exchange-energy functional. Explicit small Fock-space evolution independently checks the covariance, current and pairing conventions.

The interacting control is the two-site Hubbard model at two electrons:

$$H = -t\sum_\sigma (c_{0\sigma}^\dagger c_{1\sigma} + \mathrm{h.c.}) + U\sum_{a=0}^{1} n_{a\uparrow}n_{a\downarrow}.$$

In its singlet/doubly-occupied symmetric sector the matrix is $\left(\begin{smallmatrix} 0 & -2t \\ -2t & U \end{smallmatrix}\right)$; the triplet energy is zero. Diagonalizing gives the exact exchange gap $J = (\sqrt{U^2 + 16t^2} - U)/2$, approaching $4t^2/U$ for large positive $U$. The full six-state two-electron sector supplies density, spin-swap and current trajectories;

a separate16-state Jordan–Wigner calculation checks them. This finite dimer is not a thermodynamic phase classification.

For nuclei, velocity Verlet uses actual MMFF forces and masses, with explicit time/energy units, finite energy drift and momentum checks. A Grad-only RDKit loop initially reused stale distance information. Refreshing energy at the same coordinates before the gradient restored the expected second-order step-size behavior in independent refinement tests. Failed trajectories are retained and excluded. The finite isolated trajectories are not equilibrium sampling. MMFF charges remain fixed. The separate uncalibrated $\pi$ graph has its own dimensionless electronic time; it is not self-consistent TDDFT coupled to the nuclear femtosecond trajectory.

### S21.6 Source-backed peptides and genuine boundaries

Public wwPDB coordinates are supplied under CC0 [102]. The source packet includes Trp-cage1L2Y, villin1VII, historical Trp-zipper1HRW (superseded by 1LE0, with the actual file HEADER identity retained), pancreatic polypeptide1PPT, azurin1AZU, crambin1CRN, ubiquitin1UBQ and engrailed homeodomain1ENH. The full source file, checksum, DOI, first-model/chain selection and bounded residue-subsystem mapping remain explicit. An NMR ensemble's model order is not physical time. Azurin's electron-transport classification does not calibrate our coarse contact Hamiltonian or turn it into an electronic-structure result.

The CISS review and peptide-bridge charge-transfer preprint motivate future spin, environment and lead benchmarks [103, 104]; the source review is focused, not an independent reproduction of their experiments. A spinless contact graph cannot supply their spin polarization, electron-transfer rates or measured quantum-biological mechanism. Synthetic strain, vacancy, mirror, coupling and conformer variants have their own provenance and share the parent group. They are additional training candidates, not newly observed systems.

### S21.7 Circuits and geometric visualization

For a normal one-particle unitary $U$, adjacent row elimination gives Givens matrices $G_k$ and diagonal phases $D$ with $U = G_1^* \cdots G_m^* D$. For $a, b$ in one column, the eliminating block is

$$G = \frac{1}{\sqrt{|a|^2 + |b|^2}} \begin{pmatrix} \bar{a} & \bar{b} \\ -b & a \end{pmatrix}.$$

Direct multiplication gives $G(a, b)^{\mathsf{T}} = (r, 0)^{\mathsf{T}}$ and $G^* G = I$. A two-mode second-quantized gate acts as 1 on the vacuum, as its one-particle block on the single-occupation sector and as the determinant on double occupation. The exported circuit verifies these matchgates and numerical recomposition. Browser replay applies the actual gates to the saved initial correlation matrix. BdG pairing needs a different Bogoliubov compiler; it is not passed by a number-conserving Givens circuit. The Hubbard example instead includes Jordan–Wigner parity and measured product-formula convergence.

Actual molecular structures use CPK colors and explicit atom/bond order, with SDF/XYZ views, ball-and-stick, sticks and space-filling representations. Effective material-site layouts are labelled schematic. Electron-colored Gaussian splats are declared kernels on discrete site values, not ab-initio volumetric density. For the symmetric graph Laplacian $L$, localized wavelets are $sLe^{-sL}\delta_i$; since $\mathbf{1}^{\mathsf{T}} L = 0$, they have zero sum. Their signed colors are not probabilities. Time as a fourth projection coordinate is a visualization operation, not a new spatial dimension.

The dataset page links the checked public releases and source code. The full new probabilistic/homological/physical interfaces are not yet packaged in Lean; the existing 360 printed reports

retain their earlier scope. Neither a simulated transport law, dataset split certificate nor a compiled finite circuit closes the arithmetic comparison required for GRH.

## S22 Molecular structure, complex transport and learned chemistry

The complementary-data laboratory now separates deposited protein structure, actual finite quantum transport and learned conformation generation. Covalent bonds use explicit molecular graphs or qualified PDB templates, with depth ordering and visible atom/bond scales. The ribbon view reads deposited `HELIX`/`SHEET` records; short synthetic peptide backbones remain unassigned rather than acquiring invented secondary structure.

Five additional finite propagation examples compare matched SSH/QWZ topological and trivial controls and an SSH domain wall. Complex phase, probability, current, coherence and selected eigenstates are distinct views. Analytic hydrogen volumes illustrate nodes, quantum beats and a phase vortex; they are not molecular electronic-structure calculations. Eight further CPU runs train classical and two-qubit hybrid score/flow models on periodic MMFF butane/pentane torsion spaces. All checkpoints, losses, controls and positive noise endpoints are explicit. These examples neither establish quantum advantage nor close the independent arithmetic bridge for GRH.

## S23 Geometry, orbitals and generative time

### S23.1 Chemical connectivity and source secondary structure

An atomic coordinate cloud does not determine chemical bonds merely by looking connected. SDF graphs retain their explicit bond indices. For the PDB source view we select the first model, blank/A alternate locations and positive occupancy; solvent is omitted from the display. Standard amino-acid templates, explicit `CONECT`, qualified ligand/hydrogen proximity and geometrically valid peptide C–N links have separate provenance. The complete original file remains available. Display connectivity is not a newly measured bond label.

The parser follows the wwPDB3.3 record definitions [105]. A class-one `HELIX` is an alpha helix; class five is a $3_{10}$ helix, not another alpha helix. `SHEET` retains strand identity, sheet membership and parallel/antiparallel sense. Ribbon patches follow the actual CA trace and carbonyl orientation, with strand arrows and narrower coils. A peptide bond must connect neighboring residues before a patch or spline-control neighbor is allowed: spatially nearby disconnected fragments and TER-separated chains must not acquire a phantom ribbon. Regression tests include these controls. Backbone recognition in a short synthetic SDF peptide supplies a trace, not a DSSP assignment or evidence of a stable secondary structure.

The full-source geometry is distinct from the capped effective contact graph. In the full-source view, only mapped modeled CA sites carry its electronic field. Nuclear display interpolation adds no new integration steps. An NMR model index, a ribbon parameter and a generative time are not physical time.

### S23.2 What the complex transport animation computes

For a finite Hermitian matrix $H = V\,\mathrm{diag}(E_k)V^*$ and a normalized initial vector $\psi_0$, the continuously evaluated browser state is

$$\psi_j(t) = \sum_k V_{jk} e^{-itE_k} (V^*\psi_0)_k.$$

Unitarity gives $\sum_j |\psi_j(t)|^2 = 1$. Differentiating gives $\dot{n}_i = 2\,\mathrm{Im}\sum_j H_{ij}\psi_j\bar{\psi}_i$ and hence $\dot{n}_i + \sum_j J_{ij} = 0$ for $J_{ij} = -2\,\mathrm{Im}(H_{ij}\psi_j\bar{\psi}_i)$. The actual complex one-particle coherence is $\rho_{ij} = \psi_i\bar{\psi}_j$. A common phase cancels from both current and coherence; node colors therefore use a stated phase reference. The probability height is a display encoding, not another physical coordinate. Current arrows are fluxes, not classical electron paths.

Two48-site SSH controls use the same initial endpoint orbital and exchanged strong/weak hoppings. Their periodic bulk winding is respectively zero and one for $q(k) = t_1 + t_2e^{ik}$, with gap $2|t_1 - t_2|$. A separate61-site domain-wall chain starts and ends with strong bonds and has two weak bonds adjacent to its central even site. Its sublattices have sizes31 and30. At zero energy, set the odd-site amplitudes to zero and solve the tridiagonal recurrence on even sites. The amplitudes decay away from the wall by the weak/strong ratio. The nonzero off-diagonal entries determine this vector up to scale, so the nullspace is one-dimensional. This proves a finite interface zero mode; it does not identify an arithmetic Hilbert–Pólya spectrum.

The QWZ examples have a $12 \times 10$ rectangle and two orbitals per site. They use the same unfiltered corner spinor for masses $-1$ and $-3$. Their bulk Chern numbers are $-1$ and0 in the existing link-product convention; 32/64 meshes and gauge controls agree. Finite boundary probability, spectral weights and coherence remain measurements of the particular finite state, not definitions or proofs of that bulk invariant. Independent matrix-exponential checks reproduce the five propagated states. The hydrogen phase winding below, chemical torsion torus, dessin genus and persistent-homology classes are also different invariants.

### S23.3 An analytic orbital benchmark with actual nodes

In atomic units, take $h = -\frac{1}{2}\Delta - r^{-1}$ on the usual hydrogen domain. The displayed normalized eigenfunctions are

$$\phi_{1s} = \pi^{-1/2}e^{-r}, \qquad \phi_{2s} = \frac{(2-r)e^{-r/2}}{4\sqrt{2\pi}}, \qquad \phi_{2p_j} = \frac{x_j e^{-r/2}}{4\sqrt{2\pi}}.$$

Their energies are $E_1 = -1/2$, $E_2 = -1/8$. The angular integrals and $\int_0^\infty r^n e^{-ar}\,dr = n!a^{-n-1}$ give their unit norms; opposite angular parity gives orthogonality of $1s$ and $2p_z$. Consequently

$$\psi(t) = \sqrt{1-a}\,\phi_{1s} + \sqrt{a}\,e^{-3it/8}\phi_{2p_z}, \qquad 0 \le a \le 1,$$

has unit norm after removing its unobservable common ground-state phase. Its density has the interference term $2\sqrt{a(1-a)}\phi_{1s}\phi_{2p_z}\cos(3t/8)$: this is the actual quantum beat shown in the volume. The $2s$ radial node is $r = 2$; the $2p_z$ nodal plane is $z = 0$.

The state $(\phi_{2p_x} + i\phi_{2p_y})/\sqrt{2}$ is proportional to $(x + iy)e^{-r/2}$. On a circle avoiding the nodal axis its phase winds once, and $L_z\psi = \psi$. Its current is $\mathrm{Im}(\bar{\psi}\nabla\psi)$ and has an azimuthal component. This stationary density with nonzero current need not show a moving classical particle. A finite-radius ray-marched volume and its opacity/exposure curve are rendering choices, distinct from the analytic density and normalization. These are nonrelativistic isolated-hydrogen benchmarks, not many-electron molecular orbitals or ab-initio peptide density.

### S23.4 A useful periodic chemical target

Butane has one selected torsion; pentane has two. All other internal coordinates are held at the recorded MMFF reference geometry. The finite grids have72 and $24^2$ points, respectively. At300K define

$$p_k = Z^{-1}\exp[-(E_k - E_{\min})/(k_BT)], \qquad q_0(\theta) = \sum_k p_k\,K_{\sigma^2/2}(\theta - \theta_k), \quad \sigma = .05,$$

where the product wrapped heat kernel is used on $\mathbb{T}^d$ and $d = 1, 2$. Thus $q_0$ is an explicitly softened finite-grid Gibbs target. It is positive, smooth and normalized; it is not a full solvent or molecular-equilibrium distribution. No affinity, redox, efficacy or electronic-density label is inferred. The molecular animation rotates downstream atoms about each actual bond axis; its poses agree with independent RDKit dihedral assignments to numerical precision and preserve the bond graph and fixed bond lengths.

### S23.5 Score diffusion on the torus

For the unit circle,

$$K_\tau(x) = \sum_{m\in\mathbb{Z}} \frac{e^{-(x+2\pi m)^2/(4\tau)}}{\sqrt{4\pi\tau}}, \qquad \partial_\tau K_\tau = \partial_x^2 K_\tau.$$

The product kernel gives $q_\tau = q_0 * K_\tau$ and $\partial_\tau q_\tau = \Delta q_\tau$. A forward sample is $X_\tau = \mathrm{wrap}(X_0 + \sqrt{2\tau}\,\xi)$. Conditioning the wrapped Gaussian mixture shows

$$\nabla \log q_\tau(x) = \mathbb{E}[-\xi/\sqrt{2\tau} \mid X_\tau = x].$$

Indeed, differentiate each image term and divide by the mixture density; the Gaussian score is the stated latent-noise target. Conditional least squares therefore identifies the marginal score. The implemented weighted DSM loss uses weight $2\tau$, not an unlabelled Euclidean proxy on cut angles. The periodic-input network is trained over the full normalized time interval.

With reverse clock $u$, the ideal SDE is $dY_u = 2\nabla \log q_{T-u}(Y_u)\,du + \sqrt{2}\,dW_u$. Its Fokker–Planck equation at density $q_{T-u}$ is $-2\Delta q + \Delta q = -\Delta q$, as required. This connection to score SDEs is the standard method [106]; the periodic mixture here supplies an explicit control. Initialization is sampled from the actual $q_T$, with $T = 2$, rather than silently assuming it is uniform. Sampling stops at $\tau = .03$, so the ideal endpoint is $q_{.03}$, whose Gaussian component variance is $\sigma^2 + .06$. It is not $q_0$. The initial run exposed this distinction; its source and outputs are preserved. The corrected run uses the proper endpoint reference and no excluded small-time training interval.

Exact wrapped-mixture scores, finite-difference checks, nested64/128-step Brownian controls and stored learned trajectories accompany the experiment. A finite Euler diagnostic is neither a convergence theorem nor a guarantee that a learned score recovers equilibrium. Remaining noise and learned-field error are shown separately from the MMFF target.

### S23.6 Wrapped-geodesic flow matching

Let $X_0$ be uniform on $\mathbb{T}^d$, independently let $X_1 \sim q_0$, put $D = \mathrm{wrap}(X_1 - X_0)$ and $X_t = \mathrm{wrap}(X_0 + tD)$. The cut locus has probability zero under these continuous laws. For every smooth periodic test function $f$,

$$\frac{d}{dt}\mathbb{E}f(X_t) = \mathbb{E}\langle \nabla f(X_t), D\rangle = \int \langle \nabla f(x), v_t(x)\rangle \rho_t(x)\,dx, \quad v_t(x) = \mathbb{E}[D \mid X_t = x].$$

This is the weak continuity equation with the correct uniform and $q_0$ endpoints. Writing $B = [-\pi, \pi]^d$, the conditional formulas are explicitly

$$\rho_t(x) = (2\pi)^{-d} \int_B q_0(x + (1-t)D)\,dD,$$
$$\rho_t(x)v_t(x) = (2\pi)^{-d} \int_B D\, q_0(x + (1-t)D)\,dD.$$

Because $q_0$ is smooth and strictly positive on the compact torus, these integrals are jointly smooth through both endpoints and $\rho_t \geq \min q_0 > 0$. Division gives a smooth bounded periodic $v_t$ with a uniform spatial Lipschitz bound. In particular $v_1 = 0$ by symmetry. The usual unique flow of this globally Lipschitz time-dependent field transports the stated marginals. Conditional regression targets $D$ and learns an approximation to $v_t$, following the flow-matching and manifold formulation [107, 108]. Neither the regression loss nor a visualization replaces endpoint-error measurement.

### S23.7 Physical state preparation, RG and phase-conditioned design

The earlier seven-spin dessin experiment supplies an actual quantum Gibbs state and its partial trace; its learned circuits approximate specified state/field targets. The exact coarse action in the RG chapter is produced by integrating out physical spins. Here the chemical learner transports a classical probability law on internal coordinates. A quantum channel, an RG partial trace and a score/noise schedule are three different constructions, with different states, clocks and acceptance tests.

A concrete materials-design constraint already appears in the QWZ controls. Any continuous mass path from $m = -3$ to $m = -1$ passes through $m = -2$. The explicit bulk gap formula then gives zero gap. Thus such a generator cannot both change between these two Chern sectors and preserve the gap throughout that path. Sector-conditioned sampling, explicit transition states and gap/invariant checks are meaningful design tasks. Merely drawing a continuous latent interpolation does not meet them. The finite transport controls and the earlier learned thermal-state experiments provide distinct physics benchmarks for these requirements, rather than unverified predictions of a synthesized topological material.

### S23.8 What the quantum hybrid actually contributes

Each hybrid has two re-uploading two-qubit layers: periodic $R_z$ angle/time encoding, trainable $R_y$ rotations and CZ, followed by eight actual Pauli expectations and a two-layer64-unit tanh classical readout. The angles and readout are trained together. Integer coordinate frequencies preserve $2\pi$ periodicity of the measured features, and composition preserves it in the vector field. These are classically simulated quantum-feature hybrids, not coherent quantum chemical evolution or quantum-assisted measurements. The matched classical controls use periodic first/second harmonics and the same readout width. Parameter counts and individual seeds remain explicit.

Eight2400-update CPU runs retain heldout objectives, checkpoint weights, periodicity errors, independent Kronecker/Pauli calculations, parameter finite differences, exact-score comparisons and actual endpoint histograms. The limited runs do not establish quantum advantage. Flow and diffusion have different positive-noise endpoint laws and must not be ranked as though their targets were identical. Generative interpolation time is not molecular time.

The chemical torus is a concrete internal-coordinate space. Drawing it next to a dessin, a noncommutative torus or a Fukaya construction does not construct a functor or prove homological mirror symmetry. Likewise none of these finite learners constructs the missing arithmetic self-adjoint determinant bridge. Adding heat/noise is also distinct from integrating out physical degrees of freedom in the earlier actual RG chapter; a noise schedule alone is not an RG construction. New source, operator-domain, arithmetic and Lean obligations remain explicit.

# S24 Learning objectives with explicit analytic guarantees

A useful learned contribution must distinguish approximation of known $L$-functions from proof of their zero locations. We propose holomorphic coefficient/readout learning, certified contour losses, arithmetic trace and functional-equation residuals, Schatten-controlled operator refinement, certificate search for the original signed moment, and exact-algebra representation constraints. Each has a different acceptance test.

For an analytic target on $|s-c| \le r$, let $|F| \le M_r$ there and choose $R < \rho < r$. A degree-$N$ holomorphic readout trained on $m > N$ equally spaced points of the radius-$\rho$ circle has the explicit uniform error budget

$$\|p_k - F\|_{|s-c|\le R} \le E_{\mathrm{opt}}(k) + E_{\mathrm{oracle}} + E_{\mathrm{alias}} + E_{\mathrm{Taylor}}.$$

The following proofs give all four terms, a primitive-Dirichlet arithmetic bound for $M_r$, and an infinite-feature kernel limit with its actual rates. Thus known analytic families can be approximated to arbitrary requested accuracy with independently certified evaluation error. This does not prove GRH. Uniform approximation by an already-real-zero determinant class is a different, potentially GRH-equivalent requirement; it must not be assumed through an alleged universal-approximation theorem.

# S25 A certifiable analytic learning lane

## S25.1 Targets, fields and arithmetic scope

The output is a complex holomorphic function, not an unconstrained real MLP on $\mathbb{R}^2$. Fix a center $c$ and a finite family of targets $F_\chi$ analytic on a neighborhood of the closed radius-$r$ disk. Certified values of $M_{r,\chi} \ge \sup|F_\chi|$ and of an evaluation error $\eta$ are inputs to the quantitative theorem. They must be obtained without GRH or the desired arithmetic positivity. A finite conductor family uses the maximum of its independently bounded constants; this is not a uniform theorem over all conductors. Arithmetic character data are exact discrete inputs, not rounded embedding labels.

For primitive nonprincipal Dirichlet characters, the completed function is entire and a direct bound is given below. Pole-cleared zeta/Dedekind targets and known holomorphic Hecke or elliptic targets require their own explicit normalization and growth bounds. General Artin holomorphy is not assumed. Approximating on a compact domain avoiding possible poles does not prove that those poles are absent. Real/complex analytic norms, finite-field algebra and tropical semirings must likewise retain their distinct meanings.

## S25.2 A finite holomorphic readout theorem

Write $F(c+z) = \sum_{n\ge 0} a_n z^n$ and choose $0 < R < \rho < r$, integers $m > N \ge 0$ and sample angles $\theta_j = 2\pi j/m$. Let $|y_j - F(c+\rho e^{i\theta_j})| \le \eta$, including all quadrature and rounding errors. The model and empirical objective are

$$p_b(c+z) = \sum_{n=0}^{N} b_n (z/\rho)^n, \qquad \mathcal{L}(b) = \frac{1}{2m}\sum_{j=0}^{m-1}\left|\sum_{n=0}^{N} b_n e^{in\theta_j} - y_j\right|^2.$$

This is a fixed holomorphic feature layer and a learned complex linear readout. It is a genuine convex learning problem; no nonlinear feature-learning claim is attached to its guarantee.

**Proposition S25.1** (Four-part uniform error bound)**.** *Put $u = R/\rho$, $v = R/r$ and $w = (\rho/r)^m$. Let $b_n^* = m^{-1}\sum_j y_j e^{-in\theta_j}$, and take gradient descent with $0 < \gamma < 2$. After $k$ steps from $b^0$, on $|s - c| \le R$,*

$$|p_{b^k}(s) - F(s)| \le \frac{|1-\gamma|^k \|b^0 - b^*\|_2}{\sqrt{1-u^2}} + \eta\frac{1-u^{N+1}}{1-u} + M_r\frac{w}{1-w}\frac{1-v^{N+1}}{1-v} + M_r\frac{v^{N+1}}{1-v}.$$

*The terms are optimization, certified sample error, Cauchy aliasing and analytic truncation, respectively.*

*Proof.* Discrete Fourier orthogonality gives $A^*A/m = I$ for $A_{jn} = e^{in\theta_j}$, since $m > N$. In real and imaginary coordinates the gradient is $b - b^*$; hence $b^k - b^* = (1-\gamma)^k(b^0 - b^*)$. Cauchy–Schwarz bounds its evaluation by $\|b^k - b^*\|_2(\sum_{n=0}^N u^{2n})^{1/2}$, giving the first term. The displayed iteration is in exact arithmetic. A numerical FFT/GD implementation must also bound coefficient-rounding error; a coefficient error of norm $d_k$ adds at most $d_k/\sqrt{1-u^2}$ to the evaluation budget. This is convergence to the projected least-squares optimum; the empirical residual need not vanish when the labels have components outside the model.

For exact samples, the Fourier coefficient divided by $\rho^n$ is $a_n + \sum_{\ell\ge1} a_{n+\ell m}\rho^{\ell m}$. Uniform convergence on the sampling circle permits interchanging sums. Cauchy's estimate $|a_j| \le M_r r^{-j}$ bounds the alias by $M_r r^{-n} w/(1-w)$. Sample errors contribute at most $\eta\rho^{-n}$. Multiply by $R^n$, sum the two finite geometric series and add $\sum_{n>N} M_r(R/r)^n$. These are exactly the last three terms. □

Choose $N$, then $m > N$, then $\eta$ and the optimization tolerance so that each displayed term is at most $\varepsilon/4$. All four can tend to zero; thus any fixed compact subdisk and finite independently bounded target family have arbitrary-accuracy approximation. This claim needs a certified oracle bound, not an empirical test-set MSE. The present high-precision demonstrator is explicitly numerical and does not relabel its floating evaluations as interval certificates.

### S25.3 The infinite-width limit and its actual rate

Choose $R < \rho < r_0 < r$ and use features $\phi_n(z) = (z/r_0)^n$. Their kernel has the compact-uniform limit

$$K_N(z,w) = \sum_{n=0}^N \phi_n(z)\overline{\phi_n(w)} \longrightarrow K(z,w) = \frac{1}{1 - z\bar{w}/r_0^2}.$$

The geometric tail proves this limit directly; positivity follows because each quadratic form is a sum of squared feature combinations. The readout space is the sequence space $\ell^2$ with bounded evaluation on $|z| < r_0$. On the radius-$\rho$ circle its population coefficient Gram operator is diagonal with eigenvalues $\lambda_n = (\rho/r_0)^{2n}$.

**Proposition S25.2** (Infinite-feature readout convergence)**.** *Suppose the initial coefficient error $e(0)$ is in $\ell^2$, as for zero initialization and the analytic target above. Population gradient flow has $e_n(t) = e^{-\lambda_n t}e_n(0)$ and converges in $\ell^2$, hence uniformly on every $|z| \le R < r_0$. If $\beta > 0$ and $\sum_n \lambda_n^{-2\beta}|e_n(0)|^2 \le B_\beta^2$, then*

$$\|p_t - F\|_{|z|\le R} \le \frac{B_\beta}{\sqrt{1-(R/r_0)^2}}\left(\frac{\beta}{et}\right)^\beta, \qquad t > 0.$$

*For zero initialization this source condition holds whenever $(r_0/r)(r_0/\rho)^{2\beta} < 1$, with $B_\beta \le M_r[1 - ((r_0/r)(r_0/\rho)^{2\beta})^2]^{-1/2}$.*

*Proof.* The diagonal loss gives the scalar ODE in each coordinate. Dominated convergence, with majorant $|e_n(0)|^2$, proves norm convergence. Evaluation is bounded by the geometric-series factor. Under the source condition, $\|e(t)\|_2 \le B_\beta \sup_{\lambda>0} \lambda^\beta e^{-t\lambda}$; the maximum occurs at $\lambda = \beta/t$. Cauchy's coefficient estimate gives $|a_n r_0^n| \le M_r (r_0/r)^n$, and summing the resulting geometric series proves the explicit source bound. □

There is no width-uniform exponential rate: $\lambda_n \to 0$. At finite degree the slowest mode has rate $(\rho/r_0)^{2N}$. Fourier whitening recovers the identity Gram matrix in Proposition S25.1, but its unbounded infinite-dimensional preconditioner must not be called a free uniform rate. Ridge adds a positive rate floor and a calculable bias; its coefficient bias factor is $\alpha/(\lambda_n + \alpha)$, and $\alpha$ must tend to zero for unbiased convergence. A standard Gaussian-process initialization need not lie in the kernel's RKHS; the stated $\ell^2$ hypothesis cannot be silently applied to it. These exact fixed-feature statements are distinct from the nonlinear NTK limits of deep networks [110] and from the trained few-qubit feature-learning circuits.

### S25.4 An unconditional primitive-Dirichlet size oracle

For a primitive nonprincipal character of conductor $q$, parity $a \in \{0, 1\}$ and root number $|\epsilon_\chi| = 1$, put $d = \pi/q$ and $\theta_\chi(t) = \sum_{n\ge1} \chi(n) n^a e^{-dn^2 t}$. The standard primitive theta transformation and Mellin identity give

$$\Lambda(s,\chi) = \int_1^\infty \theta_\chi(t) t^{(s+a)/2-1}\, dt + \epsilon_\chi \int_1^\infty \theta_{\bar\chi}(t) t^{(1-s+a)/2-1}\, dt.$$

The character conjugation and root number are retained. These standard functional/analytic inputs are unconditional in the stated Dirichlet scope [111]; this text does not claim a new Lean proof of Poisson summation. Exponential decay makes both integrals entire in $s$.

For $|s - c| \le r$, choose an integer $k \ge \max(0, (1 + |\operatorname{Re} c| + r + a)/2 - 1)$ and define

$$Q_k(x) = \sum_{j=0}^{k} \frac{k!}{(k-j)! x^{j+1}}.$$

Integration by parts gives $\int_1^\infty t^k e^{-xt}\, dt = e^{-x} Q_k(x)$. Consequently $M_r = 2\sum_{n\ge1} n^a e^{-dn^2} Q_k(dn^2)$ is a valid finite upper bound. For a computable finite version, choose $n_0 \ge 1$ and sum through $n_0 - 1$; the remaining one-integral sum is at most

$$C_a Q_k(dn_0^2) \frac{e^{-dn_0^2/2}}{1 - e^{-dn_0}}, \qquad C_0 = 1, \quad C_1 = (de)^{-1/2}.$$

Indeed $n^a e^{-dn^2/2} \le C_a$, $Q_k$ decreases, and $(n_0 + j)^2 \ge n_0^2 + 2n_0 j$. Twice this tail covers both Mellin terms. The same bound controls theta truncation, with no assumption about zeros.

After truncating the $n$ sum and the $t$ integral, elementary interval exponential, logarithm, sine and cosine evaluation on a compact interval, with a bounded-derivative quadrature remainder, gives an arbitrary requested certified $\eta$. The explicit exponential tails control both truncations. This proves computability of the oracle; it does not assert that an ordinary high-precision special-function call is already a validated implementation. For a finite conductor cutoff, take the maximum of the computed bounds.

## S25.5 Additional objectives and what they can certify

### S25.5.1 Contour-aware approximation and active learning.

Augment the coefficient objective with independent interval enclosures of $|p - F|$ on contours, derivative-weighted losses and adaptive samples near small $|p|$. If $\sup_C |p-F| < \inf_C |p|$ on a Jordan contour, Rouché's theorem transfers the exact interior zero count of $p$ to $F$. An interval root count and a strict margin are acceptance conditions. Sampling finitely many heights alone proves no exhaustion of the off-critical domain.

### S25.5.2 Arithmetic and functional-equation residuals.

Use exact character masks, conductor dependence, primitive Euler factors, gamma normalization and the conjugate functional equation as structural features or independently verified residuals. On $\operatorname{Re} s \geq 1 + \delta$, with $\delta > 0$, for $N \geq 3$ the logarithmic-derivative Dirichlet tail has the elementary bound $N^{-\delta}(\log N/\delta + 1/\delta^2)$, from $\Lambda(n) \leq \log n$ and integral comparison. This is a useful certified training region; it does not extend absolute convergence into the critical strip. A functional equation alone also does not locate zeros.

### S25.5.3 Operator and determinant refinement.

Learn compatible finite compressions, graph-norm/domain residuals, self-adjoint boundary conditions and uniform Schatten-tail bounds from the actual infinite dessin/order construction. Trace-norm convergence transfers ordinary Fredholm determinants, with the standard bound

$$|\det(I + A) - \det(I + B)| \leq \|A - B\|_1 e^{1+\|A\|_1+\|B\|_1}$$

[112]. Strong resolvent convergence alone does not supply this bound. A reciprocal critical-spectrum model typically needs a genus-one regularized determinant and Hilbert–Schmidt control; it must not silently be treated as trace class. The arithmetic determinant identification remains an independent obligation even when the learned operators are Hermitian.

### S25.5.4 Certificate-guided moment and algebra learning.

For the actual original-family low-gcd moment, use adversarial legal masks, conductors, short rows and retained II–I covariance to search for a sharp envelope or counterexample. Rational/interval or exact symbolic checking must certify a proposed inequality uniformly. A sum-of-squares or dual certificate can prove its stated parameter box; extrapolating it to all conductors needs a separate analytic argument. Preserve exact locally gentle relations, surface-order completions, determinant semi-invariant weights and module category labels in the representation architecture. Small relation loss is not an exact functor or an arithmetic comparison.

### S25.5.5 Quantum kernels, width and proof search.

Quantum feature Gram positivity is not Weil/arithmetic positivity. A proved finite Gram lower bound gives a readout rate, while growing width requires the actual limiting kernel and initialization hypotheses. Proof-guided co-training can learn proposed identities or compression maps, then export rational/algebraic candidates for independent Lean/interval validation. Validation, rather than a confidence score, decides which claims survive.

### S25.6 A decisive obstruction to a false universal guarantee

**Proposition S25.3.** *No sequence of real-rooted polynomials converges locally uniformly on $\mathbb{C}$ to $1+z^2$. Every such polynomial has error at least $7/16$ somewhere on $|z-i|=1/4$.*

*Proof.* On that circle, $|1+z^2| = |z-i||z+i| \geq (1/4)(2-1/4) = 7/16$. A smaller uniform error would, by Rouché, force a zero of the polynomial inside the disk, whereas all of its zeros are real and the disk misses the real axis. The same argument excludes local-uniform convergence. □

Thus a claim that an everywhere-real-zero/self-adjoint determinant family universally approximates the completed arithmetic target may already contain the missing GRH bridge. Hurwitz transfers zero-freeness only after independent local-uniform convergence and a nonzero limit have been proved. The analytic learner above deliberately proves approximation without asserting that spectral restriction. Its learned functions are evidence-bearing candidates, not a completed GRH proof. All new analytic/probabilistic statements here are prose proofs and scoped computations; no complete new Lean layer is claimed.

### S25.7 Exact rational constants without transcendental rounding

The next implementation step certifies growth and analytic-tail constants, while leaving sample evaluation and coefficient rounding separate. Put $\delta = 3/q < \pi/q$ and, for an integer $L \geq 1$, define

$$U_L(x) = (1+x/L)^{-L}, \qquad x > 0.$$

The binomial coefficients give $(1+x/L)^L \leq \sum_{j\geq 0} x^j/j! = e^x$, so $e^{-x} \leq U_L(x) < 1$. Every evaluation of $U_L$ at a positive rational argument is exact rational. With the same $Q_k$ as above, choose $n_0 \geq 1$ and let

$$\widehat{M} = 2\left[\sum_{n=1}^{n_0-1} n^a U_L(\delta n^2) Q_k(\delta n^2) + C_a Q_k(\delta n_0^2)\frac{U_L(\delta n_0^2/2)}{1-U_L(\delta n_0)}\right], \quad C_0 = 1, \quad C_1 = \frac{1+\delta^{-1}}{2}.$$

This rational number bounds $|\Lambda(s,\chi)|$ on the specified disk. Indeed decreasing $d$ to $\delta$ increases the positive integral majorant. For $a=1$, the maximum of $xe^{-\delta x^2/2}$ is $(\delta e)^{-1/2} \leq \delta^{-1/2} \leq (1+\delta^{-1})/2$. Split the Gaussian exponent into two halves, use $(n_0+j)^2 \geq n_0^2 + 2n_0 j$, and sum the resulting geometric tail. Replacing its numerator and ratio by $U_L$ increases the bound, since its denominator stays strictly positive. This proves the formula.

For the actual conductors $3,4,5$, center $1/2$, radius $12/5$, $n_0 = 8$ and $L = 16$, exact fraction arithmetic gives respectively $\widehat{M} < 4$, $\widehat{M} < 10$, $\widehat{M} < 6$. Using $\widehat{M}$ (or these integer upper bounds) and the rational radii $R = 7/10$, $\rho = 6/5$ makes the Cauchy-alias and Taylor terms exact rational certificates as well. Full numerator/denominator records and the executable `certify_learning_constants.py` are supplied. This certifies constants, not the complex sample oracle, floating optimizer or GRH.

## S26 Learning nonlocal strategies and checking their certificates

A two-player nonlocal game samples questions $x, y$ and scores outputs $a, b$. The players may share a state, but communicate neither questions nor outcomes during play. Our finite tensor strategies have $p(a,b|x,y) = \mathrm{Tr}[\rho(P_a^x \otimes Q_b^y)]$. Offline joint optimization is allowed and is distinct from communication during the game. We train local measurement frames and a Schmidt angle using the exact Born objective, and replay actual checkpoints, probabilities, correlations, word moments and simulated shots in the interactive laboratory.

For $0 \le \alpha \le 2$ use the tilted score

$$S_\alpha = \alpha\langle A_0\rangle + \langle A_0B_0 + A_0B_1 + A_1B_0 - A_1B_1\rangle.$$

Its classical maximum is $2+\alpha$; its quantum maximum is $\sqrt{8+2\alpha^2}$ [113]. At $\alpha = 0$ the uniform CHSH winning probability is $1/2 + S_0/8$. This conversion does not apply unchanged to the tilted objective. The three seeds for each of $\alpha = 0, 1, 3/2$ give nine trained runs. One $\alpha = 3/2$ run remains at the classical value 3.5; the other runs reach the stated quantum maxima numerically. This retained failure rules out a claim of universal convergence of the optimizer.

The magic-square task uses two Bell pairs, local four-dimensional joint measurements, row parity $+1$ and column parities $(+1, +1, -1)$. Its classical value is $8/9$ and its quantum value is 1, as proved below. Three further trained local-frame runs reach values within $3 \cdot 10^{-13}$ of one. All twelve runs are small CPU simulations with saved seeds and parameters, not hardware Bell experiments or evidence for quantum training advantage.

These examples belong to both the closure of finite tensor correlations $C_{qa}$ and the commuting-operator set $C_{qc}$. The external theorem MIP* = RE separates these sets and implies the negative answer to Connes embedding [115]. We do not reproduce that theorem. Finite learned strategies, positive state Gram matrices and finite NPA constraints do not witness its separation.

# S27 Exact game calculations and the embedding boundary

## S27.1 CHSH bounds and a constructive tilted strategy

For deterministic signs, either $B_0 = B_1$ or $B_0 = -B_1$, so the CHSH term is at most 2; choosing $A_0 = 1$ gives the tilted classical maximum $2 + \alpha$. Convex mixtures cannot improve a linear objective. For self-adjoint involutions, with operators of different parties commuting, write $\mathcal{B} = A_0(B_0 + B_1) + A_1(B_0 - B_1)$. The exact identity

$$2\sqrt{2}I - \mathcal{B} = \frac{1}{\sqrt{2}}\left[\left(A_0 - \frac{B_0+B_1}{\sqrt{2}}\right)^2 + \left(A_1 - \frac{B_0-B_1}{\sqrt{2}}\right)^2\right]$$

proves the CHSH quantum bound by positivity. Expanding the squares uses only the involution and cross-party commutation relations.

For the tilted case take $|\psi_\theta\rangle = \cos\theta|00\rangle + \sin\theta|11\rangle$, $A_0 = Z$, $A_1 = X$ and $B_{0,1} = \cos\mu Z \pm \sin\mu X$. Then $S_\alpha = \alpha\cos 2\theta + 2\cos\mu + 2\sin\mu\sin 2\theta$. Choosing

$$\sin 2\theta = \sqrt{\frac{4-\alpha^2}{4+\alpha^2}}, \qquad \tan\mu = \sin 2\theta$$

with $0 \le \theta \le \pi/4$ and $0 \le \mu \le \pi/4$, gives $\sqrt{8+2\alpha^2}$ by direct substitution (including the endpoints). This verifies attainment; the universal tilted upper bound uses the sum-of-squares theorem in [113] for $\alpha < 2$; the endpoint follows by continuity since $\|\mathcal{B}_2 - \mathcal{B}_\alpha\| \le 2 - \alpha$. This is not an unproved inference from our restricted two-qubit parametrization.

## S27.2 A full magic-square strategy

On two local qubits arrange the observables

$$\begin{pmatrix} Z\otimes I & I\otimes Z & Z\otimes Z \\ I\otimes X & X\otimes I & X\otimes X \\ Z\otimes X & X\otimes Z & Y\otimes Y \end{pmatrix} = (O_{ij}).$$

Entries commute within each row and column. Pauli multiplication gives row products $+I$ and column products $+I, +I, -I$. Alice measures a row, Bob the transpose of a column, and they share $|\Phi_4\rangle = \frac{1}{2}\sum_{k=0}^{3}|k,k\rangle$. The identity $\langle\Phi_4|A\otimes B|\Phi_4\rangle = \mathrm{Tr}(A^TB)/4$ gives correlation one at every intersection. The product identities force the required parities. Thus they win every question pair and the quantum value is one.

A classical perfect table would have total product $+1$ by multiplying the rows and $-1$ by multiplying the columns, a contradiction. At least one of the nine question pairs fails for any deterministic assignment. Conversely, Alice can output all $+1$; Bob outputs all $+1$ in the first two columns and $(-1,+1,+1)$ in the last, failing only one cell. The uniform classical value is exactly $8/9$; shared randomness cannot improve it. Our learner conjugates Alice's observables by one local unitary $U_A$ and Bob's transposed observables by $U_B$, preserving each party's parity and commutation constraints at every optimization step. The exact score is $1/2 + \sum_{ij}\mathrm{Tr}(A_{ij}^TB_{ij})/72$.

### S27.3 Which learning objectives carry proofs?

**Finite strategies and noisy experimental laws.** Unitary parametrization preserves valid projective measurements, so negative Born loss searches inside the chosen finite strategy class. Mixing the state with $I/d$ with visibility $v \in [0,1]$ preserves positivity, normalization and no-signalling. For $K$ joint outcomes, $N$ ideal independent samples and $\delta \in (0,1)$, Hoeffding plus a union bound gives maximum frequency error at most $\sqrt{\log(2K/\delta)/(2N)}$ with probability at least $1-\delta$. Browser sampling is an illustration of this conditional IID statement. With dimension fixed, compactness and explicit matrix-product Lipschitz bounds permit exhaustive finite nets for arbitrary-accuracy global value approximation. Local gradient descent alone has no such guarantee.

**Moment and dual-certificate learning.** For words $u, v$ in measurement generators, define $\Gamma_{u,v} = \langle\psi|u^*v|\psi\rangle$ (or its mixed-state trace). For every $c$, $c^*\Gamma c = \|\sum_v c_v v\psi\|^2 \geq 0$; the mixed case follows by convexity. We compute these actual finite word Gram matrices, not an optimized outer relaxation. NPA learning may propose PSD moment matrices obeying word relations, while an independently checked identity $\beta I - \mathcal{B} = \sum_j P_j^*P_j$ proves an upper bound. A small residual is not an exact certificate. Successive NPA levels converge to the commuting-operator value under the hierarchy's hypotheses [114]. Each level defines an outer relaxation; a validated optimized value or a dual-feasible certificate yields an upper bound. One feasible primal moment matrix alone does not bound the optimum and is not a realization theorem.

**Tracial moments and Connes embedding.** General vector-state moments need not be cyclic. A microstate learner for a tracial algebra must additionally preserve cyclic identities, positivity, norm bounds and compatibility across increasing word lengths, and compare with normalized matrix traces. Our displayed state moments do not fulfill that separate task merely because they are PSD. The external undecidability result [115] also precludes a single computable dimension/error schedule deciding all promised nonlocal games. It neither forbids certified solutions of specified finite games nor establishes any impossibility result for GRH. The valid connection to our program is certificate search with exact relation checks; neither Bell violation nor kernel positivity gives the missing arithmetic determinant/zero-location bridge. These new game proofs and computations are not claimed as complete Lean4 formalization.

## S28 Hubs, spokes and the topology retained by a dessin

The hubs-and-spokes construction in the HoTT book, Sections 6.6–6.8 [109], gives a useful language for attaching cells. For a loop $f : S^1 \to X$, form the homotopy pushout $X \cup_f CS^1$: add a fresh hub and paths from every $f(u)$ to that hub, continuously indexed by $u : S^1$. This attaches a disk, rather than merely a finite graph of radial edges. The book's Remark 6.7.1 explains why identifying the hub with an existing boundary point requires extra coherence; Remark 6.7.2 distinguishes propositional from judgmental computation.

For our dessins the useful distinction is geometric. A vertex star has a disk neighborhood with a specified cyclic ordering of its incident bands. Removing the disk center gives a punctured disk, which retains its peripheral loop. Removing only the vertex of a finite star instead leaves disconnected open rays. These operations cannot be interchanged. In a cellular dessin, filling a face imposes its attaching-word relation; leaving its center punctured retains a boundary component of the ribbon neighborhood.



disk
coherent circle of spokes

punctured disk
peripheral loop

The shaded regions are two-dimensional; the displayed finite spokes sample them. The annulus shown on the right is a deformation retract of the punctured disk, not a missing vertex in a one-dimensional graph.

**Proposition S28.1** (Face filling changes the covering problem)**.** *Let a connected cellular dessin be a spine of an oriented genus-$g$ ribbon surface with $n > 0$ boundary components. Its graph fundamental group is free of rank $2g + n - 1$. Capping all boundary components gives the presentation*

$$\left\langle a_1, b_1, \ldots, a_g, b_g \;\middle|\; \prod_{i=1}^{g} [a_i, b_i] = 1 \right\rangle.$$

*For the one-face clean dessins $K_{2,3}$ and $K_{2,5}$ used here, these free ranks are respectively 2 and 4, before capping.*

*Proof.* A spanning-tree collapse of the connected spine leaves $E - V + 1$ circles. The ribbon surface retracts onto that spine and has Euler characteristic $2 - 2g - n = V - E$, proving the rank formula. Cutting a surface along a standard system of handle arcs gives the boundary word $\prod_i [a_i, b_i] \prod_j c_j$. Van Kampen therefore presents its group by these generators and this one relation. With a boundary present, eliminate $c_n$, leaving the stated free group. Capping attaches disks along the $c_j$, so van Kampen sets each $c_j = 1$, giving the closed-surface presentation. For our rotations on $K_{2,m}$, both black vertices have the same cyclic order on the $m$ incident edges. Suppressing the bivalent white vertices gives dart rotations $(0 \cdots m-1)(m \cdots 2m-1)$ and involution $i \leftrightarrow i + m$. The face permutation is one $2m$-cycle for odd $m$. Thus $V = m + 2$, $E = 2m$, $F = 1$, and $g = (m-1)/2$, giving both examples. □

The once-punctured torus and closed torus both have $H_1 \cong \mathbb{Z}^2$, but their groups are $F_2$ and $\mathbb{Z}^2$; only the closed torus has $H_2 \cong \mathbb{Z}$. The analogous genus-two groups are $F_4$ and the four-generator

surface group, with $H_1 \cong \mathbb{Z}^4$ in both cases. Indeed the standard closed-surface cellular complex has one vertex, $2g$ edges and one face. The face word is a product of commutators, so its abelianized boundary is zero. Removing the face leaves the wedge of circles. This computes the stated homology and also explains why $H_1$ alone misses the face relation. The torus group is abelian while $F_2$ is not; in higher genus the closed surface has nonzero $H_2$ while the graph does not.

Consequently the graph's universal cover is a tree, whereas the universal cover of the filled genus-two surface includes lifted faces and is a disk. Collapsing a spanning tree changes the combinatorial cover graph; the Cayley tree of a free basis is not literally the original biregular dessin cover. This distinction matters when lifting strings, periodic paths and boundary conditions in the infinite locally gentle algebra.

The gluing language organizes local patches and their compatibility maps. It does not reconstruct the ribbon cyclic order, complex structure, Belyi map, quadratic zero relations, coefficient ring or polynomial/formal/operator completion from the homotopy type alone. These data are retained explicitly in the existing dessin and order constructions. Descent of modules, derived categories, Hom-control, AR meshes or Fukaya categories needs a separately defined functor and hypotheses; a homotopy pushout of underlying spaces is not such a theorem. Nor does it imply an arithmetic determinant identity.

The accompanying tool computes the same one-face genus-one/two/four dart families and draws local disk/annulus patches and their surface polygon presentations, peripheral loops and face words. These are exact finite combinatorial checks plus the topological proof above. HoTT is referenced as a conceptual foundation; no univalence or higher inductive-type axiom is added to the pinned Lean4 development. Its complete cell-attachment and categorical formalization remains an explicit task.

## S29 Coherent electron-density clouds in an explicit spatial basis

The density viewer now evolves spatial functions, including interference terms, instead of drawing independent spheres around occupied sites. Let $g_i(\mathbf{r})$ be normalized real Gaussian basis functions, let $S_{ij} = \int g_i g_j$ be their positive-definite overlap, and set $L = S^{-1/2}$. For a Hermitian orthonormal-basis Hamiltonian $h$ and a matrix $C_0$ of $p$ orthonormal occupied columns, put

$$C(t) = e^{-ith}C_0, \qquad \rho(\mathbf{r},t) = \sum_{a=1}^{p} \left| \sum_i g_i(\mathbf{r})(LC(t))_{ia} \right|^2 .$$

This density is nonnegative and has integral $p$: unitarity preserves $C(t)^*C(t) = I$, and integration yields $\mathrm{Tr}(C(t)^*L^*SLC(t)) = p$. The cross terms carry coherence. Replacing them by a sum of atomic densities generally changes the model. The displayed gain/loss field is $\rho(t) - \rho(0)$, whose full-space integral is zero; positive and negative colors indicate redistribution, not particles with negative probability. Time interpolation of saved density frames is a display operation and is not an extra time-integration step.

For fixed-nuclei $H_2^+$ at separation $2a_0$, six normalized $s$ Gaussians (exponents $0.3, 1.2, 4.8$ at each nucleus) give an explicitly integrated one-electron Coulomb Galerkin model. We start in one normalized left-center Gaussian, then evolve with its time-independent Hamiltonian. The electronic ground energy is approximately $-1.08675615$ Hartree; adding nuclear repulsion gives $-0.58675615$ Hartree. This small basis is not a chemical-accuracy or basis-convergence result. Independent Coulomb quadrature, kinetic-integral and matrix-exponential checks pass.

The second example uses the deposited Trp-cage CA geometry, a specified twenty-orbital contact Hamiltonian and ten occupied model orbitals. Its spatial Gaussians have exponent $0.25\,\text{Å}^{-2}$. The resulting field is the exact density of this finite effective basis model, not a calibrated protein electron

density, valence count or transfer-time prediction. The first example uses atomic units; the second retains its effective model time and length convention. These examples are separate from the fixed private dataset release and do not change its labels.

The viewer offers orbit/zoom, volume and isosurface rendering, time scrubbing, exposure and density/gain-loss palettes. The $64^3$ grid and 41 saved frames are encoded as 8-bit square-root-density atlases. The maximum decoded integrated-charge errors are 0.00158 electron for $H_2^+$ and 0.00141 model particle for Trp-cage; analytical full-space charge conservation is exact. Rendering thresholds, finite boxes and quantization are explicit display limitations. The separate hydrogen viewer retains exact analytic hydrogen orbitals and their coherent time-dependent superpositions.

## S30 A finite, validated approximation result

Let $\chi_3 = (0, 1, -1)$, $\chi_4 = (0, 1, 0, -1)$ and $\chi_5 = (0, 1, -1, -1, 1)$, extended periodically, with parities $a = 1, 1, 0$. We approximate the completed functions

$$\Lambda_q(s) = (q/\pi)^{(s+a)/2}\Gamma((s + a)/2)L(s, \chi_q).$$

The primitive theta/Mellin identity is the explicit classical analytic input; no hypothesis about zeros is used. These three examples do not assert a uniform result for all conductors, general Hecke functions or Artin holomorphy.

**Theorem S30.1** (Scope of the supplied finite certificate)**.** *For each $q = 3, 4, 5$, the supplied exact-rational calculation constructs a complex polynomial $P_q$ of degree 28 satisfying*

$$\sup_{|s-1/2|\le 7/10} |P_q(s) - \Lambda_q(s)| < 5 \cdot 10^{-15}.$$

*The computation includes enclosures of all 64 true Fourier-circle nodes, completed-function sample error, analytic truncation, Fourier aliasing, the exact closed-form 180-step gradient-descent iterate and coefficient midpoint rounding. The Python implementation and informal analytic proof have independent checks; this is not a kernel-checked Lean theorem.*

*Proof.* Write $F(c+z) = \sum_{n\ge 0} a_n z^n$, with $c = 1/2$, $R = 7/10$, $\rho = 6/5$, $r = 12/5$. The independently recomputed rational theta majorants are strictly below $M_q = 4, 10, 6$, respectively, on $|z| \le r$. Use true roots $\zeta_j = e^{2\pi i j/64}$ and rational labels $y_j$ with proved error $|y_j - F(c + \rho\zeta_j)| \le \eta$. Use the real Euclidean gradient on complex coefficient coordinates for

$$\mathcal{L}(b) = \frac{1}{128}\sum_{j=0}^{63}\left|\sum_{n=0}^{28} b_n\zeta_j^n - y_j\right|^2.$$

Discrete Fourier orthogonality gives the least-squares optimum $b_n^* = 64^{-1}\sum_j y_j\bar{\zeta}_j^n$, $0 \le n \le 28$, and gradient $\nabla\mathcal{L}(b) = b - b^*$. For step $1/4$ and zero initialization, the exact 180-step iterate is $b_n^{180} = (1 - (3/4)^{180})b_n^*$. The code evaluates this closed form; it does not claim an explicit180-update optimizer run.

Set $u = R/\rho = 7/12$, $v = R/r = 7/24$ and $w = (\rho/r)^{64} = 2^{-64}$. Let $\widehat{b}_n$ be an outward rectangle enclosing $b_n^*$ and $\epsilon_n$ the sum of the coordinate halfwidths of the rectangle for $b_n^{180}$. Choose each actual polynomial coefficient as that rectangle's exact rational midpoint $c_n$. The normalized basis is explicit:

$$P_q(s) = \sum_{n=0}^{28} c_n\left(\frac{s - 1/2}{6/5}\right)^n.$$

Thus the exported coefficient with index $n$ multiplies the normalized variable, rather than the raw power $(s-1/2)^n$. Its error on the compact disk is bounded by

$$E_{\mathrm{opt}} = (3/4)^{180} \sum_{n=0}^{28} |\widehat{b}_n|_1 u^n, \qquad E_{\mathrm{arithmetic}} = \sum_{n=0}^{28} \epsilon_n u^n,$$

$$E_{\mathrm{oracle}} = \eta \frac{1-u^{29}}{1-u}, \qquad E_{\mathrm{alias}} = M_q \frac{w}{1-w} \frac{1-v^{29}}{1-v},$$

$$E_{\mathrm{Taylor}} = M_q \frac{v^{29}}{1-v}.$$

Here $|\widehat{b}_n|_1$ is the sum of the maximum absolute real and imaginary coordinates, a safe upper bound for every enclosed complex modulus. Cauchy's estimate $|a_n| \le M_q r^{-n}$ and the Fourier identity $b_n^* = a_n \rho^n + \sum_{\ell \ge 1} a_{n+64\ell} \rho^{n+64\ell}$ for exact labels prove the alias and Taylor bounds. Label perturbations contribute at most $\eta$ per Fourier coefficient. The remaining two terms follow directly from the exact iterate and coefficient rectangles. Each term is rounded upward to a 180-bit dyadic rational; the reported total is the exact sum of those reported terms. The certified totals below are strictly below the asserted threshold. Detailed sample enclosures are proved in the extended edition and encoded with exact integer endpoints. □

| Conductor | Maximum sample radius | Uniform polynomial bound |
|---|---|---|
| 3 | $3.026 \times 10^{-17}$ | $1.786 \times 10^{-15}$ |
| 4 | $7.206 \times 10^{-17}$ | $4.455 \times 10^{-15}$ |
| 5 | $1.404 \times 10^{-16}$ | $2.906 \times 10^{-15}$ |

The displayed values are rounded upward; exact rational records establish the strict inequalities.

## S31 What this does and does not certify

The final JSON records exact rational centers, interval endpoints, source hashes and all five error terms. Float values are display summaries only. A high-precision comparison is an independent diagnostic, not the reason an interval contains its mathematical target. Arithmetic uses outward integer operations, and invalid float/order inputs are rejected, including through typed caches. Earlier pilots and failed validation probes are retained.

This is a fixed finite tolerance/configuration. Increasing accuracy needs new truncation and precision choices; the fixed $N = 11, T = 128$ interface is not an arbitrary-accuracy oracle. The broader approximation existence theorem is separate. The certificate says nothing about the location of zeros: $P_q$ is not required to be real-rooted or a self-adjoint spectral determinant. The independent arithmetic determinant identity, positivity and original signed short-row/covariance estimates remain open. No GRH proof or complete new Lean4 formalization is claimed.

## S32 Validated primitive operations

Real intervals are closed dyadic intervals $[l2^{-224}, h2^{-224}]$ with integer endpoints; complex intervals are Cartesian products. Addition is exact at this scale. For multiplication take the minimum/maximum of all four integer endpoint products, divide by $2^{224}$ and round down/up. For a reciprocal whose interval omits zero, reverse the endpoint order and round $2^{448}/h$ down and $2^{448}/l$ up. Negative denominators and scalar division use the same endpoint rules. Each operation encloses all represented

real values by monotonicity; induction proves inclusion for every expression built from them. Rectangular complex multiplication uses these real rules. The sum of coordinate halfwidths bounds the complex distance from its exact rational center. Binary floating arguments are never silently converted.

For $0 < x < 1$, the alternating arctangent sum and the endpoint obtained by adding the next signed term enclose $\arctan x$. Sixty terms at $1/5$ and $1/239$, together with Machin's identity $\pi = 16\arctan(1/5) - 4\arctan(1/239)$, give the interval for $\pi$. The negative coefficient reverses the second interval. The identity follows from tangent addition and the range of the two angles; no decimal value of $\pi$ is a trusted input.

For rational $x > 0$, range-reduce $x = 2^k v$, $1 \le v < 2$. With $z = (v-1)/(v+1) \le 1/3$,

$$\log v = 2\sum_{j=0}^{J-1} \frac{z^{2j+1}}{2j+1} + \mathcal{R}, \qquad 0 \le \mathcal{R} \le \frac{2z^{2J+1}}{(2J+1)(1-z^2)}.$$

The geometric tail bounds each remaining denominator below by $2J+1$. Use $J = 92$, the analogous expression for $\log 2$, and sign-aware interval multiplication by $k$. This covers negative $k$ as well.

For complex $z$, repeatedly halve until its interval $\ell^1$ magnitude bound is $A \le 1/2$. The degree 60 Taylor polynomial for $e^z$ then has remainder at most

$$\frac{A^{61}}{61!}\frac{1}{1 - A/62}.$$

Successive absolute terms have ratio at most $A/62$, so geometric comparison proves this bound. Add its symmetric interval to each coordinate, then square outward the required number of times. The identity $e^{2z} = (e^z)^2$ and interval inclusion prove the final enclosure. Caches are type-sensitive: otherwise a cached exact argument can accidentally bypass rejection of an equal-valued float. No implementation change is allowed during the recorded run; the producer verifies source bytes again before emitting its certificate.

## S33 Recomputing the compact growth constants

For the larger growth disk, $c = 1/2$, $r = 12/5$, take $k = 2$ when $a = 1$ and $k = 1$ when $a = 0$. Both real Mellin exponents on that disk are at most $k$. Define

$$Q_k(x) = \sum_{j=0}^{k} \frac{k!}{(k-j)!x^{j+1}}, \qquad U_{16}(x) = (1 + x/16)^{-16}, \qquad \delta = 3/q.$$

Integration by parts gives $\int_1^\infty t^k e^{-xt}\,dt = e^{-x}Q_k(x)$. The same Gaussian split and geometric-tail comparison used below, now with cutoff $n_0 = 8$, imply the exact rational majorant

$$\widehat{M}_q = 2\left[\sum_{n=1}^{7} n^a U_{16}(\delta n^2)Q_k(\delta n^2) + C_a Q_k(64\delta)\frac{U_{16}(32\delta)}{1 - U_{16}(8\delta)}\right], \quad C_0 = 1, \quad C_1 = (1+\delta^{-1})/2.$$

Replacing the actual $d = \pi/q$ by the smaller positive $\delta$ increases the integral majorant. Exact fraction comparison gives $\widehat{M}_3 < 4$, $\widehat{M}_4 < 10$, $\widehat{M}_5 < 6$. The producer recomputes these fractions, asserts the strict comparisons and exports them; the constants in the readout proof are thus checked rather than supplied by a floating growth estimate.

## S34 The completed-function integral and its tails

Let $d = \pi/q$ and $\theta_q(t) = \sum_{n\ge1} \chi_q(n) n^a e^{-dn^2 t}$. The standard primitive theta transformation, followed by splitting the Mellin integral at1, gives

$$\Lambda_q(s) = \int_1^\infty \theta_q(t) t^{(s+a)/2-1}\,dt + \int_1^\infty \theta_q(t) t^{(1-s+a)/2-1}\,dt.$$

The second root-number coefficient is1 for these three characters: their finite Gauss sums are $i\sqrt{3}$, $2i$ and $\sqrt{5}$, respectively. For the last, $2\cos(2\pi/5) - 2\cos(4\pi/5) = \sqrt{5}$; the first two are direct conjugate-root differences. Division by $i^a\sqrt{q}$ gives the claimed normalization. Exponential decay makes the two integrals entire. This classical theta identity, not an assumption of Artin holomorphy or desired positivity, is the analytic input of the implementation.

The actual circle nodes are enclosed by computing $e^{2\pi ij/64}$ with the validated primitives. All input boxes fit $D = [-1,2] + i[-4/3, 4/3]$. Thus both exponents $p = (s+a)/2-1$ and $p' = (1-s+a)/2-1$ satisfy $|\Re p| \le 2$, $|\Im p| \le 2/3$. The domain guard precedes all term discards. Retain $1 \le n \le 11$ and integrate $1 \le t \le 128$.

Put $\delta = 3/q < d$, $U(x) = (1+x/128)^{-128}$ and $Q(x) = x^{-1} + 2x^{-2} + 2x^{-3}$. Since $(1+x/128)^{128} \le e^x$, $e^{-x} \le U(x) < 1$ for $x > 0$. Integration by parts gives $\int_1^\infty t^2 e^{-xt}\,dt = e^{-x}Q(x)$. With $C_0 = 1$, $C_1 = (1+\delta^{-1})/2$ the omitted-$n$ tail of both integrals is

$$2C_a Q(\delta 12^2)\frac{U(\delta 12^2/2)}{1-U(12\delta)}.$$

Indeed $n^a e^{-\delta n^2/2} \le C_a$, the function $Q$ decreases, and $(12+j)^2 \ge 12^2 + 24j$. These inequalities yield a geometric bound after splitting the exponent in halves. The retained-$n$ time tail is at most

$$\sum_{n=1}^{11} 2n^a 128^3 U(128\delta n^2) Q(128\delta n^2).$$

It follows by substituting $t = 128u$ and majorizing $t^{\Re p}$ by $t^2$. The two tails are disjoint: omitted $n$ use the full integral, whereas retained $n$ contribute only $t > 128$.

## S35 High-order integration with a certified remainder

Split each dyadic interval $[2^j, 2^{j+1}]$, $j = 0, \dots, 6$, into four equal panels. For its midpoint $m$ and halfwidth $h$, $h/m \le 1/9$. For $f(z) = e^{-\lambda z} z^p$, $\lambda = \pi n^2/q$, the principal logarithm is analytic on $|z-m| \le \rho = m/2$. On that circle,

$$|f(z)| \le M_\rho = 3\max\{(3m/2)^2, (m/2)^{-2}, 1\} U(\delta n^2 m/2).$$

Here $\Re z \ge m/2$, $|\arg z| \le 1$ and $e^{|\Im p|} \le e^{2/3} \le 3$. The maximum handles both signs of $\Re p$. Cauchy's coefficient estimate and cancellation of odd powers on $[-h,h]$ therefore bound the error after degree $2K$ by

$$\frac{2hM_\rho (h/\rho)^{2K+2}}{(2K+3)(1-(h/\rho)^2)}.$$

Indeed integrate each remaining even term, replace its denominator by the first omitted denominator, then sum a geometric series. Here $K = 12$. Multiply by $2n^a$ for both Mellin terms and sum all 28 panels and retained $n$. Keeping this majorant even for later discarded terms only overcounts error.

To evaluate the polynomial stably, set $u = h/m$, $q_{-1} = 0$, $q_0 = 1$ and use

$$q_{j+1} = \frac{u(p - \lambda m - j)q_j - (\lambda m)u^2 q_{j-1}}{j+1}.$$

Multiplying $f' = (-\lambda + p/t)f$ by $t = m + x$ and matching coefficients proves this recurrence, with $q_j = f^{(j)}(m)h^j/(j!f(m))$. The integrated polynomial is $2he^{-\lambda m}m^p \sum_{j=0}^{K} q_{2j}/(2j+1)$; all its arithmetic is outward.

Fixed absolute exponential-rounding errors can be amplified by high-order coefficients in late panels. The code therefore first bounds a complete $(n, \text{panel})$ pair, including both powers, by

$$4hn^a(m+h)^2 U(\delta n^2(m-h)).$$

For real $t \geq 1$ its phase has modulus1 and $t^{\Re p} \leq (m+h)^2$, proving the bound. If it is below $2^{-90}$, skip that pair entirely and add its bound to the final radius. This is controlled truncation, not numerical underflow. The final sample radius is the interval-sum halfwidth plus the omitted-$n$, time, Cauchy and discard bounds. All are recorded separately and checked as nonnegative. No high-precision special-function routine is trusted here.

## S36 Reproduction and remaining interfaces

Run the preserved source on the existing Linux Python environment with no new packages: `python3` on `examples/analytic_oracle/theta_oracle.py`, with argument `-output NEW`. It refuses to overwrite an existing experiment directory. The final-v5 experiment contains 192 samples, three rational degree 28 polynomials, copied source and SHA-256 hashes. Independent validation checks rational endpoint arithmetic, transcendental enclosures, 1160 exact coefficient identities, high- precision completed-function comparisons and the exported component sums.

The fixed-parameter oracle solves a concrete evaluation/certification gap. It does not identify a positive arithmetic operator, force real-rooted polynomials, prove the signed original-family short-row bound or supply complete Lean4 verification. Surface/order and neural architectures may use these certificates as independent targets, but a small learned loss is not a substitute for the remaining arithmetic and categorical theorems.

## S37 Requested-error readouts from preserved certified labels

The accepted final-v5 sample certificate is preserved unchanged. For each $q = 3, 4, 5$, it supplies $m = 64$ rational complex labels $y_j$, bounds $|y_j - \Lambda_q(c + \rho\zeta_j)| \leq \eta_j$, and outward rectangles enclosing the true roots $\zeta_j = e^{2\pi ij/m}$, where $c = 1/2$, $\rho = 6/5$. The classical primitive theta/Mellin identity and accepted growth bounds $M_q = 4, 10, 6$ on $|s - c| \leq r = 12/5$ remain the analytic inputs. This extension accepts an exact rational error request $\varepsilon > 0$ on the fixed compact $|s - c| \leq R = 7/10$. It changes the finite readout, while retaining the certified sample errors and the 64-node alias budget. It is not an arbitrary-accuracy theta-sampling interface.

**Theorem S37.1** (Executed rational readout with explicit refusal)**.** *For each supported conductor and rational request, the finite-data selection below either returns a normalized rational polynomial $P$ and an executed, exactly replayable gradient-descent trace with*

$$\sup_{|s-c| \leq R} |P(s) - \Lambda_q(s)| \leq \varepsilon,$$

*or refuses the request with the exact minimum base budget over all* 64 *allowed degrees. Refusal concerns attainability by this particular fixed-data majorant; it is not an impossibility theorem for approximation.*

*Proof.* Let $B_n$ be an outward rectangle enclosing $b_n^* = m^{-1}\sum_j y_j\bar{\zeta}_j^n$, $0 \le n < 64$, computed from the preserved root rectangles. Put $t_n = \mathrm{mid}(B_n)$ and let $e_n$ be the sum of its coordinate halfwidths. These are exact rationals, and $|t_n - b_n^*| \le e_n$. For a selected $d < 64$ execute, using exact rational operations,

$$b_n^0 = 0, \qquad b_n^{k+1} = \tfrac{3}{4}b_n^k + \tfrac{1}{4}t_n, \qquad 0 \le n \le d.$$

This is genuine gradient descent with step $1/4$ for the explicitly specified real Euclidean objective $J(b) = \frac{1}{2}\sum_{n=0}^{d}|b_n - t_n|^2$, whose gradient is $b - t$. All iterates and exact objective values are retained. The output is literally

$$P_k(s) = \sum_{n=0}^{d} b_n^k \left(\frac{s-c}{\rho}\right)^n.$$

Set $u = R/\rho = 7/12$, $v = R/r = 7/24$, $w = (\rho/r)^{64} = 2^{-64}$, $S_d(x) = \sum_{n=0}^{d} x^n$ and $\eta = \max_j \eta_j$. The five budgets are

$$E_{\mathrm{opt}}(k) = \sum_{n=0}^{d} |b_n^k - t_n|_1 u^n, \qquad E_{\mathrm{oracle}} = \eta S_d(u),$$

$$E_{\mathrm{alias}} = M_q \frac{w}{1-w} S_d(v), \qquad E_{\mathrm{Taylor}} = M_q \frac{v^{d+1}}{1-v},$$

$$E_{\mathrm{arithmetic}} = \sum_{n=0}^{d} e_n u^n.$$

Here $|x + iy|_1 = |x| + |y|$. Cauchy's estimate and true-root Fourier orthogonality give the alias and Taylor terms; certified label errors give the oracle term. Triangle inequality applied to $b_n^k - b_n^* = (b_n^k - t_n) + (t_n - b_n^*)$ gives the other two. Thus their exact sum bounds the compact-uniform error.

Define $C_d = E_{\mathrm{oracle}} + E_{\mathrm{alias}} + E_{\mathrm{Taylor}} + E_{\mathrm{arithmetic}}$. Select the smallest $d < 64$ with $C_d < \varepsilon$; if none exists, report $\min_{d<64} C_d$ and refuse. For a selected degree, $E_{\mathrm{opt}}(k) = (3/4)^k \sum_n |t_n|_1 u^n \longrightarrow 0$. Therefore executing updates until $E_{\mathrm{opt}}(k) \le \varepsilon - C_d$ terminates after finitely many steps. The serialized coefficients, budgets, targets and trace use exact rational endpoints; displayed decimal approximations are not proof inputs. □

## S38 Objective semantics, precision floors and replay

The original true-circle sample-label least-squares objective is

$$\mathcal{L}_y(b) = \frac{1}{2m}\sum_{j=0}^{m-1}\left|\sum_{n=0}^{d} b_n\zeta_j^n - y_j\right|^2.$$

With $d < m$, Fourier orthogonality gives $A^*A = mI$ and the real Euclidean coordinate gradient $\nabla\mathcal{L}_y(b) = b - b^*$. The executed objective $J$ instead has the exact rational midpoint target $t$. The difference $\nabla\mathcal{L}_y - \nabla J = t - b^*$ is enclosed coefficientwise by $e_n$. Consequently this is an actual rational GD run for $J$ and a certified perturbed-gradient implementation for $\mathcal{L}_y$. It is not the exact

optimizer trajectory of the ideal-root objective, nor a replay of neural or quantum training. An interval enclosure of an ideal-root GD trajectory would be a different explicitly labelled object.

Each actual update contracts $b^k - t$ by $3/4$, so the exact stored losses satisfy $J(b^{k+1}) = (9/16)J(b^k)$. A replay recomputes every rational coordinate update, loss and weighted residual. The closed form $b^k = (1 - (3/4)^k)t$ is an independent check on the executed trace, not its producer. Exact rational loop coordinates introduce no further midpoint or floating export error. The arithmetic budget covers the difference between the certified Fourier rectangles and their midpoint targets, including the inherited root enclosure and outward interval operations. A subsequent float conversion requires its own exact conversion-error bound.

The normalized polynomial convention is essential: a stored coefficient with index $n$ multiplies $((s - 1/2)/(6/5))^n$, not $(s - 1/2)^n$. The input certificate is pinned by SHA-256, including its source hashes, all 192 sample enclosures, root rectangles and analytic radius components. Producing source is captured at entry, checked unchanged before emission, and copied into a fresh output directory. An existing output is never overwritten. An independent replay should also check the fixed input hash, all dimensions, degree choice, the first stopping iterate, all five exact terms and their literal sum against the requested rational tolerance.

The attainable boundary of this majorant is

$$C_{\min,q} = \min_{0 \le d < 64} C_{q,d}.$$

Every request $\varepsilon > C_{\min,q}$ has strict budget slack for some degree and the proved finite stop rule. At or below that boundary this interface refuses. The degree minimizing $C_d$ need not be 63: Taylor error decreases with degree while the oracle and arithmetic sums increase. In particular $C_d \ge \eta$ and $C_d \ge M_q 2^{-64}/(1 - 2^{-64})$. Increasing the number of optimization steps does not lower these fixed-data budgets. Approximate boundary values on the stated compact are $7.2929 \cdot 10^{-17}$, $1.7370 \cdot 10^{-16}$ and $3.3741 \cdot 10^{-16}$ for $q = 3, 4, 5$; only the exact exported rational minima are used for acceptance. These are certificate floors, not lower bounds on the actual errors.

A full adaptive theta oracle would separately need proof-preserving choices of omitted-$n$ cutoff, time cutoff, quadrature order, whole-panel discard budget, primitive series lengths, interval precision and serialized-bound precision. The present $N = 11$, $T = 128$, 224-bit arithmetic, 60/92/60 primitive series lengths, $2^{-90}$ discard threshold and 180-bit rounded budgets cannot support every positive request. Increasing a nominal bit count alone leaves the fixed primitive remainders and analytic tails in place. The preserved labels are not recomputed here. The classical identity, finite arithmetic and informal proof are not new Lean declarations. No zero-location, all-conductor, positive arithmetic determinant, quantum-advantage or GRH conclusion follows.

The interactive replay is available at the certified readout laboratory. All decimal graphics are illustrative views of the linked exact records.

## S39 Requested-error theta samples for three fixed conductors

For the primitive characters $\chi_3 = (0, 1, -1)$, $\chi_4 = (0, 1, 0, -1)$, $\chi_5 = (0, 1, -1, -1, 1)$ with parities $a = 1, 1, 0$, retain the classical primitive theta/Mellin identity and root number one already stated above. Put $D = [-1, 2] + i[-4/3, 4/3]$ and $\delta = 3/q$. The following adaptive extension is implemented with independent interval and sample checks; it is an informal analytic theorem, not a Lean result.

**Theorem S39.1** (Finite adaptive sample specification)**.** *For every supported conductor, exact rational $s \in D$, and rational $\eta > 0$, the choices below terminate and give a rational complex midpoint $y$ and exact rational radius $R \le \eta$ with $|y - \Lambda_q(s)| \le R$. The same conclusion holds for a specified true circle node when its root and node enclosures are refined on every precision retry.*

*Proof.* For $x \geq 0$ define the exact rational bound $E(x) = 2^{-\lfloor x \rfloor}$. Since $e > 2$, $e^{-x} \leq E(x)$; this bound is nonincreasing and tends to zero. Let $Q(x) = x^{-1} + 2x^{-2} + 2x^{-3}$ and $C_0 = 1$, $C_1 = (1 + \delta^{-1})/2$. For $N \geq 1$, $n_0 = N + 1$, $J \geq 1$, $T = 2^J$, use

$$B_n = 2C_a Q(\delta n_0^2) \frac{E(\delta n_0^2/2)}{1 - E(\delta n_0)},$$

$$B_t = \sum_{n=1}^{N} 2n^a T^3 E(\delta n^2 T) Q(\delta n^2 T).$$

Here $\delta n_0 \geq 6/5$ ensures a positive denominator; $E(x)$ equals one for $0 \leq x < 1$, so this safeguard is essential. On the four panels of each $[2^j, 2^{j+1}]$, write midpoint $m$, halfwidth $h$, $\rho = m/2$, $v = h/\rho \leq 2/9$, and $B(m) = \max\{(m + \rho)^2, (m - \rho)^{-2}, 1\}$. Use the Cauchy remainder

$$B_Q = \sum_{\text{panels}} \sum_{n=1}^{N} \frac{2n^a \, 2h \, 3B(m) E(\delta n^2 m/2) v^{2K+2}}{(2K + 3)(1 - v^2)}.$$

Choose in turn the first $N \geq 1$, first $J \geq 1$, and first $K \geq 0$ for which their respective budgets are at most $\eta/8$. Gaussian decay, then exponential time decay, then $v \leq 2/9$ prove termination. These are the accepted tail and Cauchy arguments with each true exponential separately replaced by its upper bound $E$.

There are $P = 4JN$ potential pairs. A nonzero-character pair may be omitted only if its entire two-power contribution has bound

$$b = 4hn^a (m + h)^2 E(\delta n^2 (m - h)) \leq \eta/(8P).$$

Add every omitted bound to $B_{\text{discard}}$, giving $B_{\text{discard}} \leq \eta/8$. Evaluate all other integrated Taylor polynomials by the stated exact coefficient recurrence using outward rectangles. Refine both interval precision and all primitive series remainders until the final rectangle's sum of coordinate halfwidths satisfies $R_{\text{arith}} \leq \eta/4$. For this fixed finite expression all input and primitive intervals shrink; continuity and outward arithmetic imply $R_{\text{arith}} \to 0$. Precision doubling therefore terminates, and the sum of the five raw components is at most $3\eta/4$. Choose an integer output scale $B$ such that $5\,2^{-B} \leq \eta/8$ and round each component upward by $\text{up}_B(x) = 2^{-B}\lceil 2^B x \rceil$. The reported radius is their literal exact sum. With $\mathbf{b} = (R_{\text{arith}}, B_n, B_t, B_Q, B_{\text{discard}})$, it obeys

$$|y - \Lambda_q(s)| \leq \sum_{i=1}^{5} \text{up}_B(b_i) \leq \frac{3\eta}{4} + 5\,2^{-B} \leq \frac{7\eta}{8} < \eta.$$

The output retains exact rational midpoints, rounded components, $B$, raw interval radius, parameters, input metadata and source hashes. The raw analytic bounds are exactly reconstructible; they need not be serialized as potentially enormous fractions. The output precision varies with the request and introduces no fixed reporting floor. □

This fixed-conductor sample result uses no hypothesis on zeros. It does not remove a downstream fixed-node Fourier alias floor, supply an all-conductor theorem or a positive arithmetic determinant, or prove GRH.

## S40 Adaptive sampling parameters and precision termination

The primitive theta/Mellin identity with root number one yields

$$\Lambda_q(s) = \sum_{n\ge 1} \chi_q(n) n^a \int_1^\infty e^{-\pi n^2 t/q} \big(t^{(s+a)/2-1} + t^{(1-s+a)/2-1}\big)\, dt.$$

On $D = [-1,2] + i[-4/3, 4/3]$, both exponents $p$ satisfy $|\Re p| \le 2$ and $|\Im p| \le 2/3$. For exact $x \ge 0$, put $E(x) = 2^{-\lfloor x \rfloor}$. The elementary inequality $e > 2$ gives $e^{-x} \le e^{-\lfloor x \rfloor} \le E(x)$ and $E(x) \le 2e^{-x\log 2}$. Use this solely as a pointwise nonincreasing upper bound; it is not multiplicative and has no asserted exponential integral identity.

Integration by parts gives $\int_1^\infty t^2 e^{-xt}\, dt = e^{-x} Q(x)$ for $Q(x) = x^{-1} + 2x^{-2} + 2x^{-3}$. For $a = 0$ the bound $n^a e^{-\delta n^2/2} \le C_0 = 1$ is immediate. For $a = 1$, $n \le (1+n^2)/2$ and $n^2 e^{-\delta n^2/2} \le 2/(e\delta) \le 1/\delta$ give $C_1 = (1 + 1/\delta)/2$. For $n = n_0 + j$, the inequality $n^2 \ge n_0^2 + 2n_0 j$ therefore bounds the omitted-$n$ sum by

$$2C_a Q(\delta n_0^2) \frac{e^{-\delta n_0^2/2}}{1 - e^{-\delta n_0}} \le 2C_a Q(\delta n_0^2) \frac{E(\delta n_0^2/2)}{1 - E(\delta n_0)} = B_n.$$

Take $n_0 = N + 1 \ge 2$; then $\delta n_0 \ge 6/5$ and the latter denominator is positive. Substituting $t = Tu$ in each retained-$n$ time tail gives

$$B_t = \sum_{n=1}^{N} 2n^a T^3 E(\delta n^2 T) Q(\delta n^2 T).$$

The two tails are disjoint because the omitted-$n$ tail uses the whole integral and the time tail uses only retained $n$.

For $j = 0, \dots, J-1$ and $\ell = 0, \dots, 3$ take

$$m = 2^j \left(1 + \frac{2\ell + 1}{8}\right), \qquad h = \frac{2^j}{8}, \qquad T = 2^J.$$

These panels cover $[1, T]$ and have $h/m \le 1/9$. On $|z - m| = \rho = m/2$, principal log is analytic, $\Re z \ge m/2$, and $|\arg z| \le 1$. For $f(z) = e^{-\pi n^2 z/q} z^p$, the angular multiplier is at most $e^{2/3} \le 3$, whence

$$|f(z)| \le 3B(m) E(\delta n^2 m/2), \qquad B(m) = \max\{(3m/2)^2, (m/2)^{-2}, 1\}.$$

Cauchy's coefficient estimate, cancellation of odd terms, and geometric summation yield the $B_Q$ in Theorem S39.1. Including zero-character or later discarded terms in that sum is harmless. Gaussian, time, and geometric decay prove termination of the sequential first-success searches with $B_n, B_t, B_Q \le \eta/8$. For fixed q and $L = \max\{1, \log(1/\eta)\}$ their asymptotic scales can be chosen as $N = O(\sqrt{L})$, $T = O(L)$, $J = O(\log L)$, and $K = O(L)$. Polynomial prefactors in $N, T, J$ contribute only $O(\log L)$ to the logarithm of the error estimates; this is a truncation-size statement, not a rational-arithmetic bit-complexity bound.

For real $t \in [m-h, m+h]$, $t \ge 1$ and $t^{\Re p} \le (m+h)^2$. The bound for a whole pair, including both powers, is therefore $4hn^a(m+h)^2 E(\delta n^2 (m-h))$. There are $P = 4JN$ potential pairs; summing only bounds at most $\eta/(8P)$ gives $B_{\text{discard}} \le \eta/8$. No partial recurrence or isolated exponential is discarded. For active pairs retain the coefficient recurrence

$$r_{-1} = 0, \quad r_0 = 1, \quad r_{j+1} = \frac{(h/m)(p - \lambda m - j) r_j - \lambda m (h/m)^2 r_{j-1}}{j+1}, \qquad \lambda = \pi n^2/q.$$

Coefficient comparison in $tf' = (-\lambda t + p)f$ proves that the integrated Taylor polynomial is

$$2he^{p\log m - \lambda m} \sum_{j=0}^{K} \frac{r_{2j}}{2j+1}.$$

Outward arithmetic encloses its exact value, including the signed character weight and both powers.

Use an isolated p-bit dyadic context, with all primitive remainders refined as p increases. Put $\delta_p = 2^{-p-16}$. Each arctangent series separately stops when its next-term bound is at most $\delta_p$, giving total weighted analytic tail width at most $20\delta_p$ in Machin's formula, before outward dyadic rounding. Each reduced logarithm series separately has tail at most $\delta_p$; for $x = 2^k v$, the combined bound is $(1 + |k|)\delta_p$, before rounding. For the complex exponential, after reduction to $A \le 1/2$, choose the Taylor order so the bound

$$\frac{A^{H+1}}{(H+1)!} \frac{1}{1 - A/(H+2)}.$$

is at most $\delta_p$. Choose H by finite exact search; its denominator is positive. Add a symmetric exponential error enclosure to each coordinate before outward squaring. The fixed guard16 is not itself an acceptance bound: the final complete rectangle radius must satisfy $R_{\text{arith}} \le \eta/4$. All interval caches must include p and preserve exact input-type guards. The successful p and retry precisions/radii are recorded. Individual series lengths are determined by these, the exact inputs and the pinned source and are reconstructible by replay; they are not explicitly stored.

For fixed N,J,K and exact target point, the expression has finitely many operations and only nonzero fixed integer denominators. Each primitive input is bounded for sufficiently large p; exponential halving counts are consequently bounded. Exact primitive tails and dyadic rounding widths tend to zero. Induction through the finite continuous expression, including finitely many outward squarings, gives $R_{\text{arith}} \to 0$. Doubling p after a failed comparison terminates. The argument also covers true roots and nodes regenerated at every p, since their widths tend to zero. A fixed nondegenerate s rectangle has no corresponding shrinking-input claim.

**A growing-order primitive estimate.** To justify convergence despite a growing exponential Taylor order, write $\epsilon_p = 2^{-p}$, $\delta_p = 2^{-p-16}$, and let $d(X)$ be the sum of a rectangle's coordinate widths and $m(X)$ its $\ell^1$ magnitude bound. The rounded reduced input $W$ satisfies $a = m(W) \le 1/2 + 2\epsilon_p < 3/5$ and $d(W) \le 2^{-k} d(Z) + 4\epsilon_p$; the remainder is at most $2^{-H}$, so the first accepted order satisfies $H \le p + 18$. For the computed term boxes, outward multiplication and integer division give $m_j \le a m_{j-1} + 6\epsilon_p$ and $d_j \le a d_{j-1} + m_{j-1} d(W) + 12\epsilon_p$. Thus $m_j \le a^j + 15\epsilon_p$ and $\sum_{j=0}^{H} d_j \le (25/4) d(W) + 75H\epsilon_p$; adding the remainder increases width by at most $4(\delta_p + \epsilon_p)$. The pre-squaring width is therefore $O(d(Z) + p2^{-p})$ and its magnitude is bounded (indeed below 3 for $p \ge 32$). Each outward square has width at most $2m(X)d(X) + 8\epsilon_p$; the bounded number of squarings consequently preserves convergence to zero. Finally, exact rational partial sums and endpoint rounding give Machin width at most $20\delta_p + 40\epsilon_p$, and logarithm width at most $(1 + |k|)(\delta_p + 2\epsilon_p)$ for fixed $x = 2^k v$. These estimates justify the primitive-width step without assuming a fixed Taylor order or asserting a global arithmetic complexity bound.

For serialization, initialize

$$B = \max\{32, \text{bitlen}(\text{den}\,\eta) - \text{bitlen}(\text{num}\,\eta) + 8\}$$

and increase B until $5\,2^{-B} \le \eta/8$. Write $\widehat{x} = 2^{-B}\lceil 2^B x \rceil$ for each of the five raw components. Their exact reported sum is at most $3\eta/4 + 5\,2^{-B} \le 7\eta/8$, and every component remains an upper bound. The output records B, the raw interval radius and complete rectangle, N/J/K, discard threshold/pair

counts, exact target metadata and source hashes. These reconstruct all raw analytic bounds and pair sets without serializing huge exact tail denominators. For the practical requests below B is87 and114, respectively. Exact endpoints remain rational; the mathematical specification assumes an integer representation sufficient for the requested size. Dynamic rounding removes the old fixed reporting floor. For rational s, validate its exact coordinates in closed D before conversion: an outward dyadic enclosure of $\Im s = 4/3$ extends beyond D, but the analytic bounds concern the true point. Circle nodes satisfy the strict exact bounds $-7/10 \le \Re s \le 17/10$, $|\Im s| \le 6/5$.

The exact parameter-design checker gives

| $\eta$ | $q$ | $N$ | $J$ | $T$ | $K$ | active pairs |
|---|---|---|---|---|---|---|
| $10^{-24}$ | 3 | 12 | 7 | 128 | 19 | 71 |
| $10^{-24}$ | 4 | 14 | 8 | 256 | 20 | 65 |
| $10^{-24}$ | 5 | 16 | 8 | 256 | 20 | 111 |
| $10^{-32}$ | 3 | 14 | 7 | 128 | 25 | 84 |
| $10^{-32}$ | 4 | 16 | 8 | 256 | 26 | 72 |
| $10^{-32}$ | 5 | 18 | 8 | 256 | 26 | 129 |

with exact rational budgets and no special-function oracle. The arithmetic precision is selected from the actual interval radius, not inferred from this table. This extension retains the existing classical theta identity; it adds no zero-location, positive determinant, all-conductor, Lean or GRH claim. The following end-to-end adaptive readout supplies its own node-count, alias, degree, Fourier and optimization choices.

## S41 An end-to-end adaptive analytic readout

The new driver changes the sample truncations, interval precision, circle size and degree; it does not reuse the frozen labels to assert a smaller error. Its scope is still the three stated primitive quadratic characters and the fixed compact $|s - 1/2| \le 7/10$.

**Theorem S41.1** (Requested-error construction in the stated finite family)**.** *Given a positive rational $\varepsilon$ and $q \in \{3, 4, 5\}$, the specified exact-arithmetic algorithm terminates in the mathematical unbounded-resource model and returns a normalized rational polynomial $P$ with*

$$\sup_{|s-1/2|\le 7/10} |P(s) - \Lambda_q(s)| \le \varepsilon.$$

*Its sample enclosures, rational optimizer updates and all error budgets are explicit. This is an approximation theorem, with the stated classical theta identity as an input; it supplies no zero-location or positive determinant conclusion and is not a Lean theorem or a practical runtime bound.*

*Proof.* Put $u = 7/12$, $v = 7/24$. Choose $d \ge 0$ so $10v^{d+1}/(1 - v) \le \varepsilon/8$, and choose $m > d$ so

$$10\frac{2^{-m}}{1 - 2^{-m}}\frac{1}{1 - v} \le \varepsilon/8.$$

The driver rounds $m$ upward to a multiple of eight, which only decreases this bound. Set the requested sample error $\eta = \varepsilon(1 - u)/8$. The adaptive sampler gives all $m$ true circle nodes and labels with error at most $\eta$. Form their outward Fourier rectangles through degree $d$, and refine the nodes and finite evaluation arithmetic if their weighted midpoint radii exceed $\varepsilon/8$. This refinement terminates: for these fixed finite $d, m$, node uncertainty and every finite-expression arithmetic width

tend to zero with precision. Merely lifting fixed boxes would not suffice, so the driver regenerates the nodes and samples on this retry.

Choose rational midpoint targets $t_n$ and execute the previously specified quadratic-objective updates until $\sum_{n=0}^{d} |b_n^k - t_n|_1 u^n \le \varepsilon/4$. Their geometric contraction proves finite termination. The oracle, aliasing, Taylor and Fourier-midpoint terms are each at most $\varepsilon/8$, and the optimization term is at most $\varepsilon/4$. Hence the raw five-term sum is at most $3\varepsilon/4$. Choose a serialization precision $B$ with $5\,2^{-B} \le \varepsilon/8$, round each term upward, then add the five reported rationals exactly. The printed bound is at most $7\varepsilon/8 \le \varepsilon$. The polynomial uses $((s-1/2)/(6/5))^n$, and the preceding Fourier/Cauchy proof gives its asserted compact-uniform bound. □

The trace is losslessly encoded by the fixed target vector and a rational scale extracted from each *executed* coefficient state, together with its objective, weighted residual and full-state hash. Every coefficient can be recovered exactly; a separate replay expands the states and checks every update. The producer does not generate its trace from a closed form. The saved objective remains the rational Fourier-midpoint quadratic, with its perturbation from ideal-root least squares explicitly budgeted.

## S42 Termination, compression and the boundary of the guarantee

The degree and circle-size choices use only rational comparisons. Since $0 < v < 1$, the Taylor test succeeds at finite $d$; since $2^{-m} \to 0$, the alias test succeeds at finite $m > d$. The common growth bound10 is safe for all three families; the certificate also recomputes the previously proved exact conductor-specific majorants below $4, 10, 6$.

At fixed $d, m, \eta$, the analytic truncation parameters are fixed while precision is doubled. The computed sample centers stay bounded: their underlying finite expression is fixed and the true completed-function values are bounded on the sample circle. Its arithmetic widths tend to zero, as do the regenerated true-root enclosures. Fourier computation is a finite composition of additions, products and division by the positive integer $m$. Its midpoint-radius sum therefore tends to zero. This proves the driver's additional precision retry, without pretending that changing a box's nominal precision shrinks its uncertainty.

The accepted sampler error may remain above its actual arithmetic width because its analytic tails and output bounds are fixed for this $\eta$. That causes no problem: the oracle allocation uses the total certified sample radius, whereas the Fourier-midpoint allocation uses only finite coefficient-rectangle width. No analytic sample error is silently moved into the latter term or removed by numerical convergence.

For the stored training trace, the exact loop gives $b^{k+1} = 3b^k/4 + t/4$ from $b^0 = 0$. Inductively every state is a rational scalar multiple of the same target vector. The producer first executes the actual vector updates, then extracts this scalar from a nonzero target coordinate, checks all coordinates, and records a canonical full-vector SHA-256 hash. If the entire target vector is zero, its zero state needs only scale0. The compressed record is thus an exact representation of the executed states, not a floating-point sketch. Hashes are integrity aids; exact vector reconstruction and the independently checked recurrence establish the mathematical claim.

Coefficient export uses exact rational numerators and denominators. A chunked decimal integer encoder/parser handles endpoints beyond Python's usual digit-conversion guard without changing global interpreter settings. No fixed 180-bit serialization floor remains. The decimal graphics are separate approximate projections, not polynomial coefficients to which the exact certificate may be applied without a conversion-error bound.

All producing source bytes are captured, copied into fresh output paths and checked unchanged. Historical fixed producers are pinned and untouched. For the new modules the run also checks the import-time source snapshots, so a stale imported module cannot be attributed to later source bytes. After output creation, a failed evaluation, final provenance check or serialization emits uncertified failure metadata. Invalid requests and stale import-time sources are refused before creating output artifacts. Old outputs are never overwritten. Extreme requests can be computationally impractical: the termination proof gives no machine-time or memory guarantee, and finite test runs establish only their reported cases.

Nothing requires the output polynomials to be real-rooted, to equal an arithmetic Fredholm determinant, or to have a self-adjoint realization. Consequently arbitrary compact approximation of these known functions does not imply GRH. Uniformity in changing conductor, general Hecke/Dedekind/elliptic/Artin scopes, the independent determinant comparison, and actual analytic/category Lean objects remain separate obligations.

## S43 An arithmetic-family moment with every original row retained

Let $K = \mathbb{Q}(\sqrt{-3})$, $\mathcal{O} = \mathbb{Z}[\omega]$, and write $N$ for the ideal or element norm. Fix a finite-order Hecke character $\nu$ and a finite set $S$ of prime ideals containing those above 6 and those dividing its conductor. For $0 \neq u \in \mathcal{O}$, let $\kappa_u$ be the character of $K(u^{1/6})/K$ defined using arithmetic Frobenius, and put $\psi_u = \mathrm{prim}(\nu\kappa_u)$. Here prim means the underlying primitive Hecke character: it is formed before introducing any divisor variable or evaluating conductor-dependent zero extensions. The rows below are all actual Eisenstein elements, including units and prime powers supported in $S$.

The external inputs are abelian Kummer reciprocity and character uniqueness, the primitive Hecke functional equation and its usual vertical growth, Stirling's formula, scalar Dedekind-zeta convexity, and Blomer–Goldmakher–Louvel's large sieve for their specified sixth-order family. We use the pinned arXiv:1112.1650v1, Theorem 1.3, Lemma 2.1 and Appendix II; its Section 5 supplies the all-ideal decomposition whose fixed-twist proof is redone here. No statement about all primitive characters of comparable conductor, Artin holomorphy, Lindelöf bounds or zero locations is assumed.

**Theorem S43.1** (The original-family numerator moment)**.** *For every $\varepsilon > 0$, uniformly for $H \geq 1$ and $t \in \mathbb{R}$,*

$$\sum_{0<Nu\leq H} |L(1/2+it,\psi_u)|^2 \ll_{\nu,S,\varepsilon} H^{1+\varepsilon}(1+|t|)^{1+\varepsilon}. \tag{8}$$

*For every fixed $0 < \delta < 1/2$, the same estimate holds with $1/2+it$ replaced by $1/2+\delta+it$, with the constant also depending on $\delta$. If the central bound has constant $C_0$ at an exponent $0 < \varepsilon_0 \leq 1/4$, an admissible off-central constant is*

$$200C_0\left(\frac{3/2+\delta}{1/2-\delta}\right)^2. \tag{9}$$

*Proof.* For the canonical BGL character $\vartheta_{\mathfrak{a}}$, Appendix II gives a radicand $\alpha_{\mathfrak{a}}$ with $(\alpha_{\mathfrak{a}})_S = \mathfrak{a}\mathfrak{g}^{-6}$. Since $\mathcal{O}_S$ is principal, comparison with any literal $u$ whose outside-$S$ ideal is $\mathfrak{a}$ gives $\kappa_u = \eta\vartheta_{\mathfrak{a}}$ for one of finitely many fixed $S$-unit Kummer twists $\eta$. This is an equality of primitive characters, established outside a finite set and extended by uniqueness. For each outside-$S$ ideal $\mathfrak{a}$ and $S$-supported integral ideal $\mathfrak{b}$, at most six elements generate $\mathfrak{a}\mathfrak{b}$. The resulting $S$-supported multiplicity sum converges at exponent $1+\varepsilon$.

For each fixed $\xi = \nu\eta$, split the canonical parameter as $\mathfrak{a} = \mathfrak{a}_0\mathfrak{a}_1\mathfrak{a}_2^6$, where $\mathfrak{a}_0$ is squarefree, $\mathfrak{a}_1$ is squarefull and sixth-power-free, and $(\mathfrak{a}_0, \mathfrak{a}_1) = 1$. The primitive conductor is at most a fixed multiple of $N(\mathfrak{a}_0\mathfrak{a}_1)$. The common-gamma approximate functional equation, with $X = H/(N\mathfrak{a}_2)^6$

and $T = 1 + |t|$, has common maximal length $B = X^{1/2}T(HT)^{\eta}$ for any sufficiently small fixed $\eta > 0$. Mellin inversion removes its row-dependent conductor weight. Split a column as $\mathfrak{b}_S\mathfrak{b}_1\mathfrak{b}_2$, with $\mathfrak{b}_S$ supported in $S$, $\mathfrak{b}_1$ squarefree and $\mathfrak{b}_2$ squarefull, the last two coprime and prime to $S$. The original primitive values at $\mathfrak{b}_S$ remain bounded row multipliers. The $S$-supported and squarefull column weights are summable at $1/2 + \eta$. Multiplying squarefree column coefficients by the fixed $\xi$ preserves their squared norm, whatever its finite order. BGL's large sieve therefore bounds the energy for fixed $\mathfrak{a}_1, \mathfrak{a}_2$ by

$$(HT)^{O(\eta)}\{M + B + (MB)^{2/3}\}, \qquad M = X/N\mathfrak{a}_1.$$

Squarefull counting and summation give $X + B\sqrt{X} + (XB)^{2/3} \ll (HT)^{O(\eta)}XT$. The $\mathfrak{a}_2$ sum converges at exponent 6. Principal rows are separated before the pole-free functional equation and bounded by scalar zeta convexity. This proves (8).

For the off-central statement divide each $L(s, \psi)$ by $(s + 1)^{5/4}$ and, on principal rows, also multiply by $(s - 1)/(s + 1)$. These are bounded analytic functions on $\operatorname{Re} s > 1/2$, continuous on the boundary. Poisson representation and Jensen's inequality transfer the central squared moment. The convolution of $P_\delta(y) = \delta/[\pi(\delta^2 + y^2)]$ with $(1 + |t|)^{-3/2+\varepsilon_0}$ is at most eleven times that weight when $0 < \varepsilon_0 \le 1/4$. Restoring the denominator and the principal pole-removal factor gives (9). The extended edition gives the detailed derivation with all primitive local values. □

**Separate fixed-unitary extension.** The moment theorem also holds for any one fixed unitary Hecke character $\nu$. For precision choose $\nu(\mathfrak{a}) = \nu_0(\mathfrak{a})(N\mathfrak{a})^{-i\tau}$, with angular parameter $m \in \mathbb{Z}$ and completed gamma factor

$$\gamma_\nu(s) = 2(2\pi)^{-(s+|m|/2+i\tau)}\Gamma(s + |m|/2 + i\tau).$$

All finite Kummer twists have this same infinity type. Use $T_\nu = 1 + |t + \tau| + |m|$, comparable to $1 + |t|$ with $\nu$-dependent constants. Pure norm rows have $L(s, \psi) = \zeta_K(s + i\tau)$ and are separated before the pole-free functional equation; their pole at $1 - i\tau$ is removed by $(s - 1 + i\tau)/(s + 1 + i\tau)$. No uniformity in $m$, $\tau$ or the conductor of $\nu$ is asserted. This extension of the moment is separate from the following finite-order type-I residue formula.

## S43.1 The exact masked remainder and the remaining loss

Retain the original zero-extended symbol $\chi_d(u) = (u/d)_6$ on good squarefree ideals. Let $\mu$ be the ideal Möbius function and set

$$\beta_U(d) = \sum_{\substack{ab=d \\ Na, Nb \le U}} \mu(a)\mu(b), \quad T(u, d) = S \cup \operatorname{supp}(u) \cup \operatorname{supp}(d).$$

Let $\psi_{u,2} = \operatorname{prim}(\psi_u^2)$, formed independently of squaring any zero sentinel. With $\mathfrak{f}_u$ and $\mathfrak{f}_{u,2}$ the primitive conductors, the full-mask series is

$$F_{u,d}(s) = \frac{L(s, \psi_u)}{L(2s, \psi_{u,2})} C_{T(u,d)}(s), \tag{10}$$

$$C_T(s) = \frac{\prod_{p \in T,\ p \nmid \mathfrak{f}_u}(1 - \psi_u(p)(Np)^{-s})}{\prod_{p \in T,\ p \nmid \mathfrak{f}_{u,2}}(1 - \psi_{u,2}(p)(Np)^{-2s})}.$$

Initially it equals $\sum_{c \text{ squarefree},\ (c,T)=1} \psi_u(c)(Nc)^{-s}$ for $\operatorname{Re} s > 1$. Every prime of $T$ remains deleted, including those forgotten by either primitive conductor.

For finite-order $\nu$, $D \geq 1$, $U \geq 1$, $U^2 \leq D$, and $W \in C_c^\infty((0,\infty))$ supported in $[1,2]$, put

$$\widehat{W}(s) = \int_0^\infty W(x) x^{s-1}\, dx, \qquad \mathcal{M}_4(W) = \sum_{j=0}^{4} \sup_x |(x\partial_x)^j W(x)|.$$

On $\sigma = 1/2 + \delta$, the exact residue-subtracted type-I remainder is

$$E_u = \frac{D^\delta}{2\pi} \int_{\mathbb{R}} \widehat{W}(\sigma + it) D^{it} \sum_{\substack{d \text{ squarefree} \\ (d,S)=1}} \frac{\beta_U(d)\nu(d)\chi_d(u)}{(Nd)^{\sigma+it}} F_{u,d}(\sigma + it)\, dt. \tag{11}$$

This definition retains the actual $\beta_U$, all phases, $d$-masks, conductors and the independently primitive square.

**Theorem S43.2** (An improved masked residual estimate)**.** *Under the preceding finite-order hypotheses, for every $\varepsilon > 0$,*

$$\sum_{0<Nu\leq H} |E_u|^2 \ll_{\nu,S,\delta,\varepsilon} \mathcal{M}_4(W)^2 D^{2\delta} U^{2-4\delta} H (HD)^\varepsilon. \tag{12}$$

*Proof.* The same deleted set in both factors of (10) gives $|F_{u,d}(\sigma + it)| \ll (Nu\, Nd)^\eta |L(\sigma + it, \psi_u)|$: the reciprocal incomplete Euler product at $2\sigma > 1$ is bounded by $\zeta_K(2\sigma)$, and its remaining finite numerator is bounded by a small power of its radical. Apply Minkowski to (11) in the exact row space and use Theorem S43.1. The actual coefficient sum is at most

$$\sum_d \frac{|\beta_U(d)|}{(Nd)^{\sigma-\eta}} \leq \left( \sum_{Na\leq U} (Na)^{-\sigma+\eta} \right)^2 \ll U^{1-2\delta+2\eta}.$$

Four Mellin integrations by parts bound the remaining height integral by a constant times $\mathcal{M}_4(W)$. Square the result and absorb the small losses using $U^2 \leq D$. □

At $H = D^{3/4}$ and $U = D^a$ with fixed $0 < a \leq 1/2$, the upper bound's excess over $H$ is $2\delta + (2 - 4\delta)a$, up to small losses; as $\delta \downarrow 0$ it still leaves $2a$. This is a limitation of the upper bound, not an energy lower bound. The extracted principal residue, the original short term at smaller scales and the signed type-II minus type-I covariance are not controlled by (12). The next arithmetic task is to estimate the complete $d$-weighted moment before its absolute coefficient sum, or establish sufficient signed covariance. No positive arithmetic determinant, zero-location theorem or GRH proof follows. The analytic argument uses the external classical inputs above; finite diagnostics and selected existing Lean local identities do not formalize the Hecke functions, approximate functional equation, Poisson transfer or these moment bounds.

# S44 Family identification, the approximate functional equation and masks

We prove the analytic statements of Section S43 from the explicit classical inputs stated there. Write $I(S)$ for the nonzero integral ideals prime to $S$, and $\mathcal{O}_S$ for the ring of $S$-integers. Constants may depend on the fixed field, $S$, $\nu$ and any declared positive small exponent. No such constant is asserted uniform in the conductor or infinity type of a varying $\nu$.

### S44.1 A finite comparison with the literal Kummer rows

Use Appendix II of BGL, with $n = 6$. Its fixed local modulus and representatives may be chosen once, since $S$ contains the primes above 6 and $\mathcal{O}_S$ has class number one. For each $\mathfrak{a} \in I(S)$ the construction writes

$$\mathfrak{a} = (x)E\mathfrak{g}^6, \qquad \alpha_{\mathfrak{a}} = xm_E, \qquad (m_E)_S = E,$$

so $(\alpha_{\mathfrak{a}})_S = \mathfrak{a}\mathfrak{g}^{-6}$. Choose $y \in K^\times$ generating $\mathfrak{g}$ as an $\mathcal{O}_S$-ideal. If $(u)_S = \mathfrak{a}$, then

$$e = \alpha_{\mathfrak{a}} y^6/u \in \mathcal{O}_S^\times, \qquad [\alpha_{\mathfrak{a}}] = [eu] \quad \text{in } K^\times/K^{\times 6}.$$

There is a precision here: Appendix II allows choosing $\mathfrak{g}$ relative to a tested denominator. Fix one admissible global decomposition first. Its radicand gives the canonical character $\vartheta_{\mathfrak{a}}$ on all primes outside the finite set $S \cup \operatorname{supp}(\mathfrak{a}\mathfrak{g})$ where that choice applies. Uniqueness of Hecke characters, equivalently Chebotarev in the finite abelian quotient, then identifies the underlying primitive characters everywhere. With matching arithmetic-Frobenius conventions,

$$\kappa_u = \kappa_{e^{-1}}\vartheta_{\mathfrak{a}}. \tag{13}$$

If the residue-symbol convention is conjugated, conjugate the canonical family and its coefficient sequence; the same large sieve follows.

The group $\mathcal{O}_S^\times/(\mathcal{O}_S^\times)^6$ is finite. Thus (13) uses one of a finite set of fixed Kummer twists $\eta$, all unramified outside $S$ and of bounded $S$-conductor. This is a comparison of primitive characters, not a pointwise multiplication rule for independently zero-extended arrays at bad primes. In particular, a conductor drop must not be replaced by a product of artificial zeros.

Since $\mathcal{O}$ is principal with six units, for fixed $\mathfrak{a} \in I(S)$ and an $S$-supported integral ideal $\mathfrak{b}$ there are at most six $u$ satisfying $(u) = \mathfrak{a}\mathfrak{b}$. For any nonnegative row quantity, the literal moment is consequently at most

$$6 \sum_{\substack{\mathfrak{b}\ S\text{-supported} \\ N\mathfrak{b} \le H}} \sum_{\eta} \sum_{\substack{\mathfrak{a} \in I(S) \\ N\mathfrak{a} \le H/N\mathfrak{b}}} \left|L(s, \operatorname{prim}(\nu\eta\vartheta_{\mathfrak{a}}))\right|^2. \tag{14}$$

An ideal bound $X^{1+\varepsilon}T^{1+\varepsilon}$ therefore transfers to all literal rows, since

$$\sum_{\mathfrak{b}\ S\text{-supported}} (N\mathfrak{b})^{-1-\varepsilon} = \prod_{p \in S} (1 - (Np)^{-1-\varepsilon})^{-1} < \infty.$$

No quotient of the original unit rows or deletion of $S$-prime powers has occurred.

### S44.2 Fixed twists, exact tame conductors and the squarefree sieve

Fix $\xi = \nu\eta$ and write $\rho_{\mathfrak{a}} = \operatorname{prim}(\xi\vartheta_{\mathfrak{a}})$. Its fixed conductor is supported in $S$ before the Kummer twist. On columns prime to $S$, $|\xi(\mathfrak{b})| = 1$. BGL Theorem 1.3 thus gives, for any fixed $\mathfrak{a}_1$,

$$\begin{aligned}\sum_{\substack{\mathfrak{a}_0 \in I(S)\ \text{squarefree} \\ N\mathfrak{a}_0 \le M}} \left| \sum_{\substack{\mathfrak{b}_1 \in I(S)\ \text{squarefree} \\ N\mathfrak{b}_1 \le B}} \lambda_{\mathfrak{b}_1} \xi(\mathfrak{b}_1)\vartheta_{\mathfrak{a}_1}(\mathfrak{b}_1)\vartheta_{\mathfrak{a}_0}(\mathfrak{b}_1) \right|^2 \\ \ll (MB)^{\varepsilon}\{M + B + (MB)^{2/3}\} \sum_{\mathfrak{b}_1} |\lambda_{\mathfrak{b}_1}|^2.\end{aligned} \tag{15}$$

The first fixed multiplier preserves the coefficient norm; the second has modulus at most one. Restricting the positive outer row sum is harmless. A coprimality condition with another fixed ideal

belongs to $\lambda$. In particular, $\nu$ need not have order dividing six and is not asserted to belong to BGL's finite ray-class character set.

There is a unique valuation decomposition

$$\mathfrak{a} = \mathfrak{a}_0\mathfrak{a}_1\mathfrak{a}_2^6, \quad \mathfrak{a}_0 \text{ squarefree}, \quad \mathfrak{a}_1 \text{ squarefull and sixth-power-free}, \quad (\mathfrak{a}_0, \mathfrak{a}_1) = 1.$$

Indeed, for $v_p(\mathfrak{a}) = 6q + r$, $0 \le r \le 5$, put the residual prime in $\mathfrak{a}_0$ if $r = 1$, and its exponent $r$ in $\mathfrak{a}_1$ if $2 \le r \le 5$. BGL Lemma 2.1 gives $\vartheta_{\mathfrak{a}\mathfrak{b}^6} = \vartheta_{\mathfrak{a}}$ and $\vartheta_{\mathfrak{a}_0\mathfrak{a}_1} = \vartheta_{\mathfrak{a}_0}\vartheta_{\mathfrak{a}_1}$ on ideals prime to $S$ in this sixth-power-free case. Those are the identities needed here; no arbitrary multiplicativity assertion is substituted.

At a prime outside $S$, tame Kummer inertia at residual exponent $r$ has order $6/\gcd(6, r)$. It is nontrivial for $r = 1, \dots, 5$ and has conductor exponent one. The fixed $\xi$ is unramified there, so cannot cancel that inertia. At primes in $S$ the local sixth-power classes are finite, hence only finitely many primitive conductor types occur after multiplication by the fixed $\xi$. Consequently

$$Q_{\rho_{\mathfrak{a}}} = |D_K| N\mathfrak{f}_{\rho_{\mathfrak{a}}} \le C_{\xi,S} N(\mathfrak{a}_0\mathfrak{a}_1), \tag{16}$$

independently of $\mathfrak{a}_2$. The equality of the conductor's outside-$S$ part with $\mathrm{rad}(\mathfrak{a}_0\mathfrak{a}_1)$ does not identify that conductor with the original larger deleted set.

### S44.3 Removing the row-dependent functional-equation weights

First take finite-order $\nu$. All characters then have trivial infinity type at the complex place. The primitive completion may be normalized as

$$\Lambda(s, \rho) = Q_\rho^{s/2}\gamma_K(s)L(s, \rho), \qquad \gamma_K(s) = 2(2\pi)^{-s}\Gamma(s), \qquad \Lambda(s, \rho) = \epsilon_\rho\Lambda(1 - s, \bar{\rho}), \quad |\epsilon_\rho| = 1.$$

For nonprincipal rows the completed function is entire. Apply this functional equation to the Mellin integral with even auxiliary factor $G(z) = e^{z^2}$ and denominator $z$, shifting across $z = 0$. At $s = 1/2 + it$ this gives two weighted Dirichlet series, truncated below to polynomials; the dual prefactor has modulus one. Principal rows are excluded from this pole-free step and treated separately below.

For fixed $\mathfrak{a}_2$, let $X = H/(N\mathfrak{a}_2)^6 \ge 1$ and $T = 1 + |t|$. An explicit normalized kernel is

$$V_t(x) = \frac{1}{2\pi i}\int_{(\eta)} e^{z^2}\frac{\Gamma(s + z)}{\Gamma(s)}T^{-z}x^{-z}\,\frac{dz}{z}, \qquad x = \frac{2\pi N\mathfrak{b}}{\sqrt{Q_\rho}T}, \tag{17}$$

with $\eta > 0$ small. Its factors combine to $(\sqrt{Q_\rho}/(2\pi N\mathfrak{b}))^z$ in the actual completed-L kernel; there is no extra factor of $T$. The denominator $z$ supplies the residue at zero. On $z = \eta + iy$, Stirling and $|t + y| \le T(1 + |y|)$ give

$$\left|\frac{\Gamma(1/2 + \eta + i(t + y))}{\Gamma(1/2 + it)T^{\eta+iy}}\right| \ll_\eta (1 + |y|)^{C_\eta}e^{\pi|y|/2}.$$

The factor $e^{-y^2}$ makes this an integrable majorant uniformly in $t$. The analogous bound holds on every fixed positive shifted line and for the dual gamma kernel. Moving far enough to the right gives arbitrary power decay beyond a common total-ideal cutoff

$$B' = X^{1/2}T(HT)^\eta, \tag{18}$$

up to fixed $\xi, S, K$ constants. For any requested $A > 0$, a sufficiently large positive kernel shift, depending also on $\eta$, makes the truncation error $O((HT)^{-A})$ uniformly in all rows in this block.

Ideal counting $O_K(Y)$ bounds the absolute tails. Such errors are negligible after the finite row sums by taking $A$ larger if needed.

Insert (17) on $\operatorname{Re} z = \eta$. The factor $(\sqrt{Q_\rho}T)^z$ is independent of the column and is a row multiplier of modulus at most $(CXT^2)^{\eta/2}$. Its imaginary part is a phase. The remaining column coefficient is $(N\mathfrak{b})^{-1/2-\eta-it-iy}$, apart from fixed constants. Minkowski or Cauchy with the integrable gamma majorant therefore removes the row-dependent weight at cost $(HT)^{O(\eta)}$ in the squared norm. No independence of the root numbers is assumed or needed.

For each column use the unique decomposition

$$\mathfrak{b} = \mathfrak{b}_S\mathfrak{b}_1\mathfrak{b}_2, \quad \mathfrak{b}_S \ S\text{-supported}, \quad \mathfrak{b}_1 \text{ squarefree}, \quad \mathfrak{b}_2 \text{ squarefull}, \quad (\mathfrak{b}_1, \mathfrak{b}_2) = 1,$$

where $\mathfrak{b}_1, \mathfrak{b}_2 \in I(S)$; a prime of outside-$S$ valuation one goes in $\mathfrak{b}_1$, and a prime of valuation at least two goes with its full exponent in $\mathfrak{b}_2$. For fixed $\mathfrak{b}_S, \mathfrak{b}_2$ retain $\rho_{\mathfrak{a}}(\mathfrak{b}_S)$ as the original primitive row value. On the good factors the exact canonical identity applies. The remaining factors involving $\mathfrak{b}_2$, including $\vartheta_{\mathfrak{a}_0}(\mathfrak{b}_2)$, are bounded row multipliers. Their moduli are at most one; only then are they discarded in an inequality. In particular the value at $\mathfrak{b}_S$ is not replaced by the product of zero-extended primitive arrays.

For the $\mathfrak{b}_1$ polynomial the coefficient sequence is independent of $\mathfrak{a}_0$; it includes the indicator $(\mathfrak{b}_1, \mathfrak{b}_2) = 1$ and the common cutoff $N\mathfrak{b}_1 \le B'/N(\mathfrak{b}_S\mathfrak{b}_2)$. Padding it by zeros up to $B'$ is legitimate. Its squared norm is at most $\zeta_K(1 + 2\eta)$. Minkowski over $\mathfrak{b}_S, \mathfrak{b}_2$ is legitimate because

$$\sum_{\mathfrak{b}_S, \mathfrak{b}_2} N(\mathfrak{b}_S\mathfrak{b}_2)^{-1/2-\eta} < \infty.$$

The finite Euler product handles $\mathfrak{b}_S$; the squarefull series converges above $1/2$. Apply (15) to each such polynomial and to the conjugate dual family. For fixed $\mathfrak{a}_1, \mathfrak{a}_2$ the resulting energy is

$$(HT)^{O(\eta)}\{M + B' + (MB')^{2/3}\}, \qquad M = X/N\mathfrak{a}_1. \tag{19}$$

The additional condition $(\mathfrak{a}_0, \mathfrak{a}_1) = 1$ simply restricts the positive row sum. The large-sieve small exponent can be chosen sufficiently small along with $\eta$ and absorbed in the indicated small loss.

### S44.4 All-ideal summation and principal rows

Every squarefull ideal $\mathfrak{c}$ has a unique expression $\mathfrak{d}^2\mathfrak{e}^3$ with $\mathfrak{e}$ squarefree. This follows primewise by using exponent zero or one in $\mathfrak{e}$ according to the parity of an exponent at least two. Therefore

$$\#\{\mathfrak{c} \text{ squarefull} : N\mathfrak{c} \le X\} \ll_K X^{1/2}, \qquad \sum_{\mathfrak{c} \text{ squarefull}} (N\mathfrak{c})^{-s} < \infty \quad (s > 1/2).$$

For the count, sum $O((X/(N\mathfrak{e})^3)^{1/2})$ and use $\zeta_K(3/2) < \infty$. Summing (19) over $\mathfrak{a}_1$ gives

$$(HT)^{O(\eta)}\{X + B'\sqrt{X} + (XB')^{2/3}\} \ll (HT)^{O(\eta)}XT,$$

since the squarefull series converges at 1 and $2/3$, $B' = \sqrt{X}\,T(HT)^\eta$, and $T^{2/3} \le T$ for $T \ge 1$. The $\mathfrak{a}_2$ sum is bounded by $H\sum_{\mathfrak{a}_2}(N\mathfrak{a}_2)^{-6} = H\zeta_K(6)$. Choosing all internal losses small for the requested final $\varepsilon$ proves the fixed-twist all-ideal central moment.

There are $O_K(H)$ ideal rows in total. On any row where the primitive character is principal its $L$-function is $\zeta_K$, and ordinary scalar convexity gives $|\zeta_K(1/2 + it)|^2 \ll_\varepsilon T^{1+\varepsilon}$. This treats those rows without a pole-free approximate functional equation. Finally (14) proves (8) for every actual nonzero Eisenstein element.

### S44.5 Poisson transfer with an explicit constant

For a nonprincipal row define

$$f_\psi(s) = \frac{L(s,\psi)}{(s+1)^{5/4}},$$

using the holomorphic logarithm on $\mathrm{Re}(s+1) > 0$. On a principal row additionally multiply by $(s-1)/(s+1)$. Analytic continuation and ordinary degree-two convexity, used separately at each fixed conductor, show that $f_\psi$ is bounded analytic on the right half-plane $\mathrm{Re}\, s > 1/2$, with continuous boundary and decay at infinity. The principal pole is removed before applying Poisson representation to its real and imaginary parts. Jensen gives

$$|f_\psi(1/2+\delta+it)|^2 \le \int_{\mathbb{R}} P_\delta(t-v)|f_\psi(1/2+iv)|^2\,dv.$$

The exact row family is finite for fixed $H$, so its sum interchanges with the integral without a limiting-family issue. The boundary pole-removal factor has modulus at most one.

Take $0 < \varepsilon_0 \le 1/4$ and $\alpha = 3/2 - \varepsilon_0 \in [5/4, 3/2)$. Since $|3/2+iv| \ge (1+|v|)/\sqrt{2}$, the central squared moment of the $f_\psi$ is bounded by $3C_0H^{1+\varepsilon_0}w_\alpha(v)$, where $w_\alpha(v) = (1+|v|)^{-\alpha}$. For $|t| \ge 2$, split $|v-t| \le |t|/2$ from its complement. The near contribution to $P_\delta * w_\alpha$ is at most $2^\alpha w_\alpha(t) < 3w_\alpha(t)$. In the far region use $P_\delta(t-v) \le 4\delta/(\pi t^2)$ and $\int w_\alpha = 2/(\alpha-1)$, giving at most

$$\frac{2\delta 3^\alpha}{\pi(\alpha-1)} w_\alpha(t) \le 8w_\alpha(t).$$

Here $(1+|t|)^\alpha/t^2 \le 3^\alpha/4$ for $|t| \ge 2$. For $|t| \le 2$, the convolution is at most one and hence at most $3^\alpha w_\alpha(t) < 6w_\alpha(t)$. Thus $P_\delta * w_\alpha \le 11 w_\alpha$ uniformly for $0 < \delta < 1/2$. Restoring the denominator costs less than $6(1+|t|)^{5/2}$, and the inverse principal factor is bounded by $(3/2+\delta)/(1/2-\delta)$. The product $11 \cdot 3 \cdot 6 = 198 < 200$ proves (9). The constant is uniform as $\delta \downarrow 0$, but not as the contour approaches the principal pole line $\mathrm{Re}\, s = 1$. No zero-free hypothesis appears.

### S44.6 The separately scoped unitary infinity type

Let $\nu(\mathfrak{a}) = \nu_0(\mathfrak{a})(N\mathfrak{a})^{-i\tau}$ be fixed and unitary, with angular parameter $m$ and gamma factor $\gamma_\nu$ stated in Section S43. The external primitive functional equation is explicitly

$$Q_\rho^{s/2}\gamma_\nu(s)L(s,\rho) = \epsilon_\rho Q_\rho^{(1-s)/2}\gamma_{\bar\nu}(1-s)L(1-s,\bar\rho), \qquad |\epsilon_\rho| = 1.$$

A constant norm-twist phase in another completion convention is absorbed in $\epsilon_\rho$. All finite Kummer twists share this infinity type. Replace $T$ in the kernel by $T_\nu = 1 + |t+\tau| + |m|$. The numerator gamma ratio now has real base $1/2 + |m|/2 > 0$ and vertical parameter $t+\tau$; its conjugate dual has parameter $-(t+\tau)$. Stirling gives the same integrable Gaussian-majorized kernel on each fixed shifted line. Constants can depend on $m, \tau$; $T_\nu$ is comparable to $T$ for this one fixed $\nu$. The finite-prime conductor analysis, modulus-one coefficient twist, all-ideal decomposition and all original primitive row values remain unchanged.

A pure norm row has $L(s,\rho) = \zeta_K(s+i\tau)$; exclude it from the pole-free approximate functional equation and use scalar convexity at $t+\tau$. In the Poisson step cancel its actual pole by

$$h_\tau(s) = \frac{s-1+i\tau}{s+1+i\tau}.$$

This factor is analytic in the relevant right half-plane. On its boundary $|h_\tau| \le 1$, while inside $|h_\tau^{-1}| \le (3/2+\delta)/(1/2-\delta)$. Hence the same transfer factor 200 applies to the now $\nu$-dependent

central constant. If the opposite norm convention is chosen, replace $\tau$ by $-\tau$ in the gamma factor and pole alike. This proves the separate fixed-unitary moment assertion; it does not claim uniformity in a family of infinity types or replace the finite-order principal-residue expression below.

### S44.7 Keeping the full type-I mask during transfer

For $\sigma = 1/2 + \delta$, define $L^T$ by removing every prime of $T(u,d)$, not just conductor primes. The identity (10) is equivalently

$$F_{u,d}(s) = \frac{L(s,\psi_u)}{L^{T(u,d)}(2s,\psi_{u,2})} \prod_{p\in T(u,d),\ p\nmid \mathfrak{f}_u} (1-\psi_u(p)(Np)^{-s}).$$

Absolute convergence at $2\sigma > 1$ bounds the reciprocal incomplete product by $\zeta_K(2\sigma)$, without requiring any zero-free theorem in the critical strip. For every fixed $\eta > 0$,

$$\prod_{p\in T}(1+(Np)^{-\sigma}) \ll_{\sigma,\eta} (N\operatorname{rad} T)^{\eta}, \qquad N\operatorname{rad} T \le C_S Nu\, Nd.$$

Indeed $1 + q^{-\sigma} \le q^{\eta}$ beyond a fixed threshold, and only finitely many prime ideals lie below it. Therefore

$$|F_{u,d}(\sigma+it)| \ll_{\nu,S,\delta,\eta} (Nu\,Nd)^{\eta}|L(\sigma+it,\psi_u)|. \tag{20}$$

This is a bounded inequality after the exact mask identity, not an identity that removes original zeros. For example, with principal $\nu$ and $u = \pi^3$ or $\pi^6$ at $N\pi = 7$, the primitive local ratio, restoration and full masked factor at $s = 1$ are respectively

| $v_\pi(u)$ | primitive ratio | restoration | masked factor |
|---|---|---|---|
| 3 | 48/49 | 49/48 | 1 |
| 6 | 8/7 | 7/8 | 1 |

The squared primitive character in the first row is unramified at $\pi$ although the original primitive character is ramified there. Squaring a zero sentinel would give the wrong denominator.

For finite-order $\nu$, the normalized type-I piece is the retained $d$ sum of the smoothed squarefree series at $Y = D/Nd$, divided by $\sqrt{D}$. Its possible pole is only at $s = 1$, on principal rows. Contour shifting with the full mask gives $I_u = P_u + E_u$, with $E_u$ exactly (11) and

$$P_u = \mathbf{1}_{\psi_u=1}\sqrt{D}\,\widehat{W}(1) \sum_{\substack{d \text{ squarefree}\\(d,S)=1}} \frac{\beta_U(d)\nu(d)\chi_d(u)}{Nd}\rho_{T(u,d)},$$

$$\rho_T = \frac{\operatorname{res}_{s=1}\zeta_K(s)}{\zeta_K(2)} \prod_{p\in T}(1+(Np)^{-1})^{-1}.$$

This does not impose positivity on $P_u$. Since $\beta_U$ is supported on $Nd \le U^2 \le D$, every inner scale $Y$ is at least one. Four integrations by parts give $|\widehat{W}(\sigma+it)| \ll_\sigma \mathcal{M}_4(W)(1+|t|)^{-4}$, which justifies the shifted integrals and their horizontal limits with the classical Hecke growth estimates.

Apply Minkowski to (11) in $\ell^2$ of the actual $u$ rows, then use (20) and the off-central moment. For sufficiently small internal losses this bounds $\|E\|_2$ by

$$CD^{\delta}H^{1/2+O(\eta)} \sum_d \frac{|\beta_U(d)|}{(Nd)^{\sigma-\eta}} \int_{\mathbb{R}} |\widehat{W}(\sigma+it)|(1+|t|)^{(1+O(\eta))/2}\,dt.$$

All original $\nu(d)\chi_d(u)$ have modulus at most one and remain inside the exact expression before this inequality. The actual truncated factorization gives, without replacing $\beta_U$ by arbitrary coefficients,

$$\sum_d \frac{|\beta_U(d)|}{(Nd)^{\sigma-\eta}} \le \left(\sum_{Na\le U} (Na)^{-\sigma+\eta}\right)^2 \ll_{\delta,\eta} U^{1-2\delta+2\eta}.$$

Choose $\eta < \delta$ if necessary and use the fixed-field ideal count. The height integral is bounded by $C\mathcal{M}_4(W)$. Squaring and absorbing the small losses in $(HD)^{\varepsilon}$, using $U^2 \le D$, proves (12). Thus the formerly conditional numerator moment input in this residual transfer is discharged, while its positive-power cutoff loss is retained explicitly.

### S44.8 What remains open and what has been verified

The new remainder envelope improves the old scalar-convexity $H$ power $3/2 - \delta + \varepsilon$ to $1 + \varepsilon$. At fixed $U$ one may choose $\delta$ for a requested small-loss exponent; for $U = D^a$ with fixed $a > 0$, the remaining excess tends to $2a$ as $\delta \downarrow 0$. Neither statement controls $P_u$, the other signed piece, nor their covariance. In the original Vaughan identity the short $2a_U$ term must be restored at smaller scales where its support can meet the annulus. Uniform $\mathcal{M}_4$ bounds do not justify interchanging a profile supremum with the row moment or assuming $U^2 \le D'$ after a scale $D'$ is reduced.

The independent prose audit is recorded in `planning/analytic-oracle/arithmetic-review/adversarial-afm-review-20261104.md`. The source-hashed finite checker tests valuation/support decompositions, squarefull coordinates, height exponents and rational Poisson envelopes. Its finite success does not prove reciprocity, analytic continuation, the large sieve, the approximate functional equation or the moments. No new Lean declarations accompany these analytic theorems. Existing Lean local Euler identities are a separately limited verification layer. The independent positive arithmetic determinant comparison, original signed low-gcd short-row estimate, all required contour uniformities and GRH remain open; Dirichlet, Hecke, Dedekind and general Artin scopes are not identified with one another.

**Pinned primary sources and external inputs.** The principal analytic family input is Blomer–Goldmakher–Louvel, arXiv:1112.1650v1, Theorem 1.3, Lemma 2.1, Appendix II and Section 5. Our fixed-$\nu$ proof changes column coefficients and redoes the functional-equation reduction; it does not claim the published corollary directly states every fixed twist. For comparison, Gao–Zhao, arXiv:2201.01885v2, Lemma 2.9 and Section 3.2, uses the sextic large-sieve/moment interface. Its GRH-conditional central-density theorem is not invoked. Abelian Kummer identification uses the classical reciprocity theorem, with the project's pinned Milne *Class Field Theory*, version 4.03 (6 August 2020), Theorem 0.8 and Chapter I, Section 1, a reference: Milne's notes. The primitive degree-one Hecke continuation and completed functional equation are explicit classical external inputs; the project's pinned Kudla notes on Tate's thesis, Theorem 4.3 and Corollary 4.4, supply a reference. Their sketch and incomplete displayed epsilon formulas are not used as independently certified constants. Standard Stirling, Mellin inversion, Poisson representation and scalar convexity complete the external analytic dependencies. None is replaced by a finite diagnostic or a GRH-equivalent assumption.

## S45 Recombining the actual weighted cutoff

Keep the finite-order Eisenstein family and original masks of Section S43. For $U \geq 1$ and a squarefree ideal $m$, put

$$A_U(m) = \sum_{d|m} \beta_U(d) = \sum_{\substack{ab|m \\ (a,b)=1,\ Na,Nb\leq U}} \mu(a)\mu(b).$$

Every summation variable here is an integral ideal. The last formula uses squarefreeness of $m$; an unconstrained square of a Möbius polynomial would also include forbidden overlapping supports.

**Theorem S45.1** (Exact recombination and an absolute-half-plane rate)**.** *Write $T_0 = S \cup \operatorname{supp}(u)$ and $J_{u,U}(s) = \sum_{d \text{ squarefree},\ (d,S)=1} \beta_U(d)\nu(d)\chi_d(u)(Nd)^{-s}F_{u,d}(s)$. For $\operatorname{Re} s > 1$,*

$$J_{u,U}(s) = \sum_{m \text{ squarefree},\ (m,T_0)=1} A_U(m)\psi_u(m)(Nm)^{-s},$$
$$|A_U(m)| \leq 3^{\omega(m)}, \qquad A_U(m) = \mu(m) \quad (Nm \leq U).$$

*For every $\kappa > 0$ and $0 < \epsilon < \kappa$, uniformly for $\operatorname{Re} s \geq 1 + \kappa$, all original rows and all $U \geq 1$,*

$$\left| J_{u,U}(s) - \frac{1}{L^{T_0}(s,\psi_u)} \right| \leq U^{-\kappa+\epsilon}\{\zeta_K(1+\epsilon)^3 + \zeta_K(1+\epsilon)\}.$$

*Here $L^{T_0}$ removes every prime of $T_0$, including conductor-drop primes; the bound is in the absolute-convergence half-plane only.*

*Proof.* On a surviving term $m = dc$, the two squarefree ideals $c, d$ are coprime, and both avoid $T_0$. The original product $\nu(d)\chi_d(u)\psi_u(c)$ is then $\psi_u(m)$. Grouping by $m$ gives $A_U$. Each prime of $m$ has three possible positions in $(a, b, m/(ab))$, so the absolute sum is at most $3^{\omega(m)}$. When $Nm \leq U$ all divisors fit both cutoffs; the primewise sum is $1 - 1 - 1 = -1$, giving $\mu(m)$.

The same bound justifies absolute convergence and regrouping for $\operatorname{Re} s > 1$, because on squarefree ideals $3^{\omega(m)} = \tau_3(m)$ and $\sum_m \tau_3(m)(Nm)^{-v} = \zeta_K(v)^3$ for $v > 1$, where this latter sum runs over all nonzero integral ideals. Subtract the absolutely convergent Möbius series for $1/L^{T_0}$. Only $Nm > U$ remain. Rankin's inequality with exponent $\eta = \kappa - \epsilon$ bounds their absolute sum by

$$U^{-\eta} \sum_m \{\tau_3(m) + 1\}(Nm)^{-1-\epsilon},$$

which proves the stated constant independently of deleted primes, row multiplicities or character phases. □

For $W$ supported in $[1, 2]$, the same exact regrouping gives the original normalized type-I polynomial

$$I_u = D^{-1/2} \sum_{m \text{ squarefree},\ (m,S)=1} A_U(m)\nu(m)\chi_m(u)W(Nm/D).$$

This formula retains all original zero sentinels. It permits a direct family large sieve before any absolute $d$ sum; its quantitative moment is developed separately below. The displayed absolute-half-plane rate alone is not that critical-strip moment.

## S46 Euler packaging, coprimality and the analytic boundary

For $p \notin T_0$, set $z_p = \psi_u(p)(Np)^{-s}$ and $t_p = z_p/(1+z_p)$. On $\operatorname{Re} s > 1$, deleting an additional surviving prime divides the squarefree Euler product by $1+z_p$. Consequently the exact retained expression is

$$J_{u,U}(s) = F_{u,1}(s)B_{u,U}(s), \qquad B_{u,U}(s) = \sum_{d \text{ squarefree},\ (d,T_0)=1} \beta_U(d) \prod_{p \mid d} t_p.$$

Original zeros enforce $(d,u)=1$ even when a sixth power drops from its primitive conductor. This factorization cannot restore a forbidden $d$ row by replacing a zero sentinel with an unramified primitive value. The equality continues wherever both sides are holomorphic.

Without the cutoffs, $\beta_\infty(d) = \mu(d)2^{\omega(d)}$ on squarefree ideals. Absolute convergence justifies

$$B_{u,\infty} = \prod_{p \notin T_0} (1-2t_p) = \prod_{p \notin T_0} \frac{1-z_p}{1+z_p}, \qquad F_{u,1}B_{u,\infty} = \prod_{p \notin T_0} (1-z_p) = \frac{1}{L^{T_0}(s,\psi_u)}.$$

Theorem S45.1 gives a quantitative rate rather than an unsupported exchange of this limit with a critical-strip contour.

**A local counterexample to the naive square.** In a one-prime projection at a good Eisenstein prime of norm 7, take the principal local phase and $s=1$. Then $z=1/7$, $t=1/8$ and, for $U \geq 7$, the exact polynomial is $B_U = 1-2t = 3/4$. The unconstrained square $(1-t)^2 = 49/64$ differs by $1/64$: it admits the forbidden pair $a=b=p$. After multiplying by $F = 1+z = 8/7$ it gives $7/8$ instead of the correct $6/7$. This is an actual local Euler calculation, not an infinite-series or mean-square counterexample. It identifies the exact coprimality condition that a proposed mollifier-square simplification must preserve.

**The exact intermediate coefficient band.** For a squarefree $m$ with $U < Nm \leq U^2$, one has $A_U(m) = -\mu(m)$. Indeed the unrestricted disjoint-assignment sum is $\mu(m)$. If $Na > U$, every $b \mid m/a$ automatically has $Nb \leq Nm/Na < U$, and $\sum_{b \mid m/a} \mu(b)$ vanishes unless $a = m$. Hence the excluded $a$ assignments contribute exactly $\mu(m)$; the excluded $b$ assignments contribute another $\mu(m)$. They cannot overlap because $NaNb > U^2 \geq Nm$ contradicts $ab \mid m$. Subtraction proves the claim. Together with the small-norm band, this gives $A_U = \mu$ for $Nm \leq U$ and $A_U = -\mu$ for $U < Nm \leq U^2$. The original smoothed columns have $Nm \geq D \geq U^2$, so the new band does not justify replacing their coefficients by $-\mu(m)$: generally they lie beyond it.

**Theorem S46.1** (A stronger cutoff convergence would require zero-freeness)**.** *Fix a nonprincipal primitive $\psi_u$. If these actual $J_{u,U}$ converge locally uniformly on $\operatorname{Re} s > 1/2$ as $U \longrightarrow \infty$, then $L(s,\psi_u)$ has no zero there.*

*Proof.* Classical abelian Hecke continuation makes the numerator of $F_{u,1} = L^{T_0}(s,\psi_u)/L^{T_0}(2s,\psi_{u,2})$ holomorphic. Its denominator is nonzero on $\operatorname{Re} s > 1/2$ by the absolute Euler product at $2s$. Each $B_{u,U}$ is a finite sum with nonzero denominators $1+z_p$ on $\operatorname{Re} s > 0$. Thus every $J_{u,U}$ is holomorphic on the asserted half-plane. A locally uniform limit $G$ is holomorphic there. On $\operatorname{Re} s > 1$, the already proved convergence gives $L^{T_0}(s,\psi_u)G(s) = 1$. The identity theorem extends this equality to the connected half-plane, precluding a zero of $L^{T_0}$. The finitely deleted factors $1-\psi_u(p)(Np)^{-s}$ do not vanish when $\operatorname{Re} s > 0$, so the same conclusion holds for the primitive $L$-function. □

This is an implication, not a proof that the stronger convergence holds, and it does not assert a converse. In particular, a family-averaged energy bound is not silently promoted to locally uniform convergence for each individual character. Principal rows have a pole at 1 and were excluded from this proposition. General Artin holomorphy is not an input.

**Reproducible exact diagnostics.** The new source `examples/arithmetic_cutoff/exact_recombination.py` uses actual split/inert prime ideals and raw sextic symbols in $\mathbb{Z}[\omega]$, with exact rational arithmetic. It checks the direct masked sum, coefficient regrouping, Euler packaging, both finite full-cutoff limits and original zeros, including third/sixth-power rows. Its overlapping- support negative controls are not tests of an infinite large sieve or of zero location. The analytic convergence argument above uses ordinary ideal Euler products and the explicitly stated Hecke continuation theorem; no new Lean declaration or complete GRH proof is claimed.

## S47 A cutoff-uniform envelope for the original masked remainder

Fix $K = \mathbb{Q}(\omega)$, $\mathcal{O} = \mathbb{Z}[\omega]$, a finite-order Hecke character $\nu$, and a finite set $S$ containing the primes above 6 and the conductor of $\nu$. All unspecified ideal sums below are over nonzero integral ideals. Let $\chi_m(u) = (u/m)_6$ be the original symbol, zero when $(m, u) > 1$, and put

$$\beta_U(d) = \sum_{\substack{ab=d \\ Na, Nb \le U}} \mu(a)\mu(b), \qquad A_U(m) = \sum_{d \mid m} \beta_U(d).$$

The columns $d, c, m$ are squarefree and prime to $S$ where indicated. Retain the primitive characters $\psi_u = \mathrm{prim}(\nu\kappa_u)$, $\psi_{u,2} = \mathrm{prim}(\psi_u^2)$, and the full deleted set $T(u, d) = S \cup \mathrm{supp}(u) \cup \mathrm{supp}(d)$. For $H, D, U \ge 1$, $U^2 \le D$, and $W \in C_c^\infty((0, \infty))$ supported in $[1, 2]$, the original normalized type-I piece is

$$I_u = D^{-1/2} \sum_{\substack{m \text{ squarefree} \\ (m,S)=1}} A_U(m)\nu(m)\chi_m(u) W(Nm/D). \tag{21}$$

Its exact principal residue is

$$P_u = \mathbf{1}_{\psi_u=1} \sqrt{D}\, \widehat{W}(1) \sum_{\substack{d \text{ squarefree} \\ (d,S)=1}} \frac{\beta_U(d)\nu(d)\chi_d(u)}{Nd} \rho_{T(u,d)},$$

$$\rho_T = \frac{\mathrm{res}_{s=1}\, \zeta_K(s)}{\zeta_K(2)} \prod_{p \in T} (1 + (Np)^{-1})^{-1}. \tag{22}$$

Define $E_u = I_u - P_u$; this is the same residue-subtracted remainder as the fully masked Mellin contour, with no cutoff limit or profile supremum.

**Theorem S47.1** (Uniformity in the actual truncated cutoff)**.** *For every $\varepsilon > 0$,*

$$\sum_{0 < Nu \le H} |E_u|^2 \ll_{\nu,S,\varepsilon} \|W\|_\infty^2 (HD)^\varepsilon \{H + DH^{1/3} + (HD)^{2/3}\}. \tag{23}$$

*The sum is over every actual nonzero Eisenstein element, including units, $S$-supported powers and sixth-power rows. The constant is independent of $U$ under $U^2 \le D$. This is an application of the classical sextic large sieve with exact original coefficients, not a zero-location theorem.*

*Proof.* On an absolutely convergent line, expand the restored squarefree series $F_{u,d}$. Its condition $(c, T(u,d)) = 1$ includes $(c,d) = 1$, and the original symbols multiply. Grouping $m = cd$ gives (21) by Mellin inversion. Each prime of $m$ may belong to the first truncated Möbius divisor, to the second, or to neither. Consequently $|A_U(m)| \le 3^{\omega(m)} \ll_\eta (Nm)^\eta$, uniformly in $U$. The length-$D$ coefficient sequence in (21) therefore has squared norm $\ll_\eta \|W\|_\infty^2 D^\eta$.

The all-ideal row norm in Section 3 of Blomer–Goldmakher–Louvel, arXiv:1112.1650v1 gives

$$B_2^1(X,D) \ll_\eta (XD)^\eta \{X + DX^{1/3} + (XD)^{2/3}\}.$$

For completeness, split $a = a_1 a_2^2 \cdots a_5^5 a_6^6$ dyadically; $a_1, \dots, a_5$ are squarefree and pairwise coprime. If their dyadic sizes are $A_j$, put $R = \prod_{j=1}^6 A_j$ and $M = \max(A_1, A_2)$. Pivot on the first or second variable, using the squarefree powered sieve proved simultaneously in that paper's Section 3.2. The other variables are bounded column multipliers. The block energy is at most

$$(XD)^\eta \{R + DR/M + D^{2/3} R/M^{1/3}\} \sum_m |\lambda_m|^2.$$

The inequalities $R \le X$, $R^3 \le XM^3$ and $R^3 \le X^2 M$ follow from $\prod_j A_j^j \le X$ and the pivot choice. Summing the logarithmically many blocks proves the displayed bound.

Finite $S$-unit and ray-class partitions convert the original literal symbol into the canonical column value $\vartheta_m(a)$ with fixed modulus-one coefficient multipliers. On noncoprime pairs both values are zero: squarefree $m$ has outside-$S$ conductor exactly $m$. This orientation preserves sixth-power zeros. At most six elements generate each outside-$S$ ideal times an $S$-supported ideal; the $S$-supported sums converge at powers 1, 1/3 and 2/3. Thus (23) holds first with $I_u$ in place of $E_u$.

A principal $\psi_u$ requires every outside-$S$ valuation of $u$ to be divisible by 6. There are $O_{\nu,S}(H^{1/6})$ such literal rows. Equation (22), $|\rho_T| \ll_K 1$, and the actual coefficient sum

$$\sum_d \frac{|\beta_U(d)|}{Nd} \le \left( \sum_{Na \le U} \frac{1}{Na} \right)^2 \ll \log^2(2U)$$

give $\sum |P_u|^2 \ll DH^{1/6} \|W\|_\infty^2 \log^4(2U)$. This is absorbed by the right side of (23), using $U^2 \le D$ and a smaller internal exponent. Finally $|E_u|^2 \le 2|I_u|^2 + 2|P_u|^2$. The extended proof specifies the canonical-family transfer and the powered-sieve input precisely. □

Combining this with the previously proved fixed-twist AFM transfer yields

$$\sum_{0 < Nu \le H} |E_u|^2 \ll_{\nu,S,\delta,\varepsilon} \mathcal{M}_4(W)^2 (HD)^\varepsilon \min\{D^{2\delta} U^{2-4\delta} H,\ H + DH^{1/3} + (HD)^{2/3}\}. \tag{24}$$

At $H = D^{3/4}$ and $U = D^a$, $0 \le a \le 1/2$, the excess over $H$ is at most $\min\{2\delta + (2-4\delta)a, 1/2\}$, up to arbitrarily small losses. The new envelope improves the old one for $a > 1/4$ as $\delta \downarrow 0$, but still leaves an exponent loss of 1/2. It does not settle the original signed short-row bound, type-II covariance, positive arithmetic determinant comparison or GRH. The principal-residue bound used here is sufficient for this envelope, not the stronger target needed to reassemble the full signed argument.

## S48 All-ideal reciprocity and the surviving block obstruction

The proof uses no new zero-free input. Its external analytic ingredient is the pinned BGL large sieve and the powered estimates established in its proof; elementary ideal counting and the already restored Mellin identity supply the remaining interfaces. All constants may depend on fixed $K$, $S$, $\nu$ and declared small exponents, never on $U$.

## S48.1 Exact recombination retains the primitive corrections

On $\operatorname{Re} s > 1$, start from the full defining series, with its original zero masks:

$$F_{u,d}(s) = \sum_{\substack{c \text{ squarefree} \\ (c,T(u,d))=1}} \nu(c)\chi_c(u)(Nc)^{-s}.$$

Its primitive ratio involves $L(s,\psi_u)$, the independently primitive $L(2s,\psi_{u,2})$ and every deleted Euler factor. Expanding this *restored* series gives the exact equality

$$\begin{aligned}\sum_{\substack{d \text{ squarefree} \\ (d,S)=1}} &\beta_U(d)\nu(d)\chi_d(u)(Nd)^{-s}F_{u,d}(s) \\ &= \sum_{\substack{m \text{ squarefree} \\ (m,S)=1}} A_U(m)\nu(m)\chi_m(u)(Nm)^{-s}. \end{aligned} \tag{25}$$

Indeed $(c,d)=1$ makes $m=cd$ squarefree, with $d \mid m$. If $m$ meets $u$, both original symbols retain their zero. Only on the absolutely convergent line is this infinite series used. Mellin inversion there gives the finite smoothed polynomial (21). Nothing evaluates (25) as a Dirichlet series on $\operatorname{Re} s = 1/2+\delta$ or interchanges a cutoff limit with continuation across unknown zeros.

For squarefree $m$, a factorization $ab = d \mid m$ assigns each prime to $a$, to $b$, or to neither. The actual inequalities $Na, Nb \le U$ select a subset of these assignments. Thus

$$|A_U(m)| \le 3^{\omega(m)} \ll_\eta (Nm)^\eta.$$

The last divisor bound follows by choosing a fixed threshold so $3 \le (Np)^\eta$ beyond it, then absorbing the finitely many smaller prime ideals in a constant. When $Nm \le U$ all assignments occur, so $A_U(m) = \prod_{p\mid m}(1-1-1) = \mu(m)$; no such simplification is made for the length-$D$ columns in general. With $\lambda_m = D^{-1/2}A_U(m)\nu(m)W(Nm/D)$, ideal counting and a smaller internal $\eta$ give

$$\sum_m |\lambda_m|^2 \ll_\eta \|W\|_\infty^2 D^\eta, \qquad \operatorname{supp}\lambda \subseteq \{D \le Nm \le 2D\}. \tag{26}$$

## S48.2 The powered squarefree input and all-ideal pivot

Let $\vartheta_a$ be BGL's canonical primitive sixth-order family. Their Section 3 defines $B_1^j$ using squarefree row/column ideals and kernel $\vartheta_a(m)^j$, while $B_2^j$ uses *all* ideal rows and kernel $\vartheta_m(a)^j$. Theorem 1.3 states the squarefree bound for $j=1$; its Section 3.2 proves the estimates simultaneously for $1 \le j \le 5$, because the recursion in Lemma 3.4 uses $j\ell$. In particular the two inputs needed here are

$$B_1^j(M,D) \ll_\eta (MD)^\eta\{M + D + (MD)^{2/3}\}, \qquad j = 1,2. \tag{27}$$

To see the final step in the published recursion explicitly, its $E_{4/3}$ estimate is $M^{4/3}+D+(MD)^{2/3}$. If $M \le D$, its first term is at most $(MD)^{2/3}$; if $M > D$, use reciprocity/duality (3.2) and the estimate with $M, D$ reversed. The finite set of powers is treated together; no inference from a fixed conductor count to an arbitrary primitive-character sieve occurs.

Write each ideal row uniquely as

$$a = a_1 a_2^2 a_3^3 a_4^4 a_5^5 a_6^6,$$

where $a_1, \dots, a_5$ are pairwise-coprime squarefree ideals and $a_6$ is arbitrary. For valuation $6q+r$, $0 \le r \le 5$, place the prime in $a_r$ if $r > 0$ and its power $q$ in $a_6$. Dyadically let $A_j \le Na_j < 2A_j$,

with $A_j \ge 1$, and put $R = \prod_{j=1}^{6} A_j$. A nonempty block satisfies $\prod_j A_j^j \le X$ when $Na \le X$. Choose $\ell = 1$ if $A_1 \ge A_2$, and $\ell = 2$ otherwise; put $M = A_\ell$. Keep every other ideal fixed. Evaluation of the single *column* character gives

$$\vartheta_m(a) = \vartheta_m(a_\ell)^\ell \vartheta_m(a/a_\ell^\ell).$$

The second factor is a bounded column multiplier, including its original zero if $m$ meets any fixed factor. It may depend on all the fixed ideals but not on the pivot row. Pairwise-coprimality and the exact row norm restriction merely select a subset of the positive pivot-row sum. Apply (27) to the first factor. There are $O_K(R/M)$ choices of the other ideals by ideal counting. The block energy is bounded by

$$(XD)^\eta \{R + DR/M + D^{2/3}R/M^{1/3}\} \sum_m |\lambda_m|^2. \tag{28}$$

All three monomial bounds used to sum these blocks are elementary. First $R \le \prod_j A_j^j \le X$. If $M = A_1$, then

$$(R/M)^3 = A_2^3 \cdots A_6^3 \le A_1 A_2^2 A_3^3 A_4^4 A_5^5 A_6^6 \le X,$$

since $A_2 \le A_1$ and $j \ge 3$ for the remaining factors. If $M = A_2$, use $A_1 \le A_2$ in the same comparison. Finally,

$$\frac{R^3}{M(\prod_j A_j^j)^2} = \frac{A_1}{M A_2 A_3^3 A_4^5 A_5^7 A_6^9} \le 1.$$

Thus $R/M \le X^{1/3}$ and $R/M^{1/3} \le X^{2/3}$. Insert these in (28) and sum the logarithmically many blocks. This proves the all-ideal estimate used in Theorem S47.1. It is precisely the mechanism of BGL Lemma 3.4 and (3.6)–(3.9), with the final powered bound inserted. The simpler fixed-squarefull-row argument would leave the weaker $D\sqrt{X}$ term.

### S48.3 Literal rows, reciprocal orientation and principal count

Let $a = (u)_S$ denote the outside-$S$ ideal. The finite $S$-unit comparison gives $\kappa_u = \eta_u \vartheta_a$ as primitive characters, with $\eta_u$ in a finite fixed set. For $(a, m) = 1$, BGL reciprocity therefore gives

$$\chi_m(u) = \eta_u(m) C([a], [m]) \vartheta_m(a). \tag{29}$$

The notation is deliberate: the left side is the project's original $(u/m)_6$, not BGL's canonical character indexed by $m$. When $(a, m) > 1$, the left side is zero. The right side is also zero, because squarefree $m$ has outside-$S$ primitive conductor exactly $m$. Equation (29) hence extends to every pair. Using $\vartheta_a(m)$ on noncoprime pairs instead would be false: primitive $\vartheta_{b^6}$ has restored primes of $b$, whereas the original symbol must still vanish there.

Partition rows by $\eta_u$ and $[a]$, both finite. On each part the factor $\eta_u(m)C([a], [m])$ is a fixed modulus-one column multiplier. A positive row sum over that part is bounded by the full $B_2^1$ row sum. For each $S$-supported ideal $b$ and outside-$S$ ideal $a$, at most six actual elements generate $ab$. Thus the required literal energy is bounded by the all-ideal norm summed over $b$, with $X = H/Nb$. The geometric products

$$\sum_{b \ S\text{-supported}} (Nb)^{-q} = \prod_{p \in S} (1 - (Np)^{-q})^{-1} < \infty, \qquad q = 1,\ 1/3,\ 2/3,$$

transfer the three terms without losing any unit or $S$-power row. Together with (26), this proves the claimed $I$ energy.

If $\psi_u$ is principal, $\nu$ is unramified outside $S$, and tame Kummer inertia at $v_p(u)$ has order $6/\gcd(6, v_p(u))$. It must be trivial, so each such valuation is divisible by 6. Writing $(u) = ba_6^6$ with $b$ supported in $S$, ideal counting gives

$$\#\{0 < Nu \le H : \psi_u = 1\} \ll_K H^{1/6} \prod_{p \in S} (1 - (Np)^{-1/6})^{-1}.$$

This is only a necessary condition, sufficient for an upper count. The exact residue formula (22) and $|\widehat{W}(1)| \le \|W\|_\infty$ then give the principal energy bound used above. No sign or positivity of that residue is assumed.

### S48.4 What the envelope still fails to prove

The $DH^{1/3}$ term is still larger than $H(HD)^\varepsilon$ at $H = D^{3/4}$. The monomial comparison itself is saturated by $A_1 = A_2 = X^{1/3}$ and $A_3 = \cdots = A_6 = 1$. This proves sharpness of that *particular comparison*, not an energy lower bound for actual arithmetic coefficients. In particular, no conclusion that the signed $A_U$ cannot cancel is justified. The next load-bearing input must exploit those actual truncated Möbius coefficients inside the mixed moment, or establish the required signed type-II minus type-I covariance. The smaller-scale short term, principal contribution, full arithmetic determinant identification and GRH remain separate obligations.

Exact finite checks use the split Eisenstein primes generated by $3 + \omega$, $4 + \omega$, $5 + 2\omega$, $6 + \omega$, of norms $7, 13, 19, 31$. Their actual residue symbols are computed in $\mathbb{Q}(\zeta_6)$, retaining sixth-power zeros. Coefficient recombination and dyadic inequalities pass; they check finite algebra and bookkeeping, not the external analytic sieve or the contour theorem. No new Lean declarations accompany this argument.

## S49 High cutoffs, actual bilinear factors and a signed covariance test

Retain the original Eisenstein family, fixed finite-order $\nu$, fixed $S$, common $W \in C_c^\infty((0,\infty))$ supported in $[1, 2]$, and every raw shared-prime zero. Put

$$C_U(m) = \sum_{\substack{abc=m \\ Na>U,\, Nb>U}} \mu(a)\mu(b), \qquad r_U(m) = \#\{(a, b) : ab = m,\ Na, Nb > U\}.$$

All factors are ideals, squarefree and pairwise coprime; on the active annulus $A_U = C_U - \mu$. A coefficient identity is different from a row-energy bound.

**Proposition S49.1** (The high-cutoff window)**.** *If $D > 1$ and $2D/7 \le U^2 \le D$, then for every surviving squarefree column $D \le Nm \le 2D$,*

$$C_U(m) = \mu(m) r_U(m), \qquad A_U(m) = \mu(m)(r_U(m) - 1), \qquad |A_U(m)| \ge 1.$$

*For each fixed nonzero $W$, the normalized squared coefficient norm satisfies*

$$D^{-1} \sum_{\substack{m \text{ sf} \\ (m,S)=1}} |A_U(m) W(Nm/D)|^2 \gg_{W,S} 1 \tag{30}$$

*for large $D$, uniformly in this window. This is not a lower bound for row energy.*

*Proof.* The least norm of a nonunit ideal outside $S$ is at least 7. A long-long term with $c \neq 1$ would have $Nm > 7U^2 \geq 2D$. Thus $ab = m$, including both endpoints. The fixed-point-free swap $(a, b) \mapsto (b, a)$ makes $r_U$ even, so $A_U$ is an odd nonzero integer. The last claim follows by counting squarefree ideals on a fixed interval where $|W|$ is bounded below; the constant and error are proved in the extended proof. □

The exact remaining polynomial is genuinely bilinear, with $(a, b) = 1$, $U < Na, Nb < 2D/U$, $D \leq Na\,Nb \leq 2D$, and $x_{ab}(u) = x_a(u)x_b(u)$, including zeros. Both factors see the same row $u = \varepsilon\alpha\beta^2$. Cauchy followed by the accepted powered BGL bound gives only

$$Q_{B_X}(C_U, C_U) \ll_{\nu,S,\epsilon} (DX)^{\epsilon}\|W\|_{\infty}^2\{\sqrt{D}\,X^2 + DX + D^{5/6}X^{5/3}\}. \tag{31}$$

At $X \asymp D^{1/4}$ the exponents are $1, 5/4, 5/4$; the earlier regrouped bound has $1/2, 5/4, 13/12$ and is stronger. No $BT_\sigma$ saving is obtained. Multiplying two marginal operator norms does not supply a coupled fourth moment.

**A finite test in the original family.** There is no automatic nonnegative signed covariance, even when $C_U(m)\mu(m) \geq 0$. Take $\nu = 1$, $S$ the primes above $2, 3$, $D = 400$, $U = 17$, and all 72 literal rows of $B_7$. Choose the common nonnegative smooth profile constructed in the extended proof, with values 10 at 439/400, 1 at 475/400, and zero at every other integer norm divided by 400. There are exactly four active squarefree ideal columns, of norms $439, 439, 475, 475$. In that order,

$$A = (1, 1, 1, 1), \quad C = (0, 0, 2, 2), \quad \mu = (-1, -1, 1, 1), \qquad \Re G_{B_7} = \begin{pmatrix} 72 & 0 & 0 & 18 \\ 0 & 72 & 18 & 0 \\ 0 & 18 & 72 & 0 \\ 18 & 0 & 0 & 72 \end{pmatrix}.$$

Exact raw-symbol enumeration, independently reproducible from the ideal and residue-field definitions, gives

$$Q(C, \mu) = -\frac{27}{25}, \qquad Q(C, C) = \frac{36}{25}, \quad Q(\mu, \mu) = \frac{864}{25}, \quad Q(A, A) = \frac{954}{25}. \tag{32}$$

The real Gram matrix is positive definite, with eigenvalues $54, 90$; its positivity does not force a cross covariance to be positive. This fixed finite example refutes that shortcut, not the asymptotic $BT_\sigma$ target or the original signed mean-square estimate.

**The two-cutoff contour remains coupled.** For $T_0 = S \cup \mathrm{supp}(u)$, $\psi = \psi_u$ and independently primitive $\psi_2 = \mathrm{prim}(\psi^2)$, put $q_p = \psi(p)(Np)^{-s}$, $\alpha_p = (Np)^{-z}$, $\beta_p = (Np)^{-w}$ outside $T_0$. The actual three-placement Euler factor is $1 + q_p - q_p\alpha_p - q_p\beta_p$. Its exact factorization is

$$\mathcal{F}(s, z, w) = \frac{L^{T_0}(s, \psi)}{L^{T_0}(s + z, \psi)L^{T_0}(s + w, \psi)} \prod_{p \notin T_0} \left(1 - \frac{q_p^2(1 - \alpha_p)(1 - \beta_p)}{(1 - q_p\alpha_p)(1 - q_p\beta_p)}\right). \tag{33}$$

Exact double Perron inversion is valid on positive auxiliary lines $c, d > 1 - \Re s$ for $1/2 < \Re s < 1$, retaining the original principal pole exclusion. These safe lines reproduce the previous $U^{2-4\delta}$ energy loss at $\Re s = 1/2 + \delta$, with the same common-profile derivative and fixed-margin dependencies. The correction product alone does not justify moving those lines through reciprocal-$L$ poles. In fact $\mathcal{F}(s, z, 0) = 1/L^{T_0}(s + z, \psi)$ as a meromorphic identity. No zero-free region, reciprocal moment, analytic saving, new Lean formalization or GRH proof is supplied by this construction.

# S50 Proof details for the high-window and two-cutoff tests

## S50.1 The coefficient lower bound and its precise limitation

The element lattice $\mathbb{Z}[\omega] \subset \mathbb{C}$ has covolume $\sqrt{3}/2$. Counting disks with boundary error $O(\sqrt{x})$ and dividing by the six units, using that this ring is a PID, gives

$$N_K(x) = \kappa x + O(\sqrt{x}), \qquad \kappa = \frac{\pi}{3\sqrt{3}}.$$

Finite inclusion-exclusion at $S$ gives $N_{K,S}(y) = \kappa y \prod_{p\in S}(1-(Np)^{-1}) + O_S(\sqrt{y})$. Insert $1_{\mathrm{sf}}(m) = \sum_{d^2\mid m}\mu(d)$. Absolute convergence at 2, the tail $\sum_{Nd>\sqrt{x}}(Nd)^{-2} = O(x^{-1/2})$, and $\sum_{Nd\le\sqrt{x}}(Nd)^{-1} = O(\log(2x))$ yield

$$N_{\mathrm{sf},S}(x) = c_S x + O_S(\sqrt{x}\log(2x)), \qquad c_S = \frac{\kappa}{\zeta_K(2)}\prod_{p\in S}(1+(Np)^{-1})^{-1} > 0.$$

These tail estimates follow by partial summation from the displayed lattice count. For a fixed nonzero $W$, choose $[a,b] \subset (1,2)$ with $|W| \ge \eta > 0$. Proposition S49.1 gives, for large $D$,

$$D^{-1}\sum |A_U(m)W(Nm/D)|^2 \ge \tfrac{1}{2}\eta^2 c_S(b-a).$$

There is no uniform lower constant over all profiles with one specified supremum norm. Nor can a coefficient lower bound be combined with an operator upper bound to produce a row-energy lower bound: no lower singular value has been established on the actual vector.

## S50.2 What legal factor Cauchy actually proves

For dyadic $Na \asymp A, Nb \asymp B$, retain the exact $(a,b)=1$ mask, hyperbolic support and $W(Na\,Nb/D)$ in

$$T_a(u) = \sum_b \mu(b)\nu(b)W(Na\,Nb/D)x_b(u).$$

Cauchy gives

$$|\sum_a \mu(a)\nu(a)x_a(u)T_a(u)|^2 \ll A\sum_a |T_a(u)|^2.$$

For fixed $a$, fixed second row factor and fixed unit/ray classes, the $b$ coefficients are independent of the pivot row and have squared norm $\ll B\|W\|_\infty^2$. The prior classical power-one/two BGL input has $\Delta(X,B) = X + B + (XB)^{2/3}$. Summing the $O(A)$ fixed columns and $O(X)$ other row factors gives

$$\mathcal{E}_{A,B} \ll_{\nu,S,\epsilon} (DX)^\epsilon \|W\|_\infty^2 \frac{A^2B}{D} X\Delta(X,B) \ll (DX)^\epsilon\|W\|_\infty^2\{AX^2 + DX + D^{2/3}A^{1/3}X^{5/3}\}.$$

In the high window $A, B \asymp \sqrt{D}$, and only boundedly many dyadic pairs occur. This proves (31); exchanging factors or row pivots leaves its $DX$ term. A product of marginal second-moment bounds supplies no estimate for the two transforms on the same $(r,s)$. Any useful saving must control that coupled fourth moment or the signed actual-coefficient covariance, with all masks retained.

### S50.3 A smooth profile and the exact covariance computation

Let

$$\phi(t)=\begin{cases}\exp(1-1/(1-t^2)), & |t|<1,\\ 0, & |t|\ge 1,\end{cases}\qquad W_x(y)=x\phi(4(400y-439))+\phi(4(400y-475)).$$

For $x\ge 0$ this is nonnegative and smooth; its disjoint supports lie strictly inside $(1,2)$. For every integer $n$, $W_x(n/400)$ equals $x$ at $n=439$, 1 at $n=475$ and zero otherwise. Thus the computation uses exact values of a genuine common $C_c^\infty$ profile, rather than claiming a finite rational weight is smooth. At $x=10$, $\|W_x\|_\infty=10$.

The primary ideal generators, in the order used above, are

$$(-23,-18),\quad (-5,18),\quad (10,-15),\quad (25,15),$$

where $(a,b)$ denotes $a+b\omega$. The first two have norm 439. The last two factor as $(-2,3)(-5,0)$ and $(-5,-3)(-5,0)$, with norms $19\cdot 25$. Since $19,25>17$, the long-factor count is 2 there and zero on the prime columns. All squarefree ideals at both selected norms are included. The complete $B_7$ uses the four good primes of norms $7,7,13,13$ for distinct ordered row factors and all six units. It has 72 distinct rows, maximum norm 2197; each is nonprincipal by its outside-$S$ Kummer valuation 1 or 2 modulo 6, whose nontrivial tame character has conductor exponent 1.

For the raw Gram $G_{ij}=\sum_{u\in B_7} x_{m_i}(u)\overline{x_{m_j}(u)}$, exact finite-field exponentiation and lattice enumeration give the displayed real matrix. The weighted real vectors are $\ell=(0,0,2,2)$, $v=(-x,-x,1,1)$, and $a=(x,x,1,1)$. Therefore

$$Q(C_U,\mu)=D^{-1}\ell^t(\Re G)v=\frac{36-9x}{50}.$$

Its diagonal contribution is $36/50>0$ and its off-diagonal contribution is $-9x/50$. For $x=10$ this is $-27/25$ and gives (32). The identity $(\Re G-72I)^2=18^2I$ proves the real eigenvalues $54,90$; these are not being asserted as the eigenvalues of the full complex Gram. The exact source is `examples/arithmetic_cutoff/signed_covariance.py`. A second test retains raw zeros on the 246 rows $0<Nu\le 89$ whose outside-$S$ part is nonunit. Its covariance at $x=10$ is $-69/50$. This subset is guaranteed nonprincipal; it is not called the entire nonprincipal family. Here $89=\lfloor 400^{3/4}\rfloor$. These are finite counterexamples to automatic covariance positivity, not asymptotic counterexamples to $BT_\sigma$ or GRH.

### S50.4 Double Perron without a divergent critical-strip series

For $T_0=S\cup\operatorname{supp}(u)$ retain $F_0(s)=L^{T_0}(s,\psi)/L^{T_0}(2s,\psi_2)$, with the independently primitive square character. Set $t_p=q_p/(1+q_p)$ outside $T_0$. The finite actual cutoff is

$$J_U(s,u)=F_0(s)B_U(s),\qquad B_U(s)=\sum_{\substack{a,b\ \mathrm{sf},\ (a,b)=1\\ (ab,T_0)=1,\ Na,Nb\le U}}\mu(a)\mu(b)\prod_{p\mid ab}t_p.$$

For $\sigma=\Re s>1/2$, the series

$$\mathcal{G}(s,z,w)=\sum_{\substack{a,b\ \mathrm{sf},\ (a,b)=1\\ (ab,T_0)=1}}\mu(a)\mu(b)\prod_{p\mid ab}t_p\,(Na)^{-z}(Nb)^{-w}=\prod_{p\notin T_0}\left(1-t_p(Np)^{-z}-t_p(Np)^{-w}\right)$$

converges absolutely on $\sigma+\Re z>1$, $\sigma+\Re w>1$, uniformly in all vertical parts at fixed positive margins. This follows from $|t_p|\le (Np)^{-\sigma}/(1-(Np)^{-\sigma})$. There $\mathcal{F}=F_0\mathcal{G}$, without evaluating an infinite $c$ series in the critical strip. Initial absolute convergence also proves (33) by the local identity

$$(1-q)(1+q-q\alpha-q\beta)=(1-q\alpha)(1-q\beta)-q^2(1-\alpha)(1-\beta).$$

The correction product is holomorphic on

$$\sigma+\Re z>0,\quad \sigma+\Re w>0,\quad 2\sigma+\min(\Re z,0)+\min(\Re w,0)>1.$$

On margins $\sigma+\Re z,\sigma+\Re w\ge\eta>0$ and last exponent $\ge 1+\epsilon$, its local deviation from 1 is at most $4(1-2^{-\eta})^{-2}(Np)^{-1-\epsilon}$. This proves normal convergence and a phase/deletion-uniform vertical upper bound. Nonvanishing of the correction is not asserted.

For $1/2<\sigma<1$, positive lines $c,d>1-\sigma$, and $V=\lfloor U\rfloor+1/2$, define

$$\mathcal{I}_{T_1,T_2}=\frac{1}{(2\pi i)^2}\int_{c-iT_1}^{c+iT_1}\int_{d-iT_2}^{d+iT_2}\mathcal{F}(s,z,w)\frac{V^{z+w}}{zw}\,dw\,dz.$$

The half-integer $V$ includes the actual equality cutoff, since $Na\le U$ is exactly $Na<V$. Put $\Delta_V=\inf_{n\in\mathbb{Z}_{\ge1}}|\log(V/n)|\ge 1/(2V+1)$, and let $\mathcal{A}=\sum_{a,b}|\mu(a)\mu(b)\prod_{p\mid ab}t_p|(Na)^{-c}(Nb)^{-d}$ with the same disjoint support. For $c,T>0$ and $v>0$, $v\ne1$, define

$$K_{c,T}(v)=\frac{1}{2\pi i}\int_{c-iT}^{c+iT}\frac{v^z}{z}\,dz.$$

Finite-height termwise integration of the absolutely convergent $\mathcal{G}$ and the one-variable tail estimate $|K_{c,T}(v)-1_{v>1}|\le 2v^c/(\pi T|\log v|)$ give

$$|\mathcal{I}_{T_1,T_2}-J_U|\le|F_0|\mathcal{A}V^{c+d}\left\{\frac{2}{\pi T_1\Delta_V}+\frac{2}{\pi T_2\Delta_V}+\frac{4}{\pi^2T_1T_2\Delta_V^2}\right\}. \tag{34}$$

Thus $\mathcal{I}_{T_1,T_2}\to J_U$ as both heights tend independently to infinity, away from the original principal pole. This is an oscillatory limit, not an absolutely integrable infinite vertical integral.

Taking $T_1=T_2=V^2$ gives the safe upper bound $|J_U|\ll(Nu)^\eta|L(s,\psi)|V^{c+d}\log^2(2V)$, with constants retaining fixed margins, $\nu,S$, and $\sigma$. Choosing $c,d$ close to $1-\sigma$ reproduces $U^{2-2\sigma+\eta}$ in amplitude and $U^{2-4\delta+\eta}$ in energy on $\sigma=1/2+\delta$. Insertion into the original common-profile Mellin argument retains $M_4(W)^2$ and all previously required height/conductor uniformities. No derivative-free improvement follows from this Perron estimate.

The correction converges even on some negative auxiliary lines, but moving there can cross $z=\rho-s$ or $w=\rho-s$, from zeros of the primitive $L$-function. The exact meromorphic slice $\mathcal{F}(s,z,0)=1/L^{T_0}(s+z,\psi)$ forbids assuming universal cancellation of these poles. At a numerator zero the slice must be understood meromorphically, not by an unjustified pointwise substitution. Sequential shifts have single boundary integrals and crossed-zero residues as well as the kernel double residue; away from $s$ zeros or poles that double residue is $1/L^{T_0}(s,\psi)$. A principal $L$ pole supplies a reciprocal zero, not a reciprocal pole. Uniform control of actual reciprocal integrals/residues or the signed covariance remains a new input. Numerator AFM alone supplies neither.

## S51 A quantitative covariance budget without a new exponent

Retain the original Eisenstein sextic symbols, fixed finite-order $\nu$, fixed $S$, one possibly complex common $W\in C_c^\infty$ supported in $[1,2]$, all squarefree columns prime to $S$, and every raw shared-prime

zero. In the accepted window $2D/7 \le U^2 \le D$, put $r_m = r_U(m)$, $\lambda_m = \mu(m)\nu(m)W(Nm/D)$ and $v_m = \overline{\lambda_m}$. Thus $C_U = \mu r$ and $A_U = \mu(r-1)$ on the active annulus. For a finite literal row block $B$, let

$$G_{mn} = \sum_{u\in B} x_m(u)\overline{x_n(u)}, \qquad R = \operatorname{diag}(r_m), \qquad M_j = \sum_m r_m^j |W(Nm/D)|^2 \quad (0 \le j \le 2).$$

The primitive $\psi_u, \psi_{u,2}$ and full deleted sets retain their previous meanings; no conductor-drop value replaces a raw zero.

**Proposition S51.1** (An actual-vector lower budget)**.** *If $0 \le G \le \Lambda_B I$, then*

$$\Re Q_B(C_U, \mu) \ge -\frac{\Lambda_B}{2D}\{\sqrt{M_0 M_2} - M_1\}. \tag{35}$$

*In this high window, uniformly in $U$,*

$$\sum_{D\le Nm\le 2D} r_m \ll_K D, \qquad \sum_{D\le Nm\le 2D} r_m^2 \ll_K D\log(2D). \tag{36}$$

*Consequently*

$$\Re Q_B(C_U, \mu) \ge -C_K \|W\|_\infty^2 \Lambda_B \sqrt{\log(2D)}.$$

The vector inequality is sharp among positive contractions with the specified $v, Rv$; no such sharpness is asserted for the arithmetic Gram. The moment bound counts actual balanced ideal factorizations, rather than assuming independence of the two row factors.

On $B_X$, inserting the previously accepted powered-BGL norm gives only

$$\Re Q_{B_X}(C_U, \mu) \ge -C_{\nu,S,\epsilon}\|W\|_\infty^2 (DX)^\epsilon \{X^2 + DX + D^{2/3}X^{5/3}\}.$$

At $X \asymp D^{1/4}$ the exponents remain $1/2, 5/4, 13/12$. At $H = D^{3/4}$ this budget still has the old $D^{1/2}$ excess over $H$; it supplies no $BT_\sigma$ saving. A negative-part budget is weaker than $2\Re Q(C_U, \mu) \ge Q(C_U, C_U) + Q(\mu, \mu) - R_\sigma$.

For $S_R = \sqrt{R}$, the exact identity

$$\tfrac{1}{2}(RG + GR) = S_R G S_R + \tfrac{1}{2}[S_R, [S_R, G]]$$

localizes the commutator to unequal factor-count levels, with entry $(\sqrt{r_m} - \sqrt{r_n})^2 G_{mn}$. If $B$ retains all six literal units, the actual unit sum further gives $G_{mn} = 0$ unless $Nm \equiv Nn \pmod{36}$. These are exact deletions, not a power saving. Even zero different-level pairings leave the nonnegative within-level energies $\sum_k (k-1)^2 Q_B(\mu \mathbf{1}_{r=k}, \mu \mathbf{1}_{r=k})$. The unequal-product coupled fourth moment and its mixed Möbius term remain unestimated. No new analytic exponent, energy lower bound, Lean formalization or GRH conclusion follows.

# S52 Proof and arithmetic boundary of the covariance budget

## S52.1 The positive-contraction calculation

Since $G$ is the exact raw Gram, it is positive semidefinite. If $\Lambda_B > 0$, put $T = 2G/\Lambda_B - I$, a self-adjoint contraction. Then

$$\Re(v^* R G v) = \frac{\Lambda_B}{2}\{M_1 + \Re(v^* R T v)\} \ge \frac{\Lambda_B}{2}\{M_1 - \|Rv\|_2 \|v\|_2\},$$

which is (35); $\Lambda_B = 0$ is immediate. The Hermitian rank-two matrix $\{(Rv)v^* + v(Rv)^*\}/2$ has possible nonzero eigenvalues $(M_1 \pm \sqrt{M_0 M_2})/2$. Its negative spectral projection realizes the lower bound among all $0 \le G \le \Lambda_B I$, a linear-algebra sharpness statement only. If $a \le r_m \le b$ on the weighted support, $a + b > 0$, the pointwise inequality $r_m^2 \le (a+b)r_m - ab$ also gives

$$\Re Q_B(C_U, \mu) \ge -\frac{\Lambda_B M_0 (b-a)^2}{8D(a+b)}.$$

For $a = b = 0$ the covariance is zero. The constant follows by maximizing $\sqrt{(a+b)t - ab} - t$ on $[a, b]$.

### S52.2 The balanced-divisor second moment

Set $T_0 = \sqrt{D}$. The high-window factor restrictions imply

$$\sqrt{2/7}\, T_0 < Na, Nb < \sqrt{14}\, T_0.$$

The elementary ideal count $\#\{a : Na \le Y\} \ll_K Y$ proves the first assertion of (36). For the second, two ordered factorizations $ab = cd = m$ have the unique form

$$a = gh, \quad c = gk, \quad b = k\ell, \quad d = h\ell, \qquad g = (a, c).$$

Squarefreeness makes $g, h, k, \ell$ pairwise coprime and squarefree. To upper-bound this positive count, enlarge to arbitrary ideals. For $M = \max(Nh, Nk)$ the individual bounds are

$$Ng \le \frac{\sqrt{14}\, T_0}{M}, \qquad N\ell \le \frac{\sqrt{14}\, T_0}{M}, \qquad 1 \le M \le \sqrt{14}\, T_0.$$

On $2^j \le M < 2^{j+1}$ there are $O_K(2^{2j})$ choices of $(h, k)$ and $O_K((T_0/2^j)^2)$ choices of $(g, \ell)$, with bounded terminal ranges absorbed in the constant. Each dyadic range contributes $O_K(D)$; there are $O(\log(2D))$ ranges. This proves the second moment with no extra logarithm from counting only $NgN\ell$. Restricting to primes outside fixed $S$ decreases the count. Hence

$$M_0, M_1 \ll_K \|W\|_\infty^2 D, \qquad M_2 \ll_K \|W\|_\infty^2 D \log(2D).$$

These constants are independent of the cutoff in the stated window. Only the common supremum norm is used, including for complex $W$.

### S52.3 What the commutator actually controls

Write

$$t_{mn} = D^{-1} \Re\{\lambda_m \overline{\lambda_n} G_{mn}\}, \qquad \delta_{mn} = (\sqrt{r_m} - \sqrt{r_n})^2.$$

The exact commutator identity proves

$$\Re Q_B(C_U, \mu) \ge Q_B(\sqrt{r}\,\mu, \sqrt{r}\,\mu) - \tfrac{1}{2} \sum_{m,n} \delta_{mn} |t_{mn}|.$$

The displayed loss retains the actual character and profile phases before taking absolute values; it is at most the phase-blind loss obtained from $|t_{mn}| \le D^{-1} |\lambda_m \lambda_n G_{mn}|$. On the accepted four-column smooth $B_7$ fixture at amplitude $x \ge 0$, the positive term is $18/25$, this coefficient-adapted loss is $9x/50$, and the lower bound is attained. Purely imaginary entries of the full complex Gram make the phase-blind loss larger. This calibrates the new budget and does not give an asymptotic obstruction.

For the unit deletion let $\zeta_6 = 1+\omega$. Every good prime norm $q$ is $1 \pmod 6$, and direct residue-field exponentiation gives $x_p(\zeta_6) = \zeta_6^{(q-1)/6}$. Multiplying over $p \mid m$ and reducing $Nm = \prod_{p|m}(1+6e_p)$ modulo 36 gives

$$x_m(\zeta_6) = \zeta_6^{(Nm-1)/6 \bmod 6}.$$

Multiplication of all literal rows by $\zeta_6$ preserves their raw zeros and proves the norm-congruence deletion in the core. The vector budget may be applied separately on these six classes. No D-dependent number of classes or conductor saving is produced.

The moment proof above counts the coefficient diagonal $ab = cd$. It has not estimated the unequal-product character quadruples in the coupled row fourth moment. Replacing the inherited $DX$ norm by $D^{1-\sigma}X$ merely because equal-level and different-class entries vanish is the unsupported step. Small negative commutator loss alone also leaves the within-level energies in the core. Principal residues are unchanged; this covariance is used on the nonprincipal blocks where the original residue is zero, not as a new centering prescription.

### S52.4 The complete near-saturating face

For a six-factor row block let $R_0 = \prod_{j=1}^{6} A_j$, $M = \max(A_1, A_2)$ and $\prod_j A_j^j \le H$. The accepted exact-mask block norm is

$$\Lambda_A \ll_{\nu,S,\epsilon} (HD)^{\epsilon}\{R_0 + DR_0/M + D^{2/3}R_0/M^{1/3}\}.$$

The column moment and covariance proofs apply to every such block. Write $x_j = \log A_j$, $h = \log H$, $c_0 = \sum_j jx_j$ and $r_0 = \sum_j x_j$. Then, for $x_1 \ge x_2$,

$$h - 3(r_0 - \max(x_1, x_2)) = h - c_0 + (x_1 - x_2) + x_4 + 2x_5 + 3x_6;$$

for $x_2 \ge x_1$ replace $x_1 - x_2$ by $2(x_2 - x_1)$. All terms are nonnegative. Thus the middle monomial is saturated on

$$A_1 = A_2, \qquad A_1A_3 = H^{1/3}, \qquad A_4 = A_5 = A_6 = 1,$$

with full capacity. The balanced $A_3 = 1$ case is only an endpoint. The $a_3$ direction remains nonprincipal by its outside-$S$ valuation $3 \pmod 6$. If $R_0/M \ge H^{1/3}D^{-\eta}$, the total displayed slack is at most $3\eta \log D$. Therefore a fixed power gain must be uniform on this full neighboring region. The proved budget still costs $DR_0/M$; it gives no global improvement over the accepted envelope.

The independent standard-library diagnostic checks nine ideal moment cases, eight raw Gram cases with 27 000 phase values and 5 184 raw zeros, exact radical commutator/vector budgets, and 227 exponent cases. Finite tables use common norm profiles realizable by smooth disjoint bumps; they are not tests of an infinite analytic estimate. Full derivation, source bindings and preserved checks are under `planning/covariance-budget-20261105/attempt/` and `outputs/checks/2026-11-05-covariance-budget/attempt/`.

## S53 The fourth-moment conductor and its full raw zero set

Retain the original Eisenstein sextic symbols $x_m(u) = (u/m)_6$, fixed finite-order $\nu$, fixed $S$, and one possibly complex common smooth $W$ supported in $[1,2]$. Put $\lambda_m = \mu(m)\nu(m)W(Nm/D)$ and $r_m = r_U(m)$. Columns remain squarefree and prime to $S$. In $2D/7 \le U^2 \le D$, the actual polynomial is

$$\mathcal{C}_U(u) = \sum_{\substack{(a,b)=1,\ Na,Nb>U \\ D \le Na\,Nb \le 2D}} \mu(a)\mu(b)\nu(ab)W(Nab/D)x_a(u)x_b(u).$$

Its squared row energy has four factor indices $a, b, c, d$, with $(a, b) = (c, d) = 1$. These internal masks are essential. At each prime,

$$e_p = \mathbf{1}_{p|a} + \mathbf{1}_{p|b} - \mathbf{1}_{p|c} - \mathbf{1}_{p|d} \quad \text{belongs to } \{0, \pm 1\}.$$

Thus, for the two squarefree products $m = ab, n = cd$, the primitive oscillatory conductor outside fixed $S$ is $q = mn/(m, n)^2$. The full raw zero-preserving period is $L = mn/(m, n)$:

$$x_m(u)\overline{x_n(u)} = \mathbf{1}_{(u,(m,n))=1}\theta_{m/(m,n),\, n/(m,n)}(u).$$

No common-prime zero is restored by taking the primitive character.

A generic fourth moment $|\sum_m \lambda_m x_m(u)|^4$ has four individually squarefree columns $m_1, \dots, m_4$ with no such internal coprimality. Its signed local exponent $e_p = \sum_{i=1}^{2} \mathbf{1}_{p|m_i} - \sum_{i=3}^{4} \mathbf{1}_{p|m_i}$ can also be $\pm 2$, giving a nontrivial cubic component. For $L = \mathrm{rad}(m_1 m_2 m_3 m_4)$, the exact row-power interface is

$$Q_j = \prod_{\substack{p|L \\ je_p \not\equiv 0 \ (6)}} p, \qquad Z_j = L/Q_j, \qquad \prod_{i=1}^{2} x_{m_i}(v^j)\overline{\prod_{i=3}^{4} x_{m_i}(v^j)} = \theta_{j,e}(v)\mathbf{1}_{(v,Z_j)=1}.$$

Here $\theta_{j,e}$ is primitive on $Q_j$, with its own zero extension. At $j = 3$ an exponent $\pm 2$ becomes primitive-principal but its raw zero remains. For the original two-squarefree-product expansion, $Q_j = q$ for all $1 \le j \le 5$, including the quadratic $j = 3$ direction. For the accepted row blocks $Nr_j \asymp A_j$, put $R_0 = \prod_j A_j$, $P = \max(A_1, \dots, A_5)$ and retain $\prod_j A_j^j \le H$. The full $A_3$ saturation face and all row-factor masks remain.

The small-conductor pair count recovers an already controlled high-gcd sector; it is not a new analytic advance. On the remaining original sector $Nq > P^2$, favorable smooth completion costs $\min(A_j, \sqrt{Nq}) = A_j$ for $1 \le j \le 5$. The literal sharp cutoff is not replaced by this smooth trial. The signed large-conductor fourth-moment and mixed Möbius sums remain open; no improved full exponent, $BT_\sigma$, new Lean theorem, zero-location result or GRH conclusion follows.

## S54 Scope of the raw fourth-moment reduction

Write $c_{a,b} = \mu(a)\mu(b)\nu(ab)W(Nab/D)$ on the exact factor support in the core. Before absolute Cauchy, the normalized energy is

$$\frac{1}{D} \sum_{(a,b),(c,d) \text{ on that support}} c_{a,b}\overline{c_{c,d}} \sum_{u \in B} x_a(u)x_b(u)\overline{x_c(u)x_d(u)}.$$

Squarefreeness of $ab$ and $cd$ forces each positive and negative prime multiplicity to be at most one, proving $e_p \in \{0, \pm 1\}$. For the mixed term against the actual $\mu$ polynomial the exponent is $\mathbf{1}_{p|a} + \mathbf{1}_{p|b} - \mathbf{1}_{p|n}$, with the same conclusion and the same full union mask. The common possibly complex $W$, $\nu$, hyperbolic support and all raw zeros stay in these identities.

In contrast, unrestricted four-column memberships $(1, 1, 0, 0)$ give $e_p = 2$, a cubic component; $(1, 1, 1, 0)$ give $e_p = 1$ although both product supports contain $p$. Neither the parity symmetric difference of four supports nor the symmetric difference of two nonsquarefree product supports gives the general conductor. For the interface in the core, its local character order is

| $j$ | $1, 5$ | $2, 4$ | $3$ | $6$ |
|---|---|---|---|---|
| $e_p = \pm 1$ | 6 | 3 | 2 | 1 |
| $e_p = \pm 2$ | 3 | 3 | 1 | 1 |

Nontrivial entries have tame conductor exponent one; trivial entries remain deleted at $p \mid v$. Multiplying the interface over all six row factors, with the fixed unit/$S$ phase, preserves every original zero, including those at sixth-power factors. All inter-factor, squarefree, ray and capacity restrictions remain external restrictions on the same row sum; this identity asserts no independence.

At a residue field of size $Q$, let $\chi$ have order six and $\rho$ be nontrivial additive. The zero-extended complete sum $G_k(t) = \sum_{z \neq 0} \chi(z)^k \rho(tz)$ satisfies

| | $t = 0$ | $t \neq 0$ |
|---|---|---|
| $k \not\equiv 0$ (6) | 0 | $\lvert G_k(t)\rvert = \sqrt{Q}$ |
| $k \equiv 0$ (6) | $Q - 1$ | $-1$. |

For $t \neq 0$, substitution gives $G_k(t) = \overline{\chi(t)^k} G_k(1)$; for nontrivial $k$, expanding the norm square gives $(Q-1) - \sum_{a \neq 0,1} \chi(a)^k = Q$. The proof applies at split and inert primes. By CRT the full period-$L$ transform vanishes if any local frequency on $Q_j$ is zero; otherwise its magnitude is

$$(NQ_j)^{1/2} \prod_{p \mid Z_j} \left| -1 + Np\, \mathbf{1}_{\text{local frequency at } p \text{ is } 0} \right|.$$

Canceled common primes thus supply Ramanujan factors.

Exact deletion, with all remaining restrictions kept in $F$, gives

$$\sum_r \theta(r) F(r) \mathbf{1}_{(r,Z)=1} = \sum_{d \mid Z} \mu(d) \theta(d) \sum_z \theta(z) F(dz).$$

For a squarefree pivot let $E$ be the radical of its fixed excluded primes, including $Z$ and the relevant other row factors, with conductor primes omitted since $\theta$ already vanishes there. The additional squarefree identity is

$$\mu(r)^2 \mathbf{1}_{(r,E)=1} = \sum_{\substack{h^2 \mid r \\ (h,E)=1}} \mu(h) \sum_{\substack{d \mid E \\ d \mid r}} \mu(d).$$

Since $(d,h) = 1$, its lifted row sum has terms $\mu(d)\mu(h)\theta(d)\theta(h)^2 \sum_z \theta(z) F(dh^2 z)$. The scaled $F$ retains the exact window, primary/ray restrictions and other masks. Divisor triangles cost at most $(HD)^\epsilon$; they do not restore a forbidden nonzero.

The previous fixed-radial-profile high-gcd proof (`research/short-row-high-gcd.md`, independently reviewed in `planning/reviews/book/short-row-formal.md`) already bounds its off-diagonal tail by $O_{\Phi,\epsilon}(D^{2+\epsilon}/G_0^2)$ for $Nh \geq G_0$. For annular $m = hr, n = ht$, $Nq \leq Y$ implies $Nh \geq D/\sqrt{Y}$, so the old smooth-scope bound is $O_{\Phi,\epsilon}(D^\epsilon Y)$. The present elementary count $O(D\sqrt{Y})$, factor multiplicities $r_U(m) r_U(n) \ll_\epsilon D^\epsilon$, and $\#B_A = O(R_0)$ give only $O(\|W\|_\infty^2 D^\epsilon R_0 \sqrt{Y})$ on a literal sharp block. This is a weaker conductor-form recovery in its own bookkeeping scope, not a new high-gcd theorem; the smooth estimate is not silently transferred to a selected sharp off-diagonal piece. The full $A_3$ face and neighborhood slack are also inherited.

Finally, primitive smooth Poisson completion has bound $O_V(\min(A_j, \sqrt{Nq}))$ on the original support. For $Nq > P^2$ and $1 \leq j \leq 5$, every scaled length $A_j/(Nd\,Nh^2)$ is below $\sqrt{Nq}$, so even this favorable trial takes its trivial length term. Its constant uses fixed smooth seminorms and need not stay bounded when a sharp boundary is approached. In the admissible subsector $(m,n) = 1$, $Nq = Nm\,Nn \asymp D^2$. A complete zero mean does not control the nonzero frequencies of the short original masked row window. The unresolved target is their joint signed sum with weights $r_U(m) r_U(n) \lambda_m \overline{\lambda_n}$ and $r_U(m) \lambda_m \overline{\lambda_n}$, before per-conductor absolute estimates. No new global analytic conclusion is claimed.

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